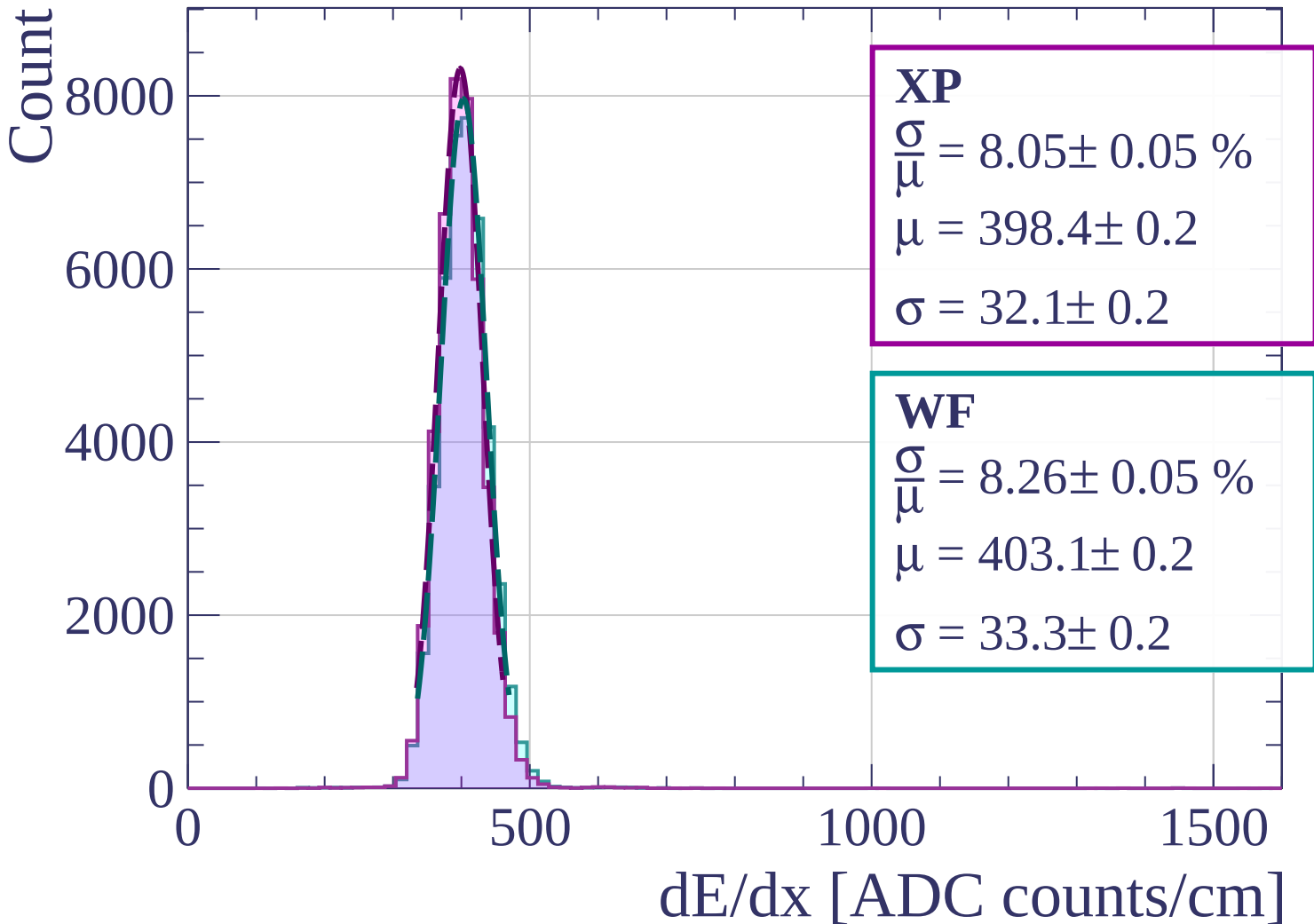
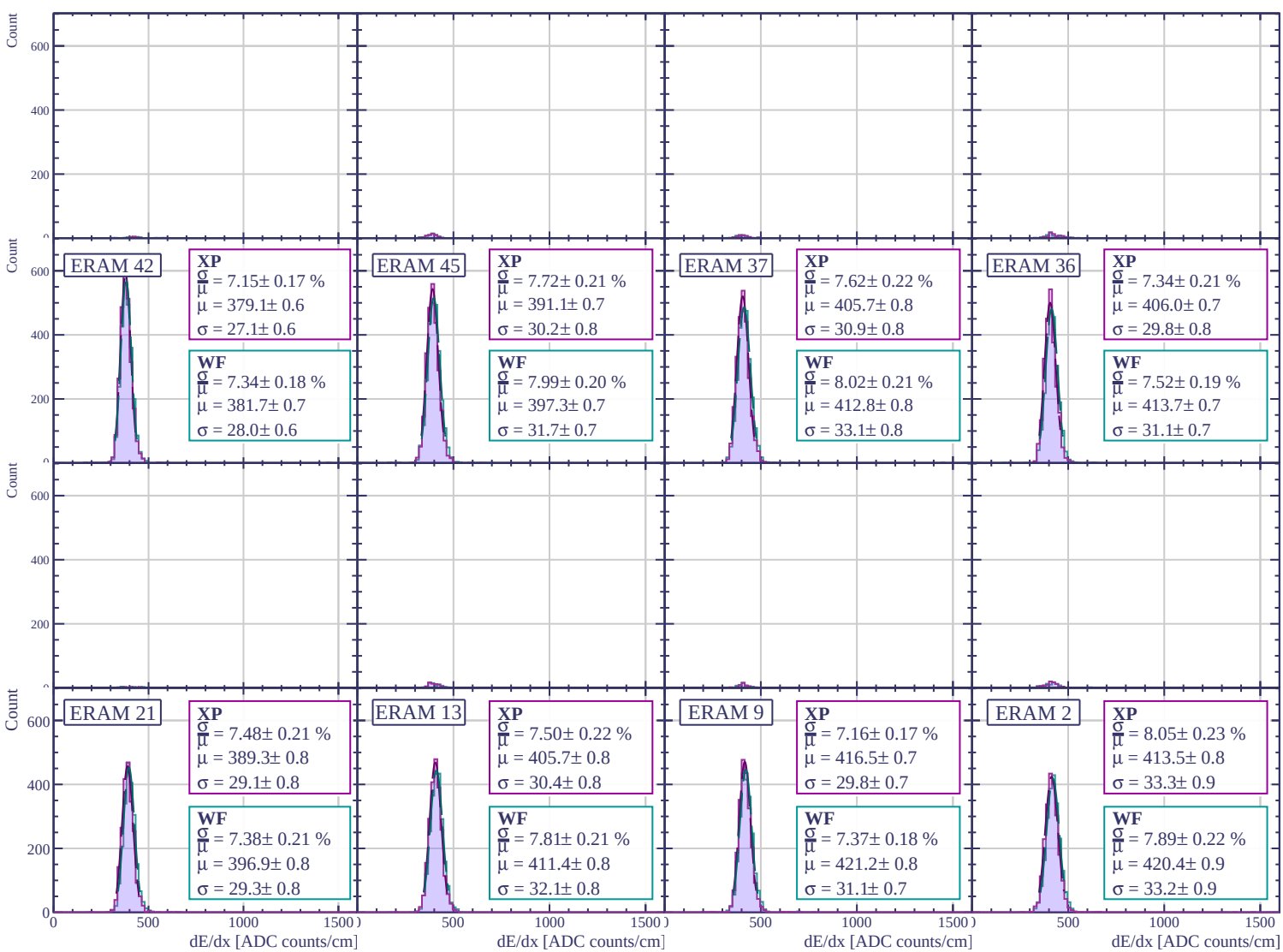
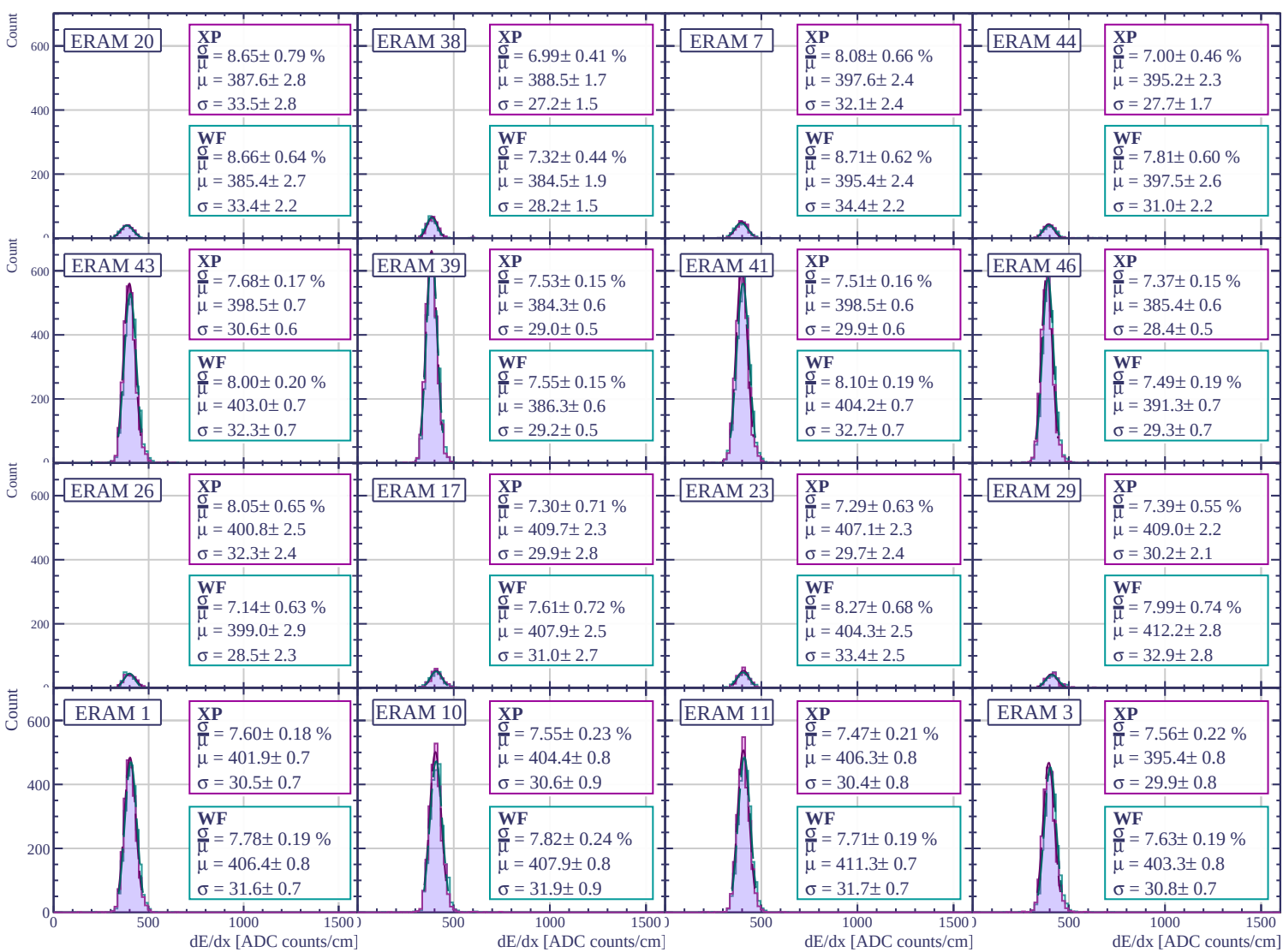
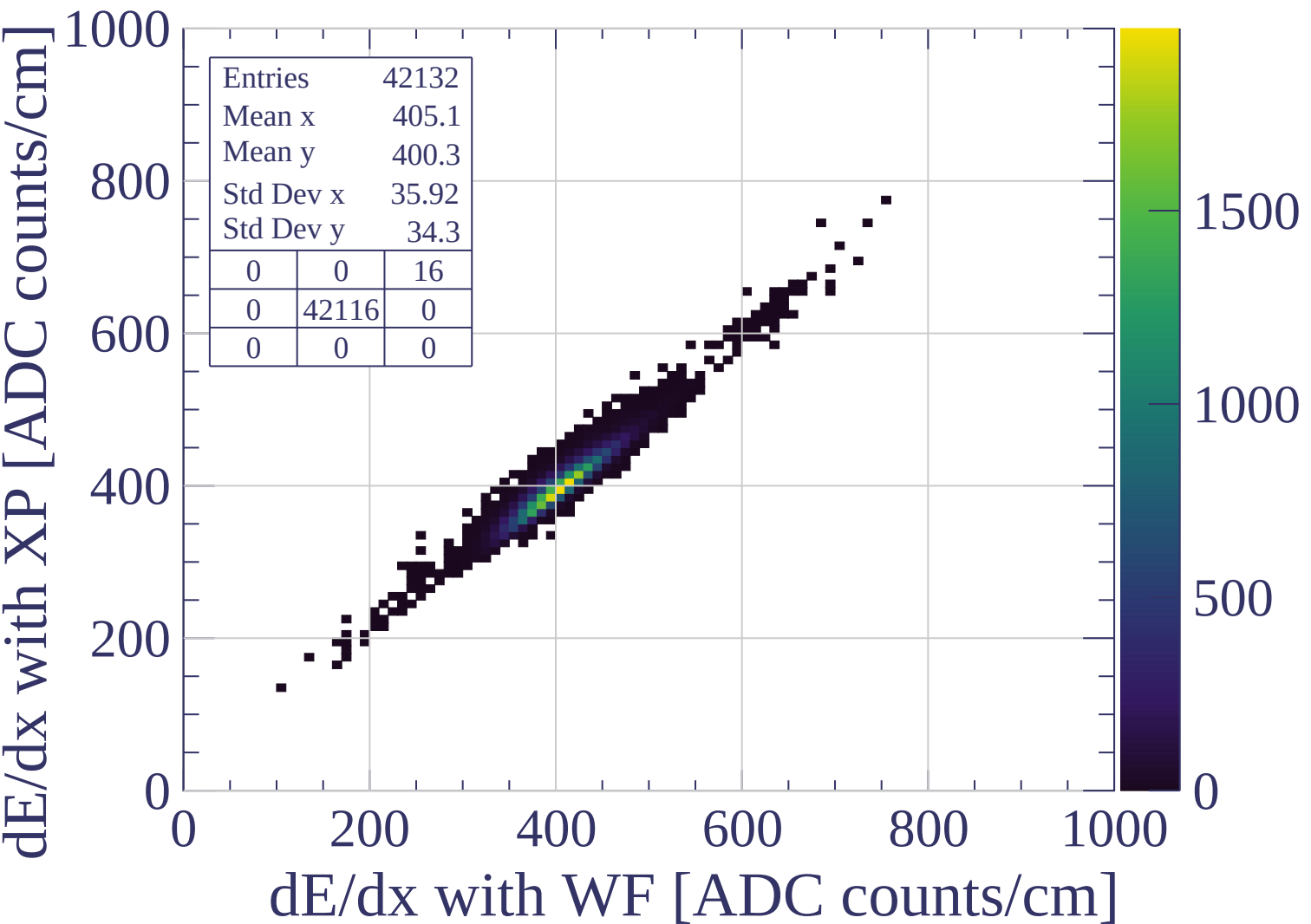
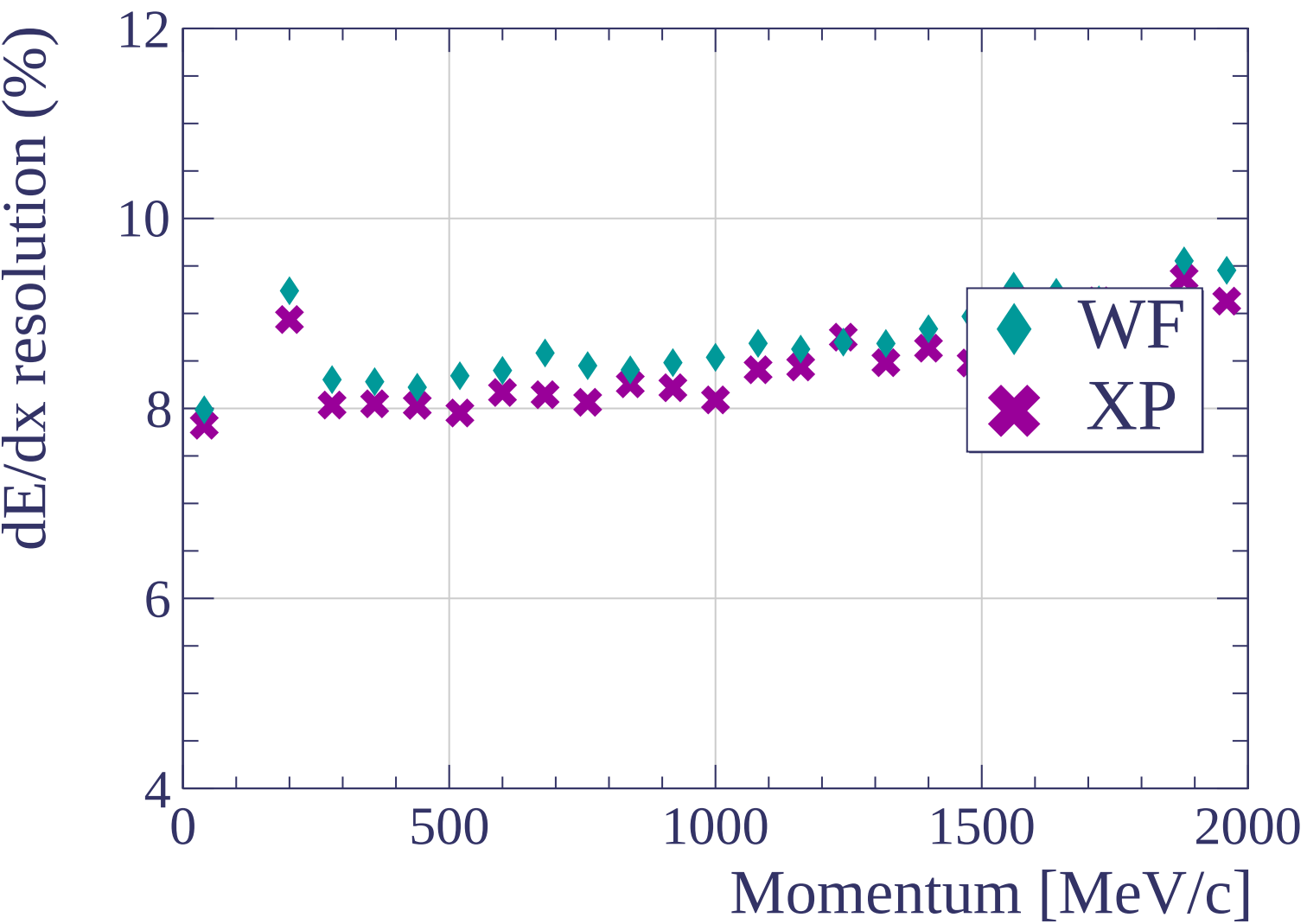


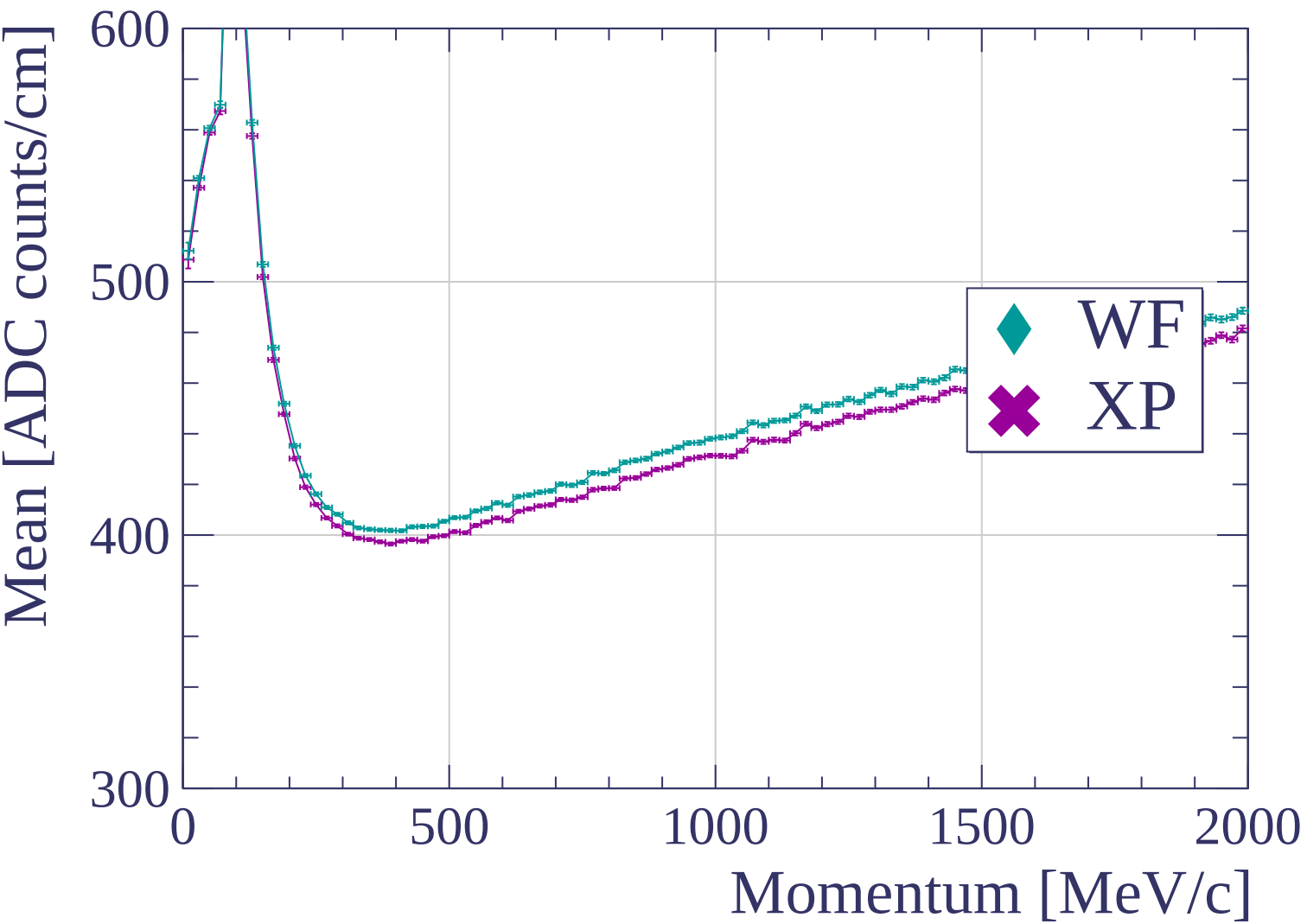


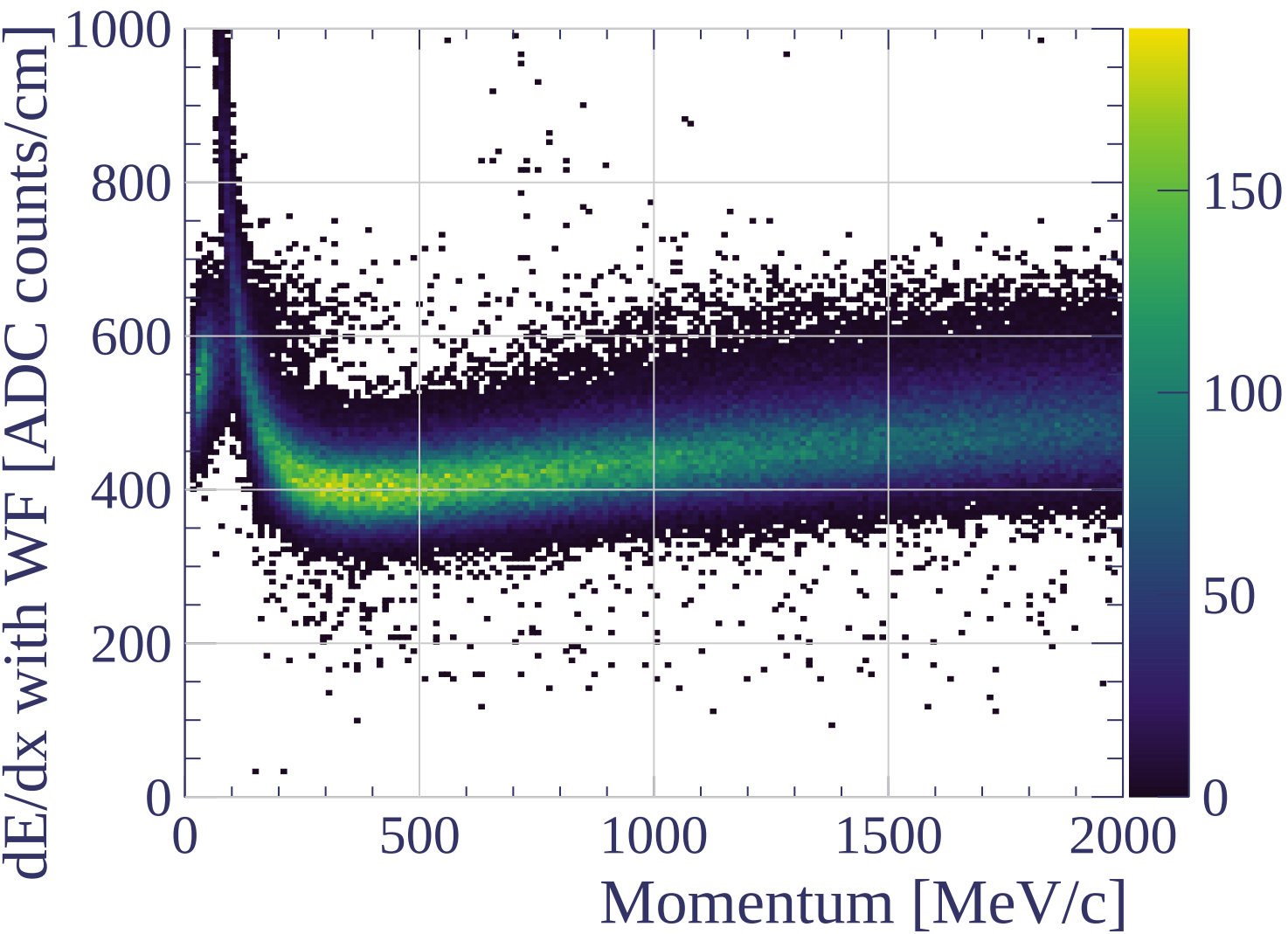


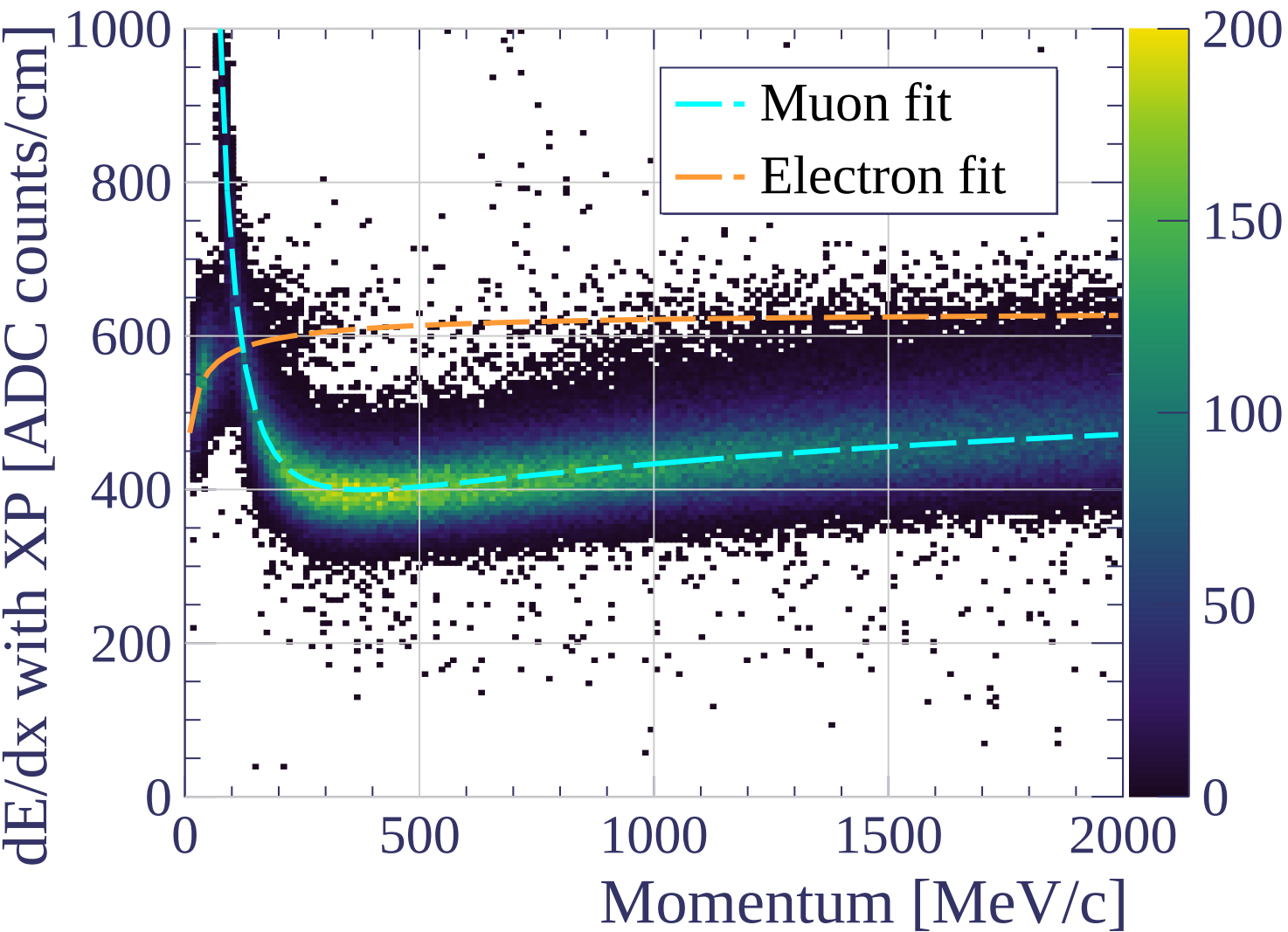




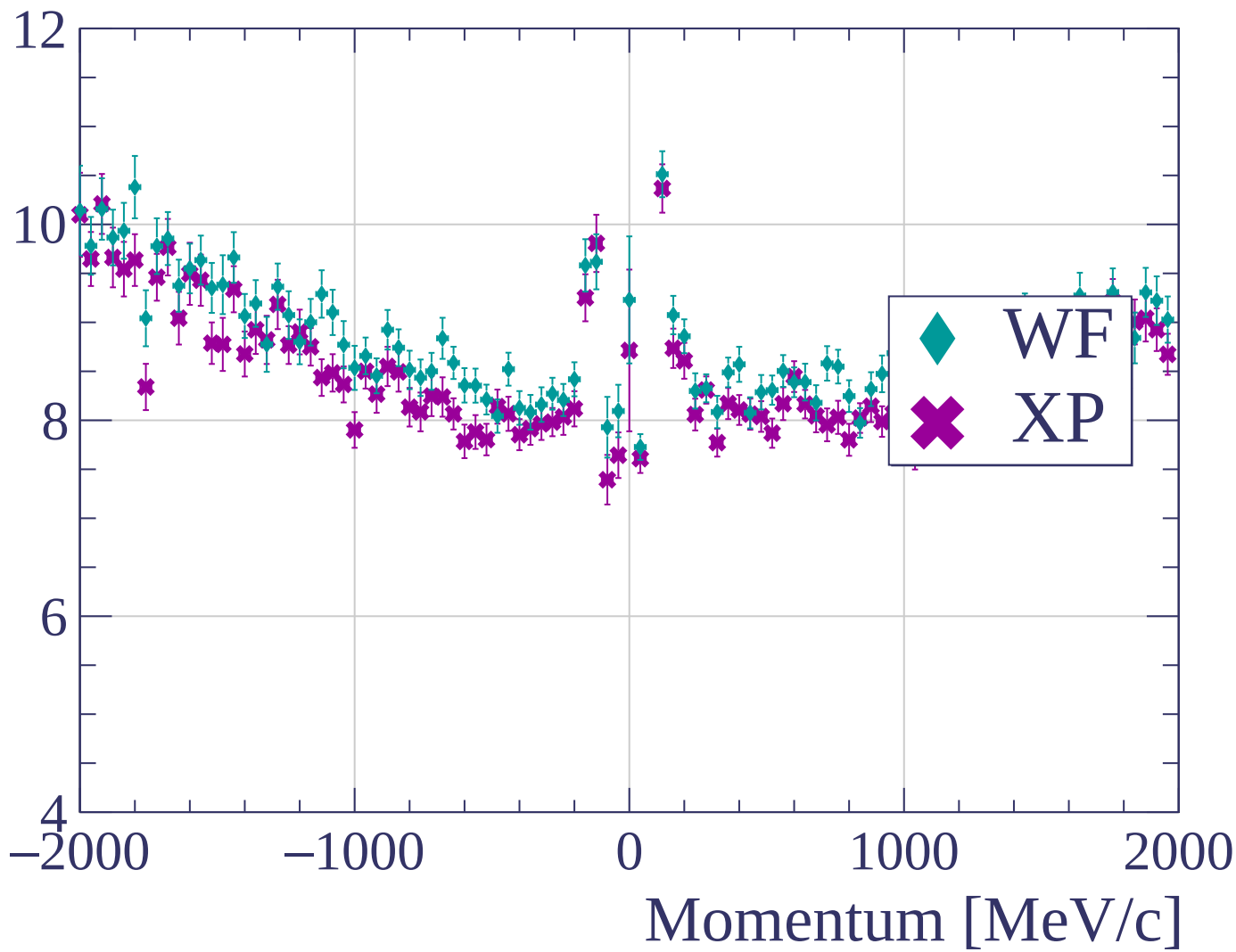


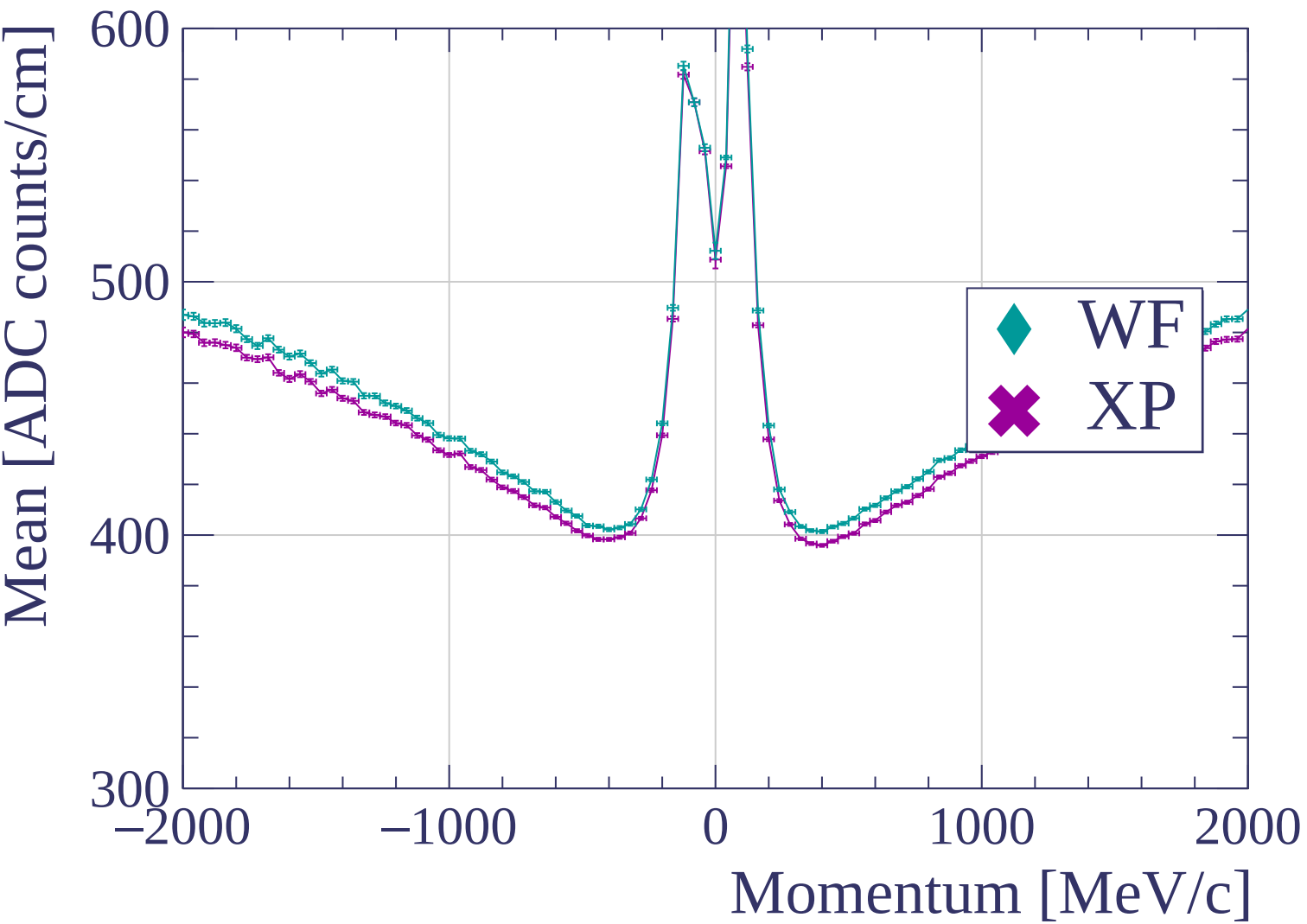


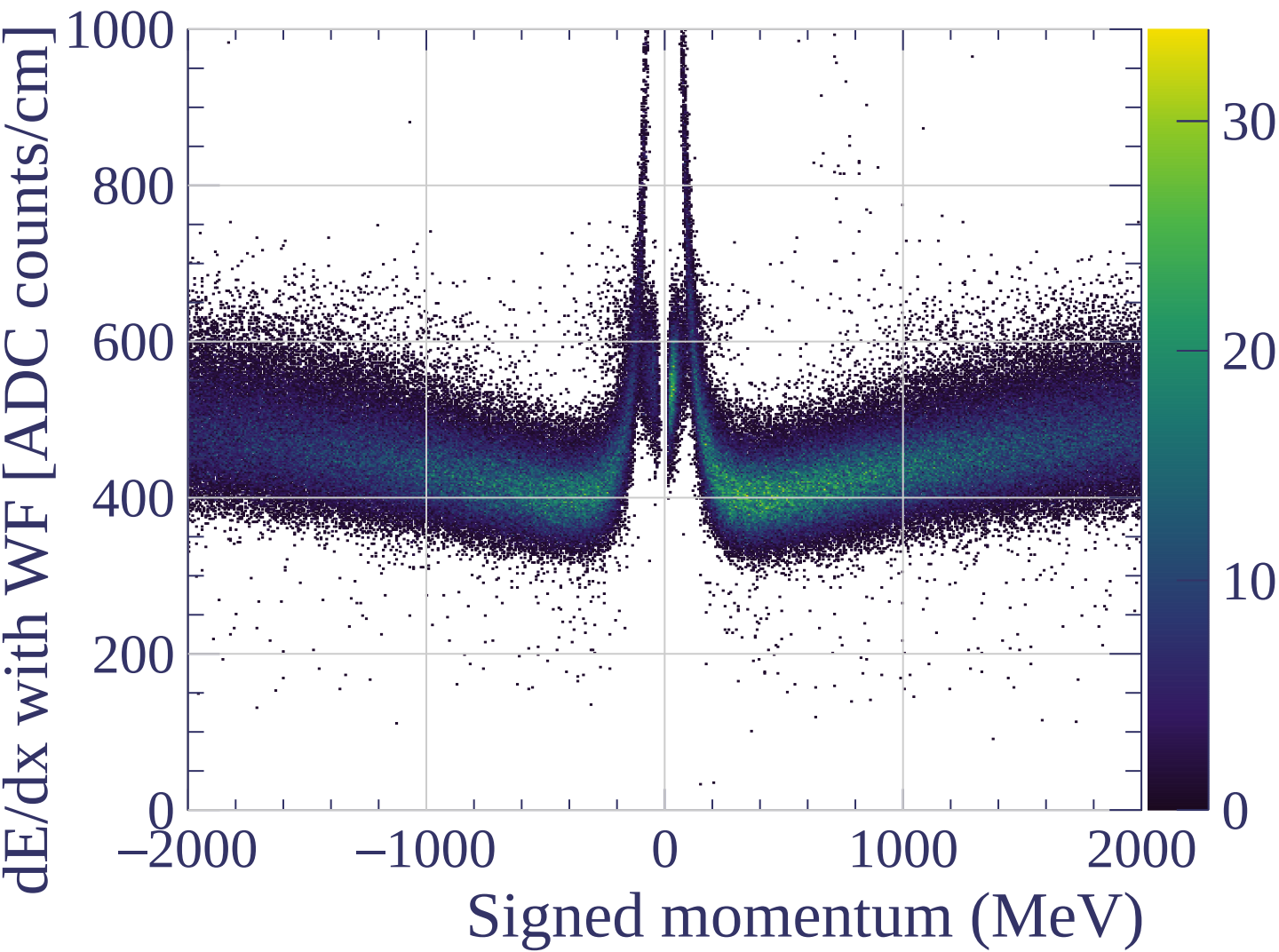


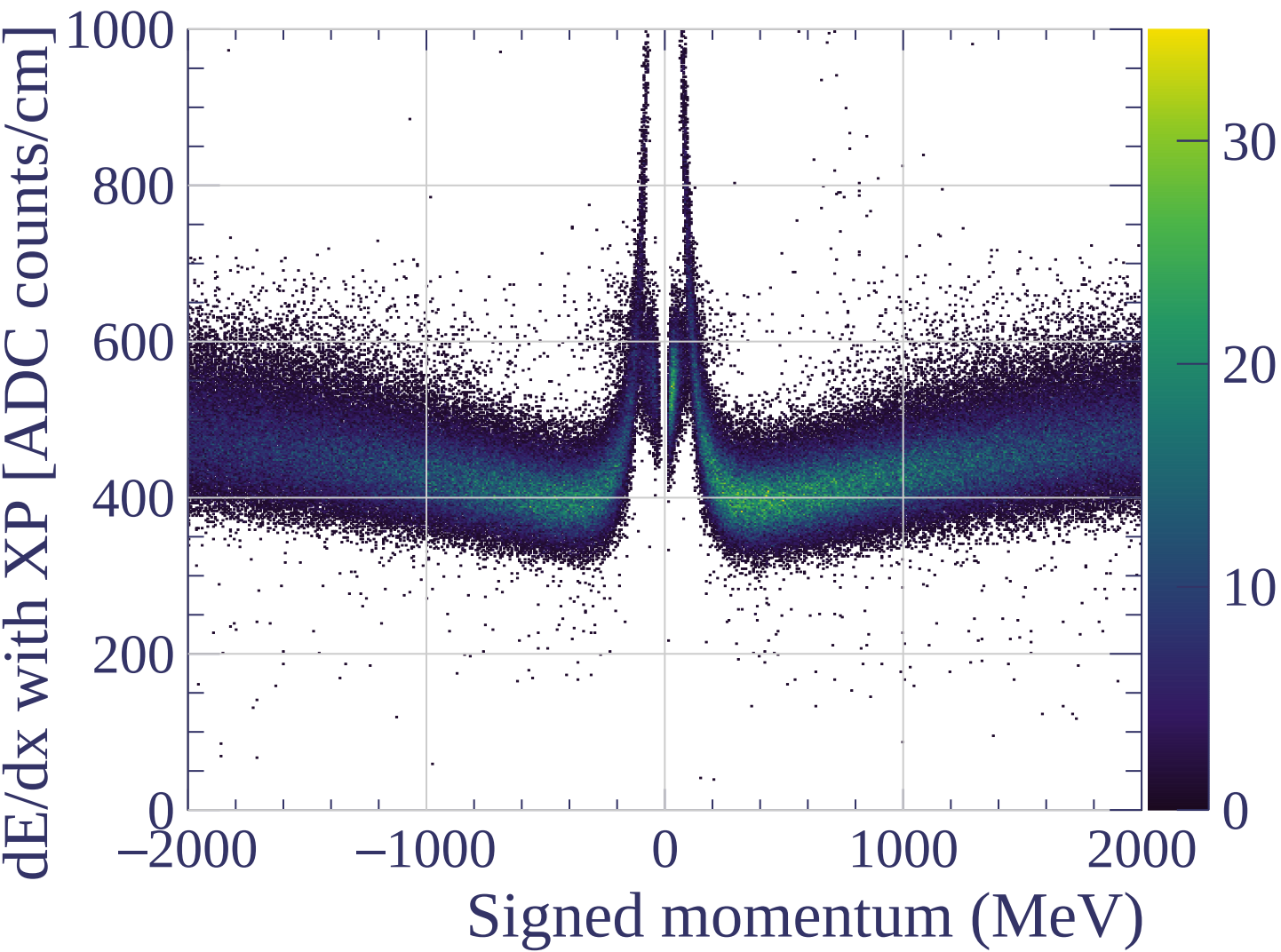


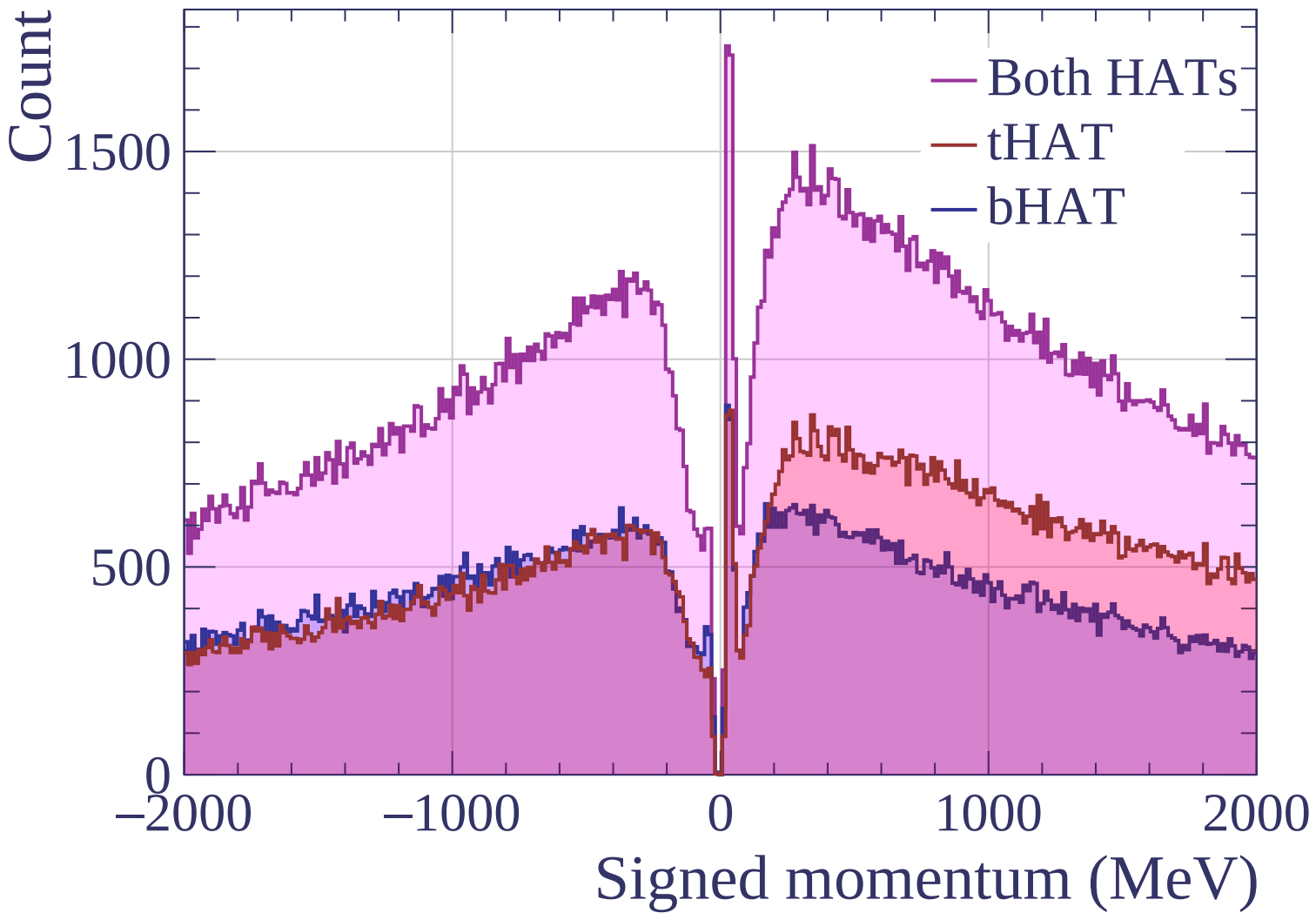
dE/dx resolution (%)

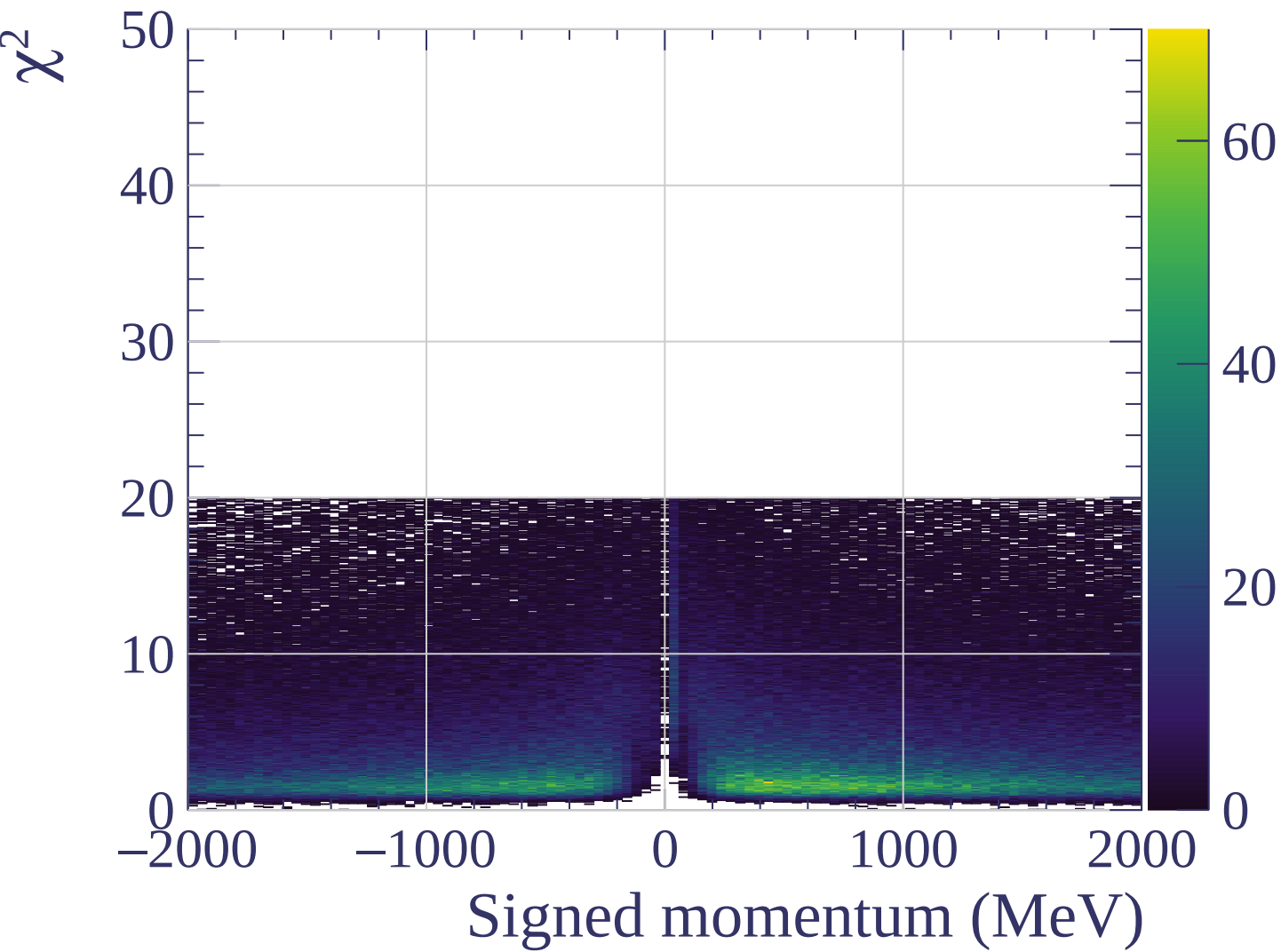


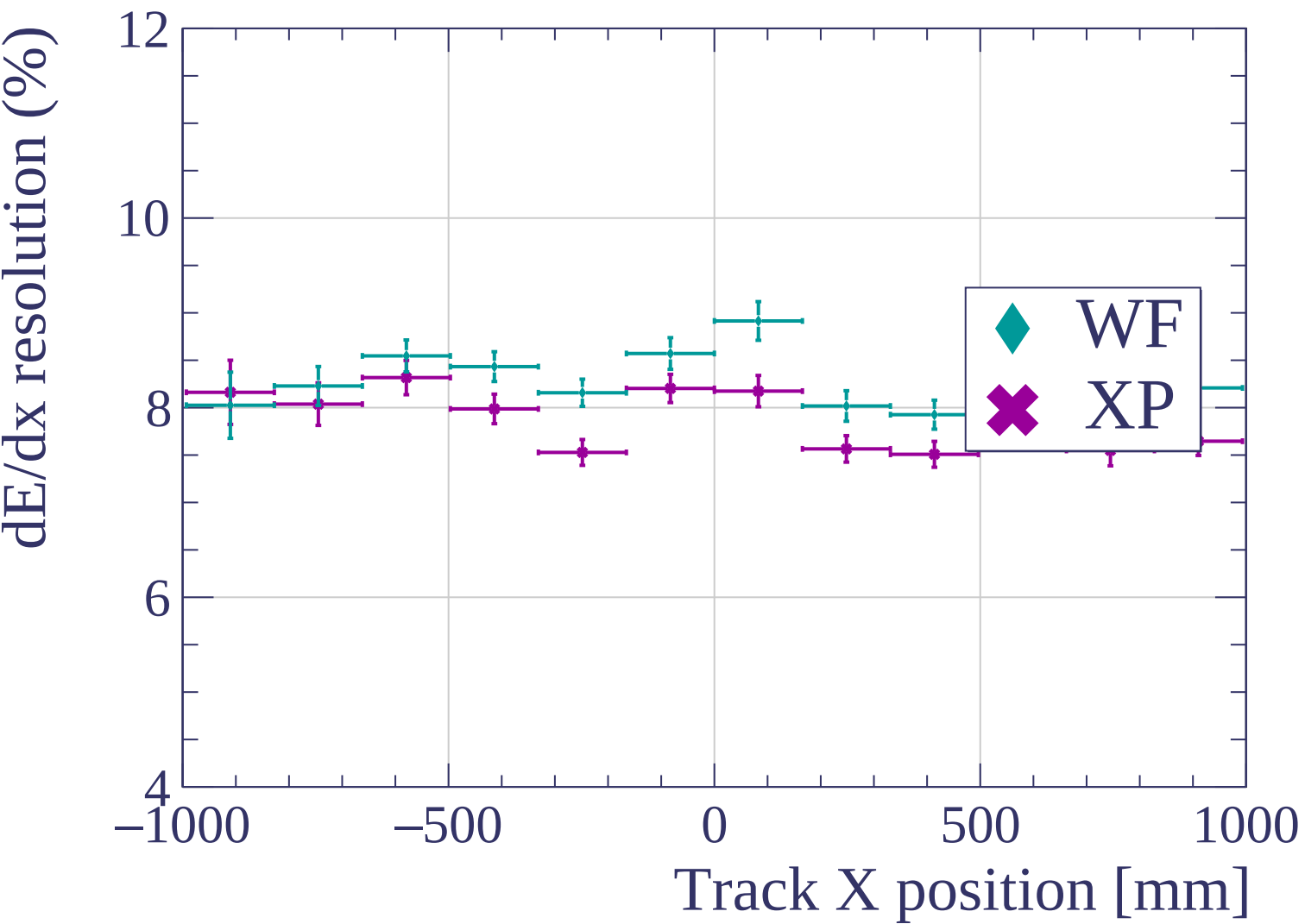


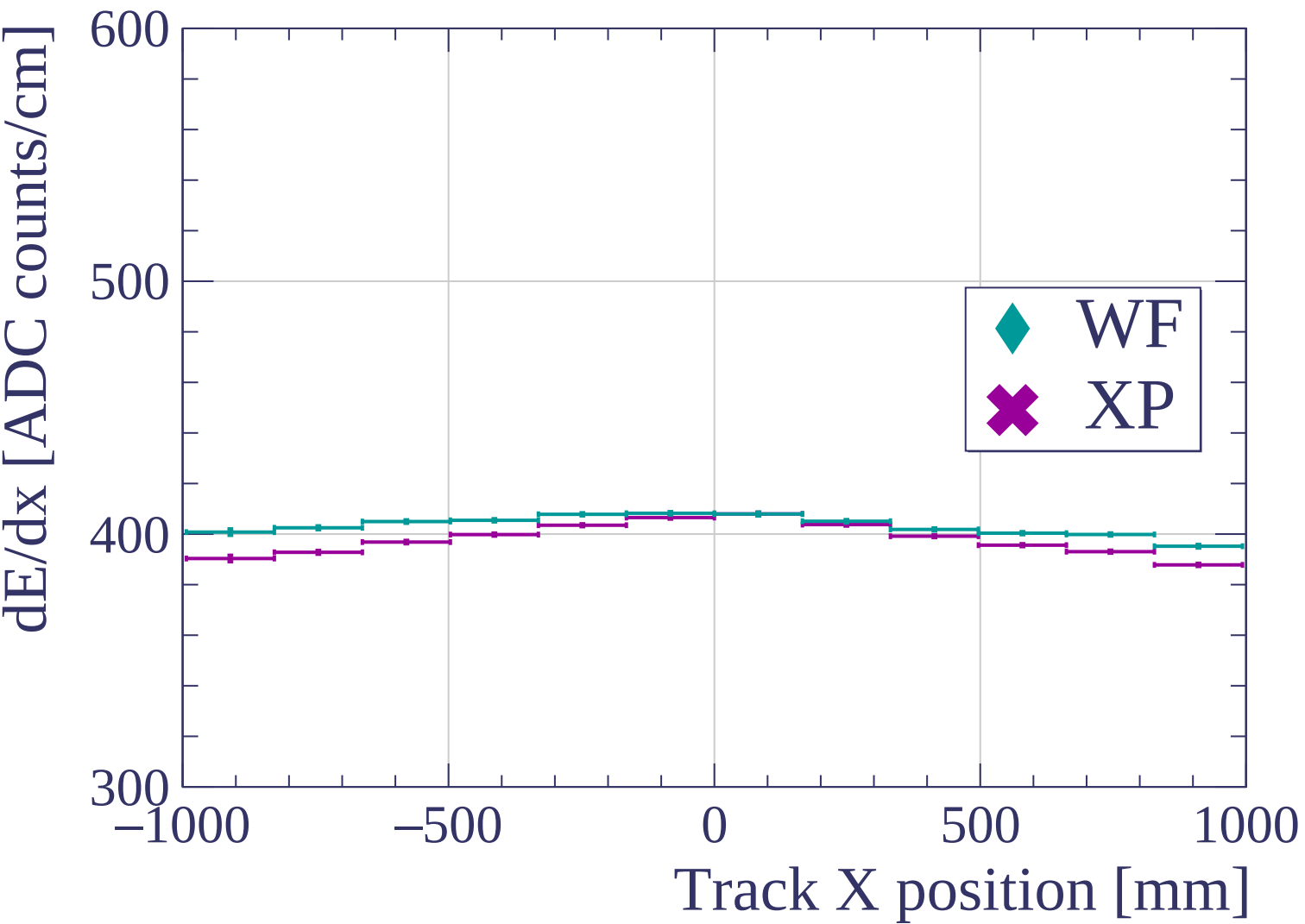


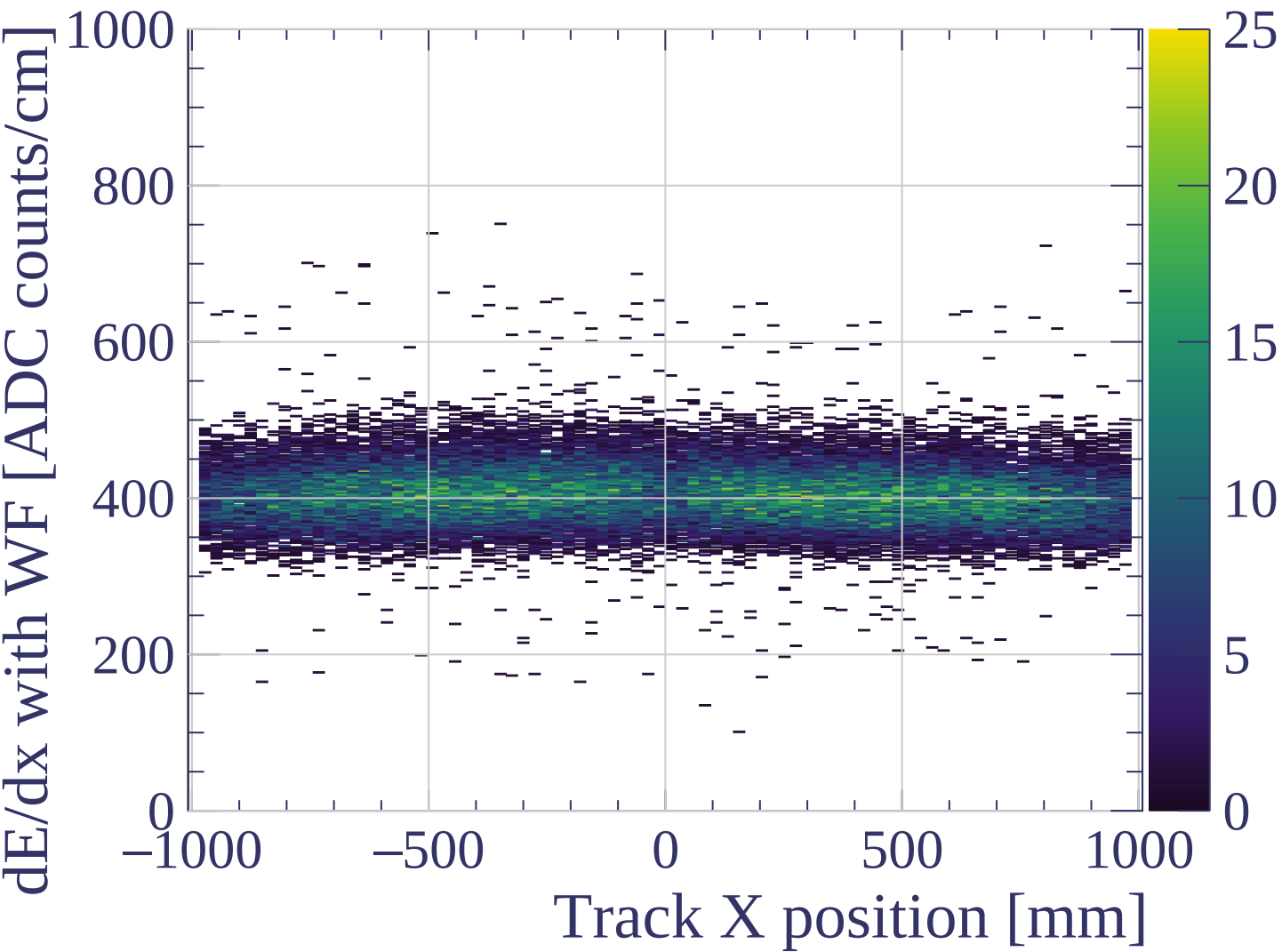


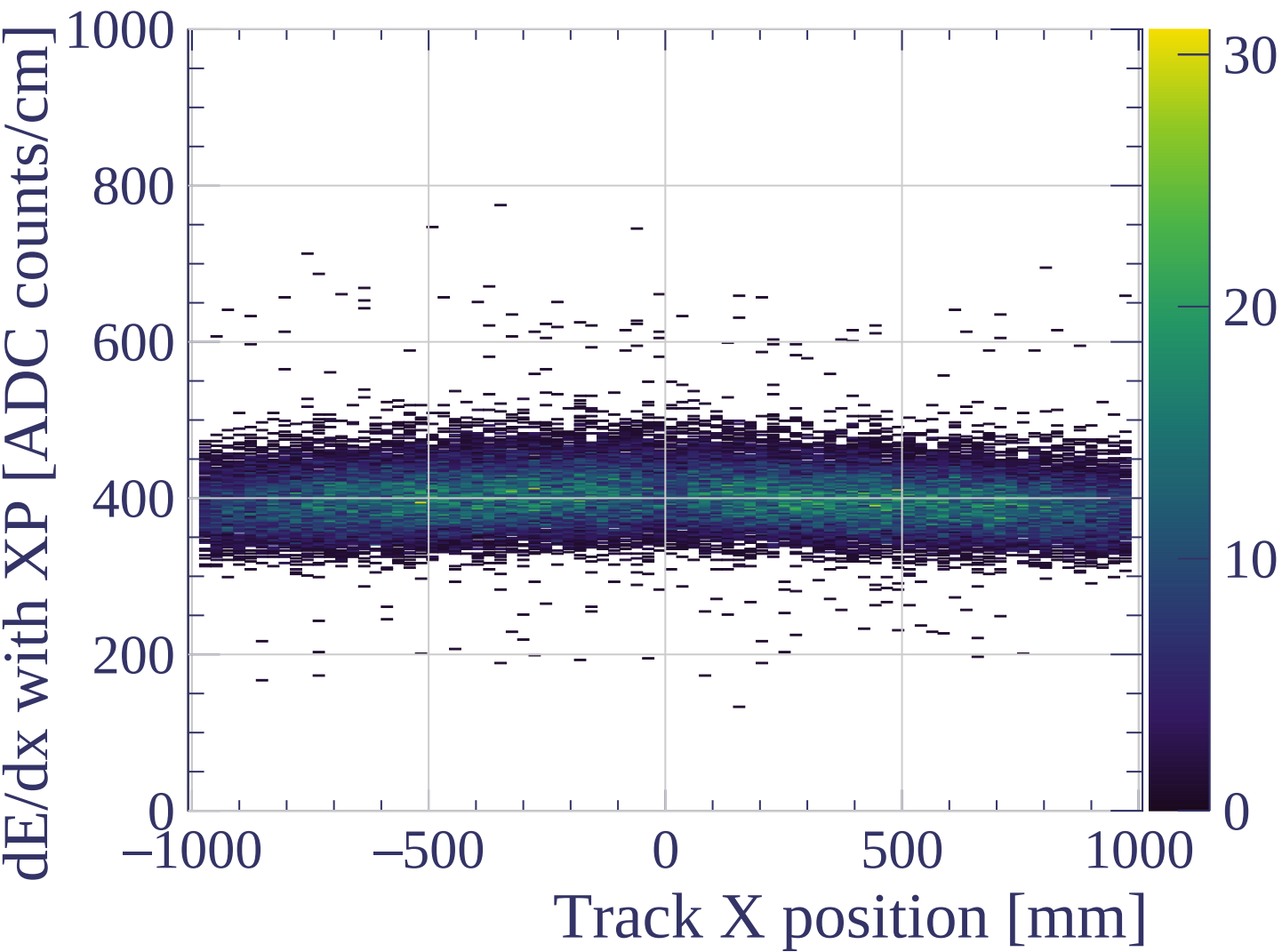


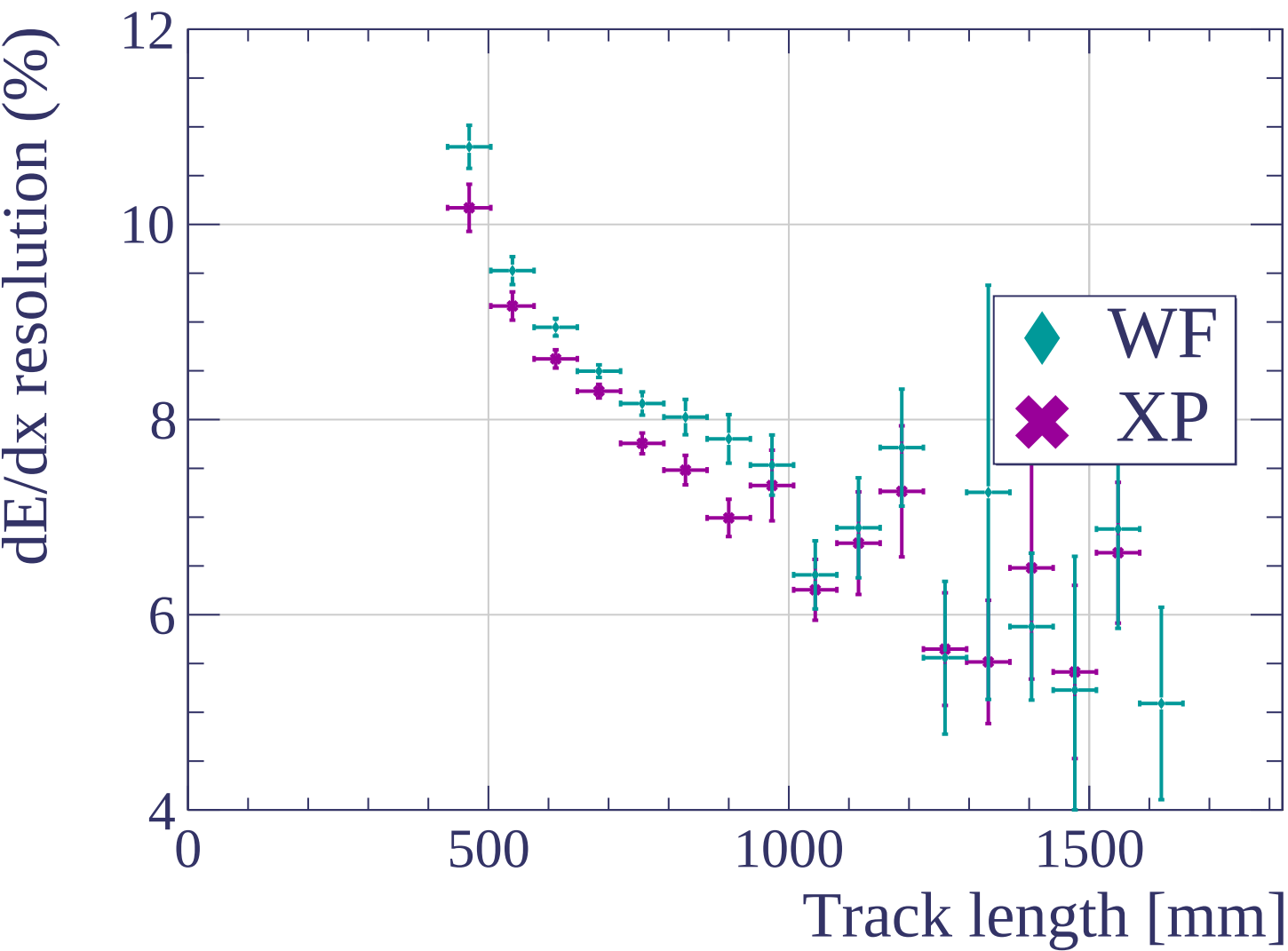


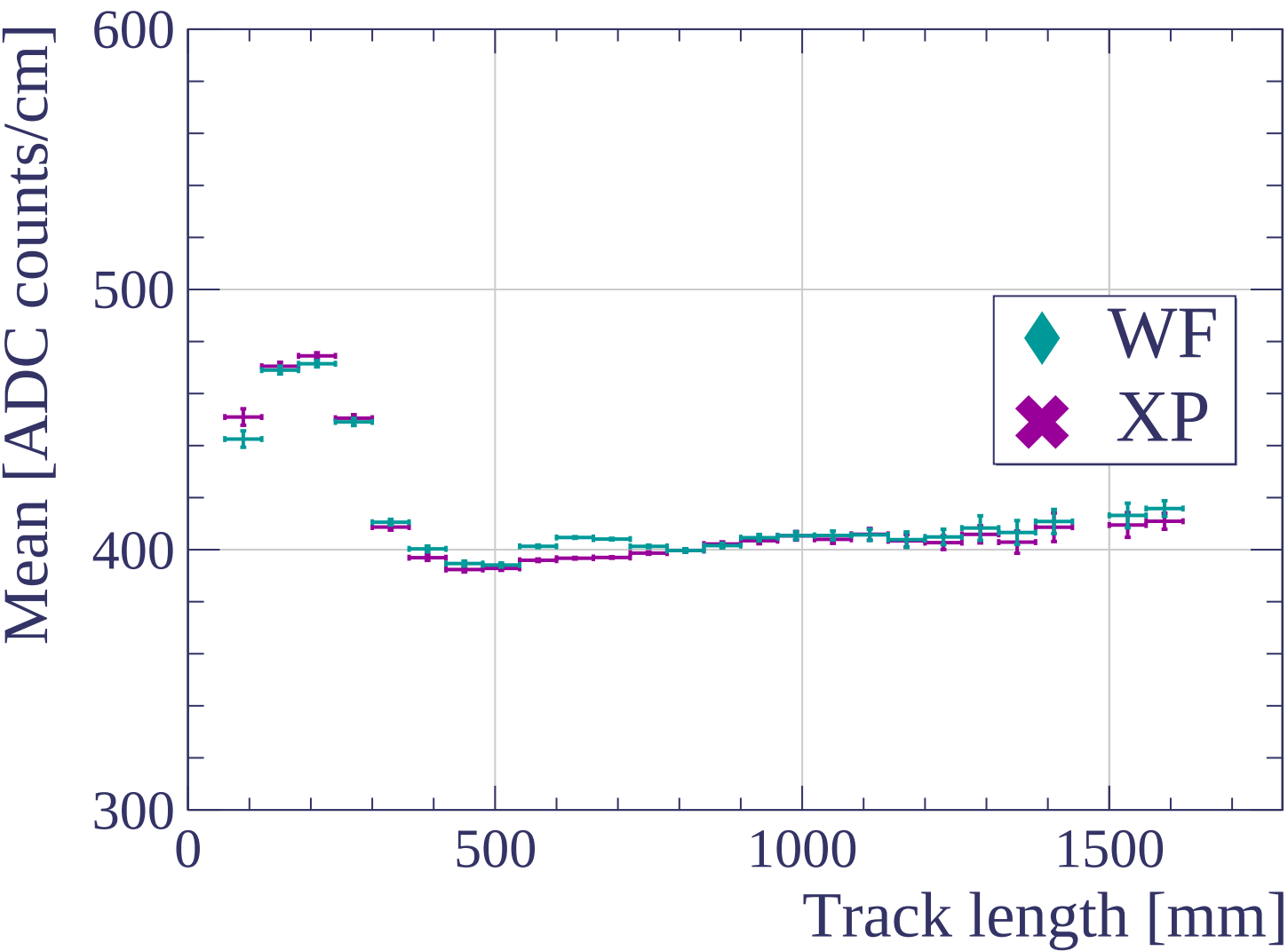


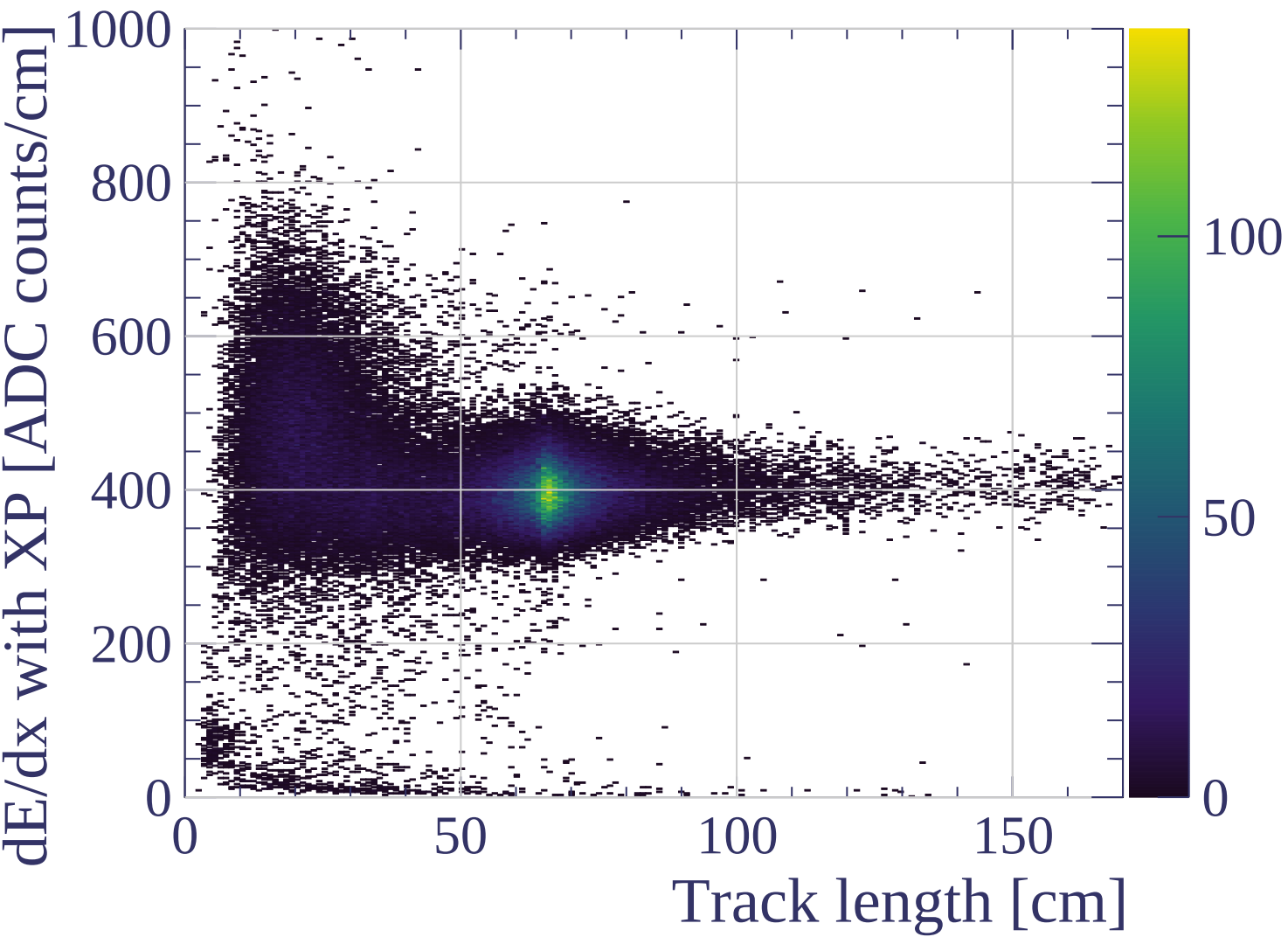


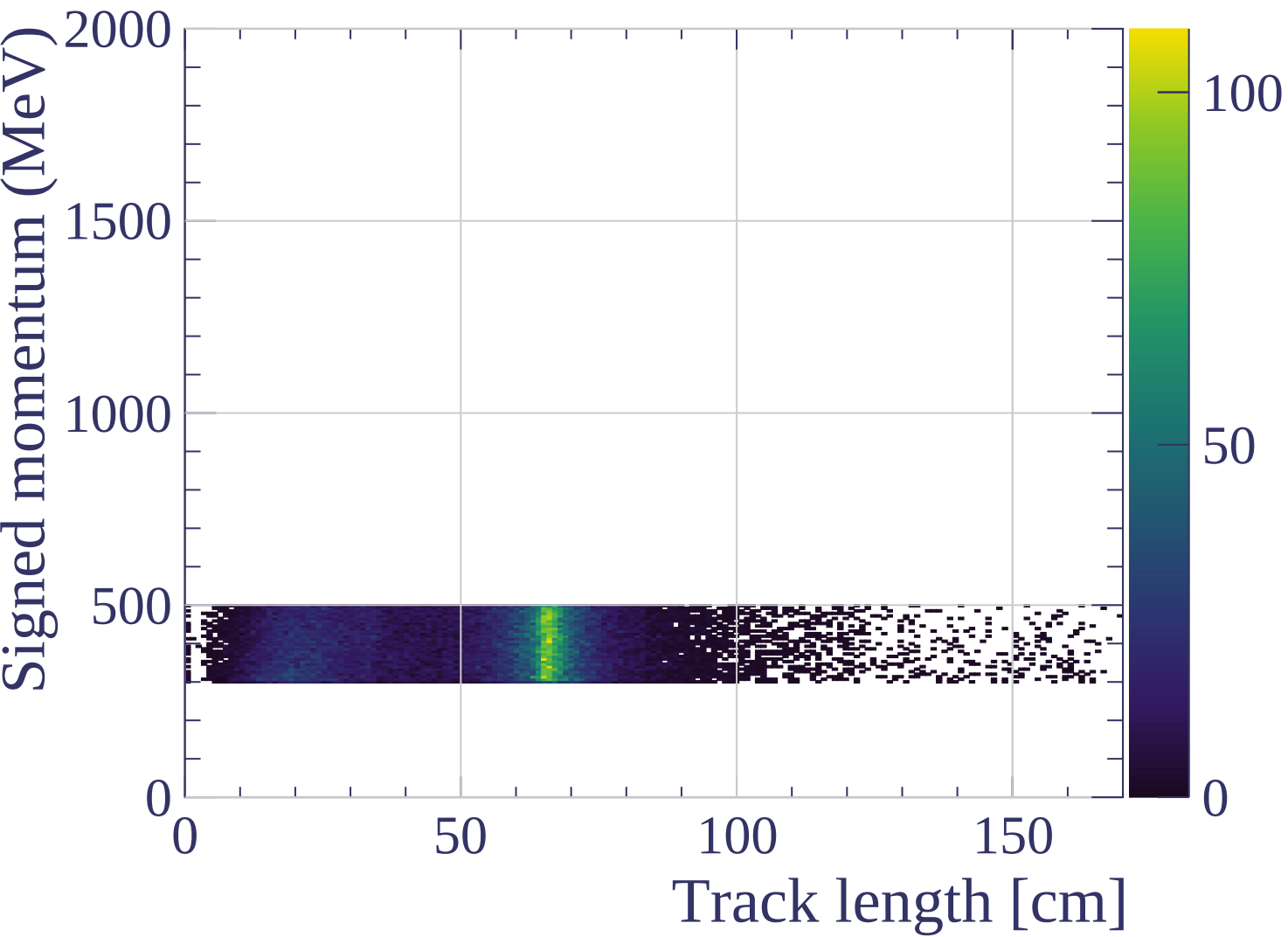


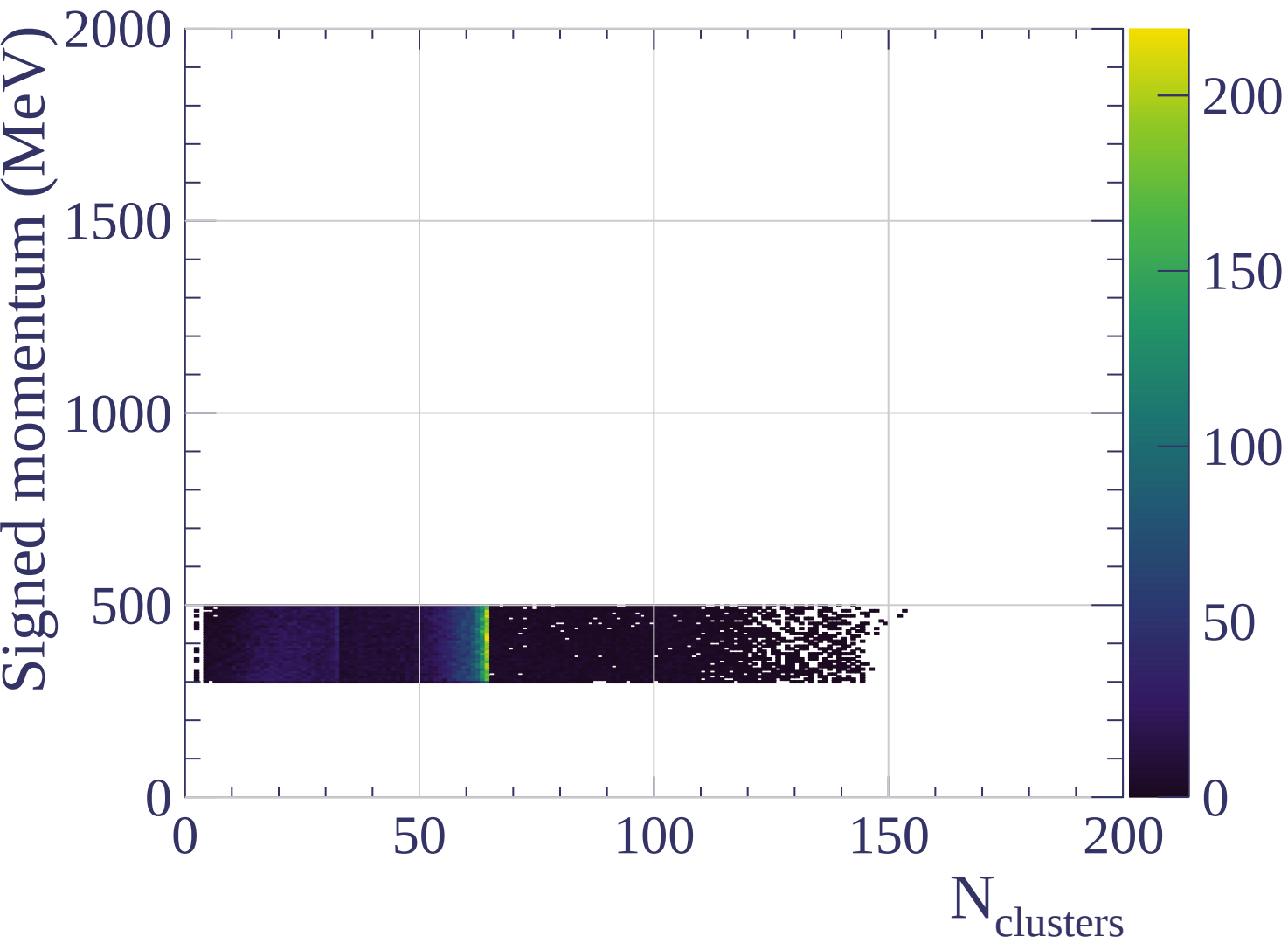


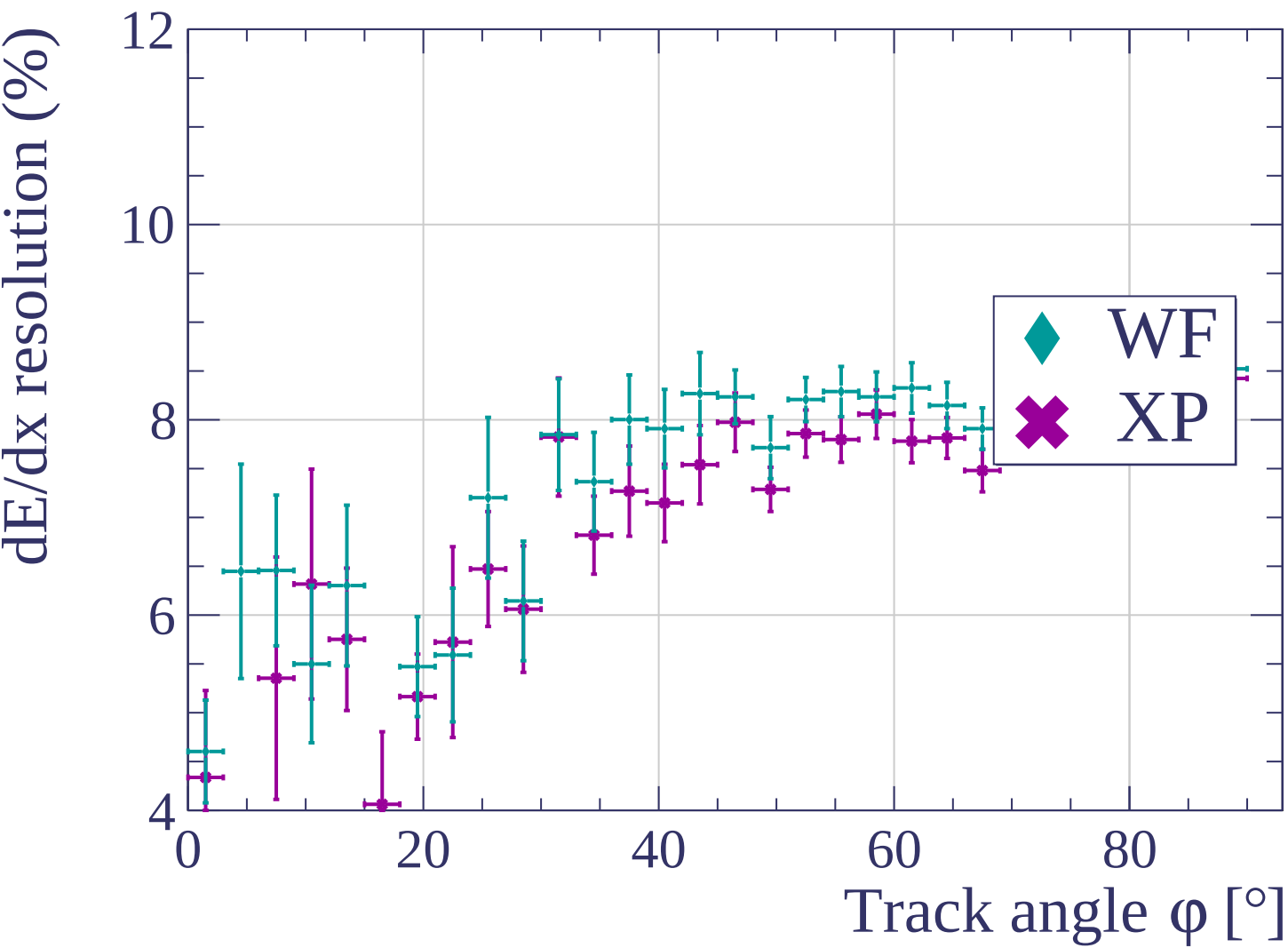


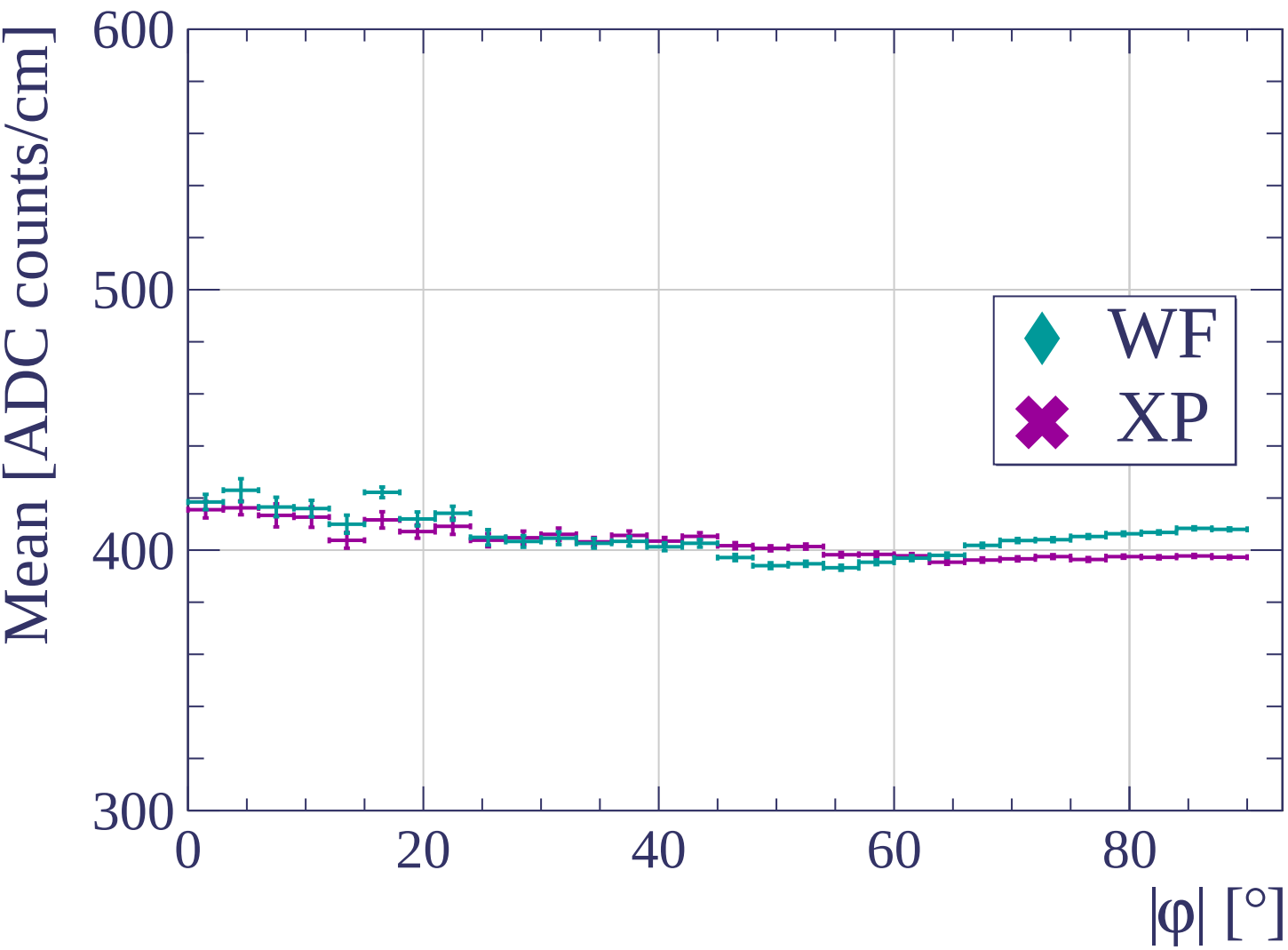


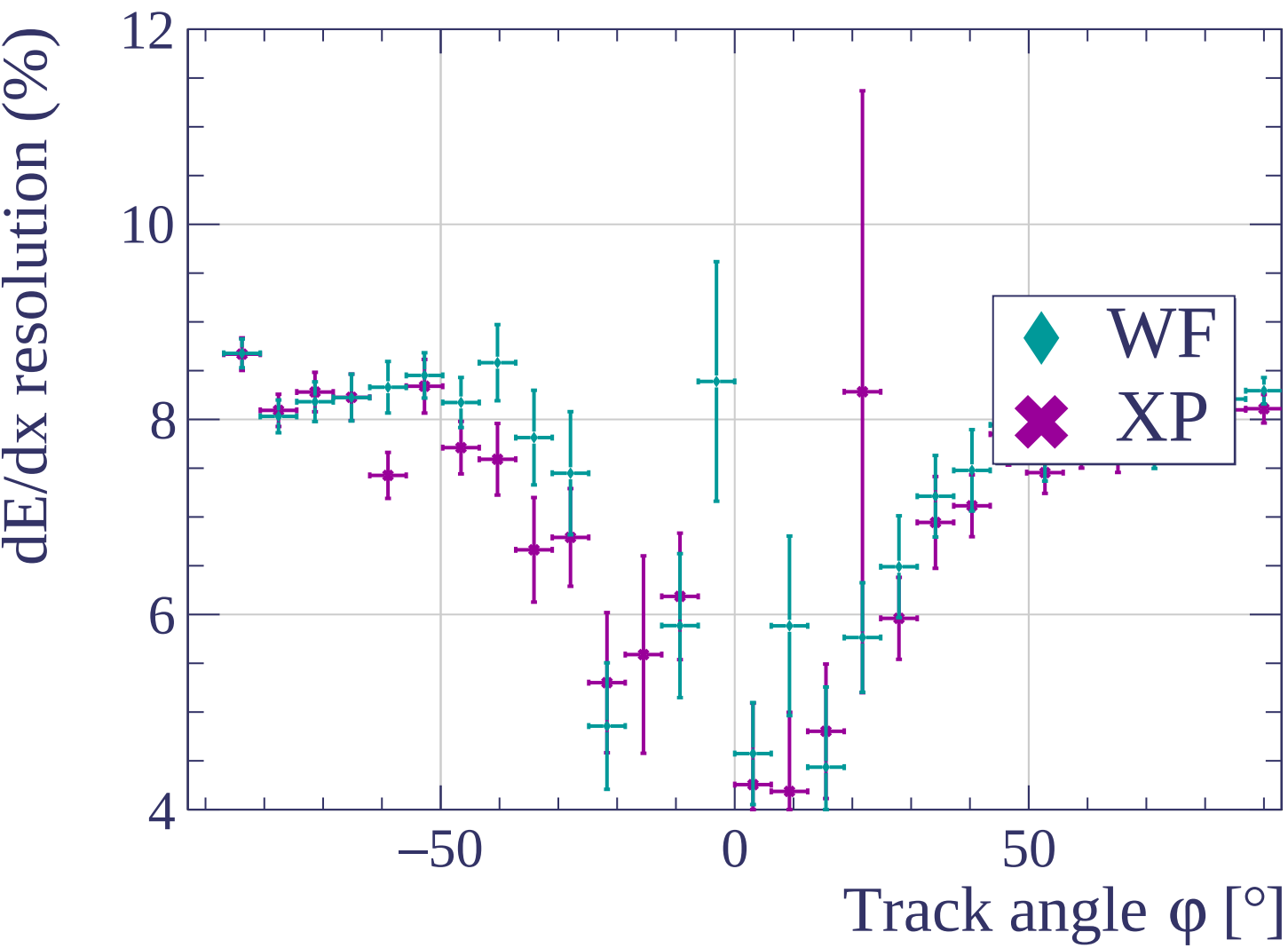


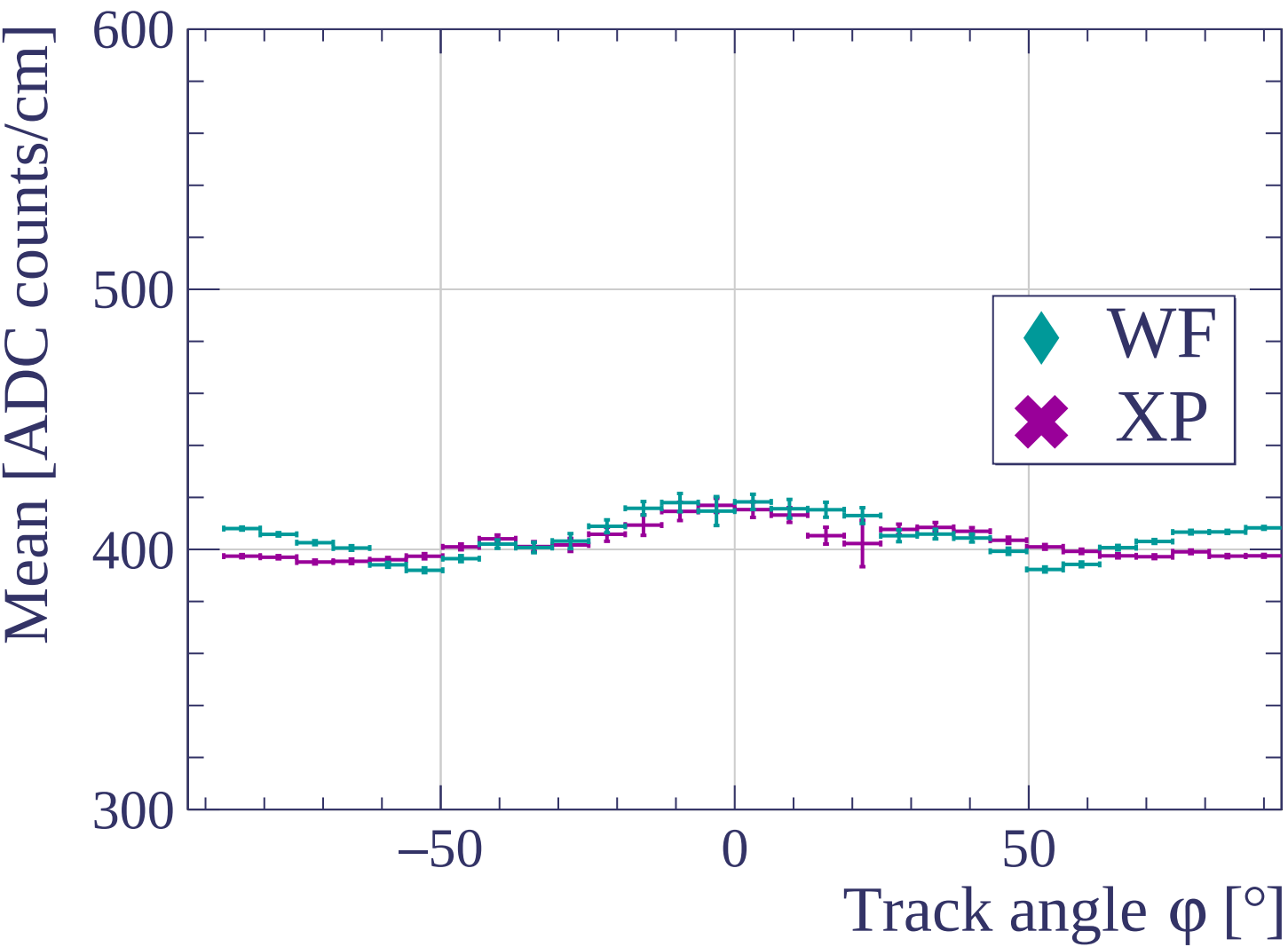


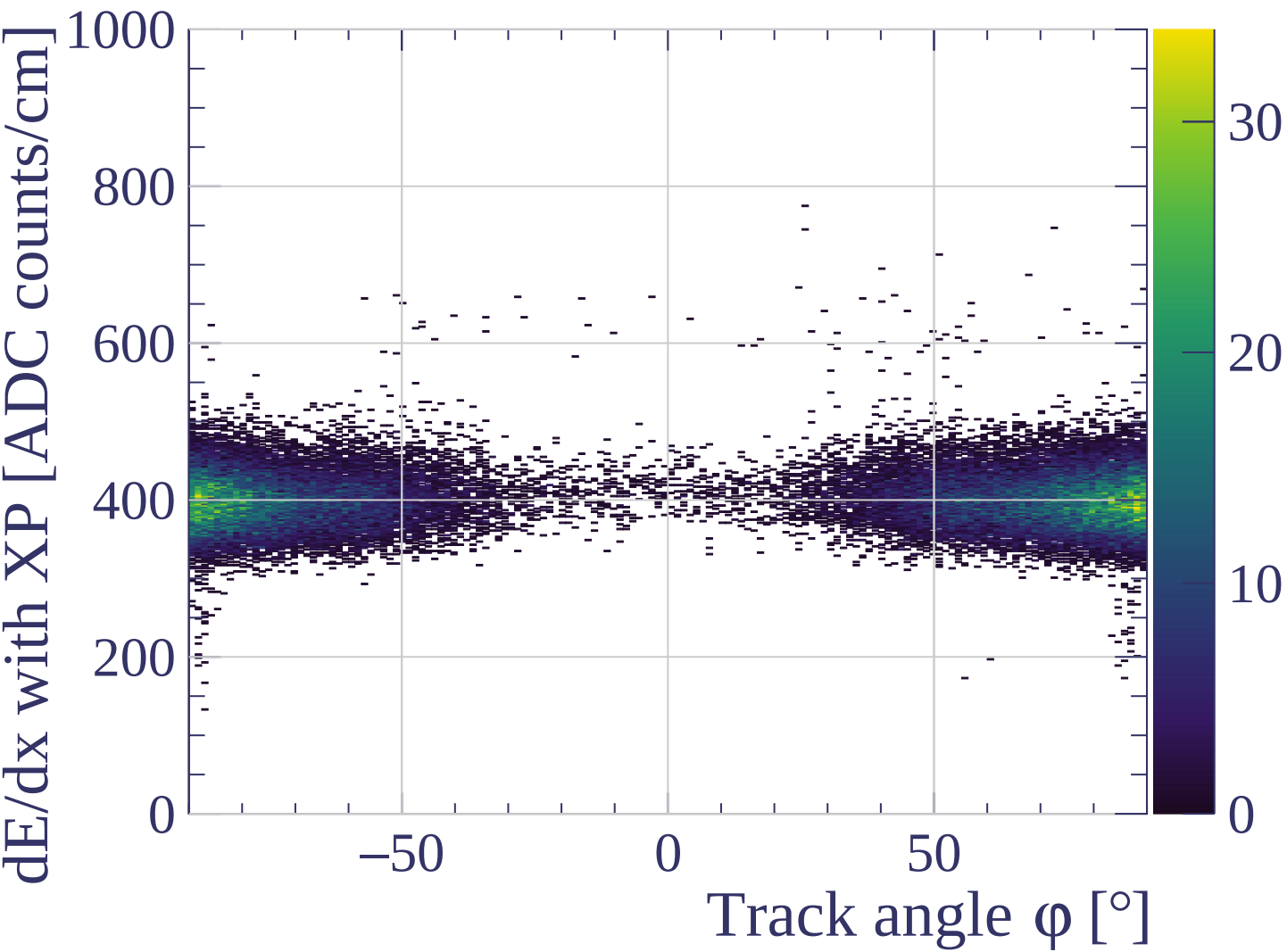




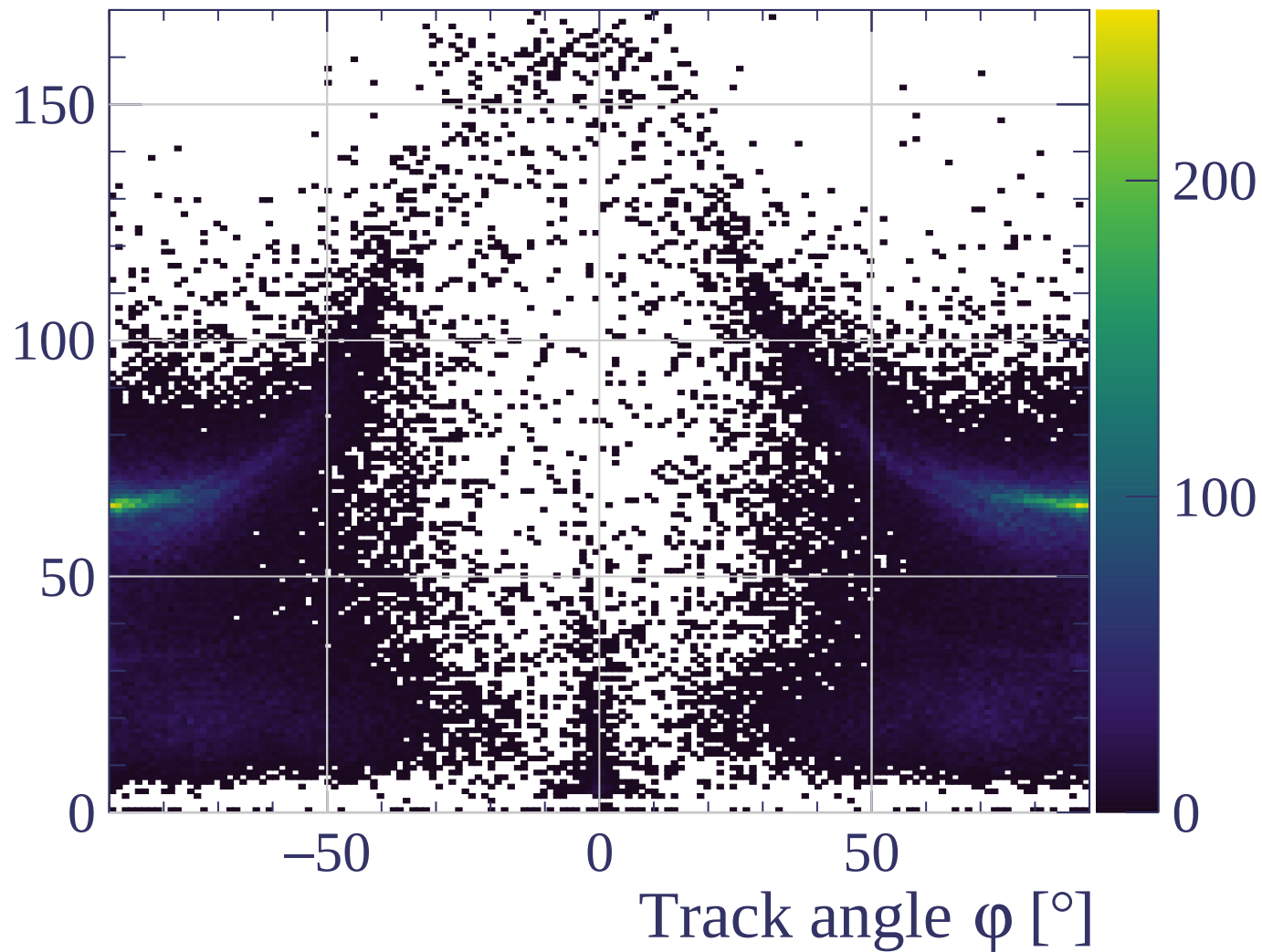


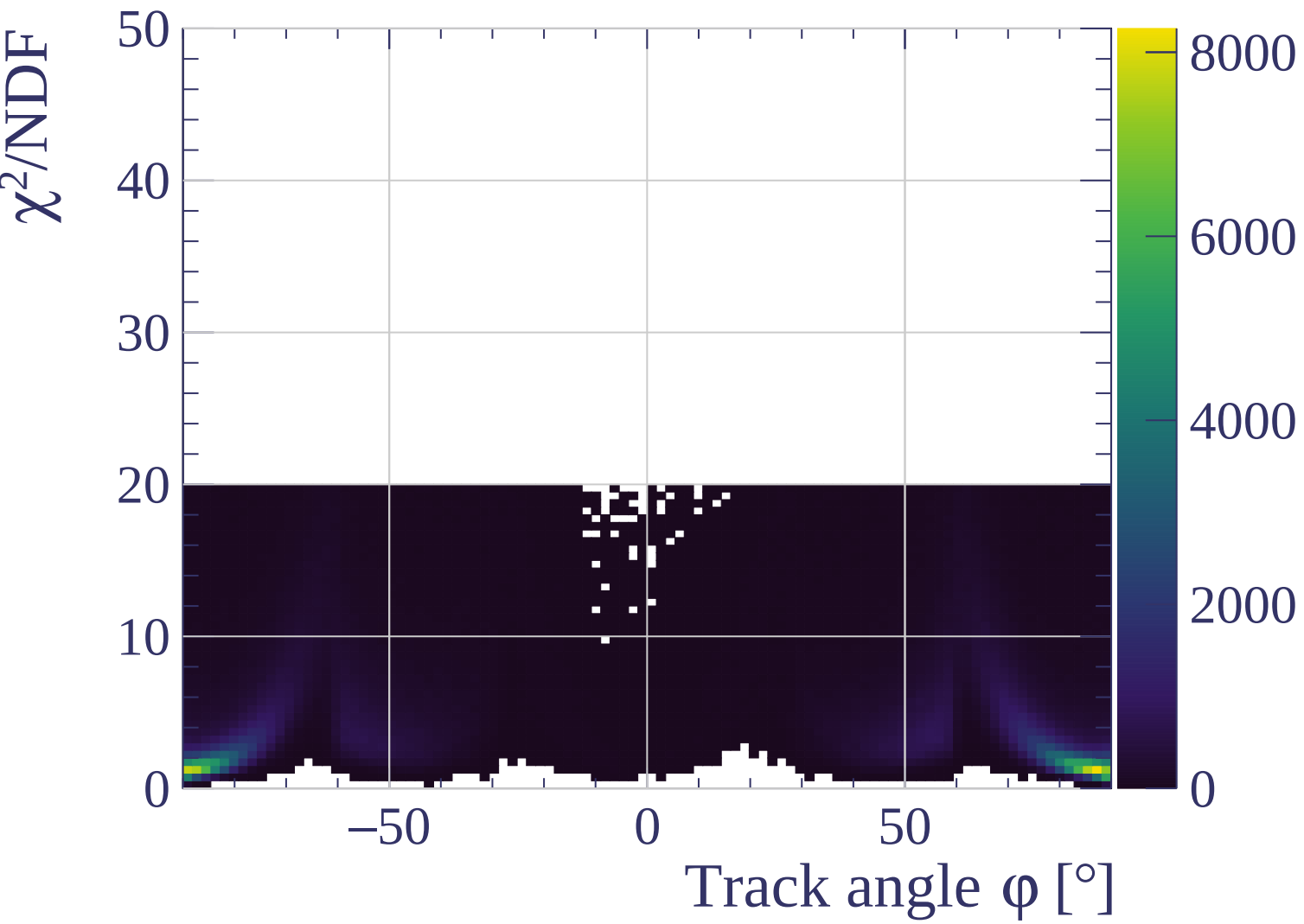


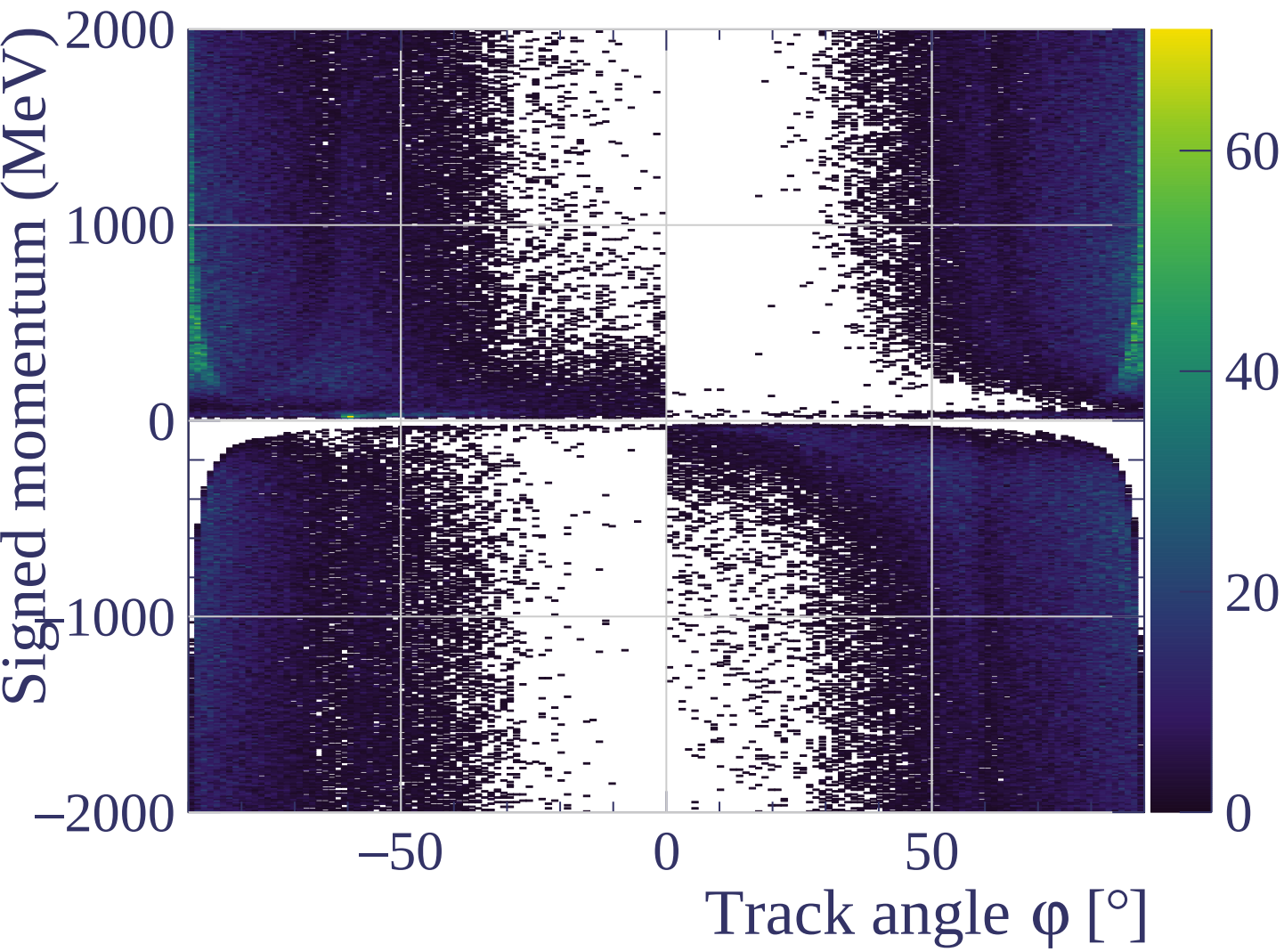


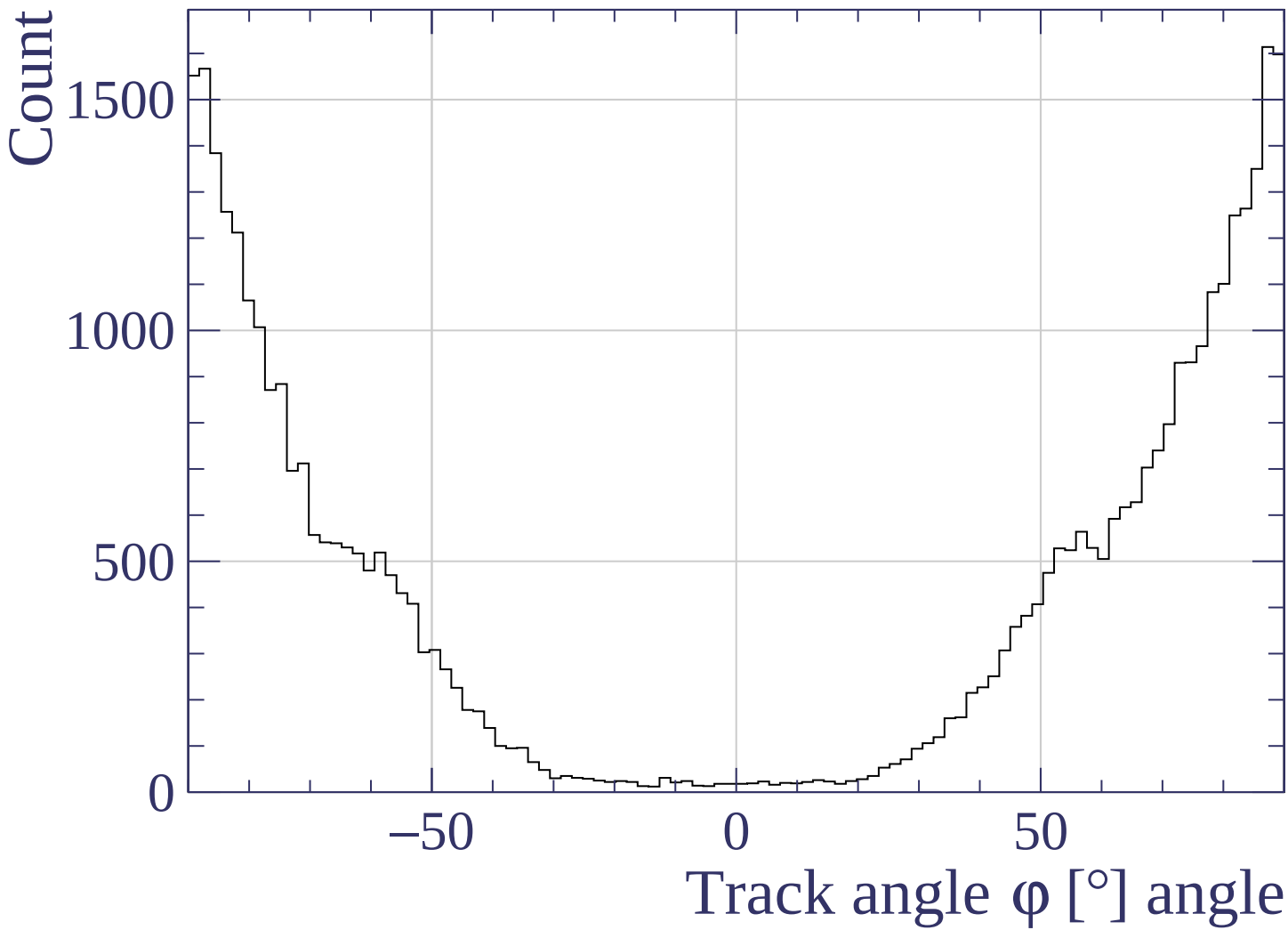


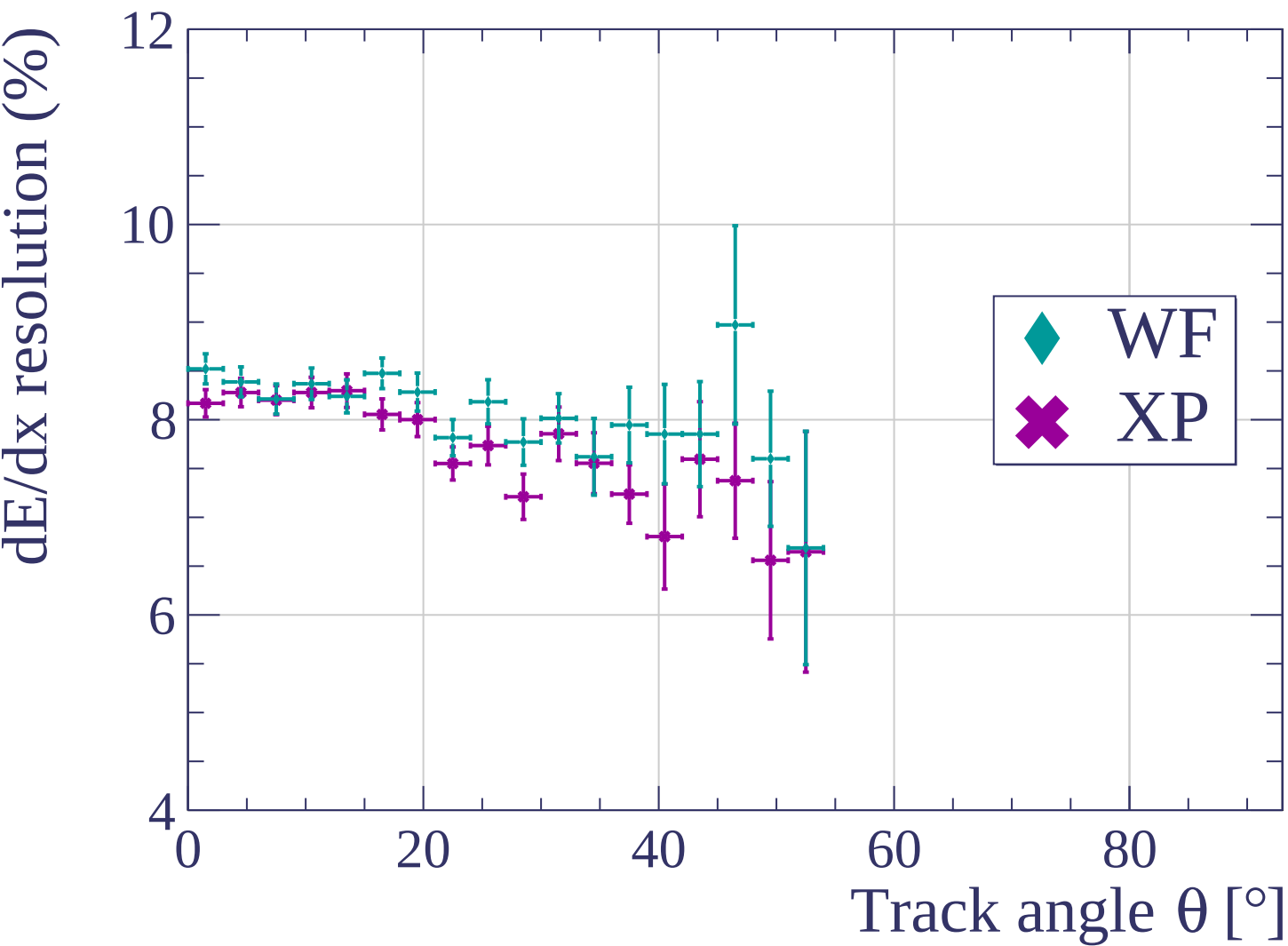
Track length [cm]

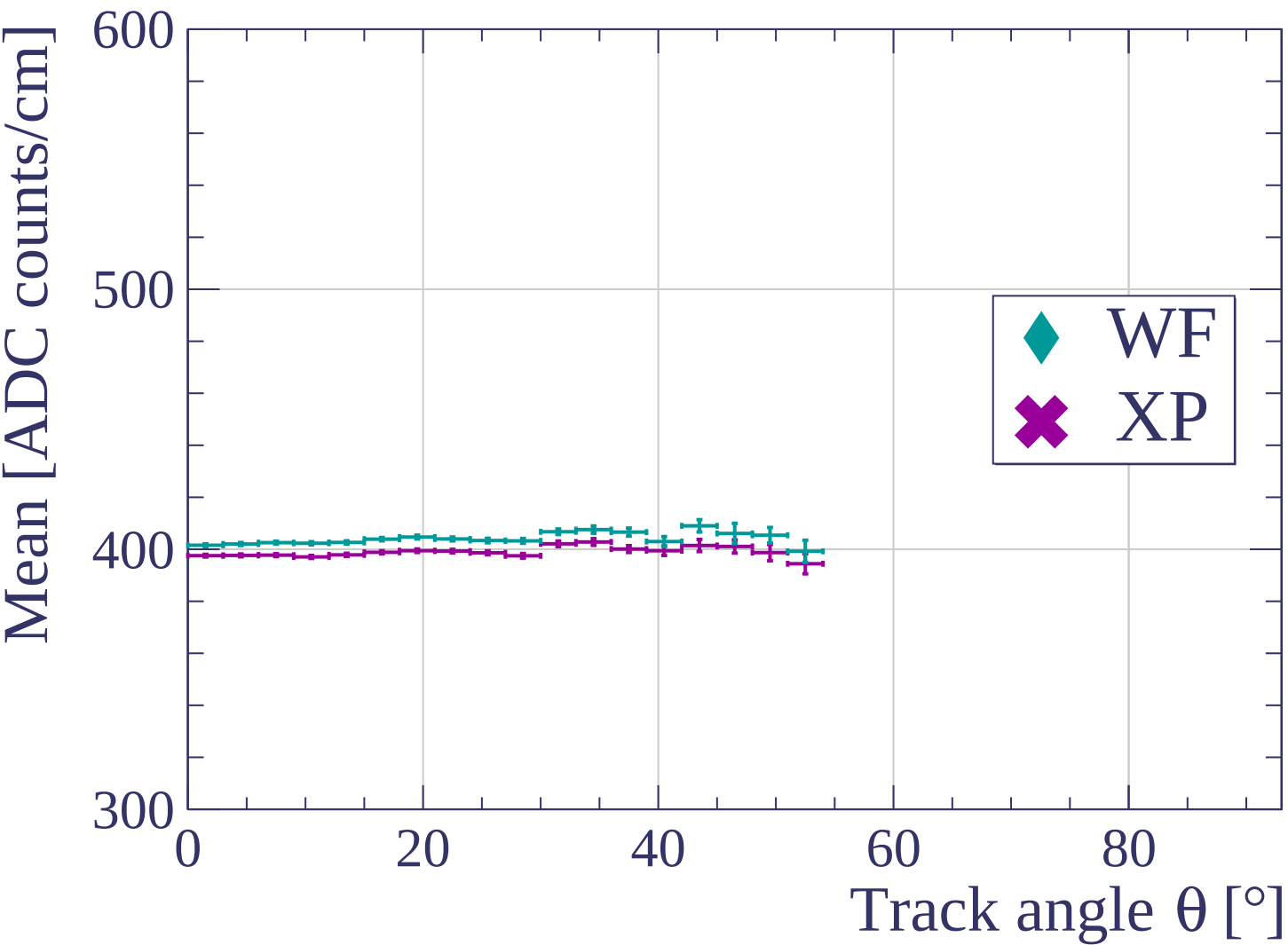


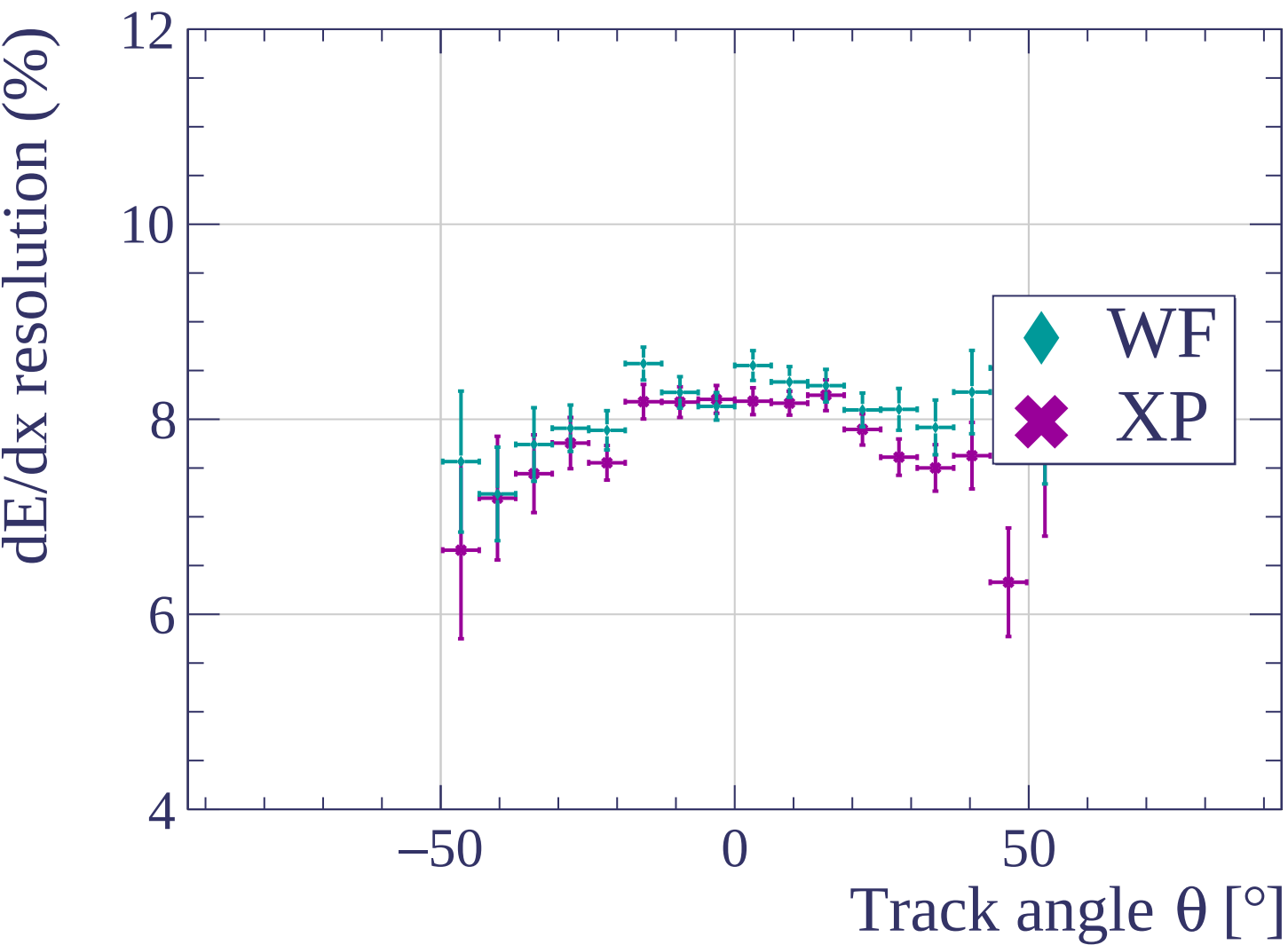


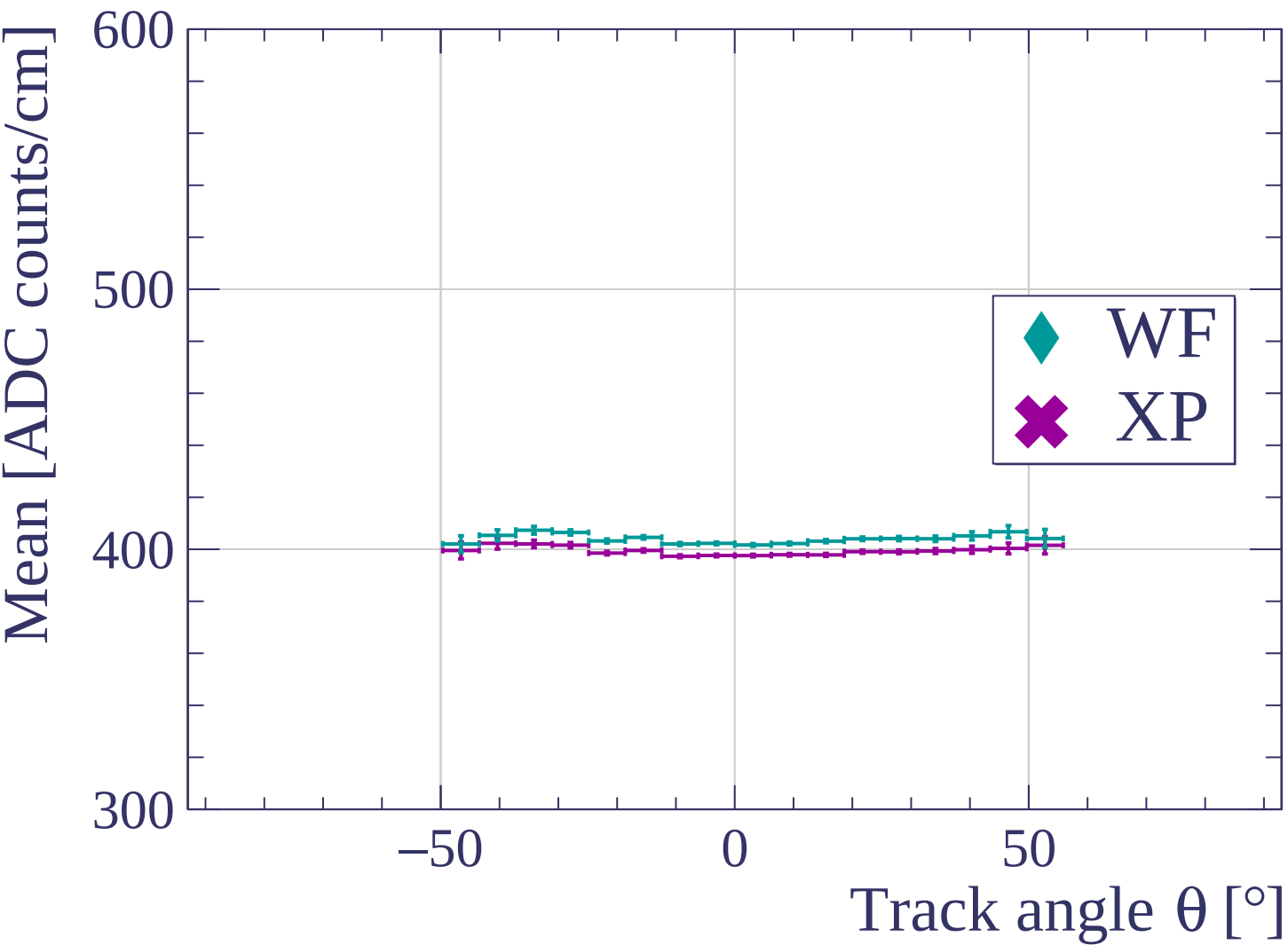


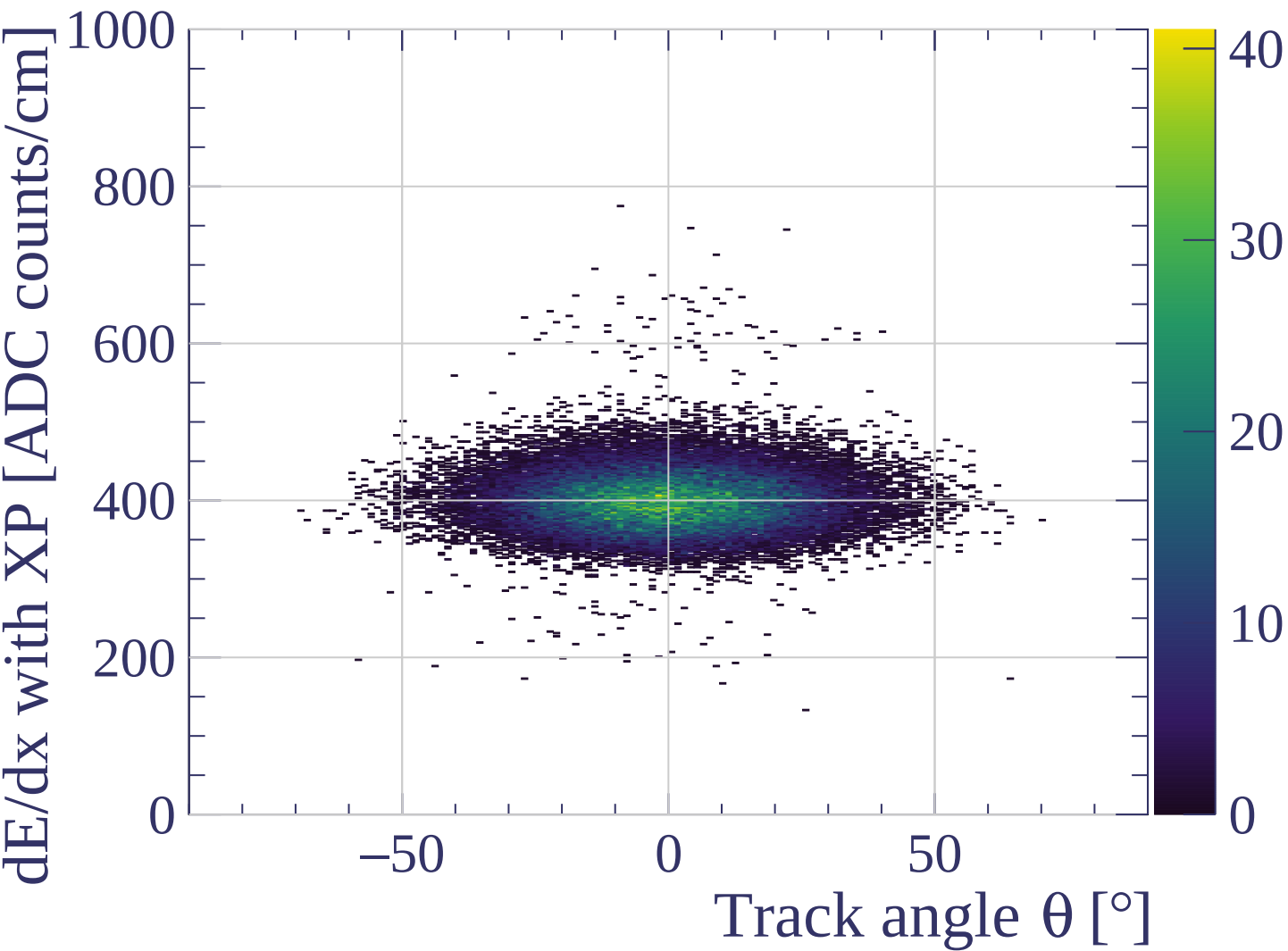




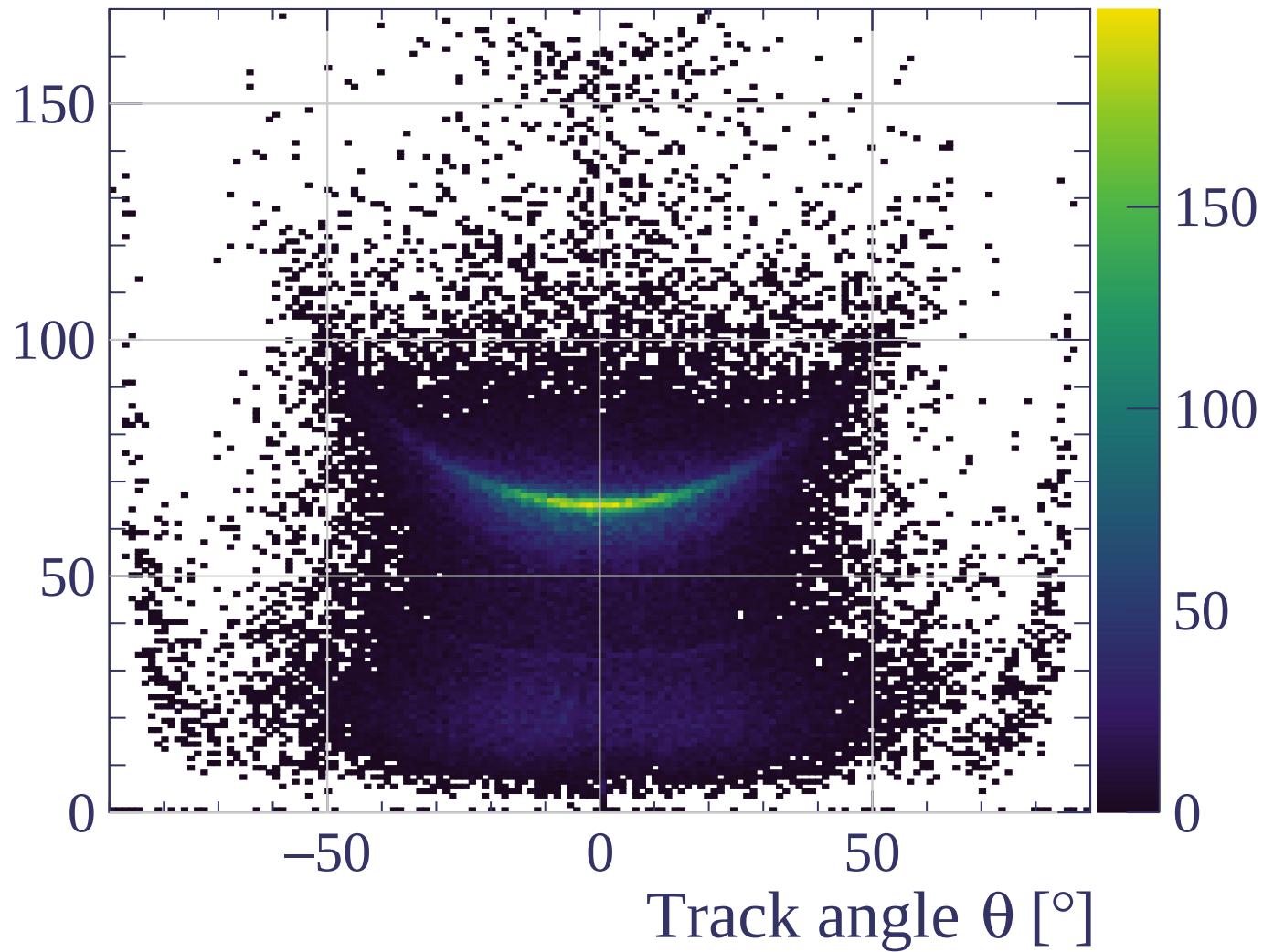


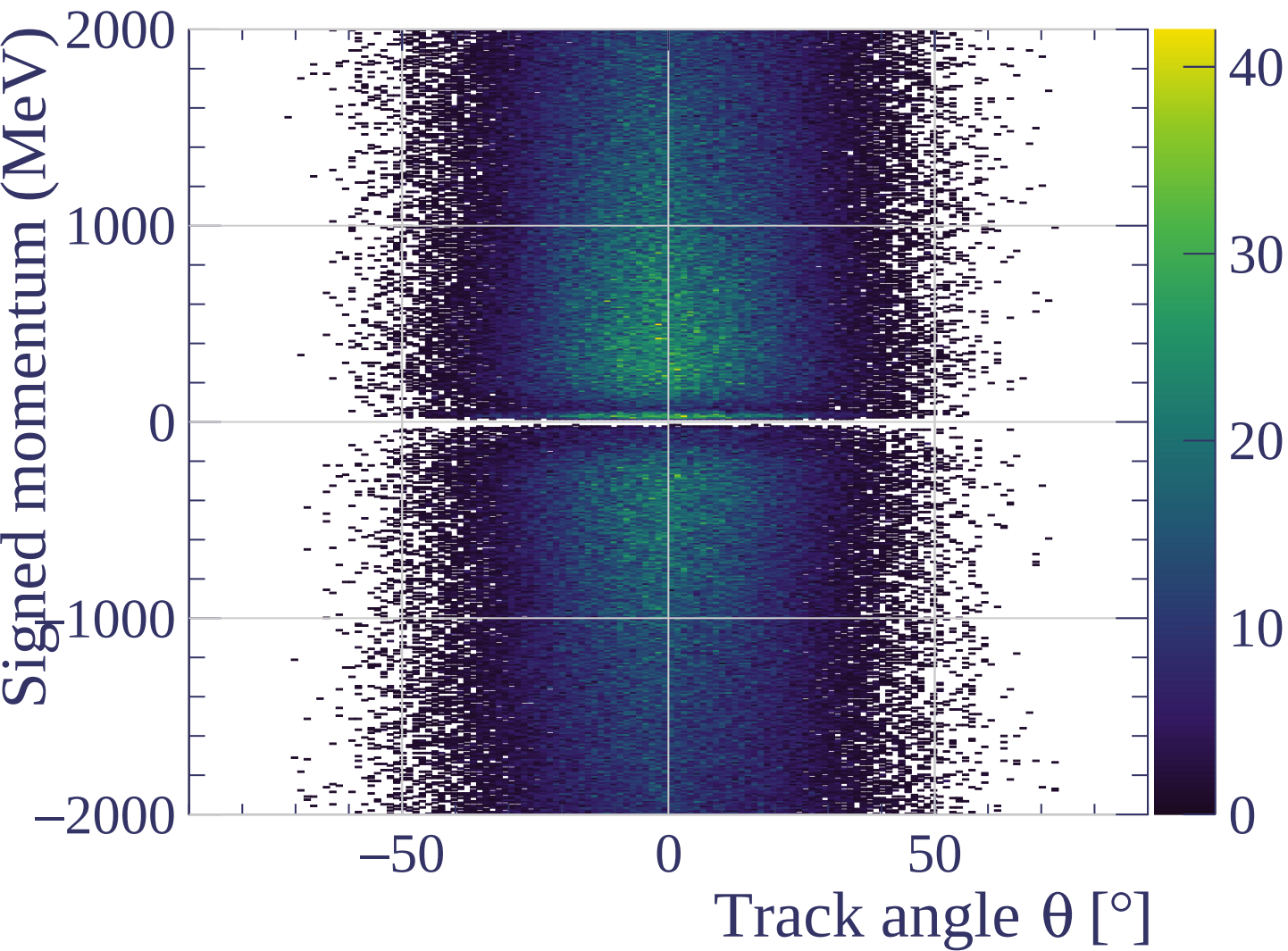


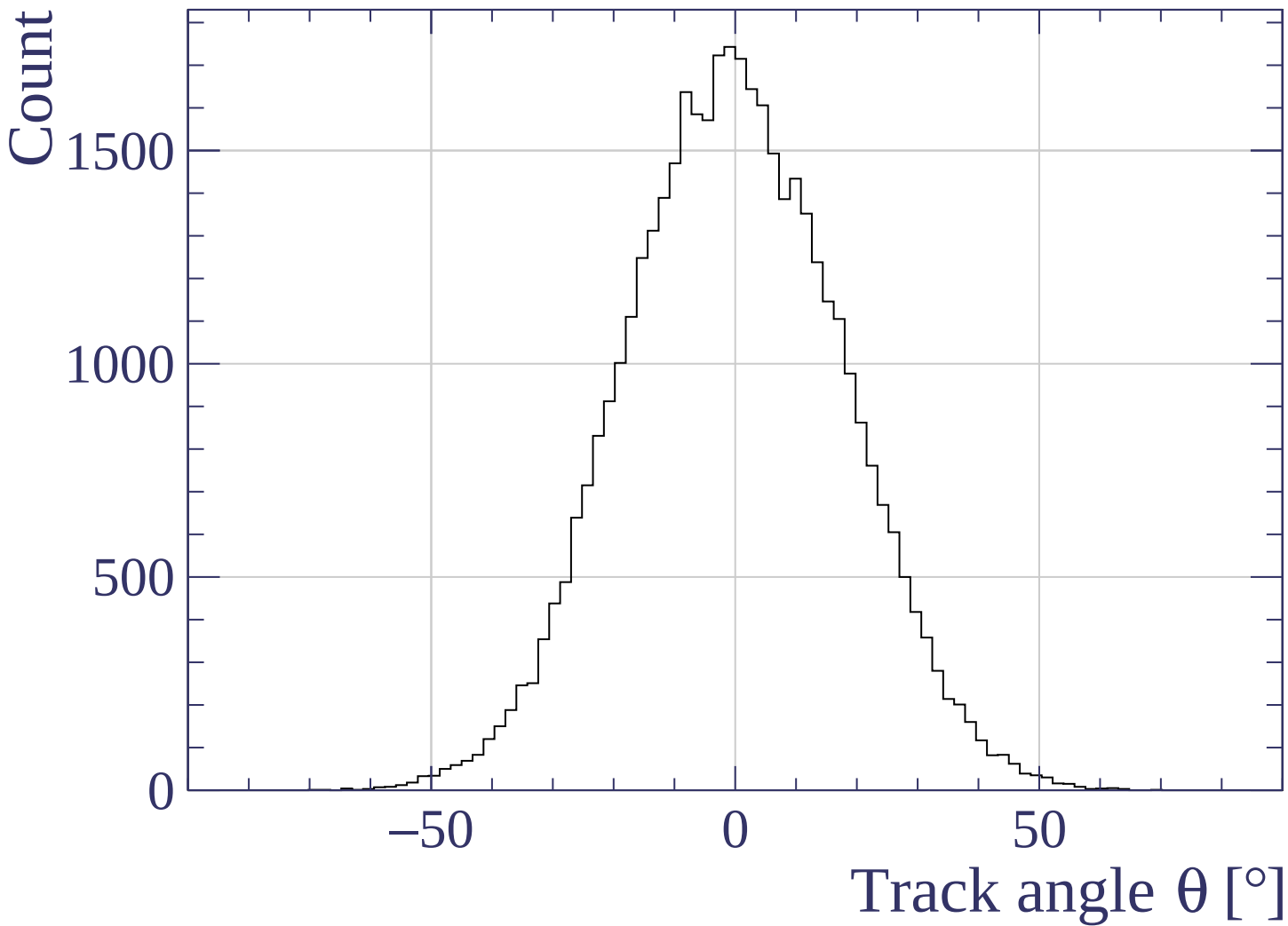




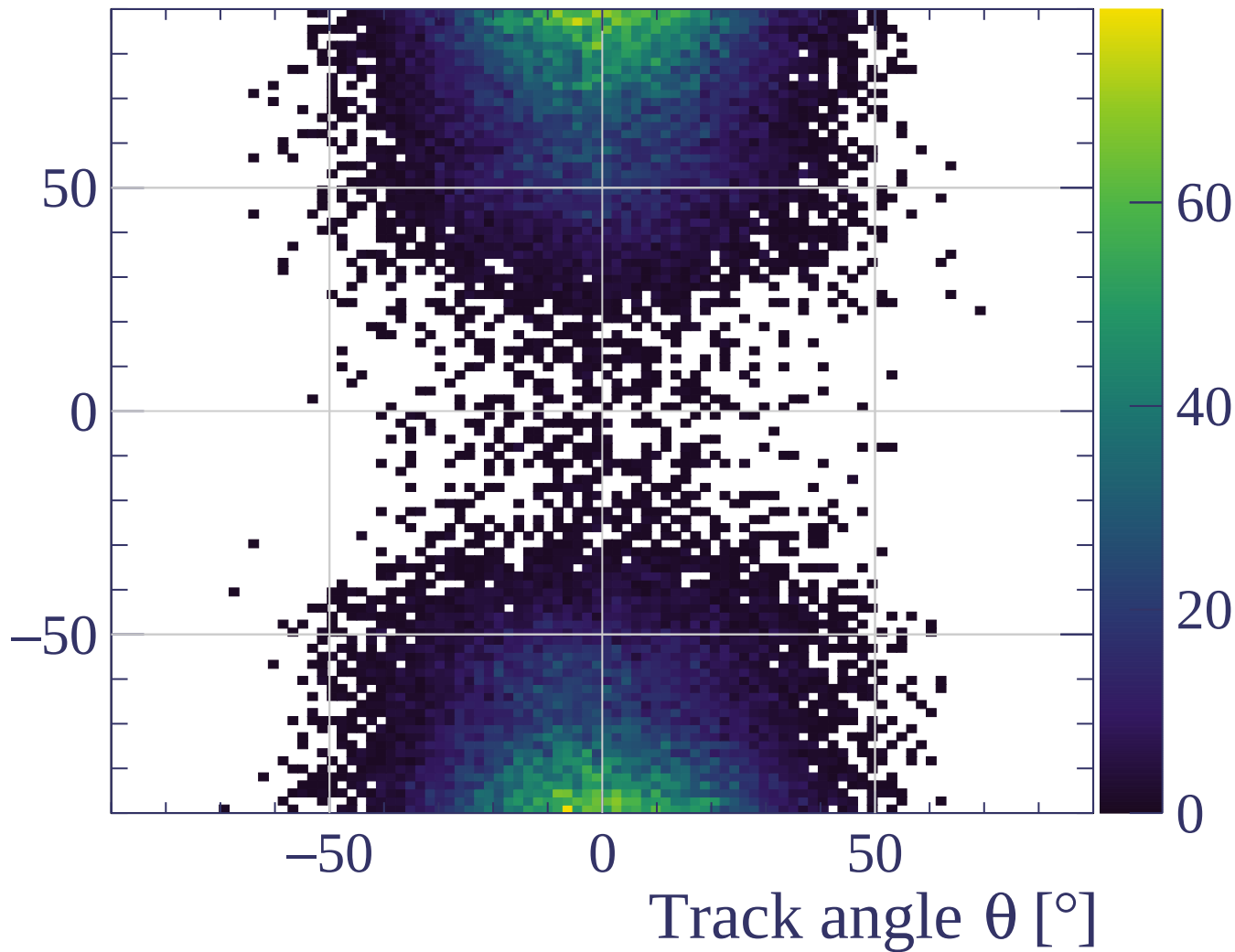
Track length [cm]



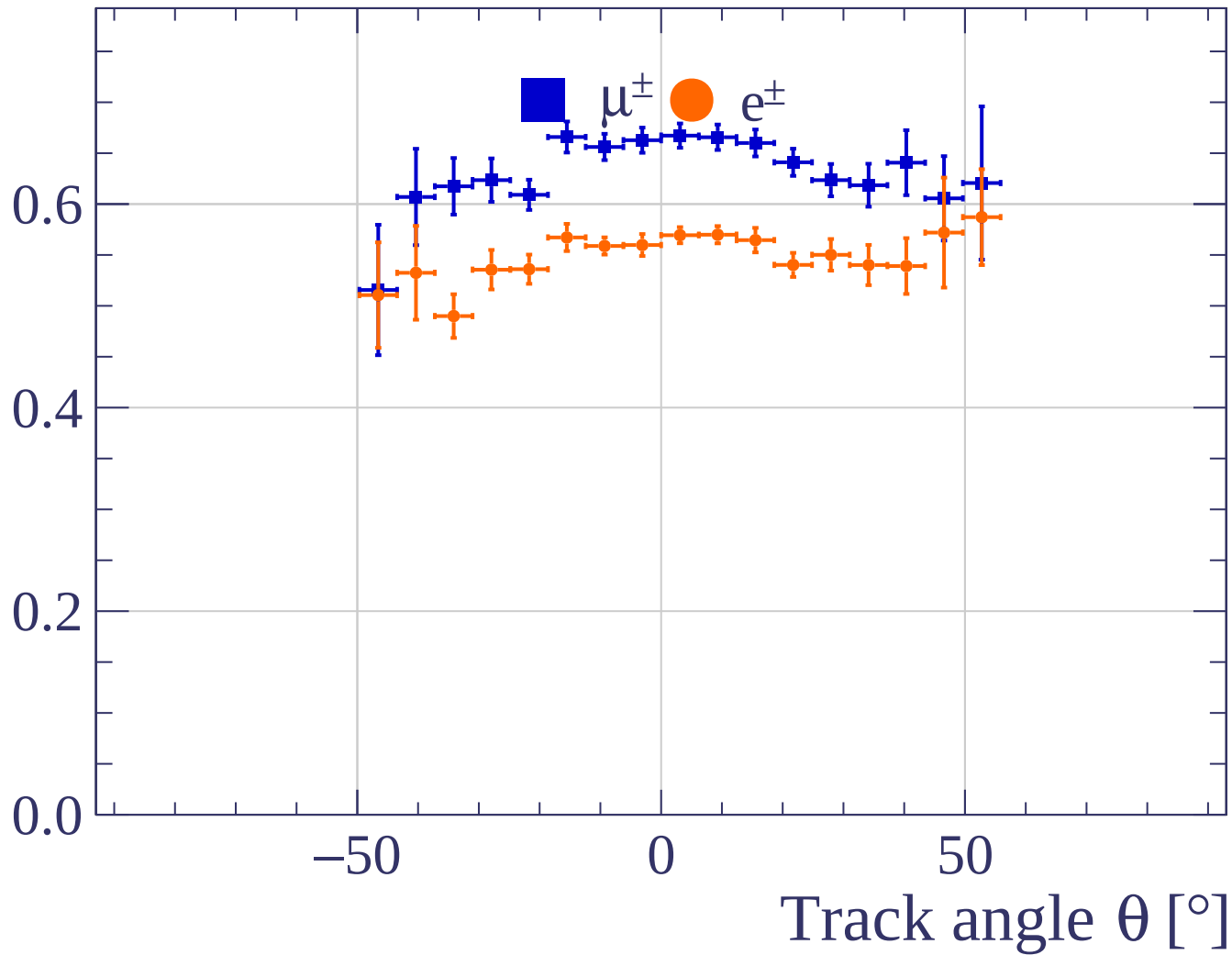




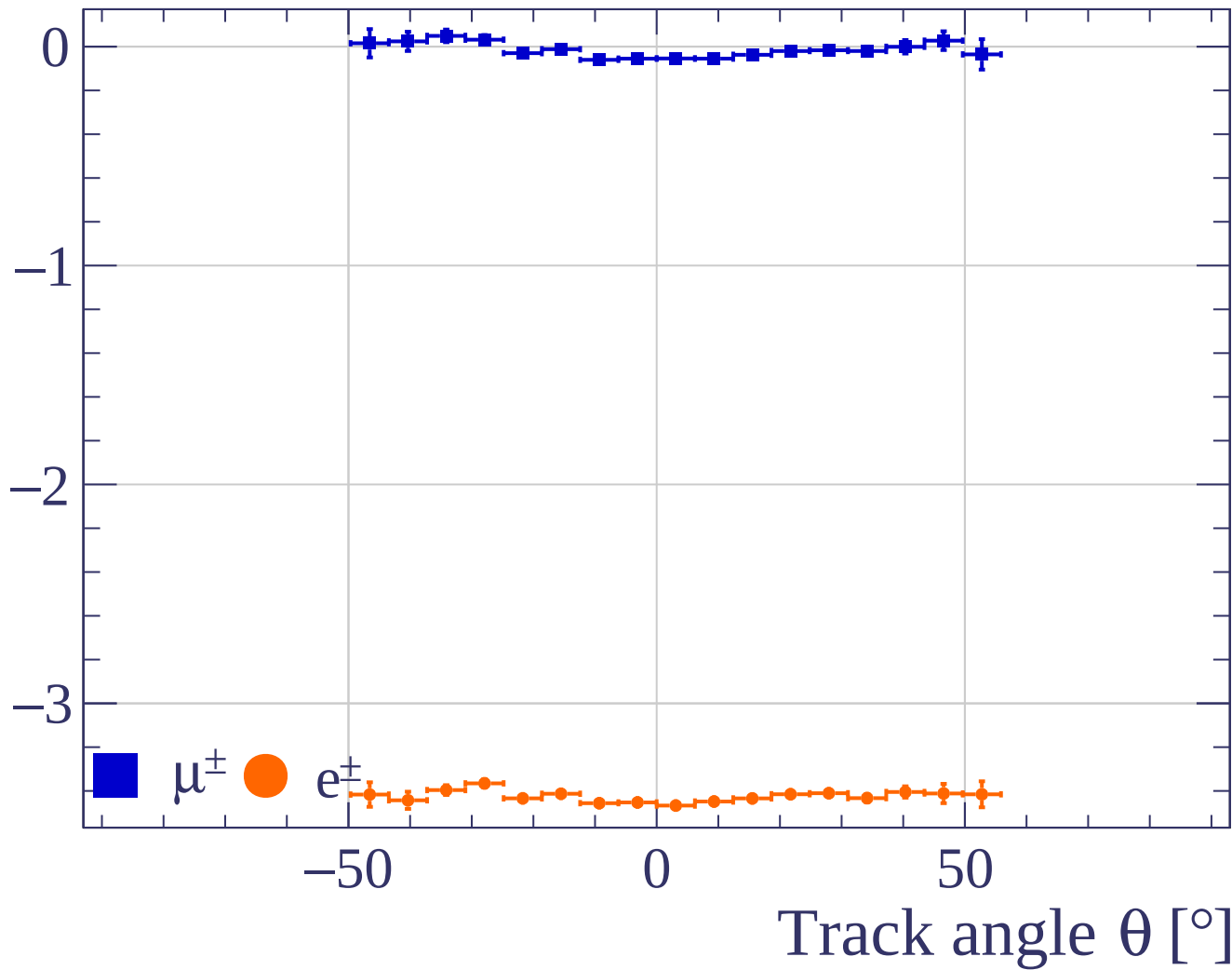
Track angle ϕ [°]



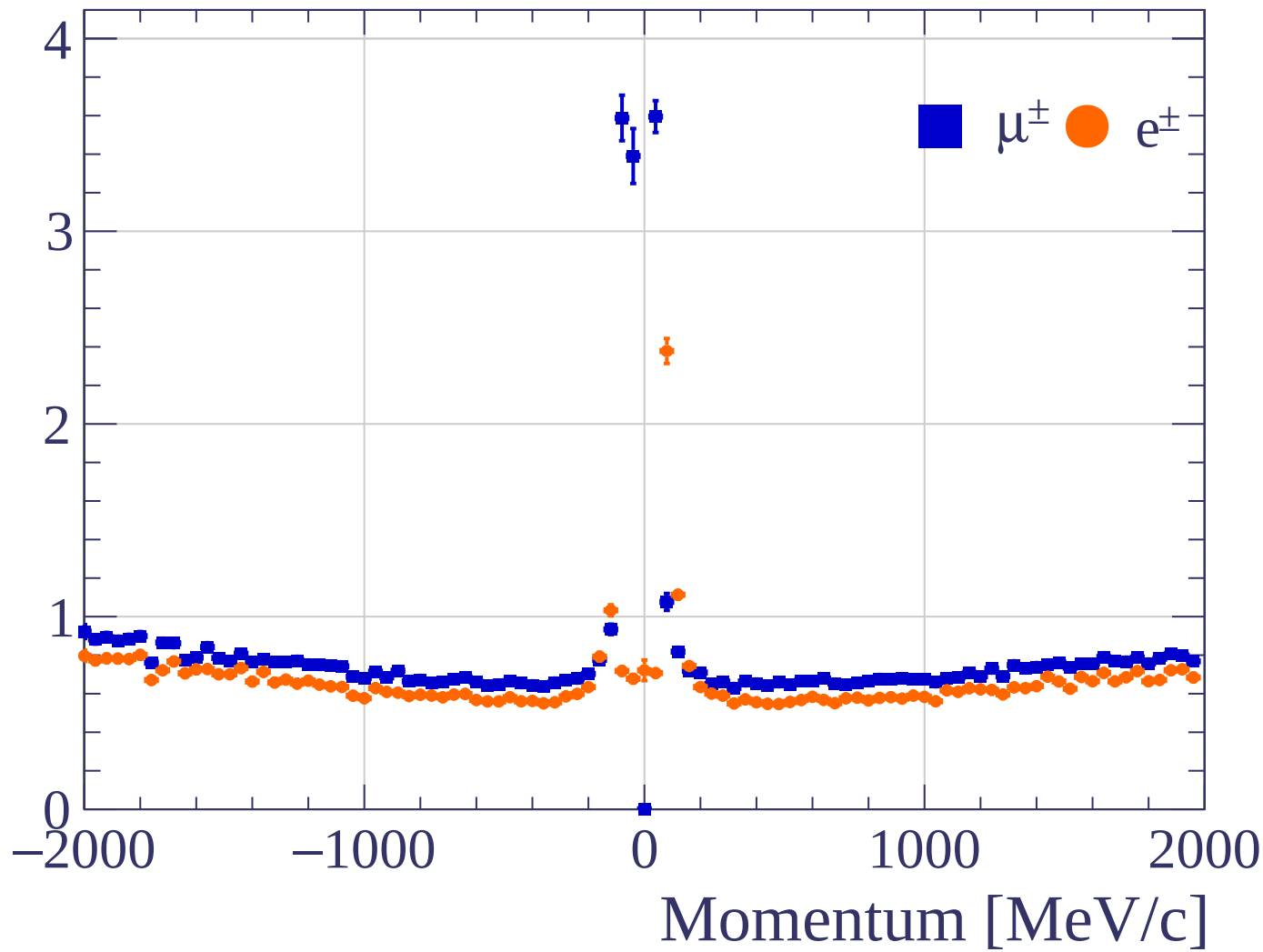
Pulls standard deviation



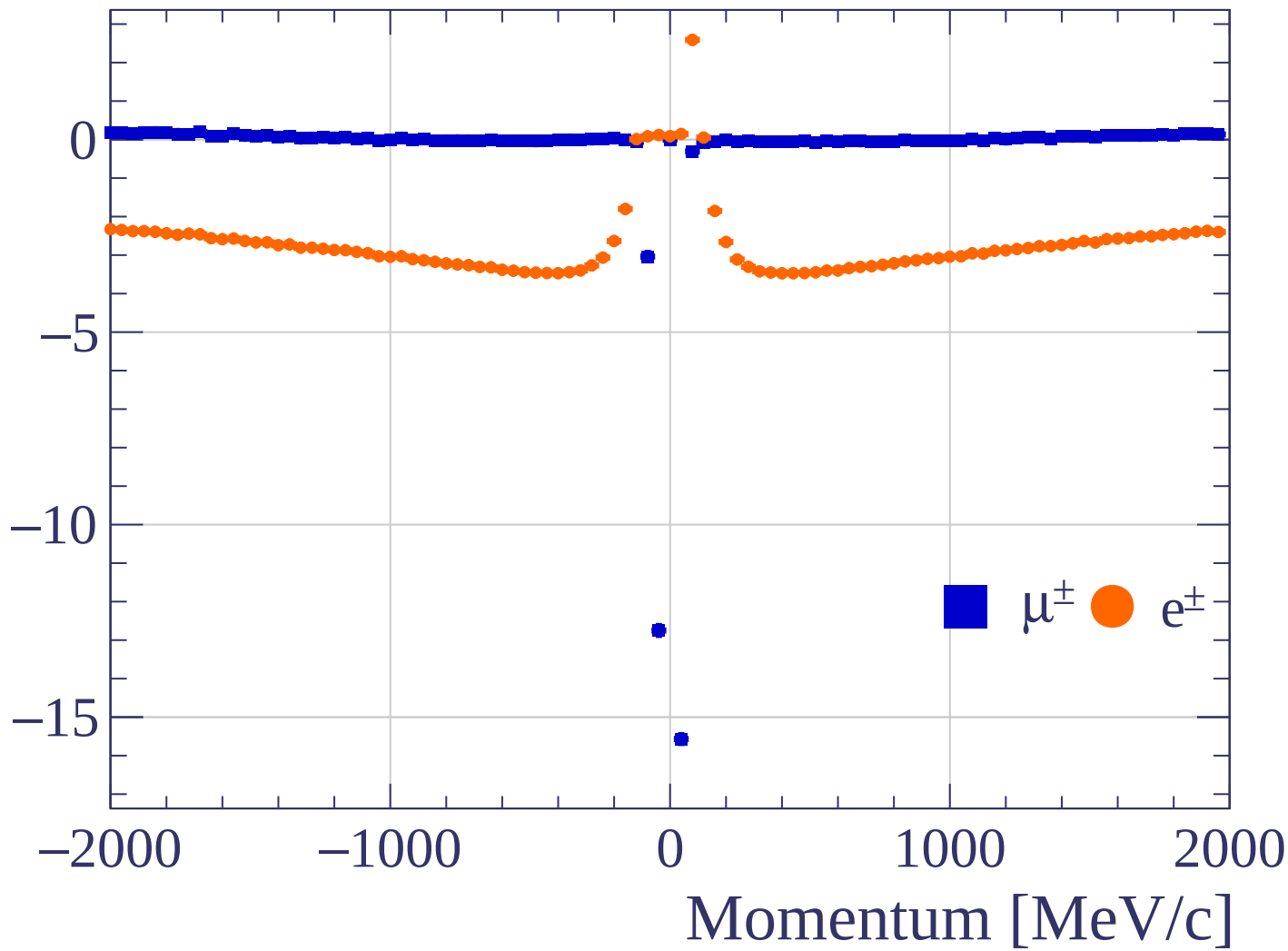
Pulls mean

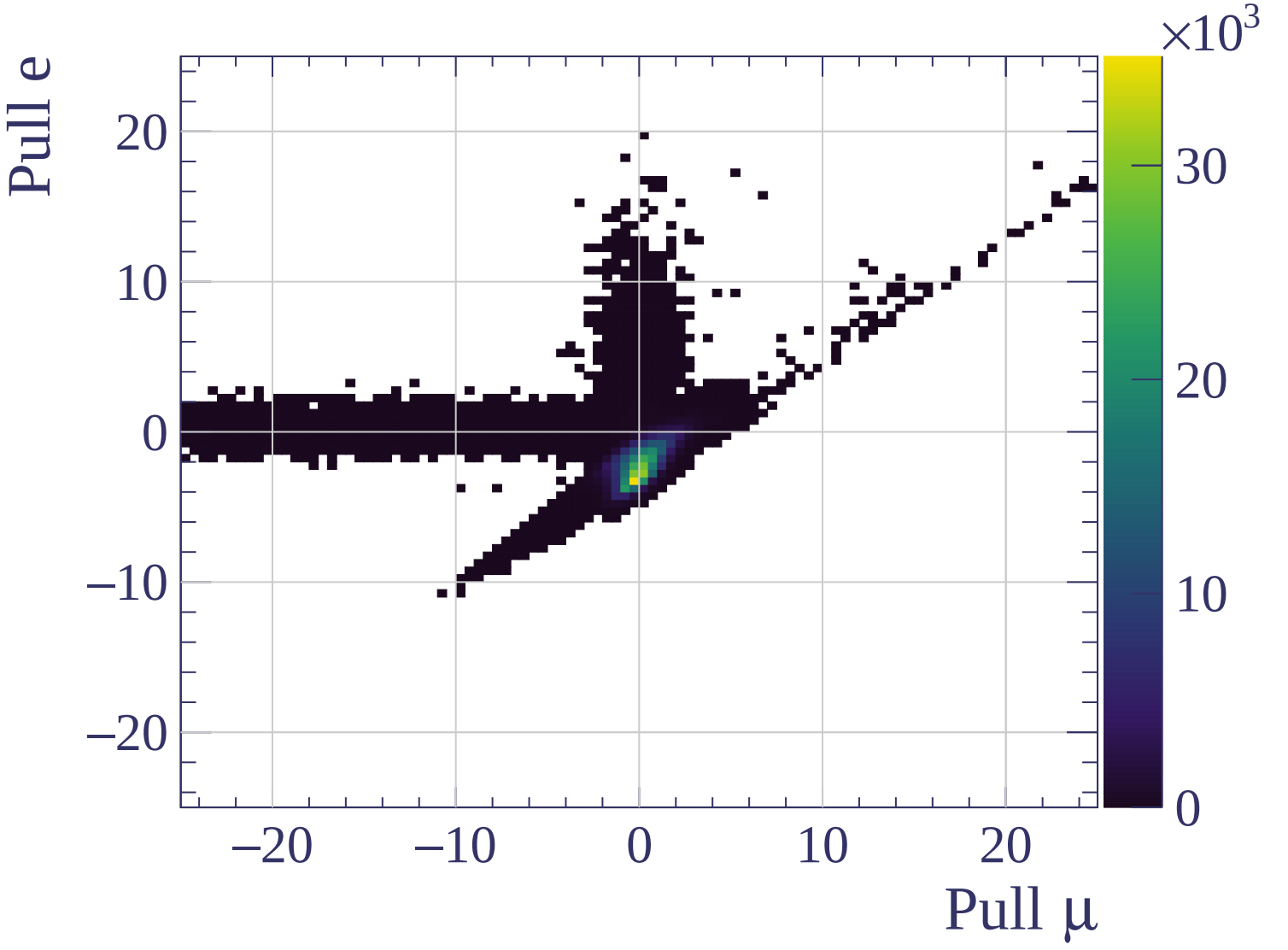


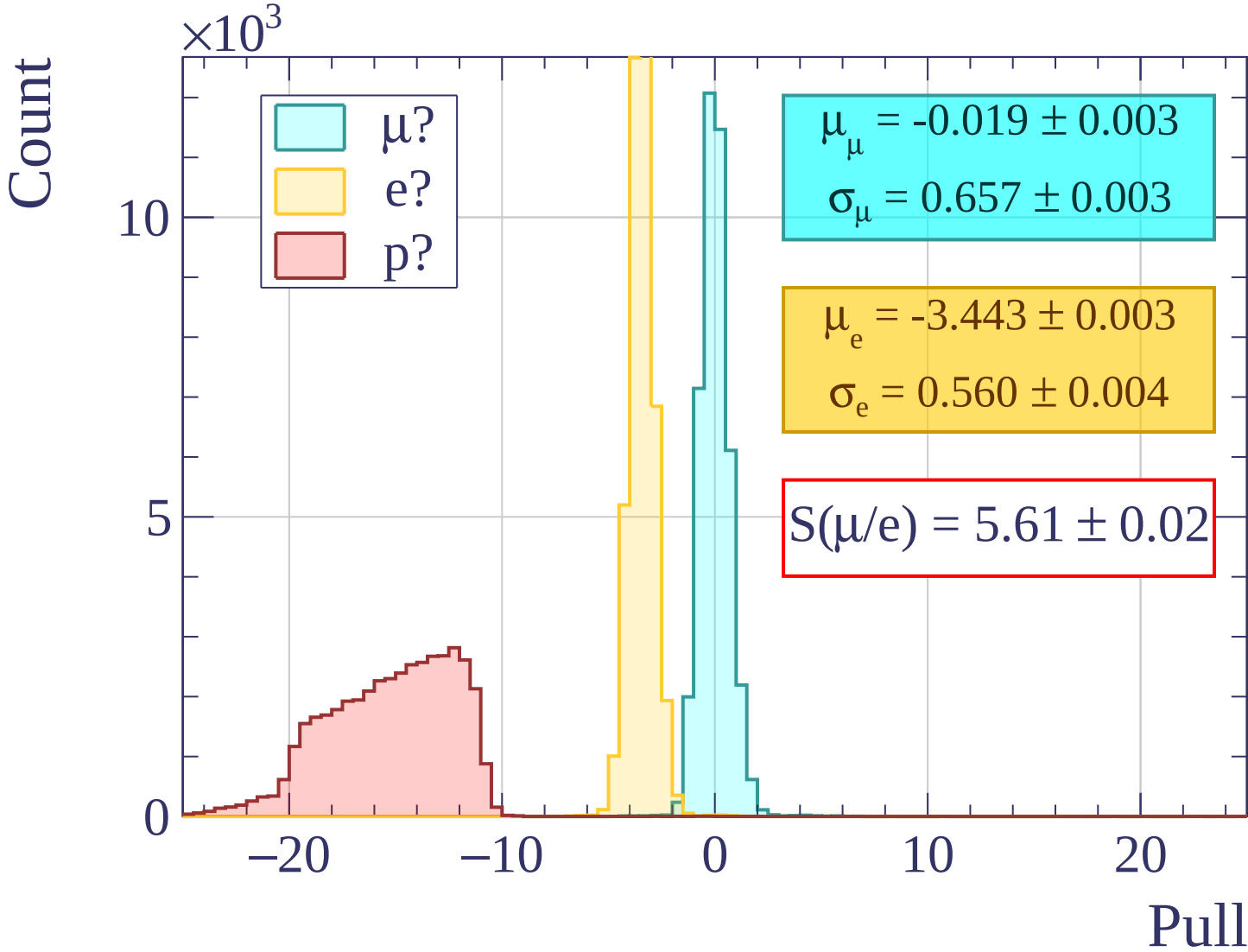
Pulls standard deviation

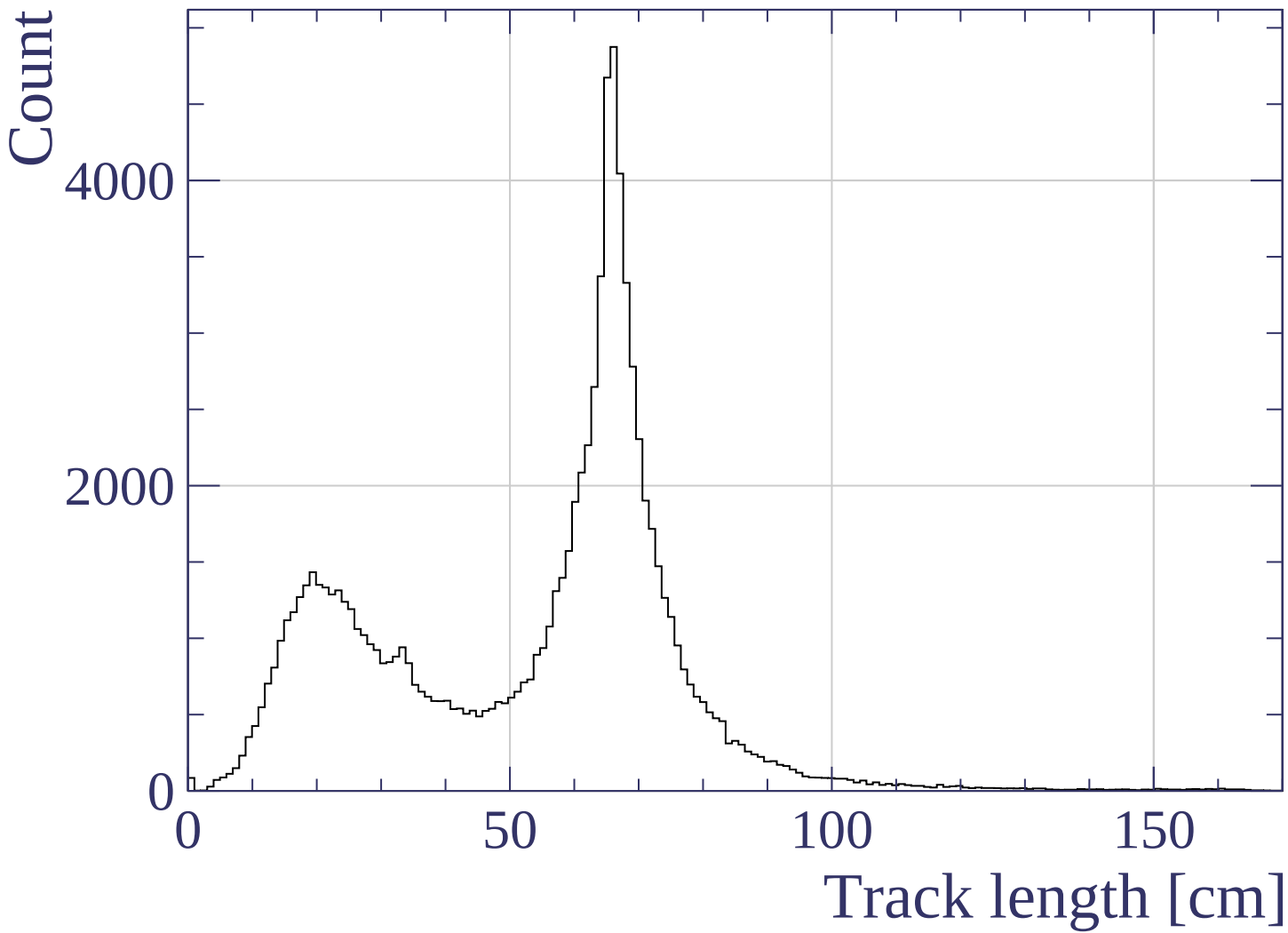


Pulls mean

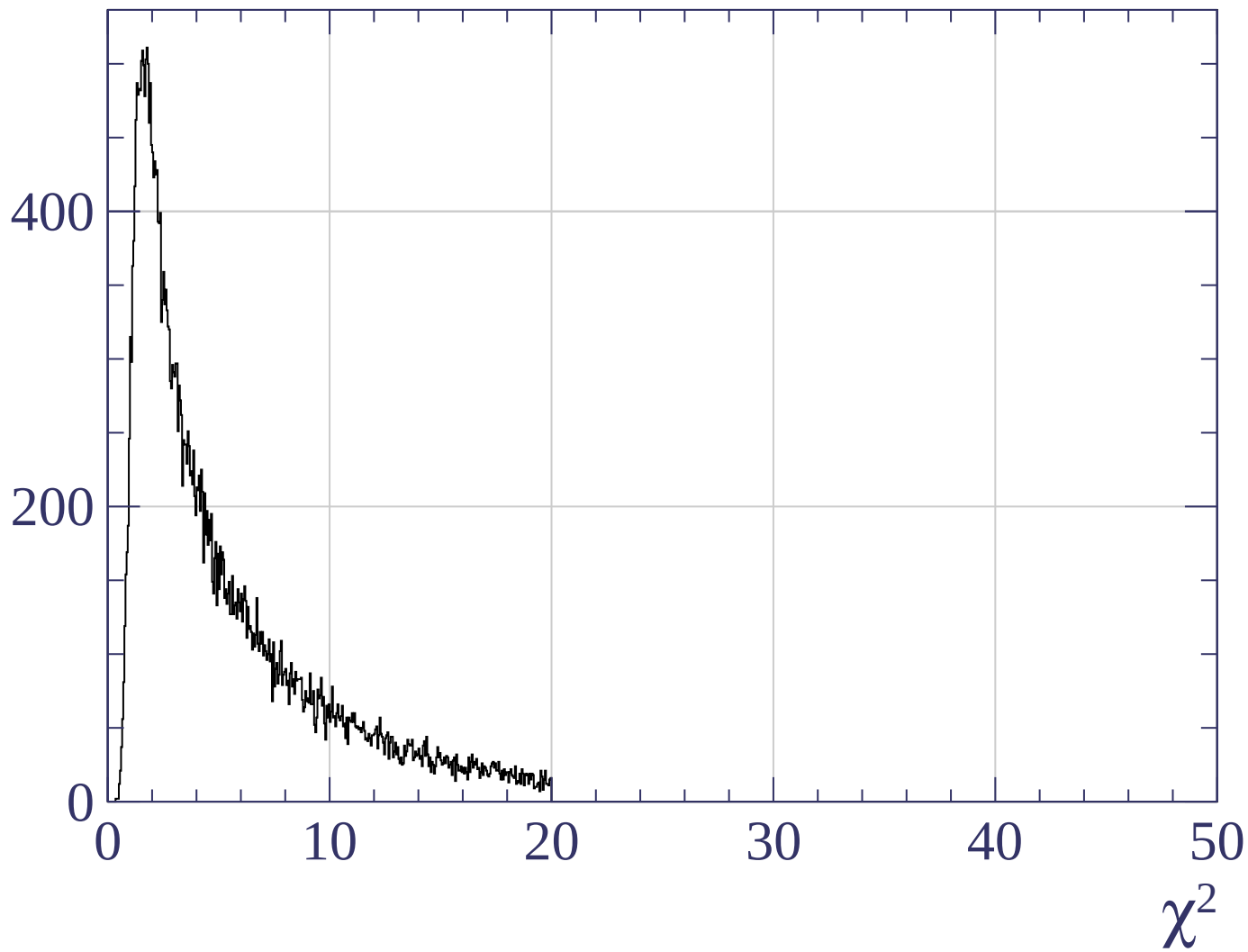




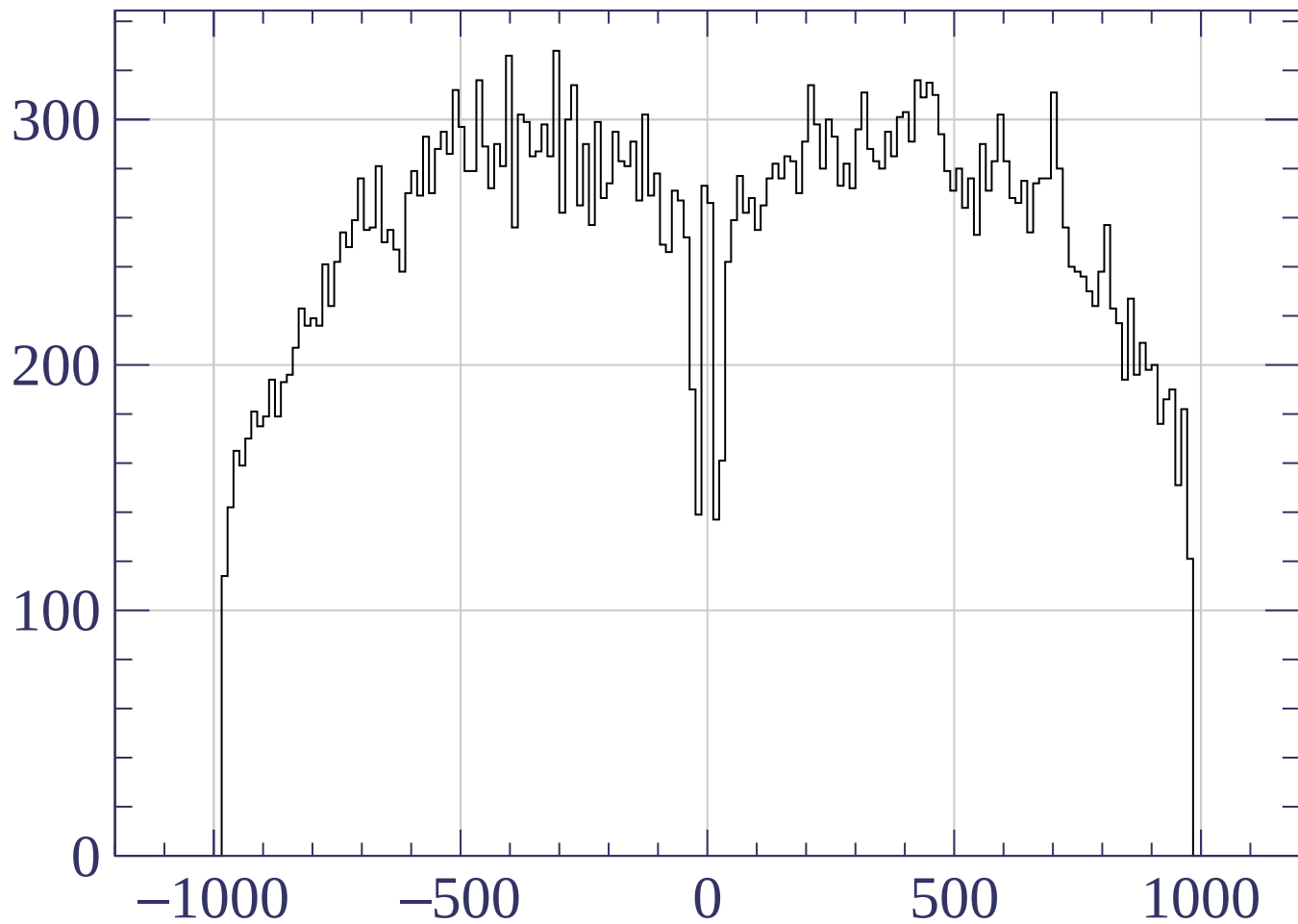




Count

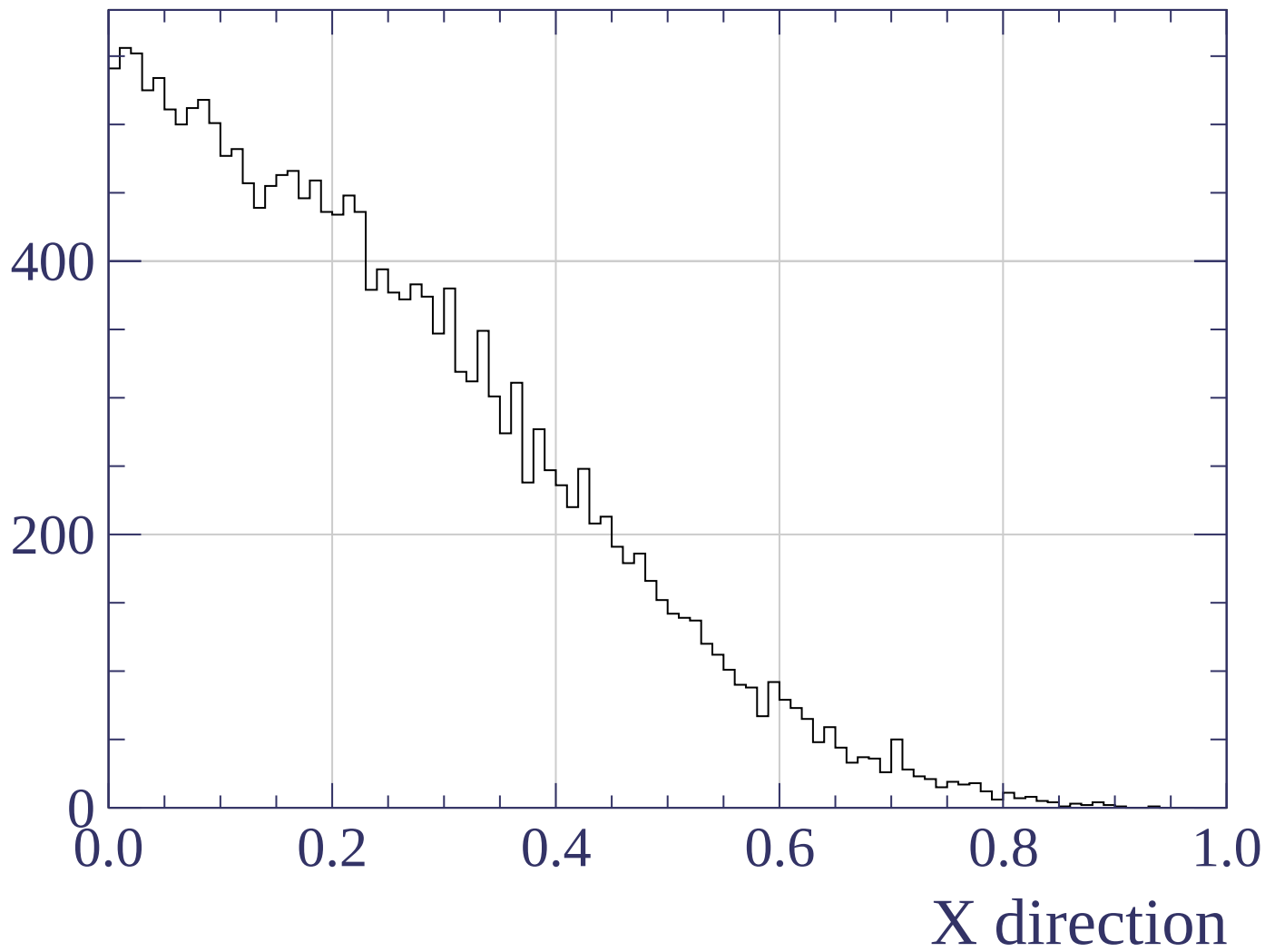


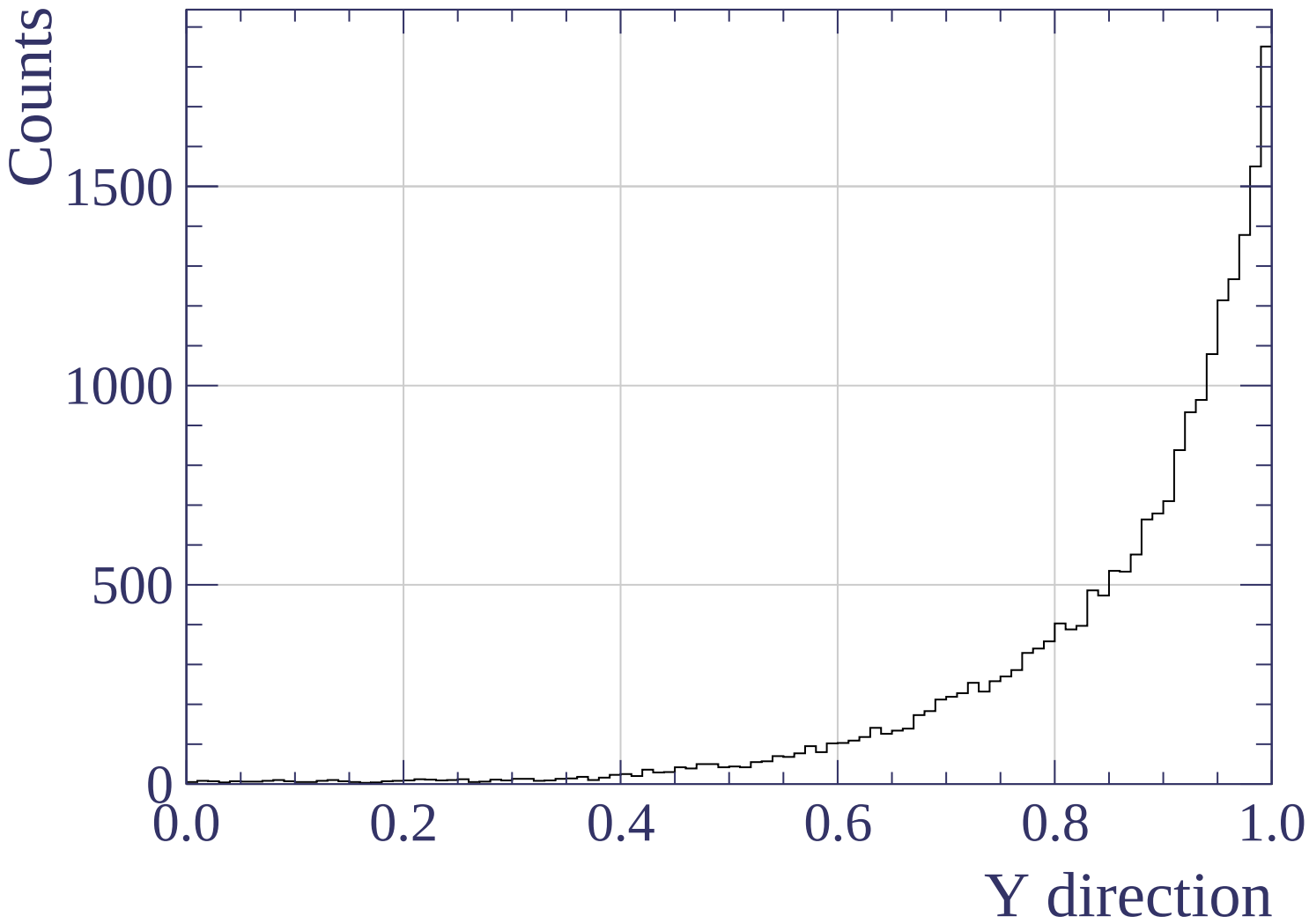
Count



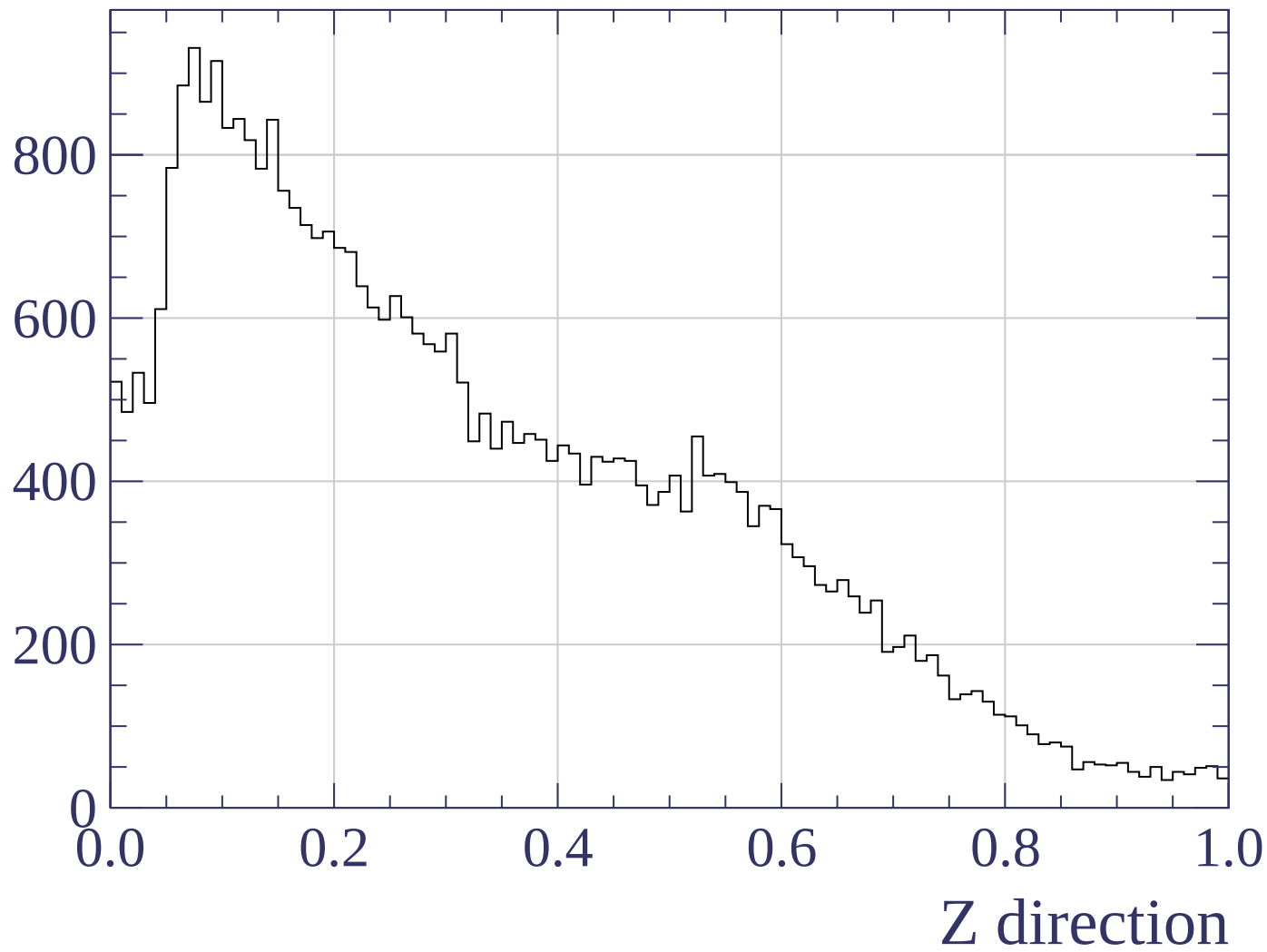
Track X position [mm]

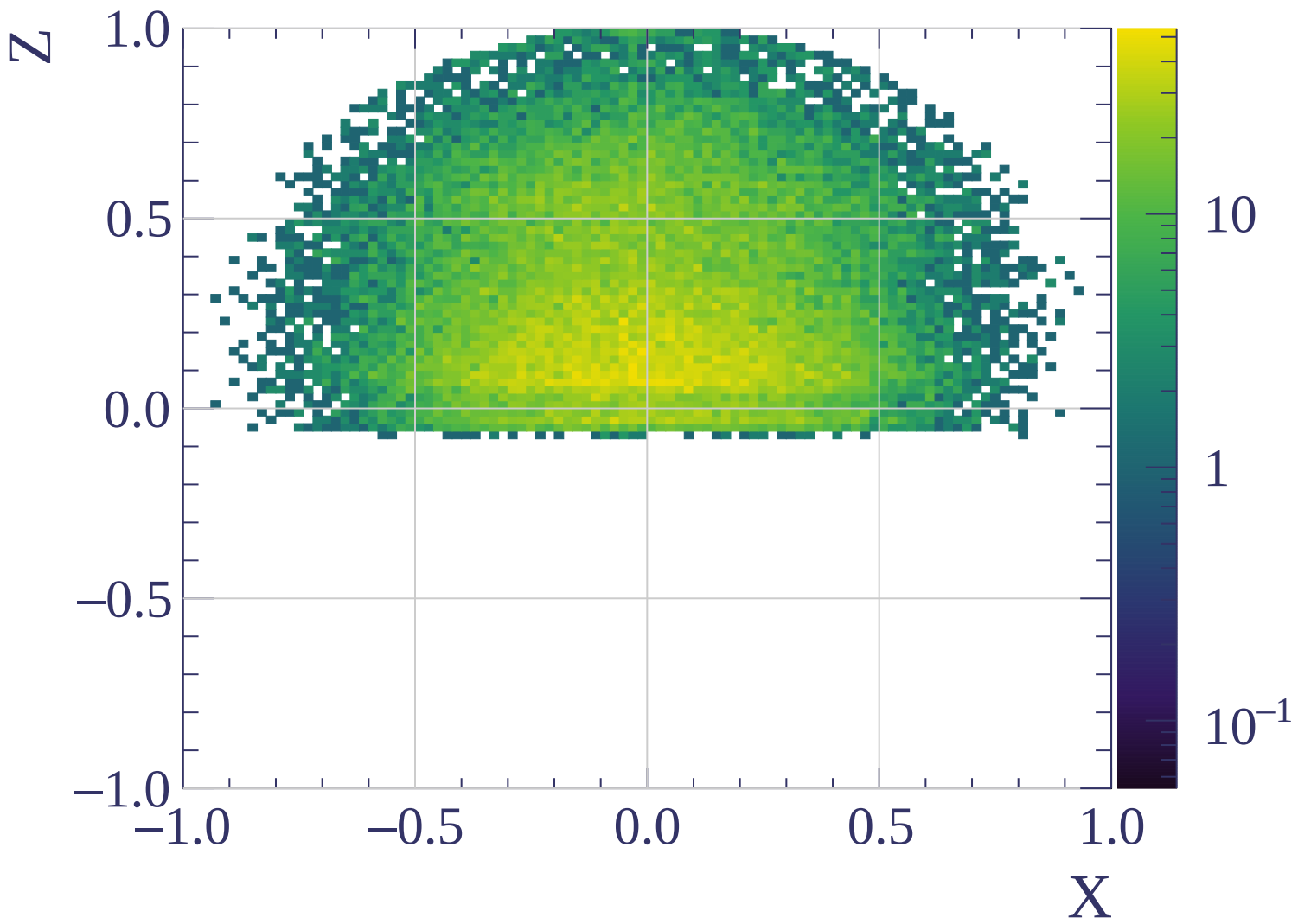
Counts

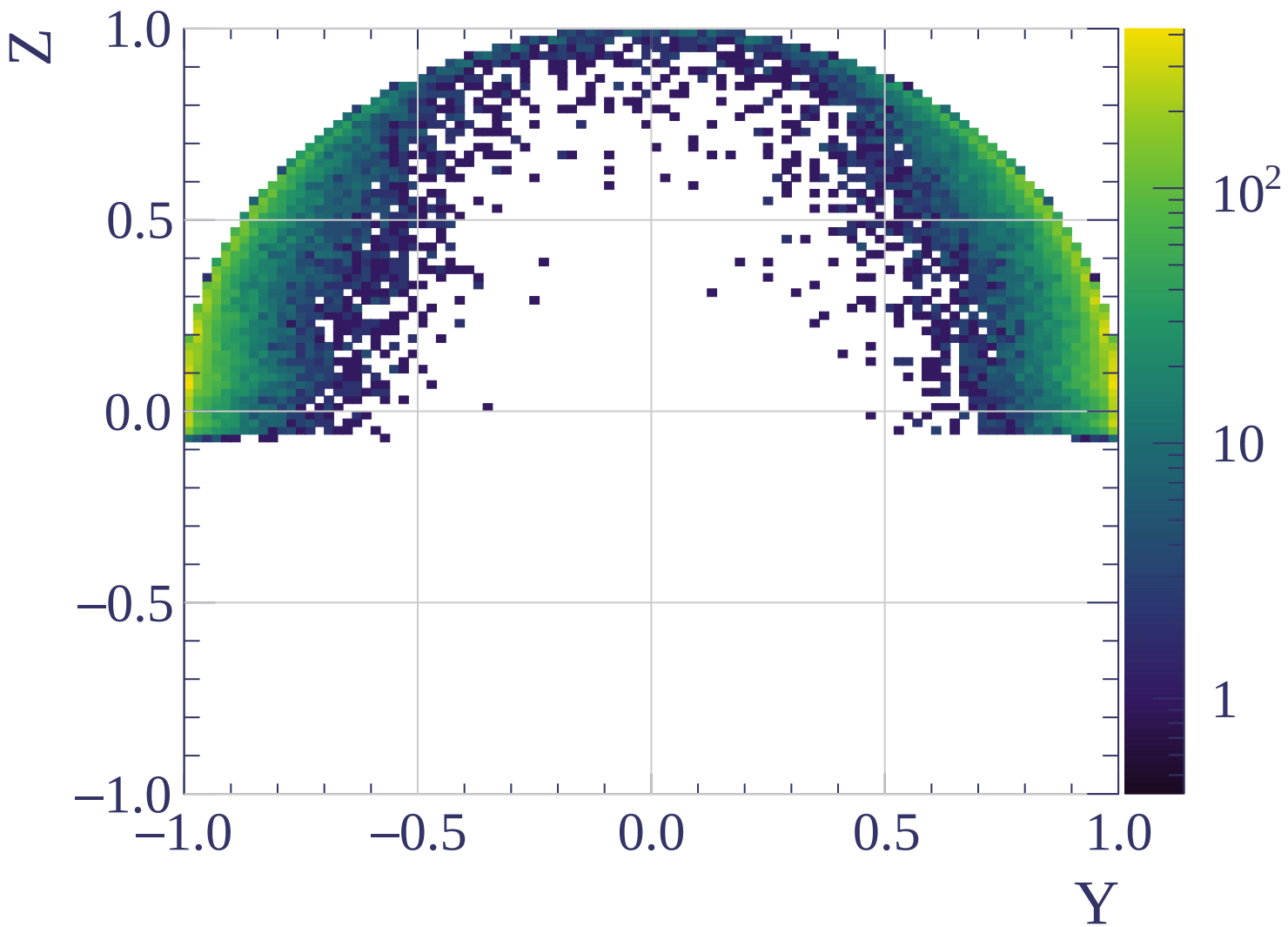


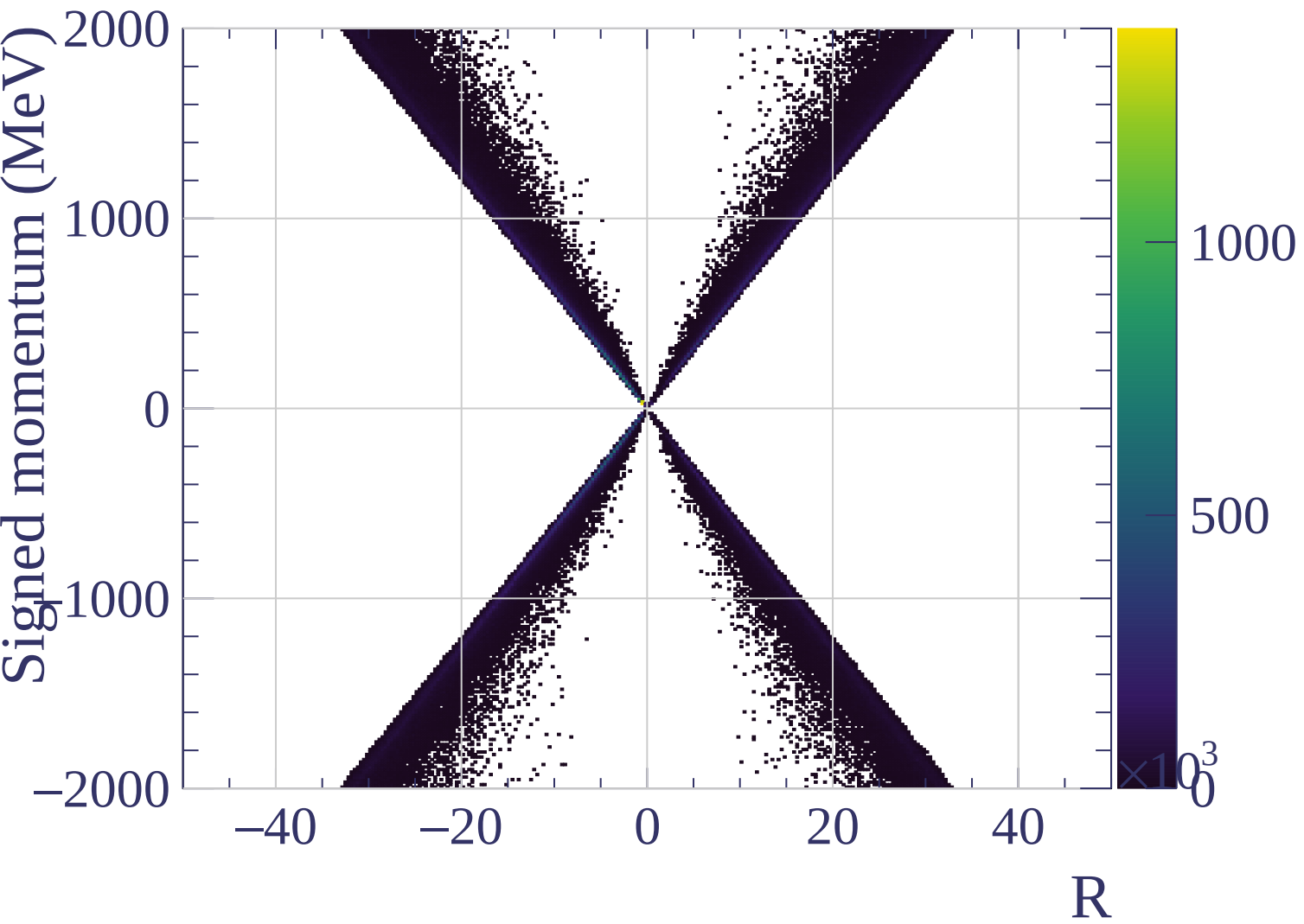


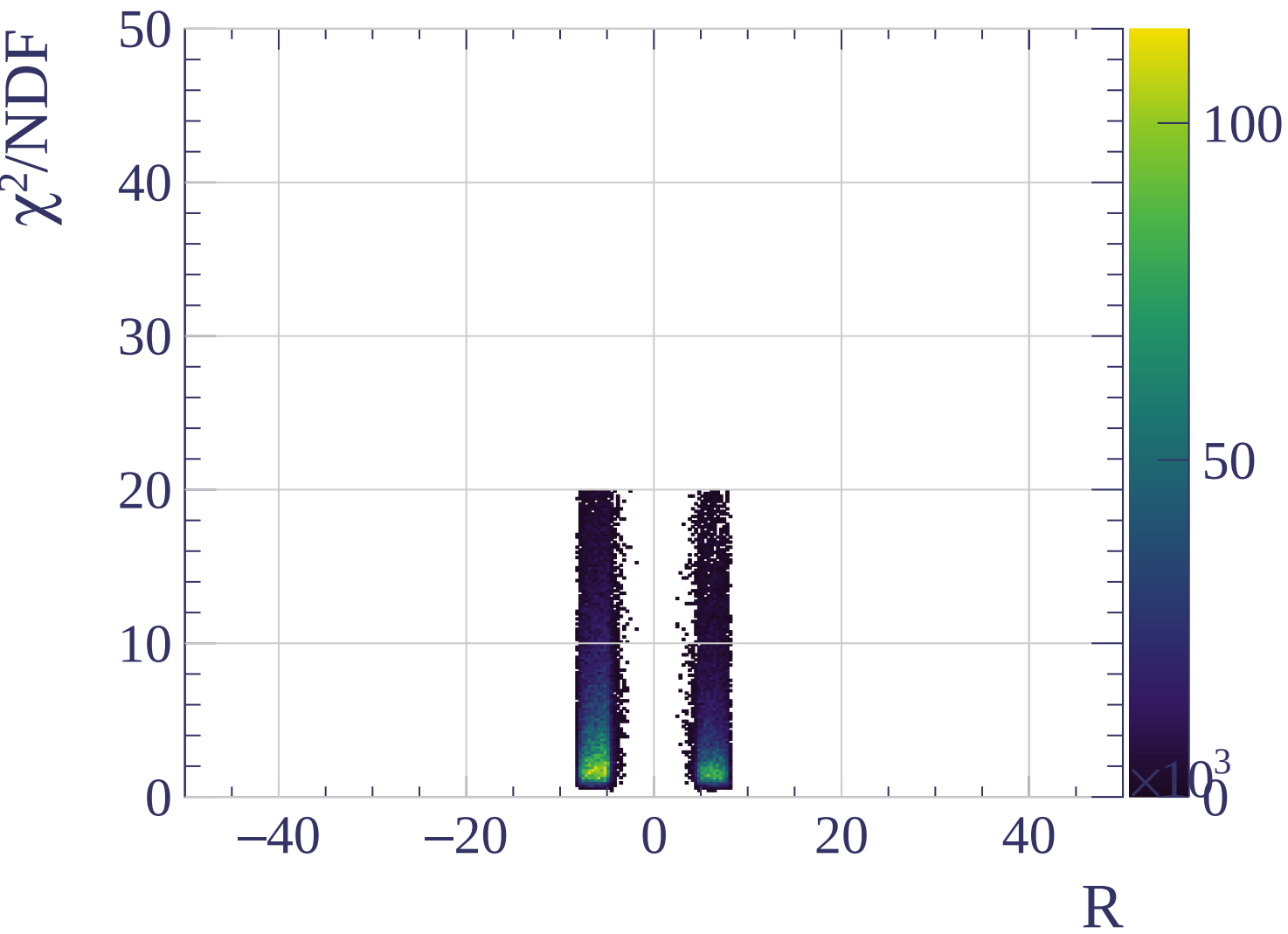
Counts



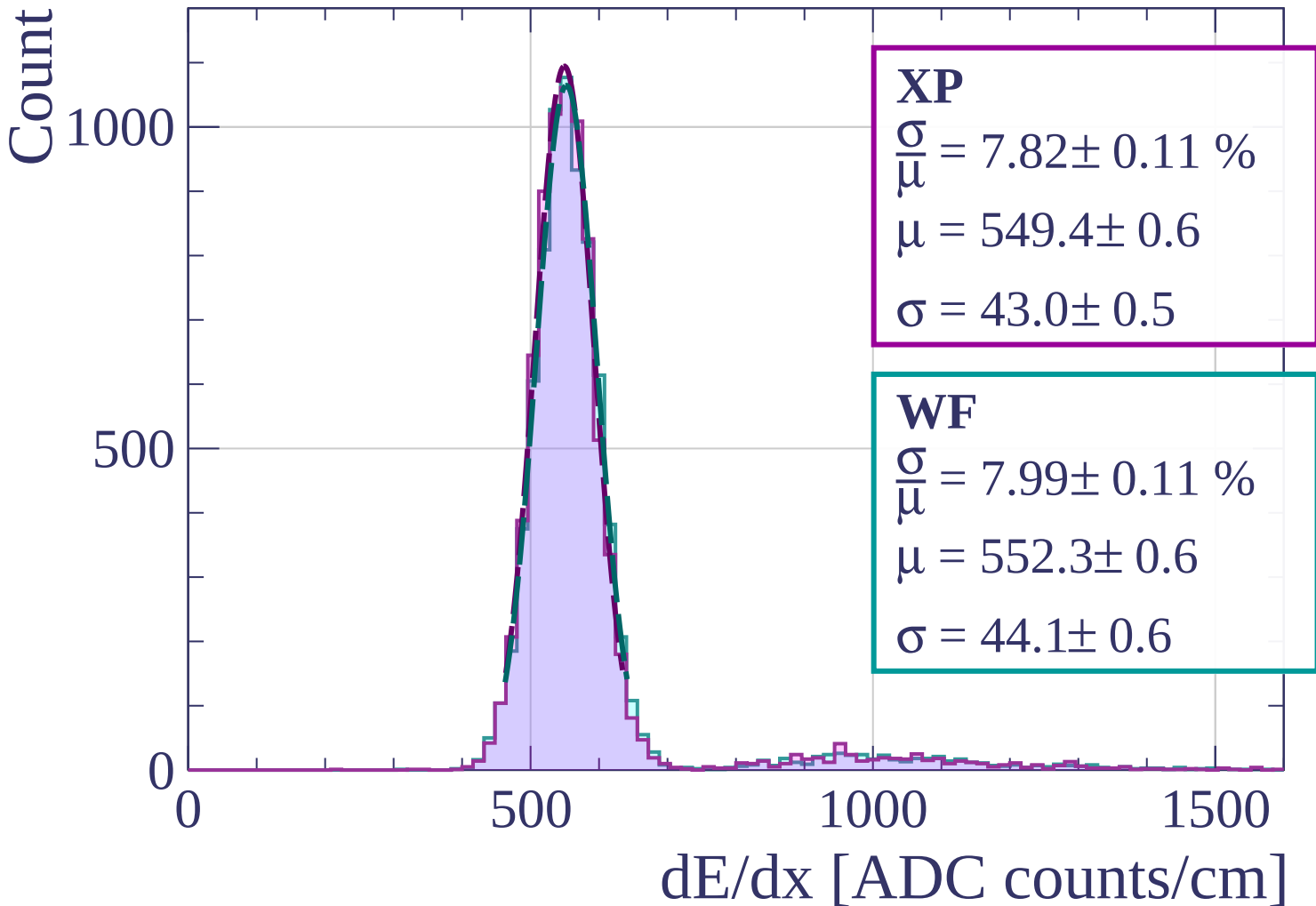






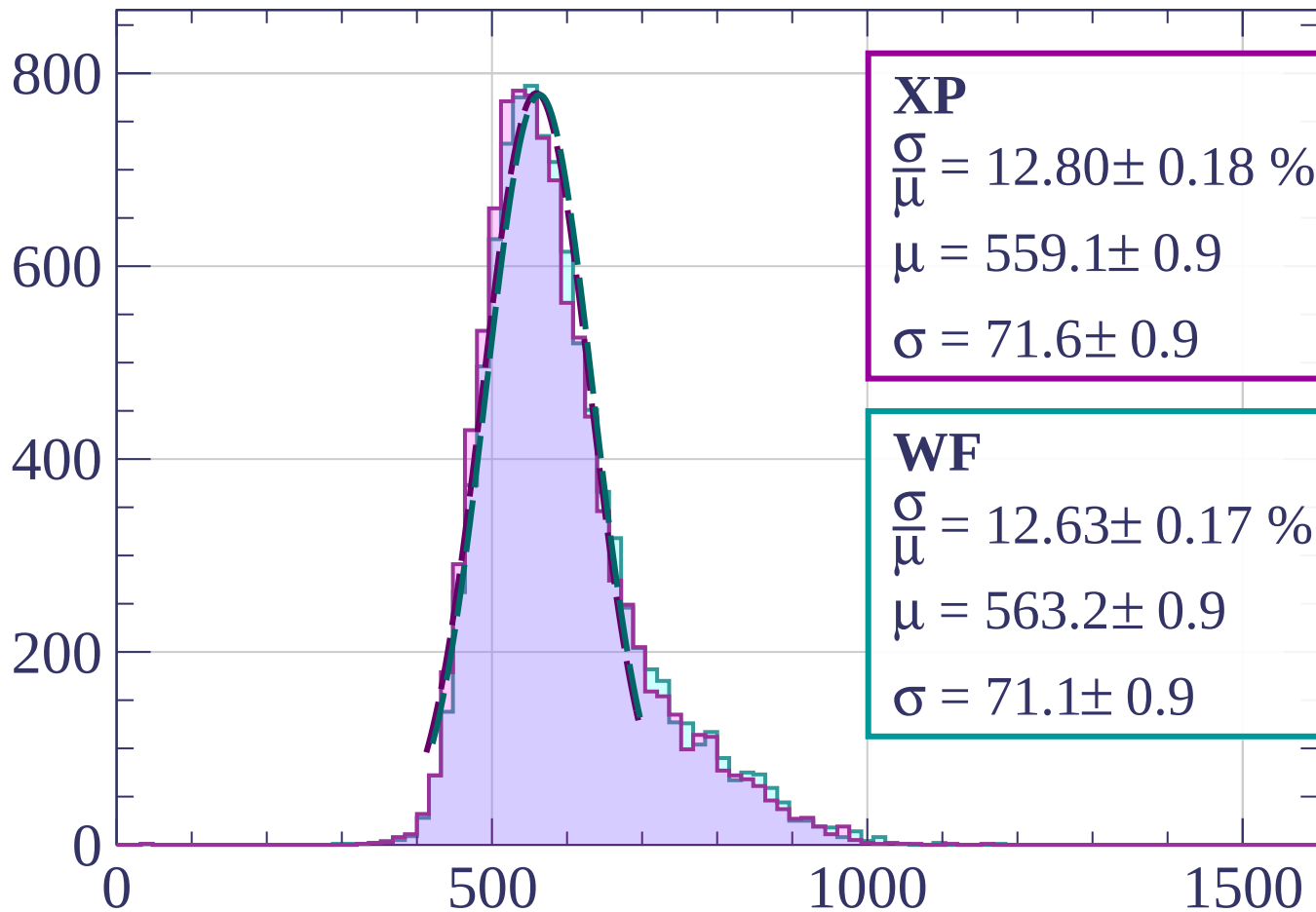


Energy loss | $0 < p < 80$



Energy loss | $80 < p < 160$

Count



XP

$$\frac{\sigma}{\mu} = 12.80 \pm 0.18 \%$$

$$\mu = 559.1 \pm 0.9$$

$$\sigma = 71.6 \pm 0.9$$

WF

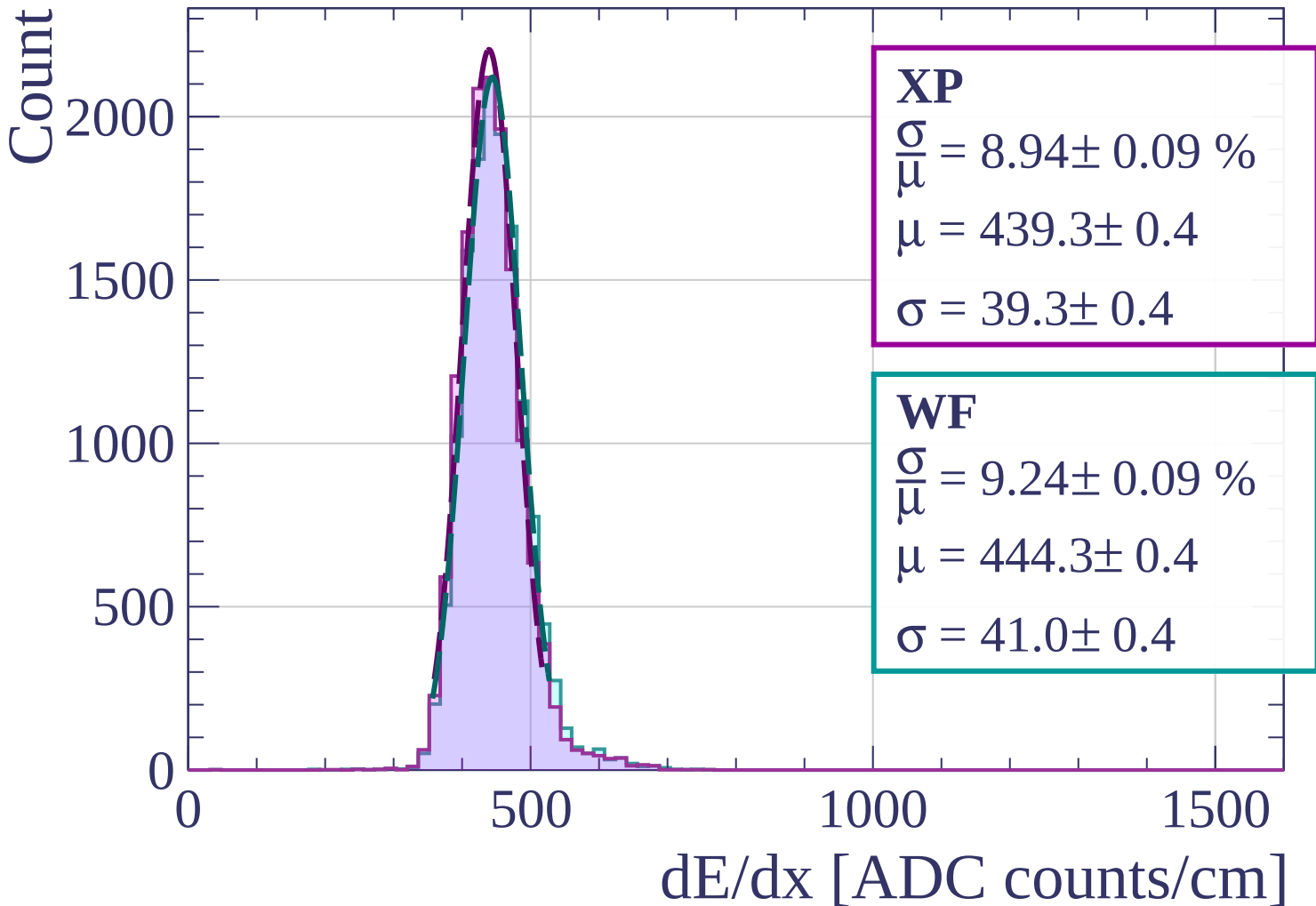
$$\frac{\sigma}{\mu} = 12.63 \pm 0.17 \%$$

$$\mu = 563.2 \pm 0.9$$

$$\sigma = 71.1 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $160 < p < 240$



Energy loss | $240 < p < 320$

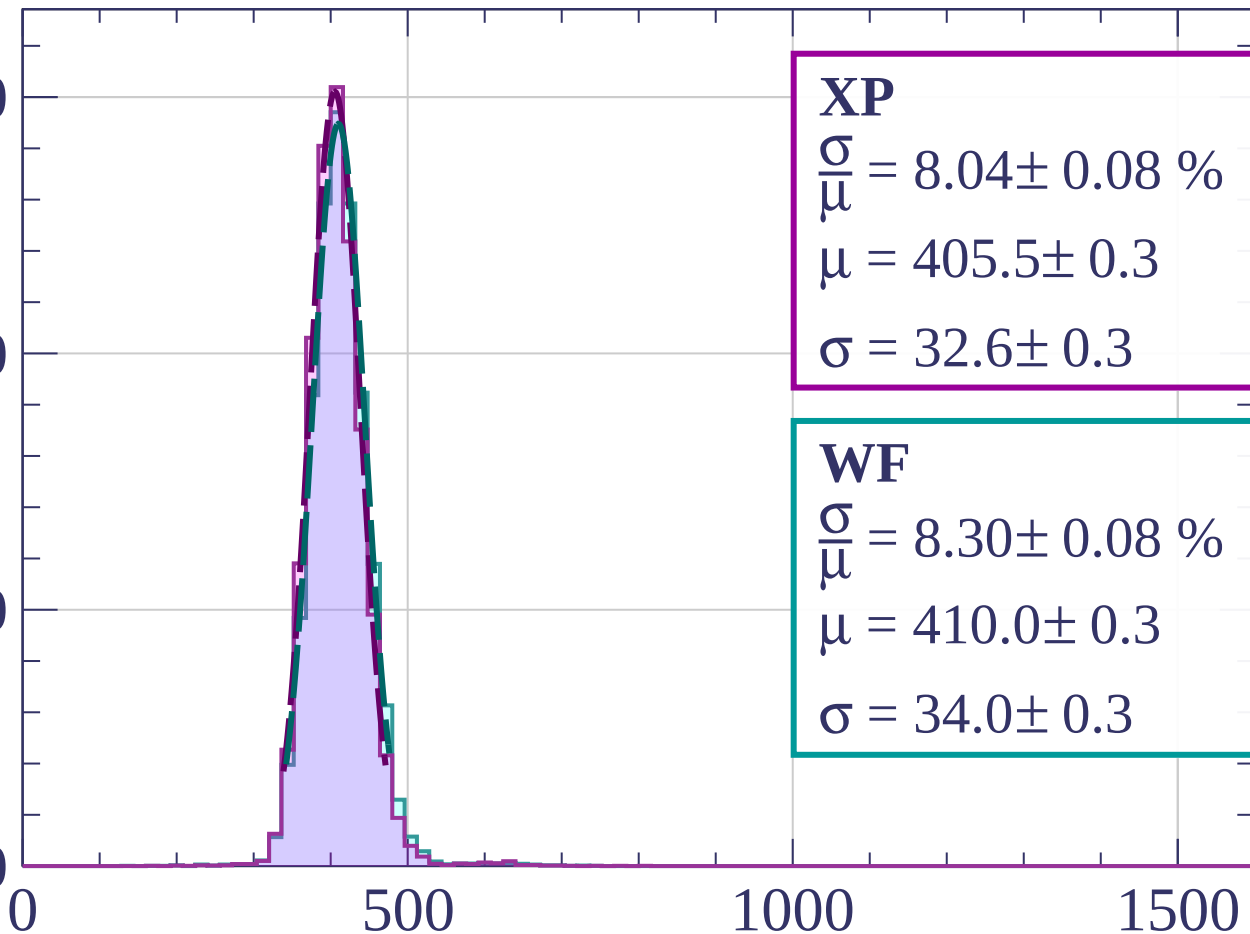
Count

3000

2000

1000

0



dE/dx [ADC counts/cm]

Energy loss | $320 < p < 400$

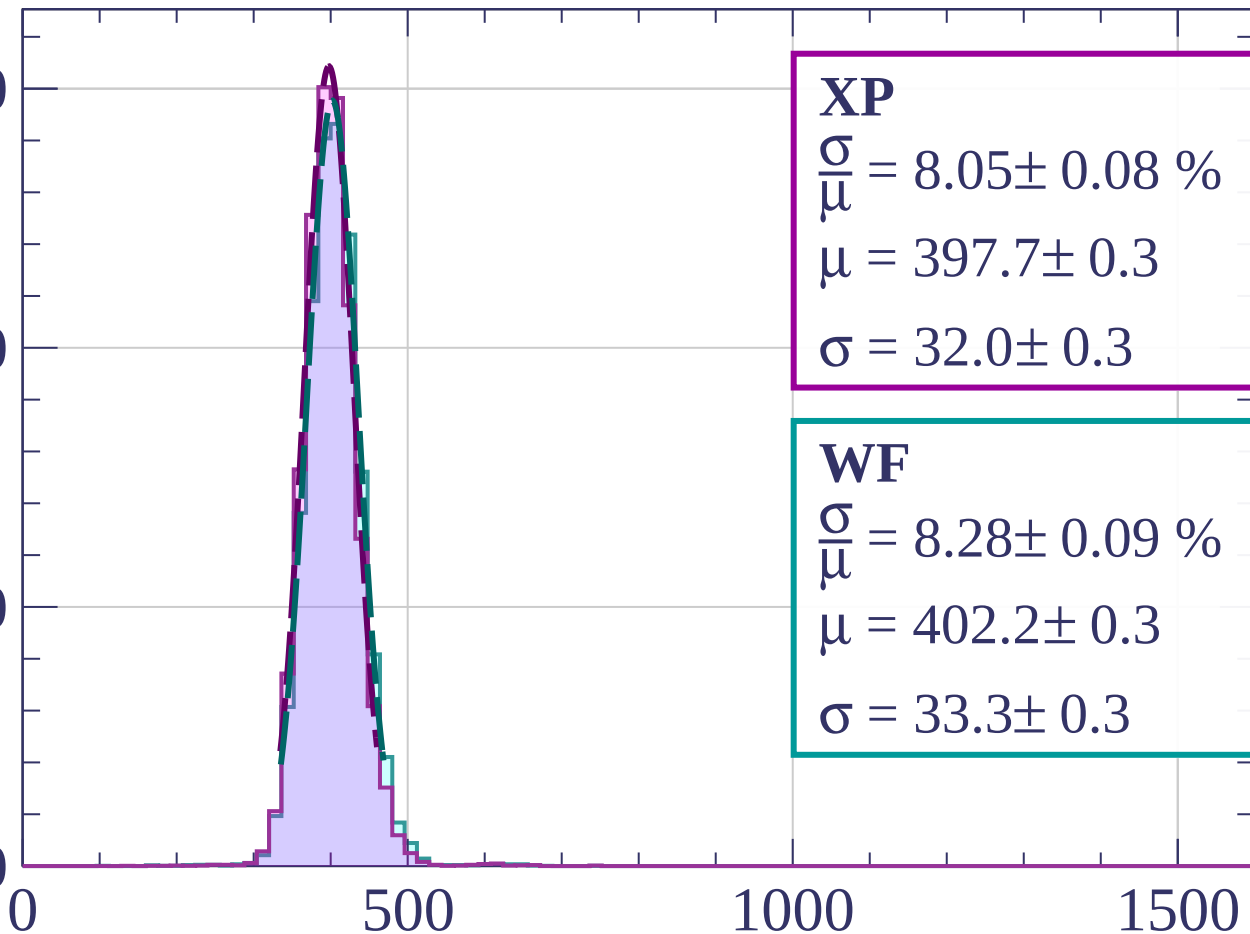
Count

3000

2000

1000

0



XP

$$\frac{\sigma}{\mu} = 8.05 \pm 0.08 \%$$

$$\mu = 397.7 \pm 0.3$$

$$\sigma = 32.0 \pm 0.3$$

WF

$$\frac{\sigma}{\mu} = 8.28 \pm 0.09 \%$$

$$\mu = 402.2 \pm 0.3$$

$$\sigma = 33.3 \pm 0.3$$

dE/dx [ADC counts/cm]

Energy loss | $400 < p < 480$

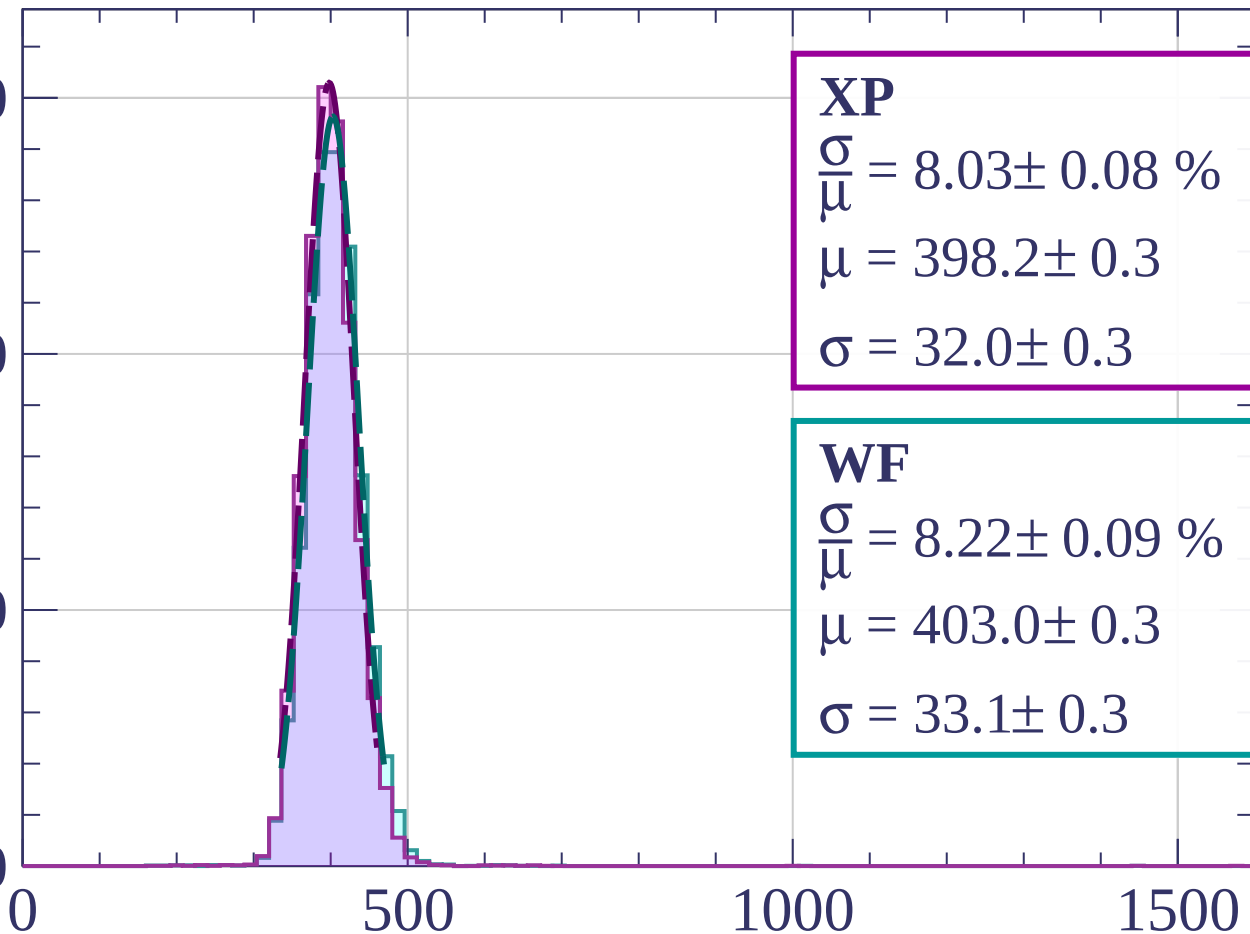
Count

3000

2000

1000

0



XP

$$\frac{\sigma}{\mu} = 8.03 \pm 0.08 \%$$

$$\mu = 398.2 \pm 0.3$$

$$\sigma = 32.0 \pm 0.3$$

WF

$$\frac{\sigma}{\mu} = 8.22 \pm 0.09 \%$$

$$\mu = 403.0 \pm 0.3$$

$$\sigma = 33.1 \pm 0.3$$

dE/dx [ADC counts/cm]

Energy loss | $480 < p < 560$

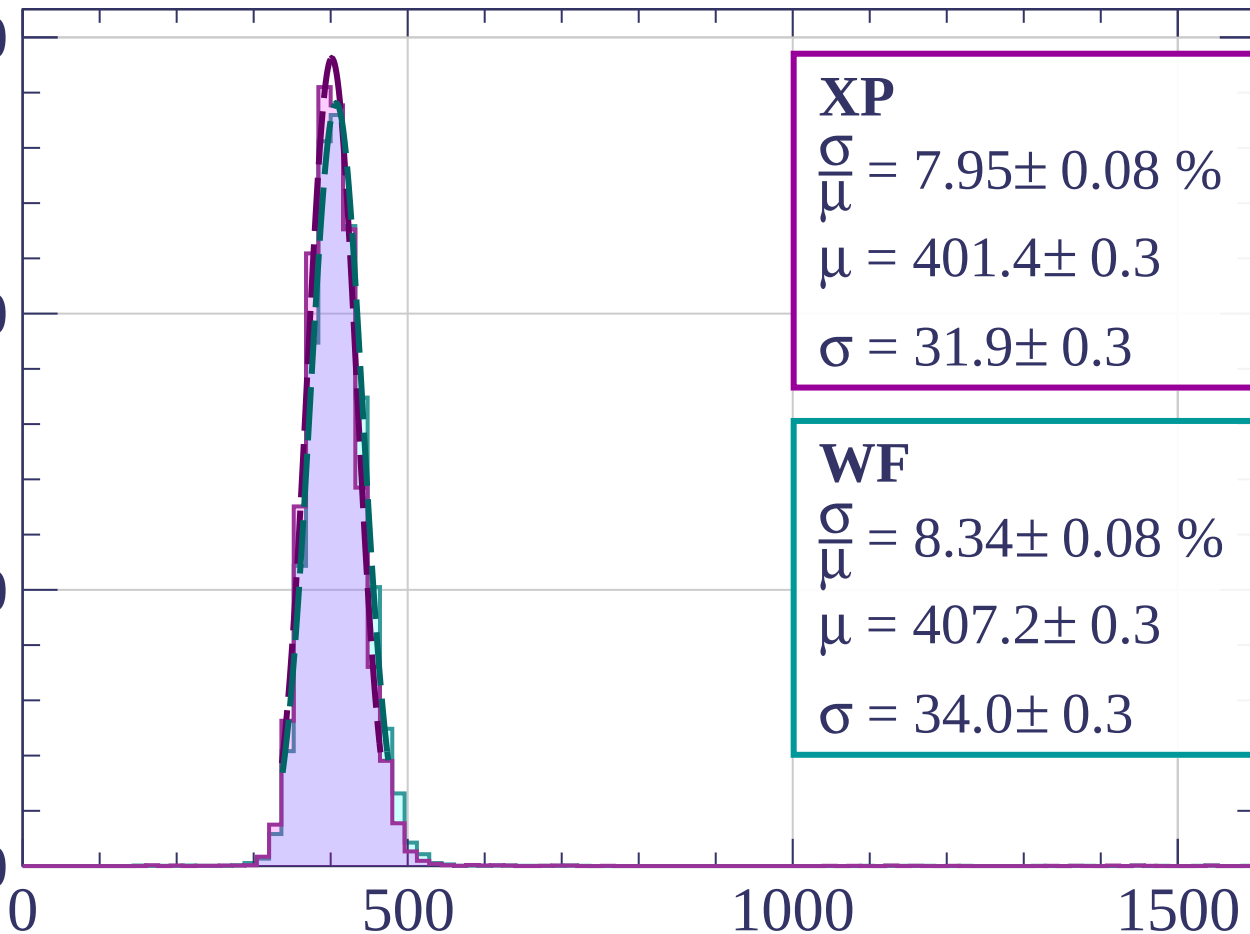
Count

3000

2000

1000

0



XP

$$\frac{\sigma}{\mu} = 7.95 \pm 0.08 \%$$

$$\mu = 401.4 \pm 0.3$$

$$\sigma = 31.9 \pm 0.3$$

WF

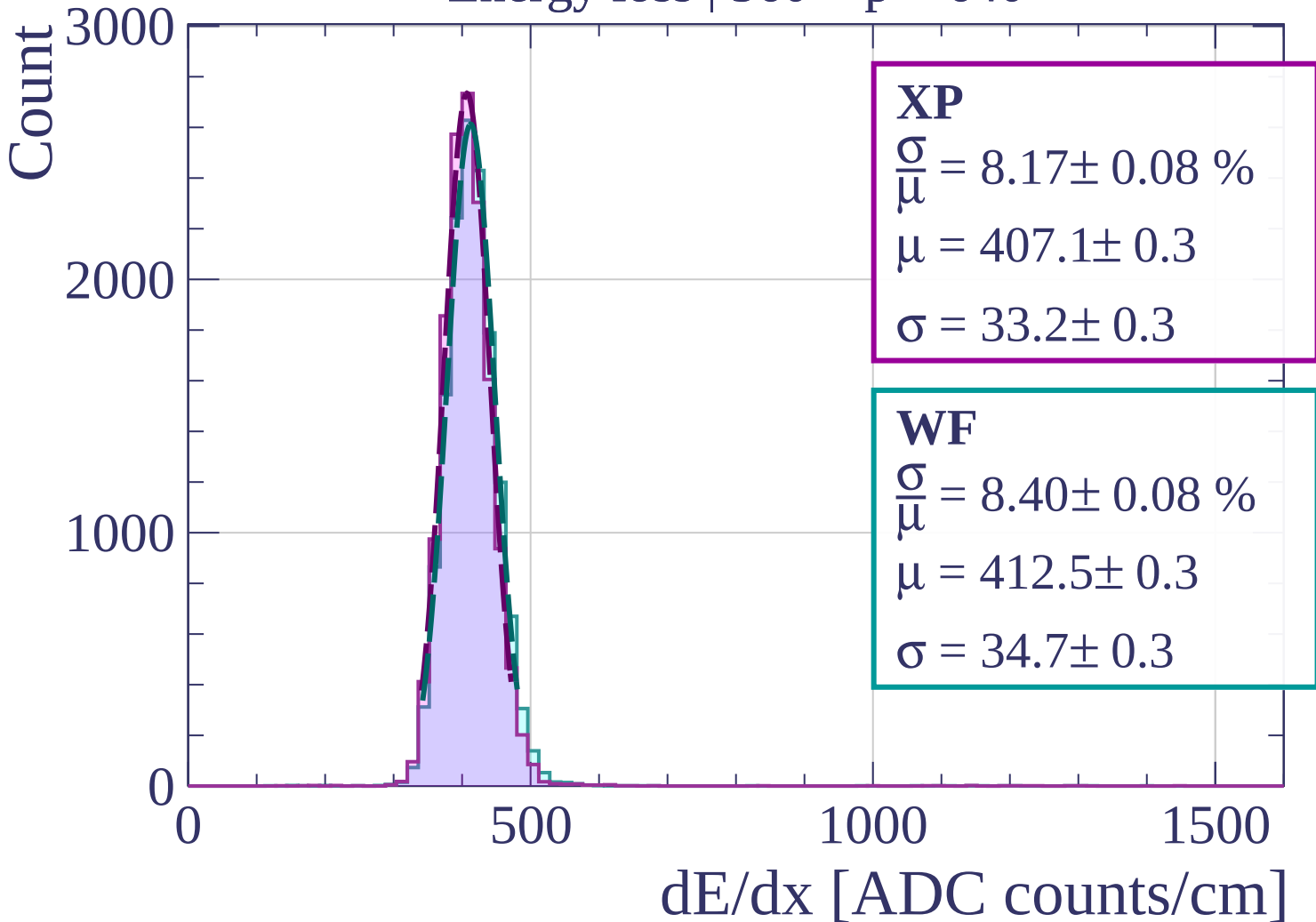
$$\frac{\sigma}{\mu} = 8.34 \pm 0.08 \%$$

$$\mu = 407.2 \pm 0.3$$

$$\sigma = 34.0 \pm 0.3$$

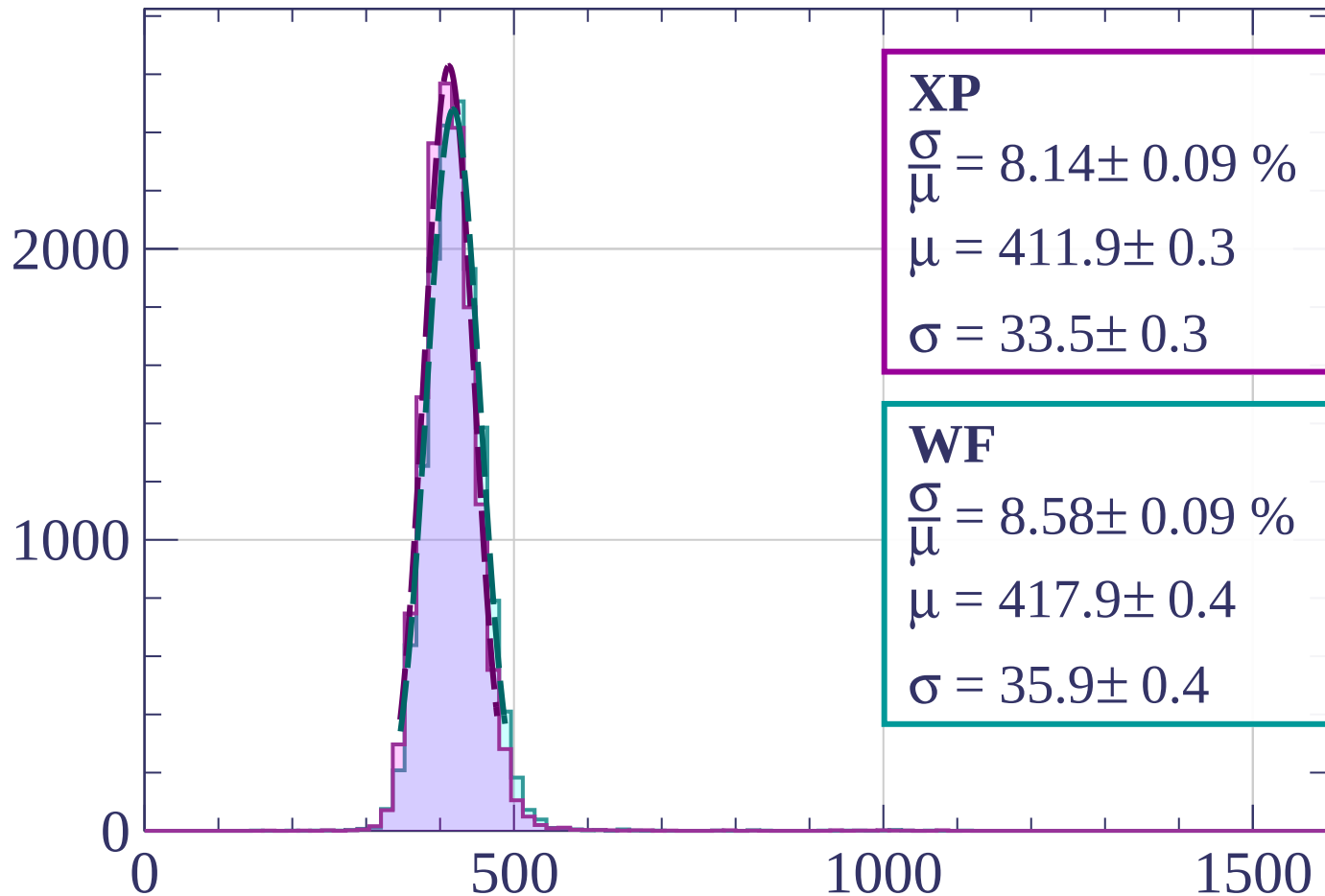
dE/dx [ADC counts/cm]

Energy loss | $560 < p < 640$



Energy loss | $640 < p < 720$

Count



XP

$$\frac{\sigma}{\mu} = 8.14 \pm 0.09 \%$$

$$\mu = 411.9 \pm 0.3$$

$$\sigma = 33.5 \pm 0.3$$

WF

$$\frac{\sigma}{\mu} = 8.58 \pm 0.09 \%$$

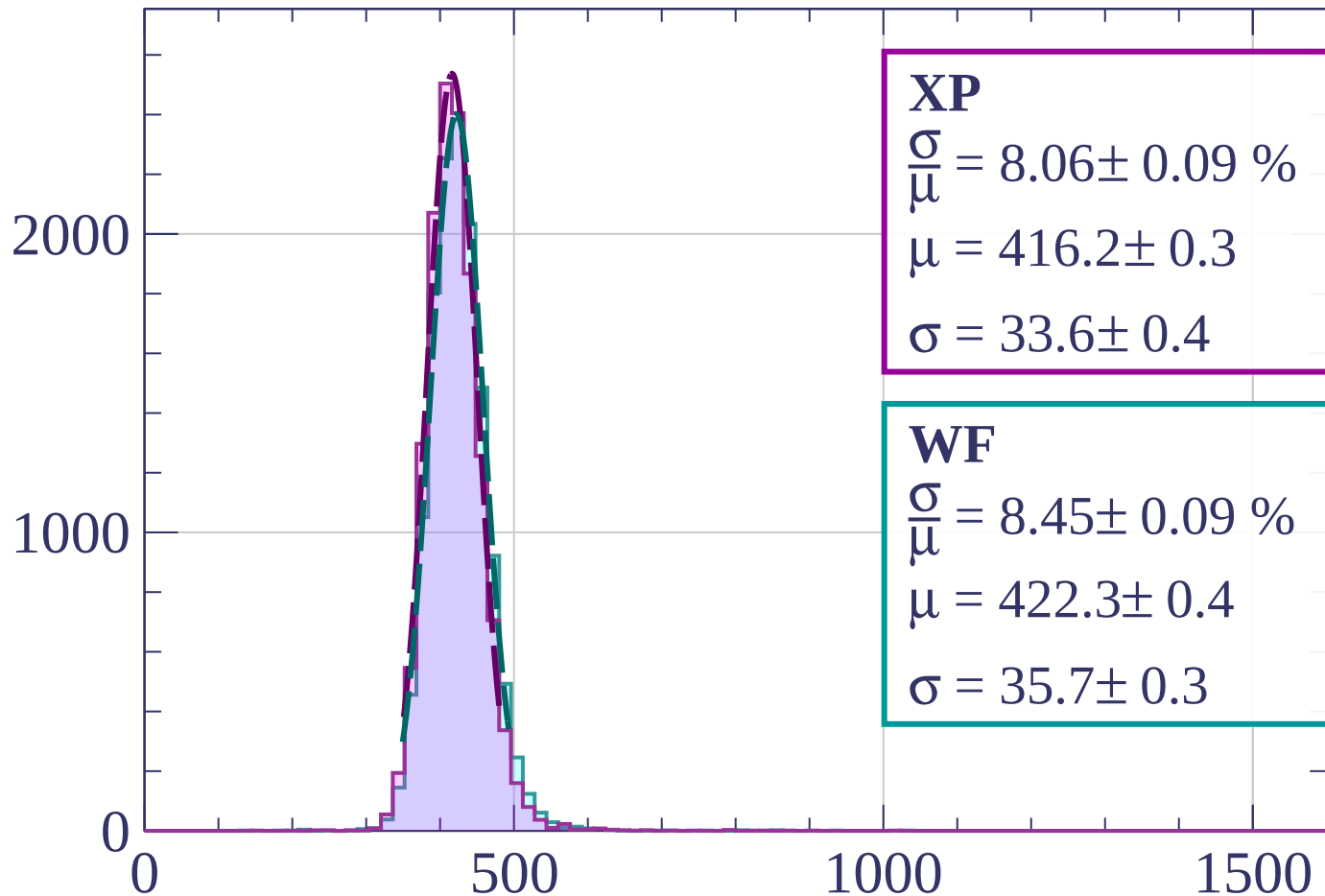
$$\mu = 417.9 \pm 0.4$$

$$\sigma = 35.9 \pm 0.4$$

dE/dx [ADC counts/cm]

Energy loss | $720 < p < 800$

Count



XP

$$\frac{\sigma}{\mu} = 8.06 \pm 0.09 \%$$

$$\mu = 416.2 \pm 0.3$$

$$\sigma = 33.6 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 8.45 \pm 0.09 \%$$

$$\mu = 422.3 \pm 0.4$$

$$\sigma = 35.7 \pm 0.3$$

dE/dx [ADC counts/cm]

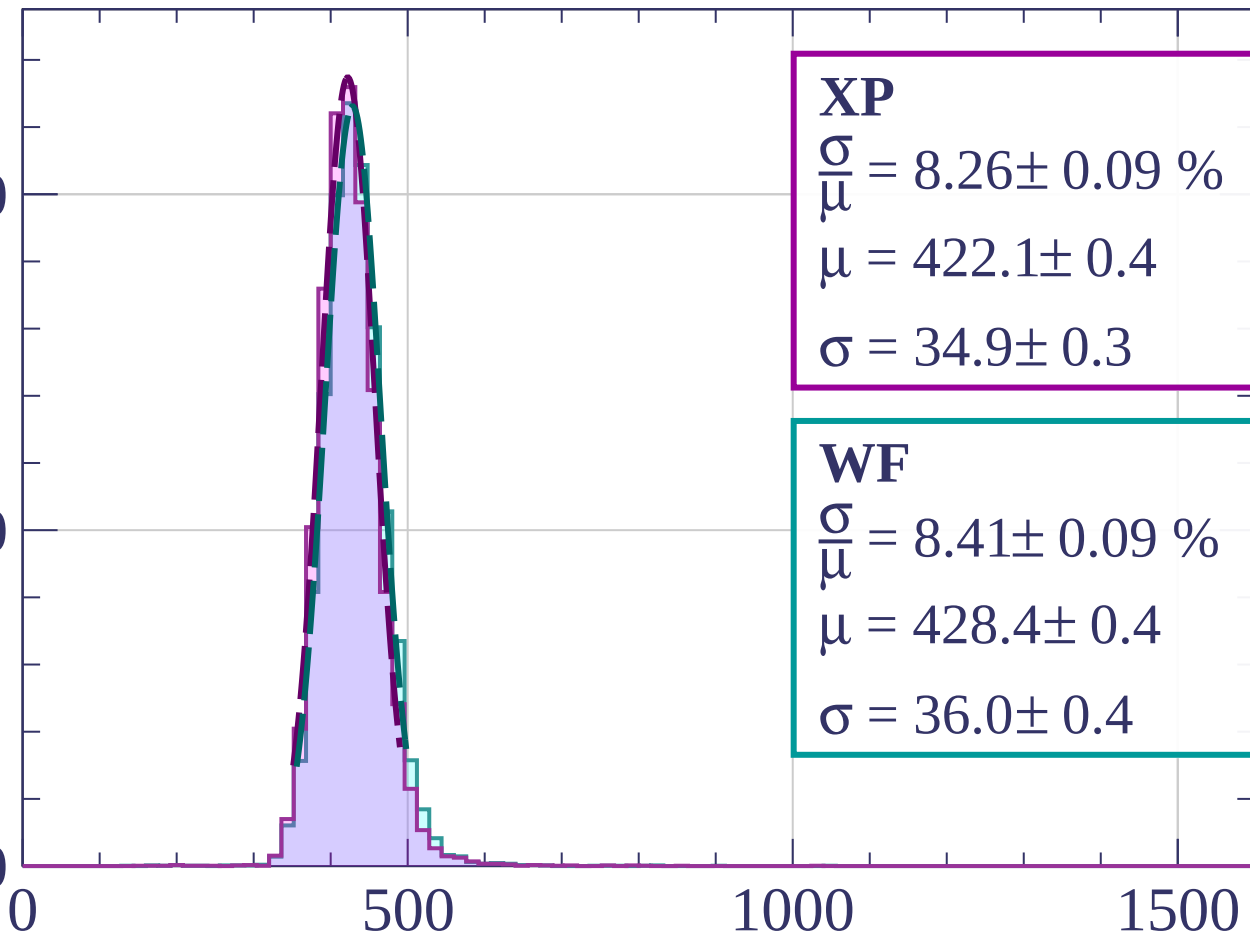
Energy loss | $800 < p < 880$

Count

2000

1000

0



XP

$$\frac{\sigma}{\mu} = 8.26 \pm 0.09 \%$$

$$\mu = 422.1 \pm 0.4$$

$$\sigma = 34.9 \pm 0.3$$

WF

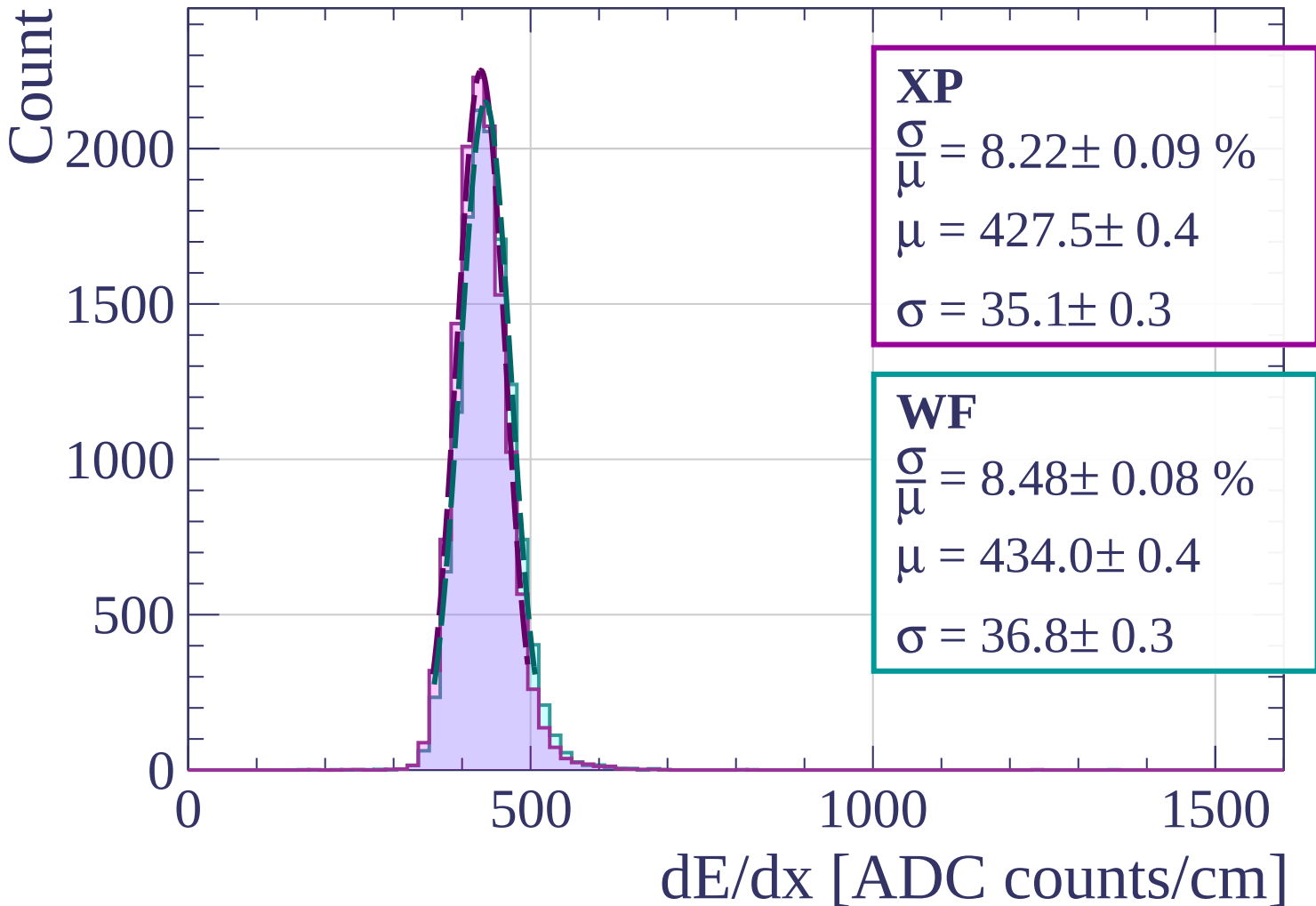
$$\frac{\sigma}{\mu} = 8.41 \pm 0.09 \%$$

$$\mu = 428.4 \pm 0.4$$

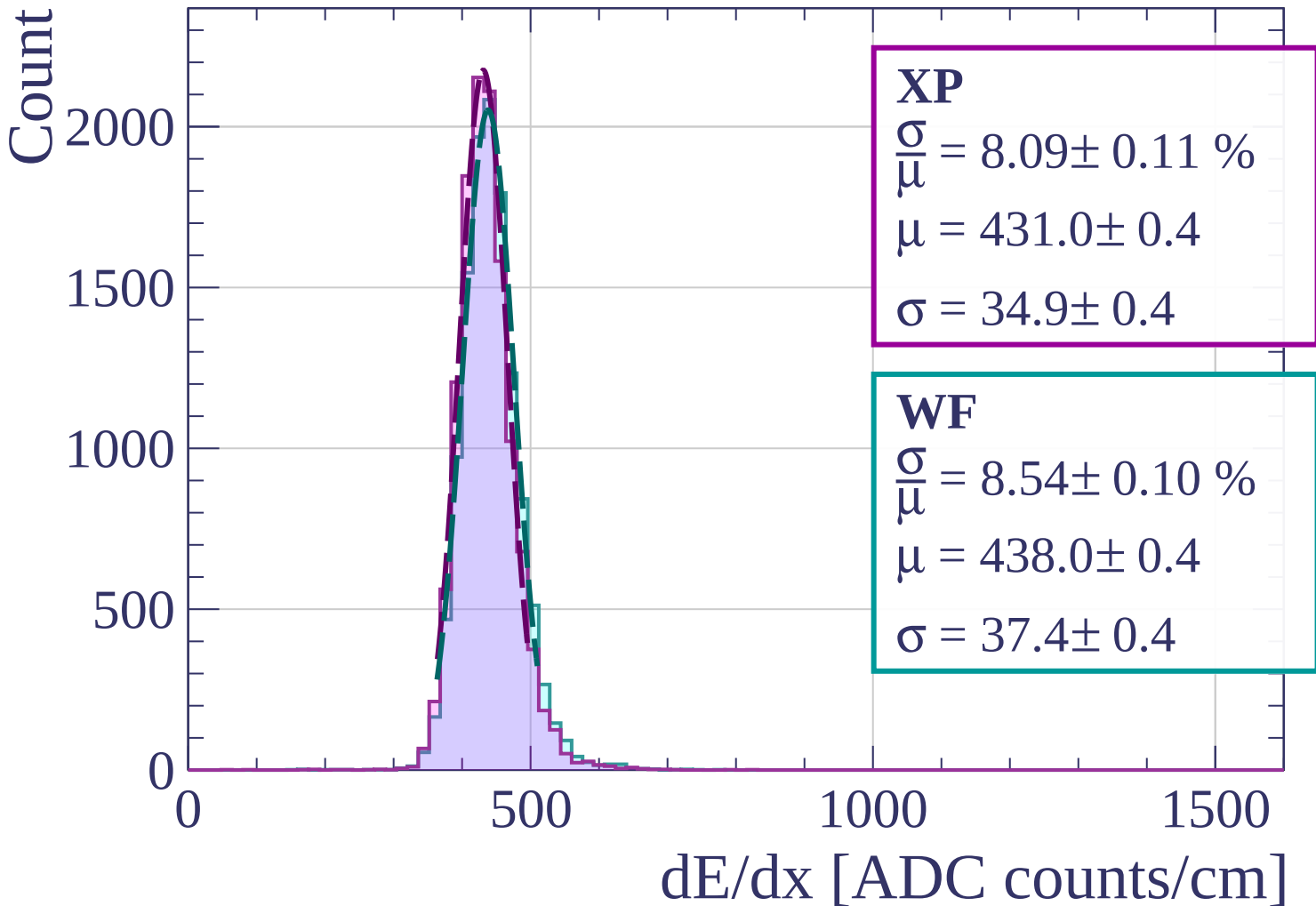
$$\sigma = 36.0 \pm 0.4$$

dE/dx [ADC counts/cm]

Energy loss | $880 < p < 960$



Energy loss | $960 < p < 1040$



Energy loss | $1040 < p < 1120$

Count

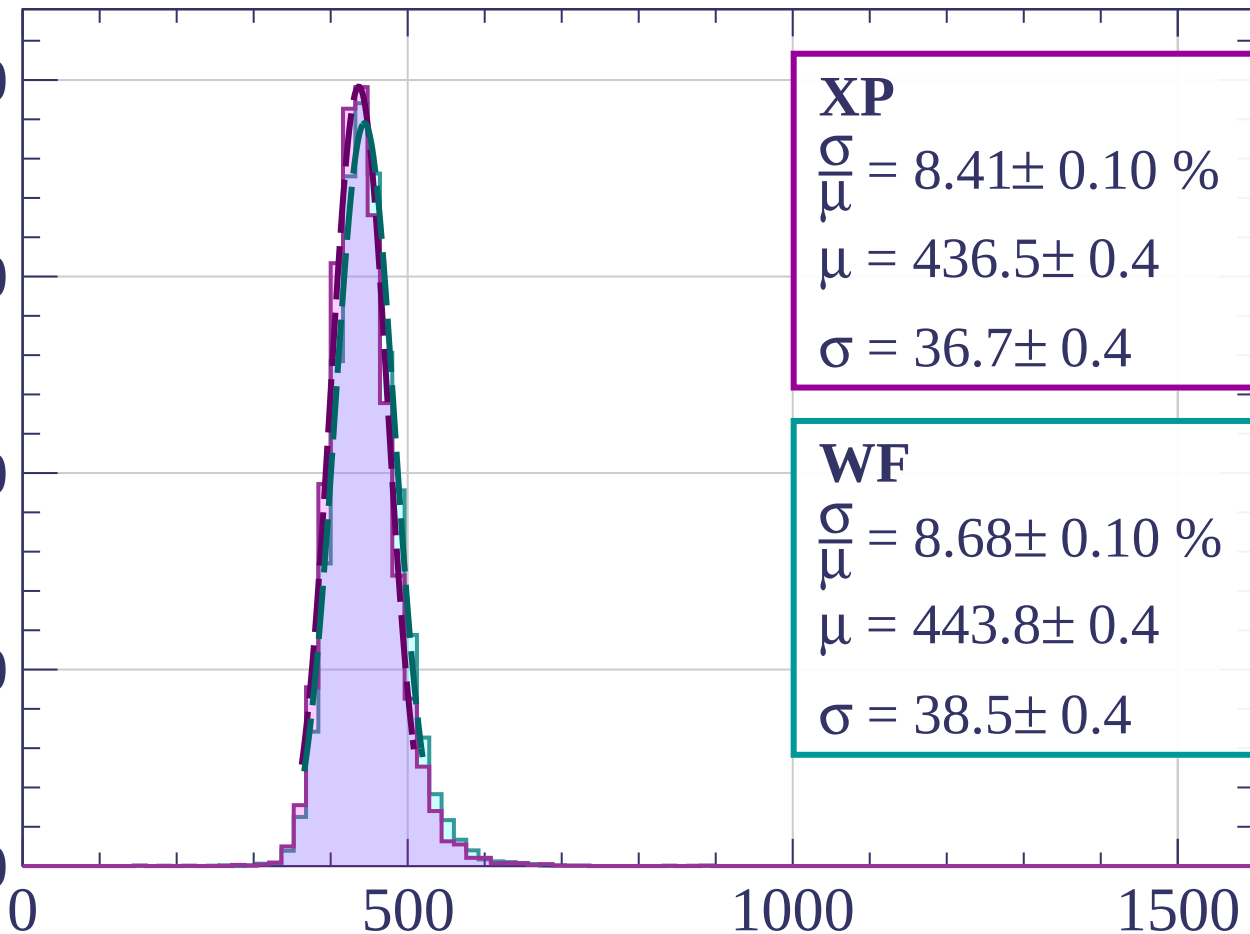
2000

1500

1000

500

0



XP

$$\frac{\sigma}{\mu} = 8.41 \pm 0.10 \%$$

$$\mu = 436.5 \pm 0.4$$

$$\sigma = 36.7 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 8.68 \pm 0.10 \%$$

$$\mu = 443.8 \pm 0.4$$

$$\sigma = 38.5 \pm 0.4$$

dE/dx [ADC counts/cm]

Energy loss | $1120 < p < 1200$

Count

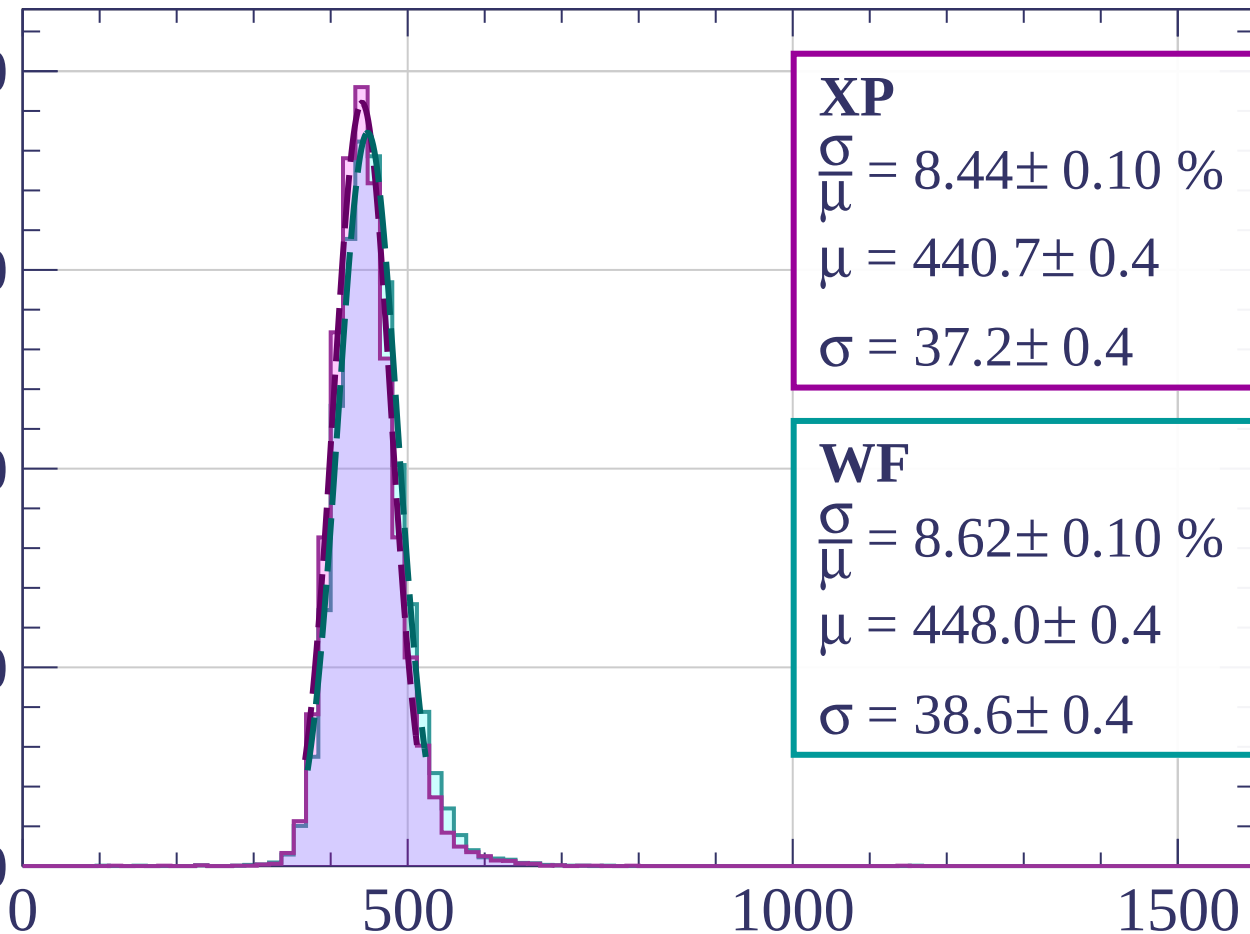
2000

1500

1000

500

0



XP

$$\frac{\sigma}{\mu} = 8.44 \pm 0.10 \%$$

$$\mu = 440.7 \pm 0.4$$

$$\sigma = 37.2 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 8.62 \pm 0.10 \%$$

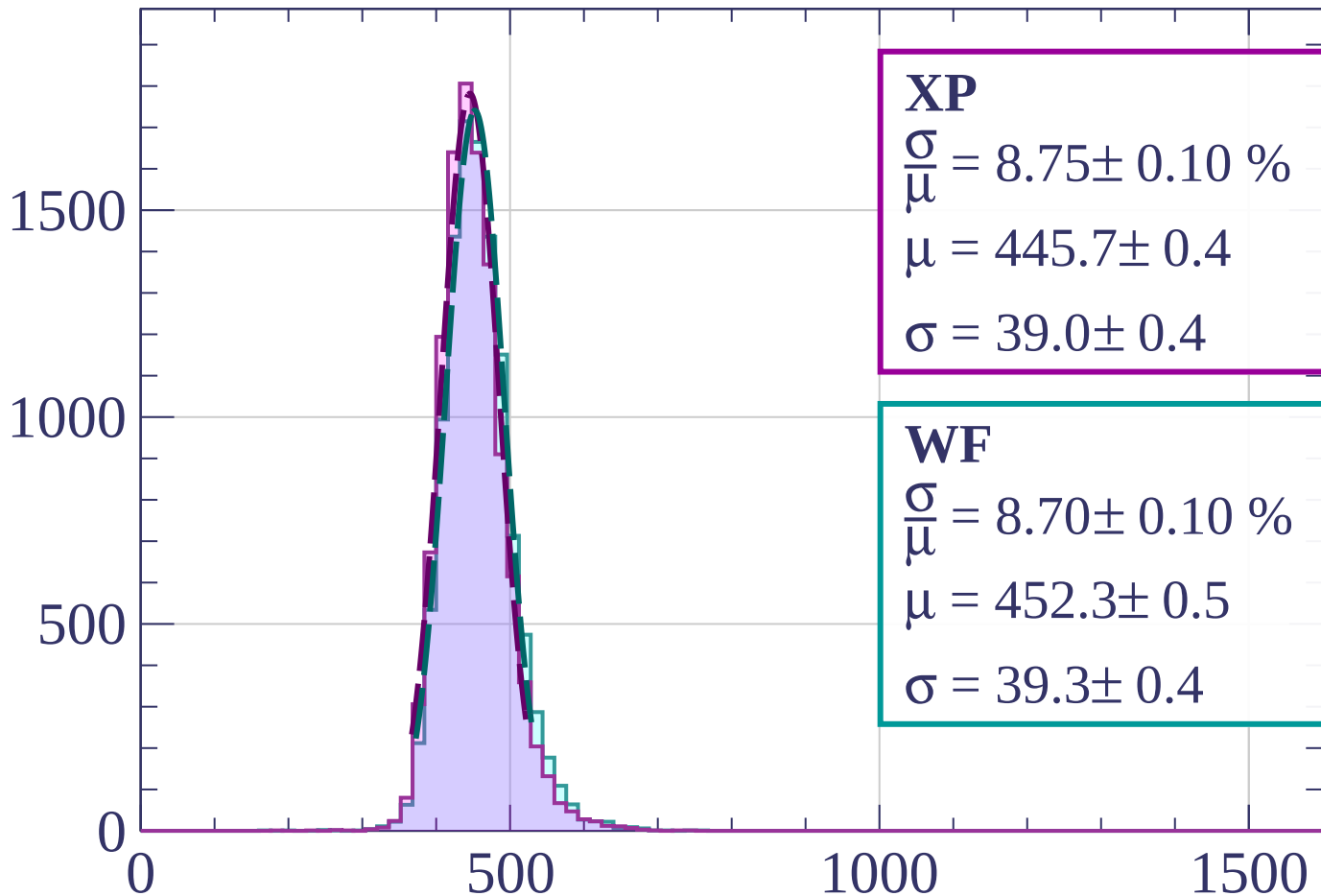
$$\mu = 448.0 \pm 0.4$$

$$\sigma = 38.6 \pm 0.4$$

dE/dx [ADC counts/cm]

Energy loss | $1200 < p < 1280$

Count



XP

$$\frac{\sigma}{\mu} = 8.75 \pm 0.10 \%$$

$$\mu = 445.7 \pm 0.4$$

$$\sigma = 39.0 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 8.70 \pm 0.10 \%$$

$$\mu = 452.3 \pm 0.5$$

$$\sigma = 39.3 \pm 0.4$$

dE/dx [ADC counts/cm]

Energy loss | $1280 < p < 1360$

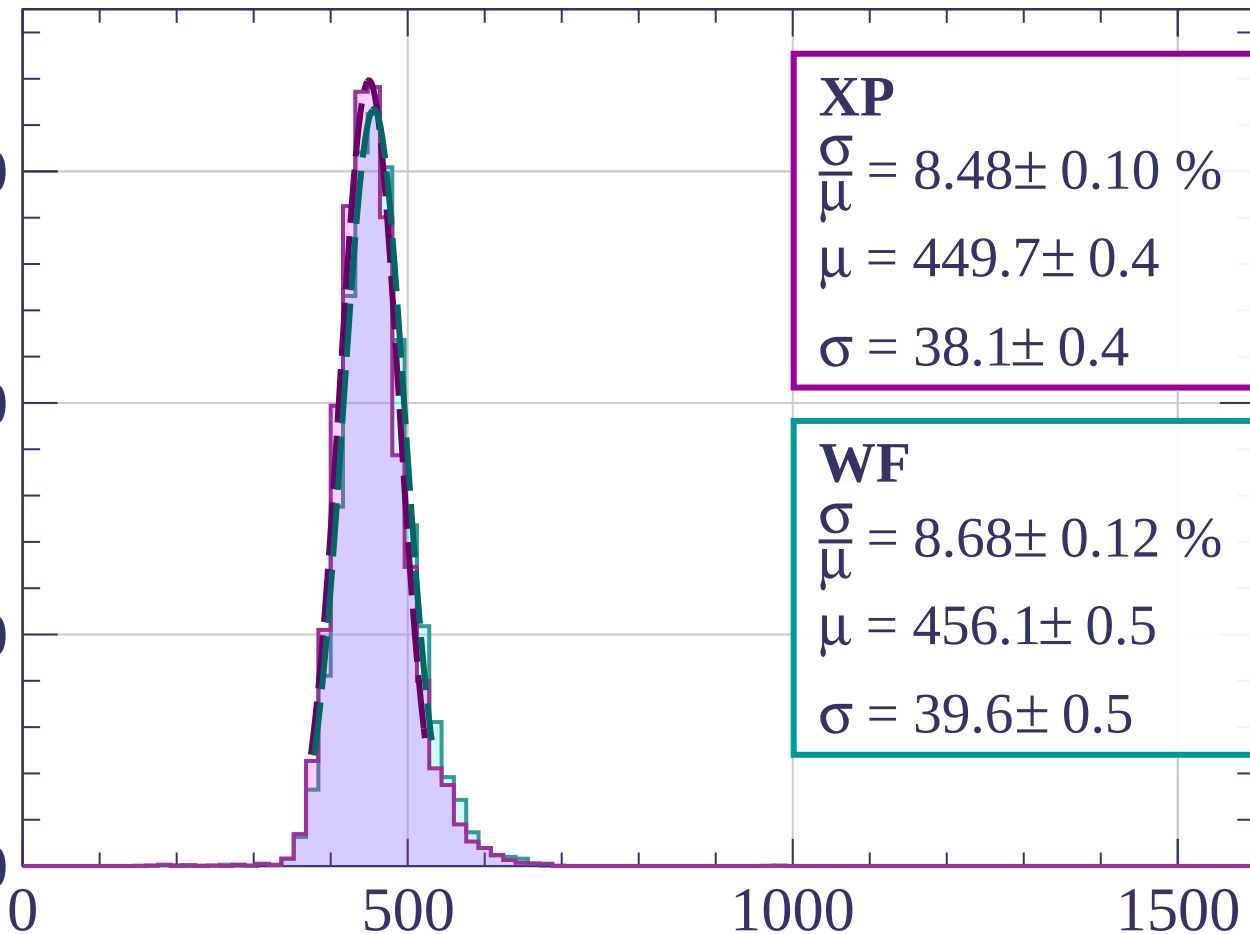
Count

1500

1000

500

0



XP

$$\frac{\sigma}{\mu} = 8.48 \pm 0.10 \%$$

$$\mu = 449.7 \pm 0.4$$

$$\sigma = 38.1 \pm 0.4$$

WF

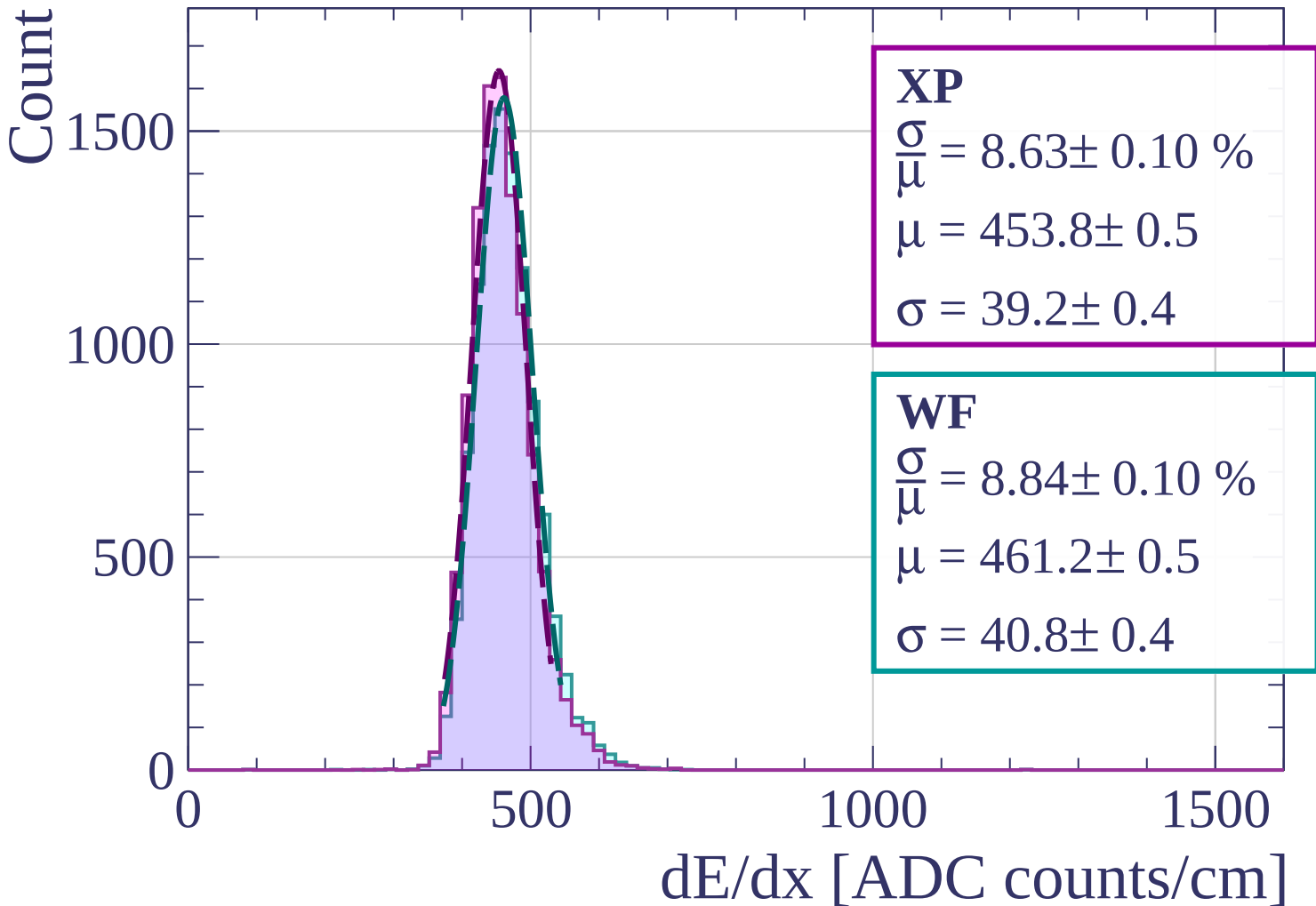
$$\frac{\sigma}{\mu} = 8.68 \pm 0.12 \%$$

$$\mu = 456.1 \pm 0.5$$

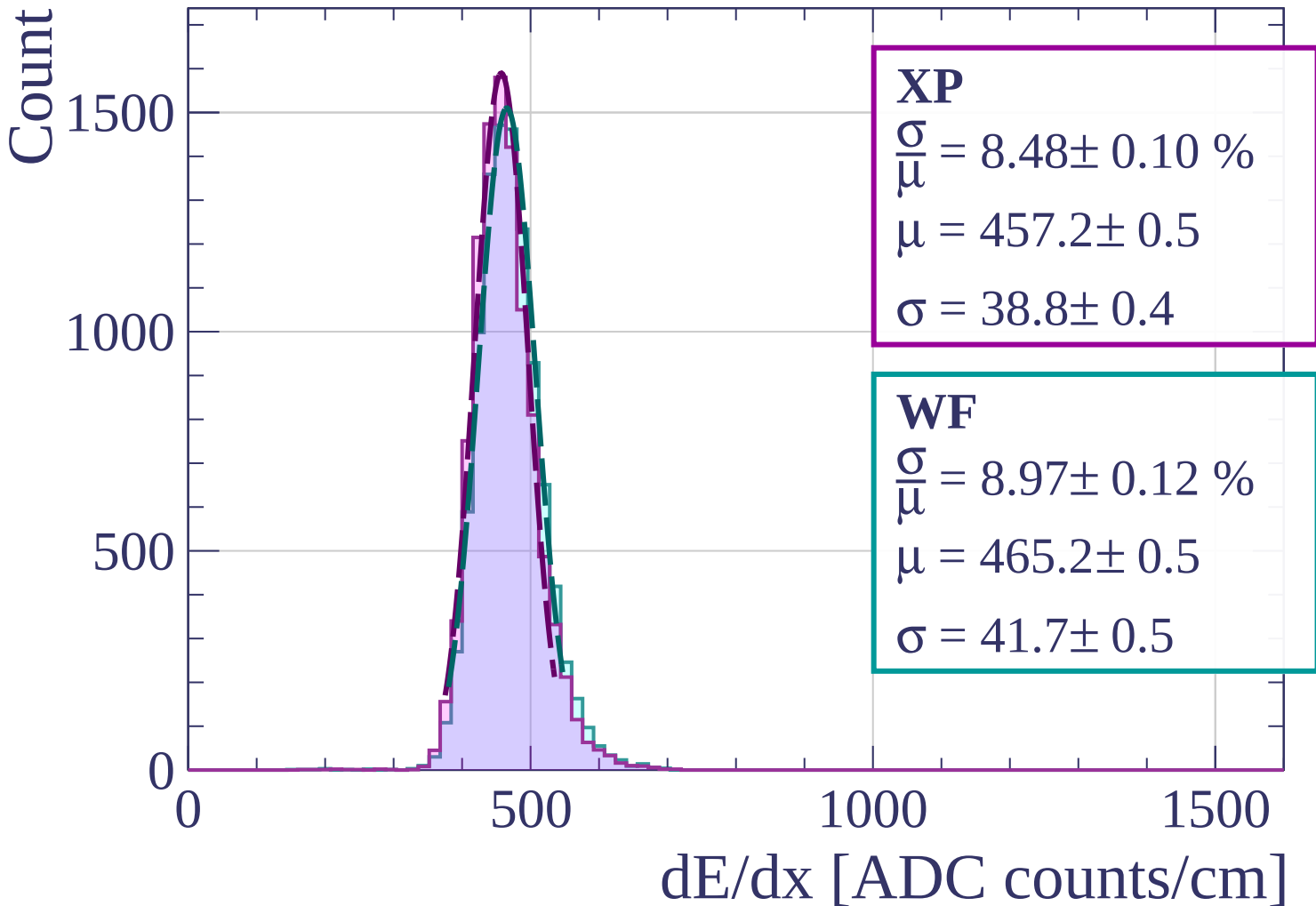
$$\sigma = 39.6 \pm 0.5$$

dE/dx [ADC counts/cm]

Energy loss | $1360 < p < 1440$



Energy loss | $1440 < p < 1520$



Energy loss | $1520 < p < 1600$

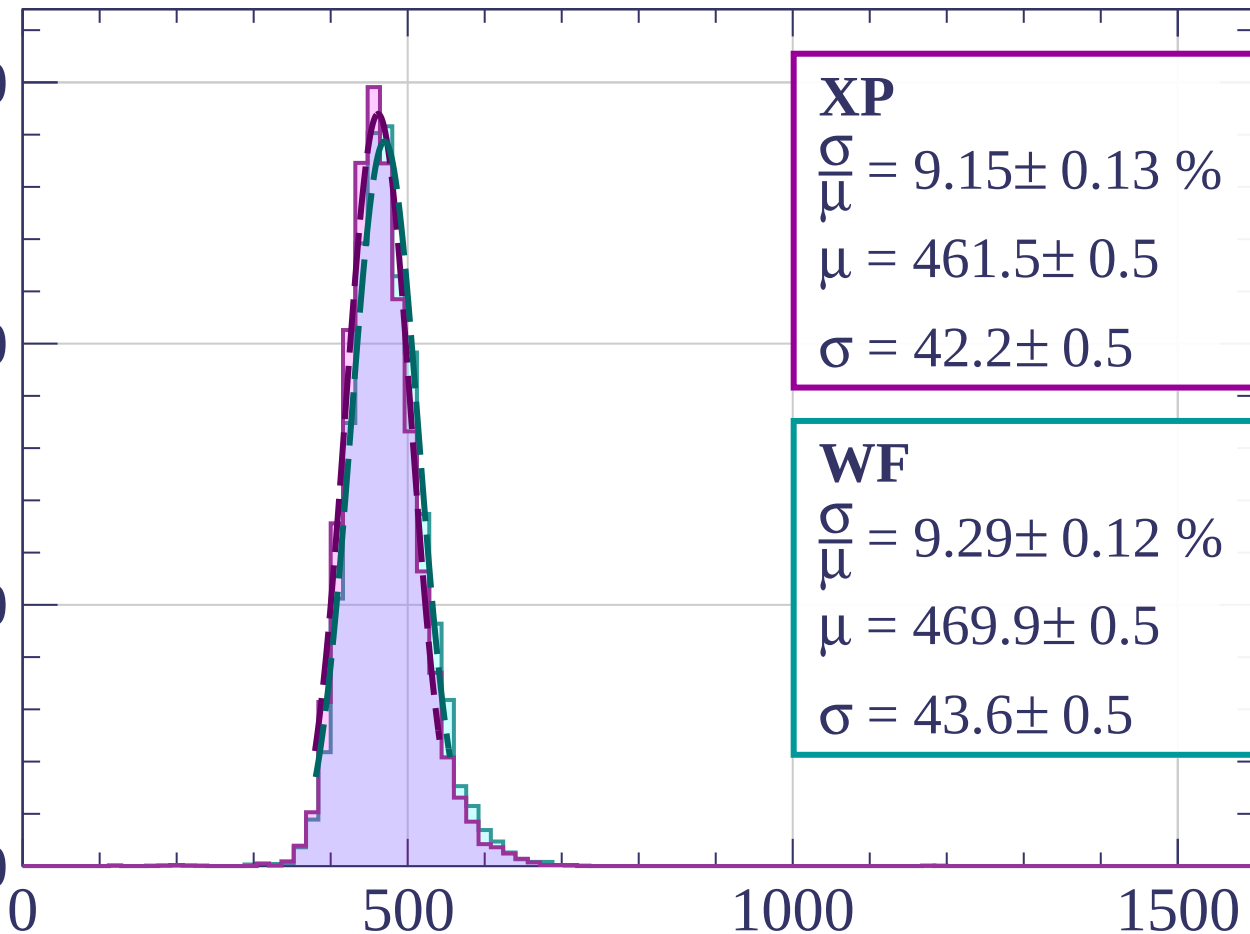
Count

1500

1000

500

0



XP

$$\frac{\sigma}{\mu} = 9.15 \pm 0.13 \%$$

$$\mu = 461.5 \pm 0.5$$

$$\sigma = 42.2 \pm 0.5$$

WF

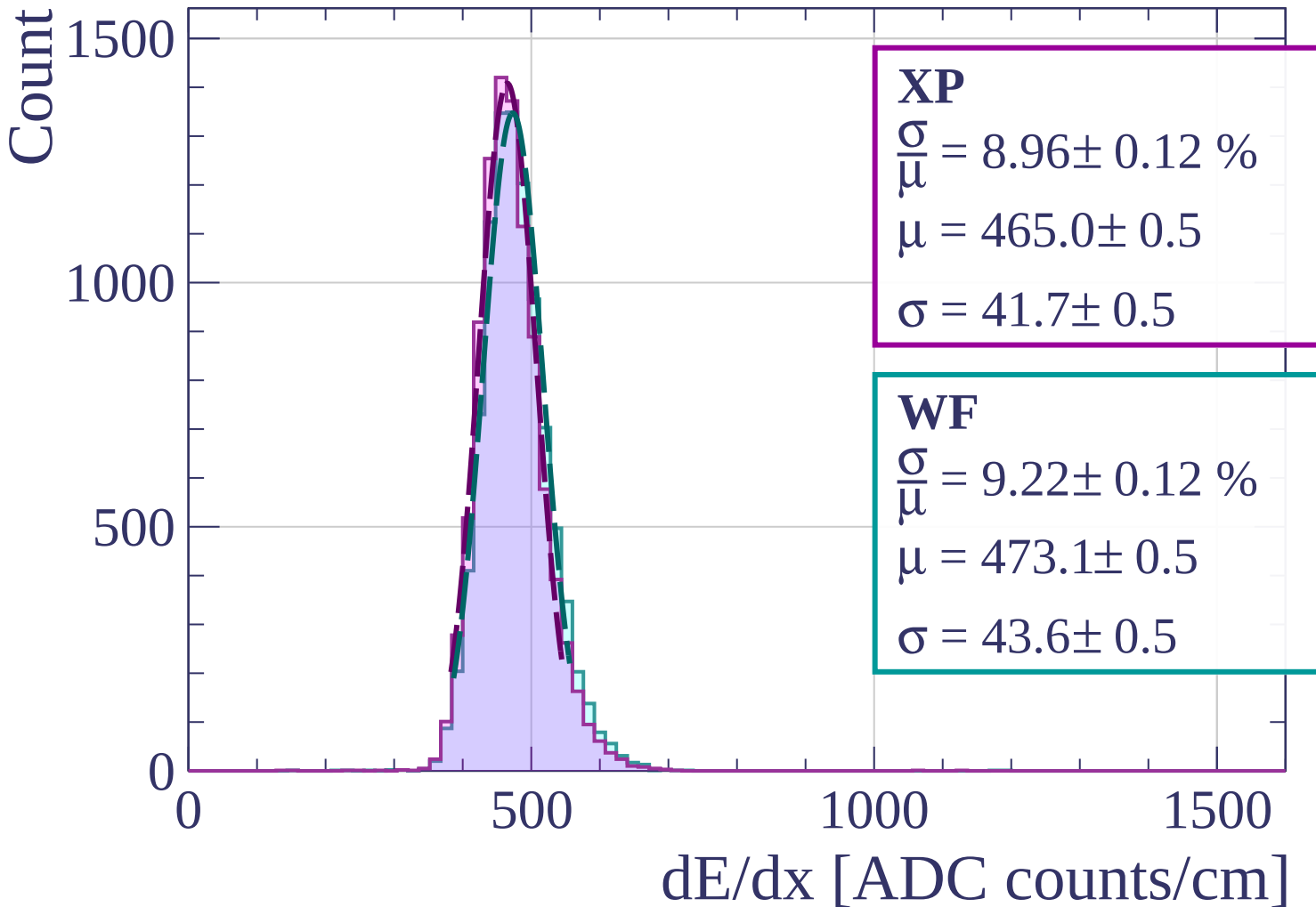
$$\frac{\sigma}{\mu} = 9.29 \pm 0.12 \%$$

$$\mu = 469.9 \pm 0.5$$

$$\sigma = 43.6 \pm 0.5$$

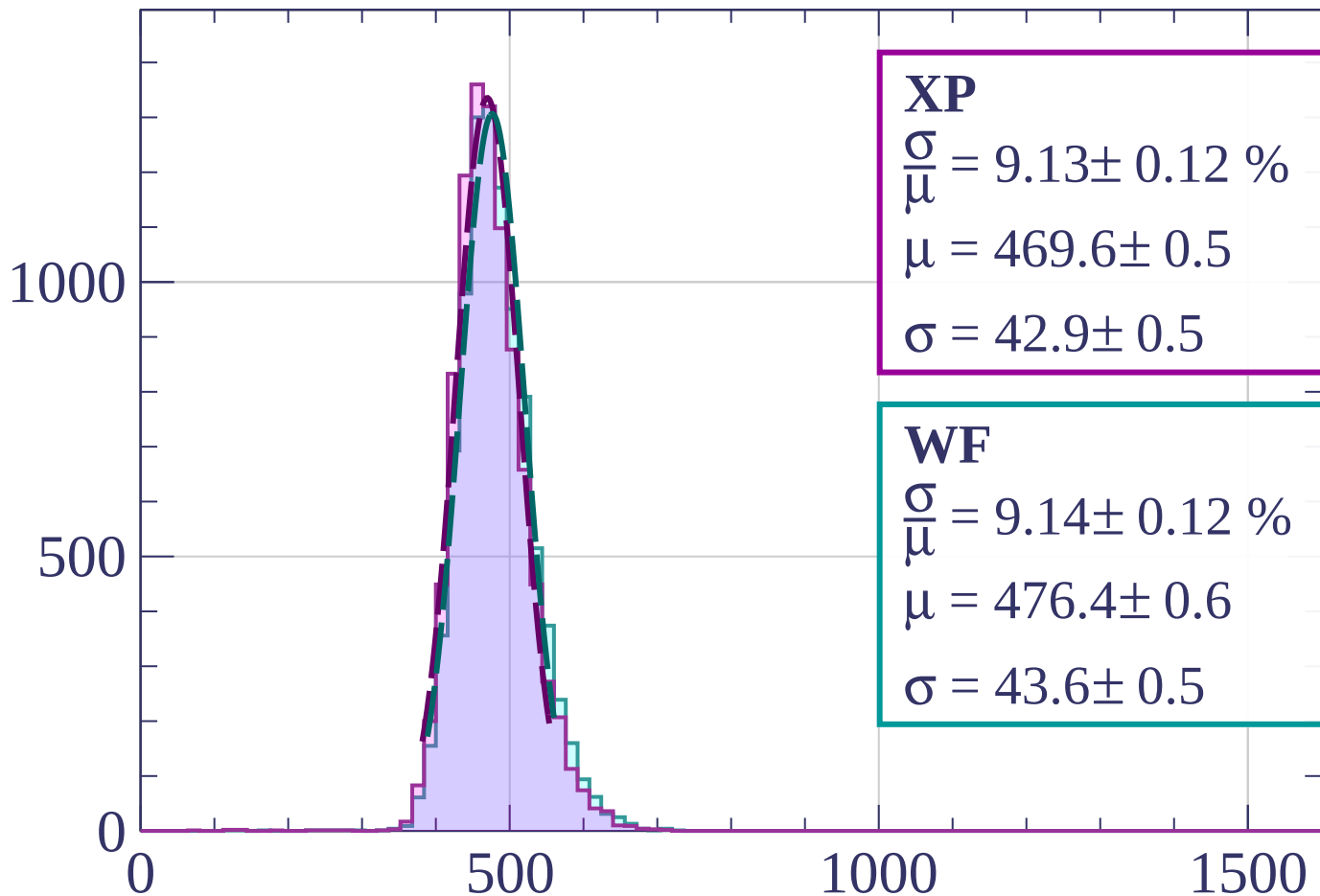
dE/dx [ADC counts/cm]

Energy loss | $1600 < p < 1680$



Energy loss | $1680 < p < 1760$

Count



XP

$$\frac{\sigma}{\mu} = 9.13 \pm 0.12 \%$$

$$\mu = 469.6 \pm 0.5$$

$$\sigma = 42.9 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 9.14 \pm 0.12 \%$$

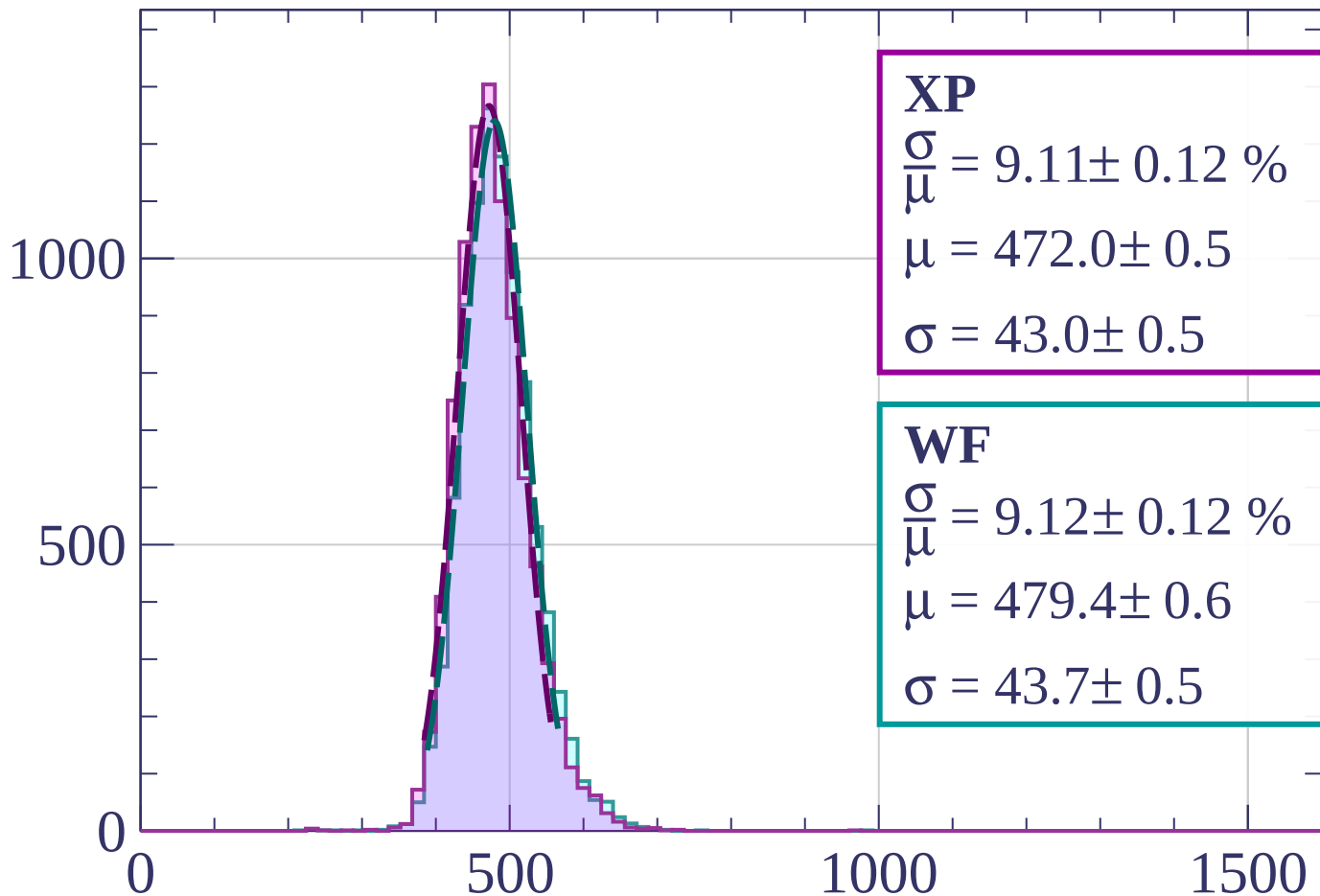
$$\mu = 476.4 \pm 0.6$$

$$\sigma = 43.6 \pm 0.5$$

dE/dx [ADC counts/cm]

Energy loss | $1760 < p < 1840$

Count



XP

$$\frac{\sigma}{\mu} = 9.11 \pm 0.12 \%$$

$$\mu = 472.0 \pm 0.5$$

$$\sigma = 43.0 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 9.12 \pm 0.12 \%$$

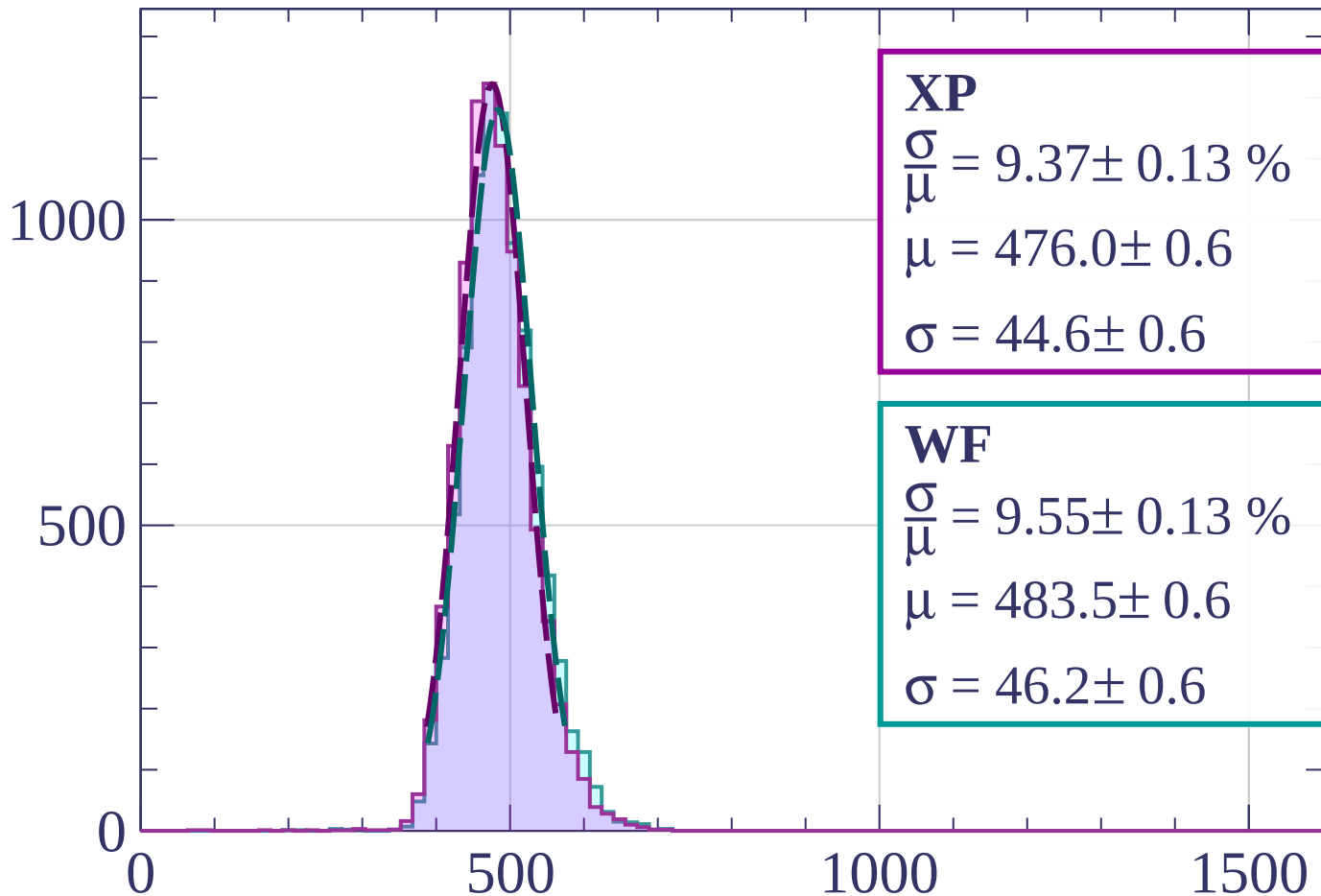
$$\mu = 479.4 \pm 0.6$$

$$\sigma = 43.7 \pm 0.5$$

dE/dx [ADC counts/cm]

Energy loss | $1840 < p < 1920$

Count



XP

$$\frac{\sigma}{\mu} = 9.37 \pm 0.13 \%$$

$$\mu = 476.0 \pm 0.6$$

$$\sigma = 44.6 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.55 \pm 0.13 \%$$

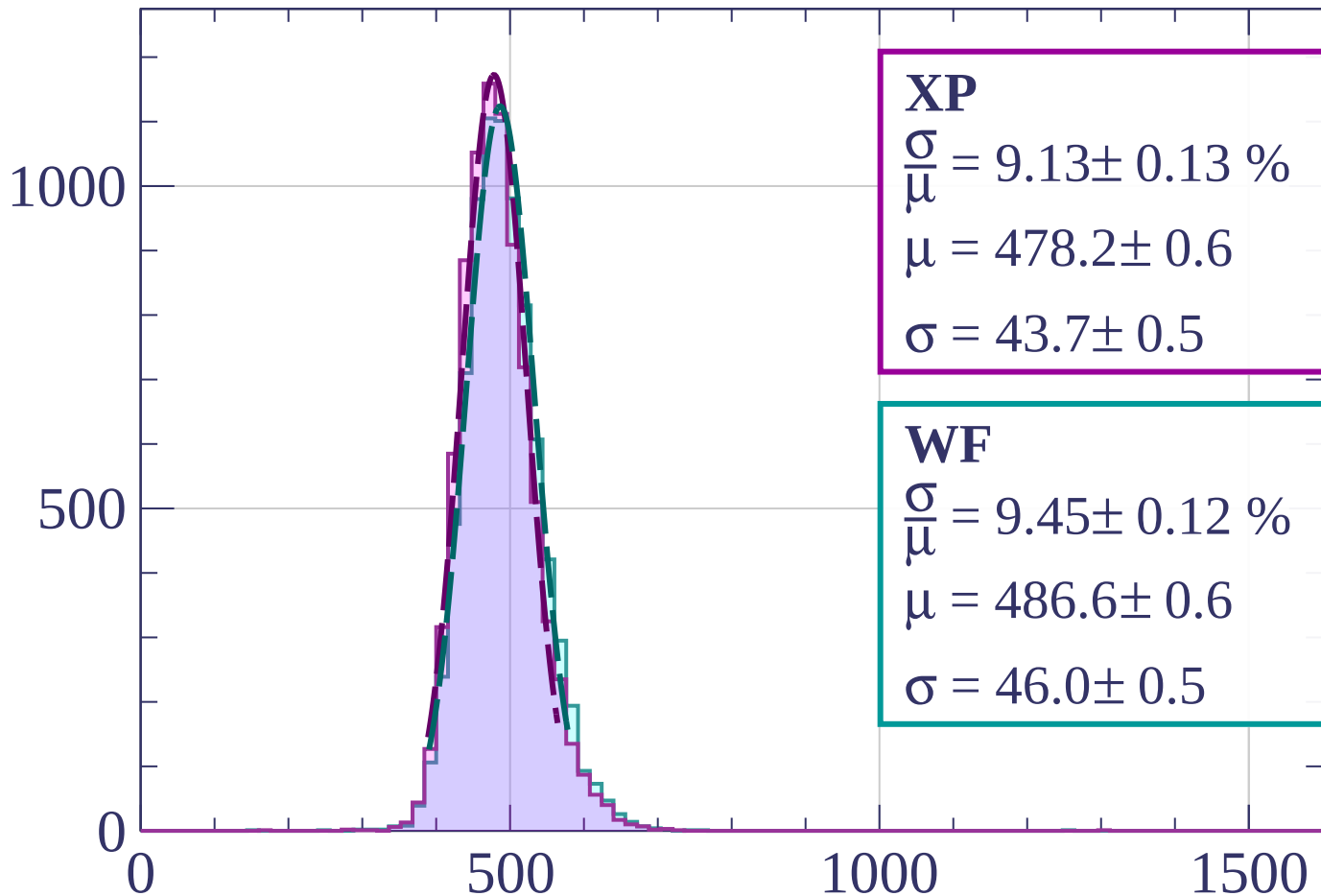
$$\mu = 483.5 \pm 0.6$$

$$\sigma = 46.2 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $1920 < p < 2000$

Count



XP

$$\frac{\sigma}{\mu} = 9.13 \pm 0.13 \%$$

$$\mu = 478.2 \pm 0.6$$

$$\sigma = 43.7 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 9.45 \pm 0.12 \%$$

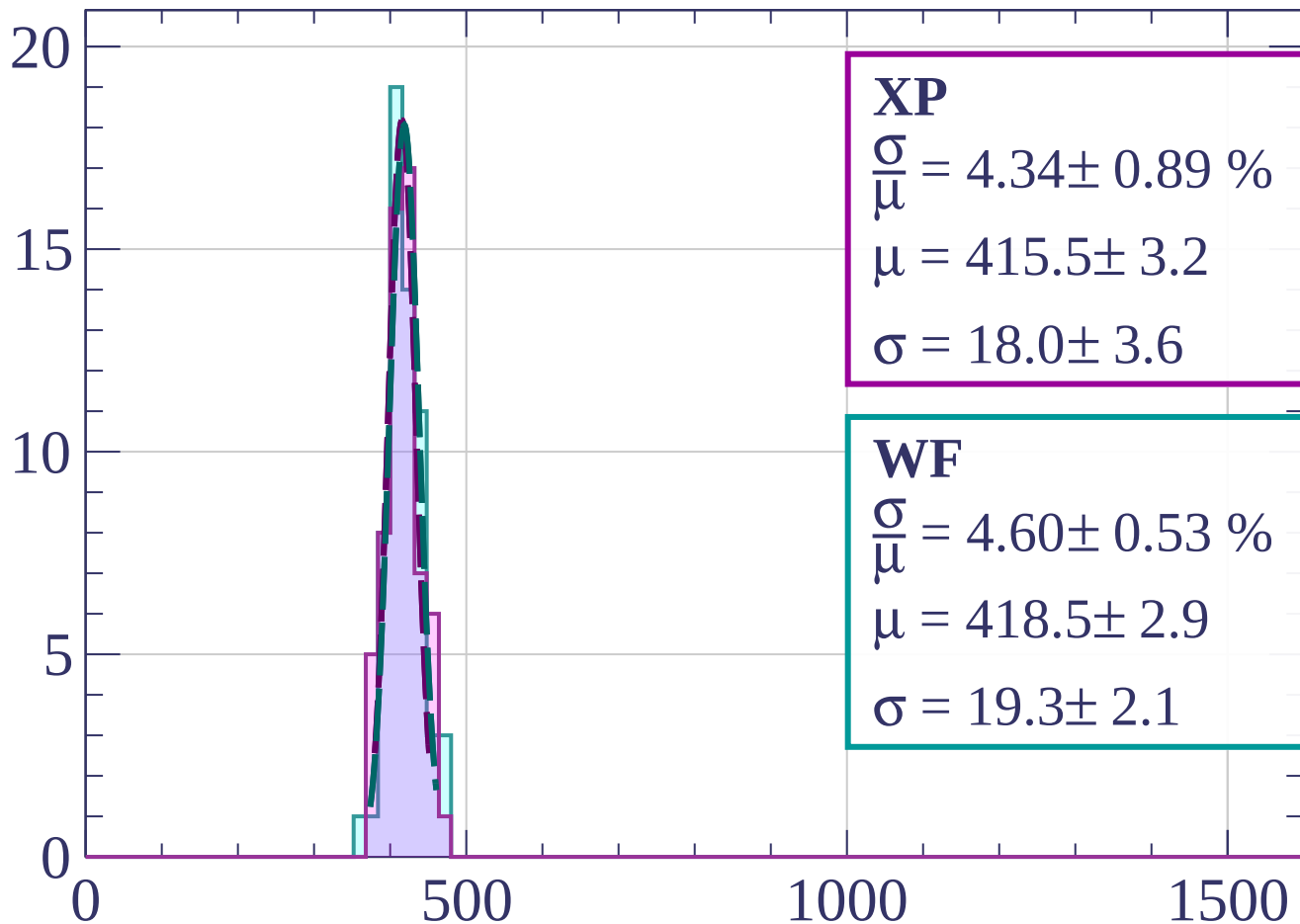
$$\mu = 486.6 \pm 0.6$$

$$\sigma = 46.0 \pm 0.5$$

dE/dx [ADC counts/cm]

Energy loss | $0 < \phi < 3$

Count



XP

$$\frac{\sigma}{\mu} = 4.34 \pm 0.89 \%$$

$$\mu = 415.5 \pm 3.2$$

$$\sigma = 18.0 \pm 3.6$$

WF

$$\frac{\sigma}{\mu} = 4.60 \pm 0.53 \%$$

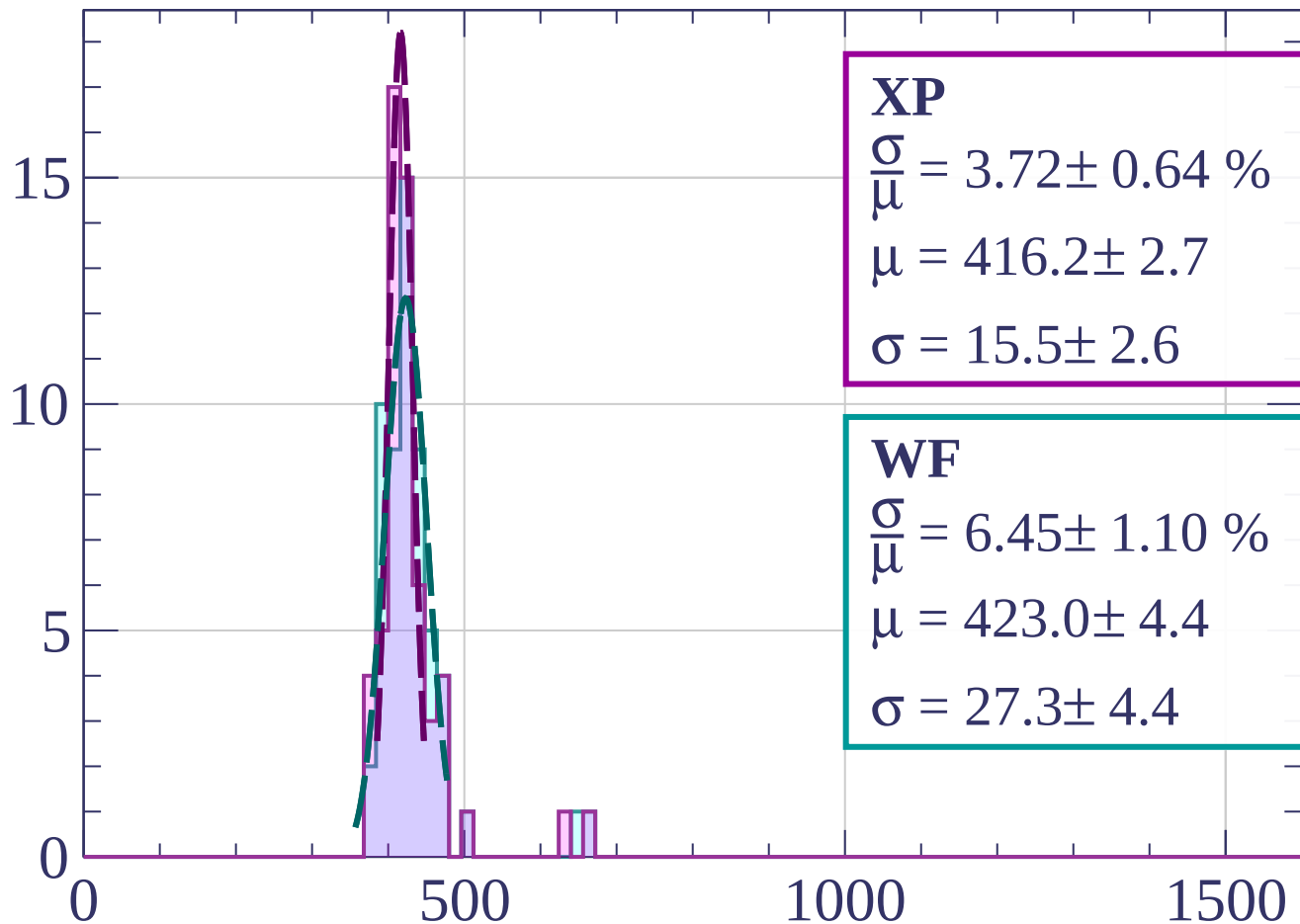
$$\mu = 418.5 \pm 2.9$$

$$\sigma = 19.3 \pm 2.1$$

dE/dx [ADC counts/cm]

Energy loss | $3 < \phi < 6$

Count



XP

$$\frac{\sigma}{\mu} = 3.72 \pm 0.64 \%$$

$$\mu = 416.2 \pm 2.7$$

$$\sigma = 15.5 \pm 2.6$$

WF

$$\frac{\sigma}{\mu} = 6.45 \pm 1.10 \%$$

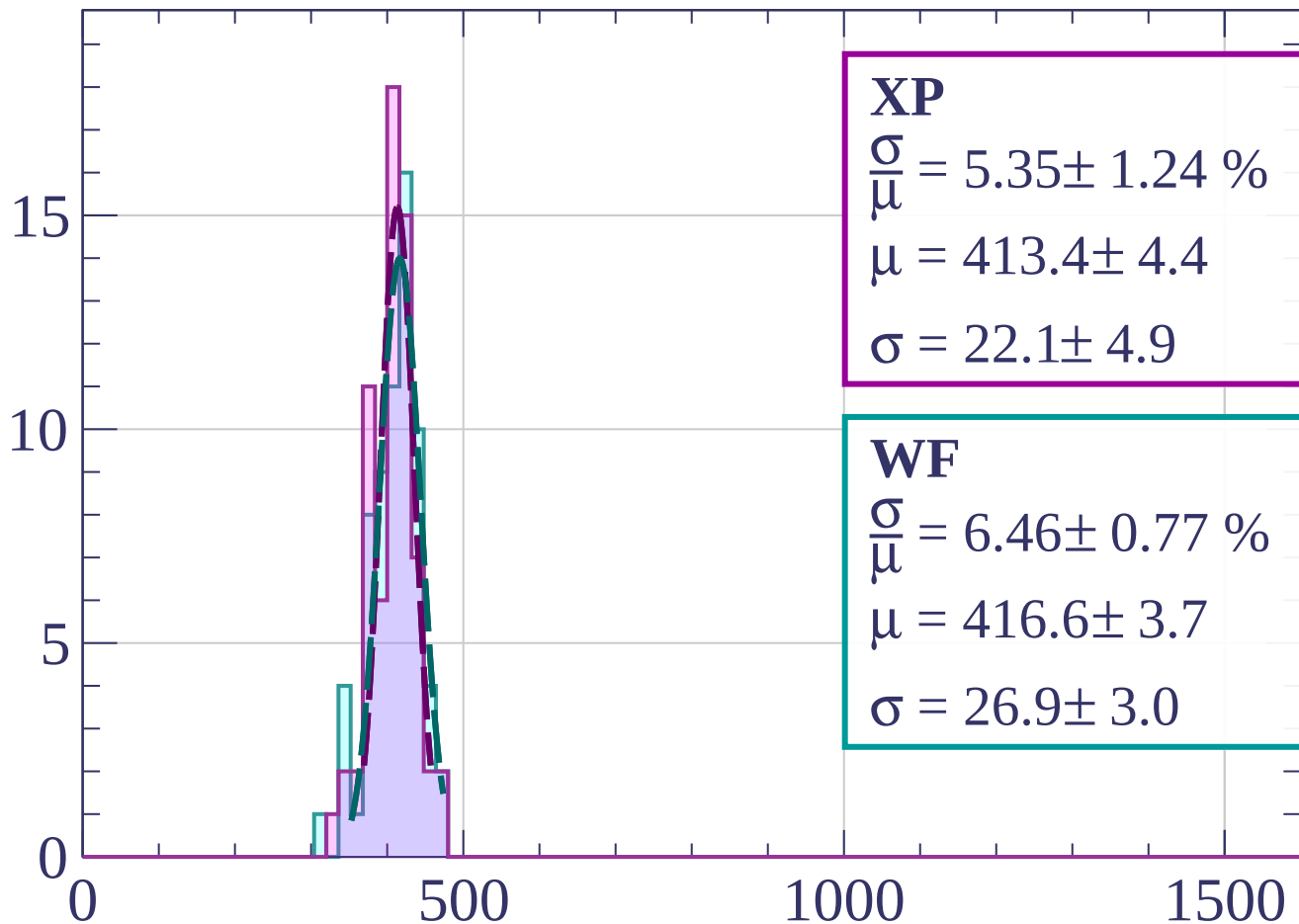
$$\mu = 423.0 \pm 4.4$$

$$\sigma = 27.3 \pm 4.4$$

dE/dx [ADC counts/cm]

Energy loss | $6 < \phi < 9$

Count



XP

$$\frac{\sigma}{\mu} = 5.35 \pm 1.24 \%$$

$$\mu = 413.4 \pm 4.4$$

$$\sigma = 22.1 \pm 4.9$$

WF

$$\frac{\sigma}{\mu} = 6.46 \pm 0.77 \%$$

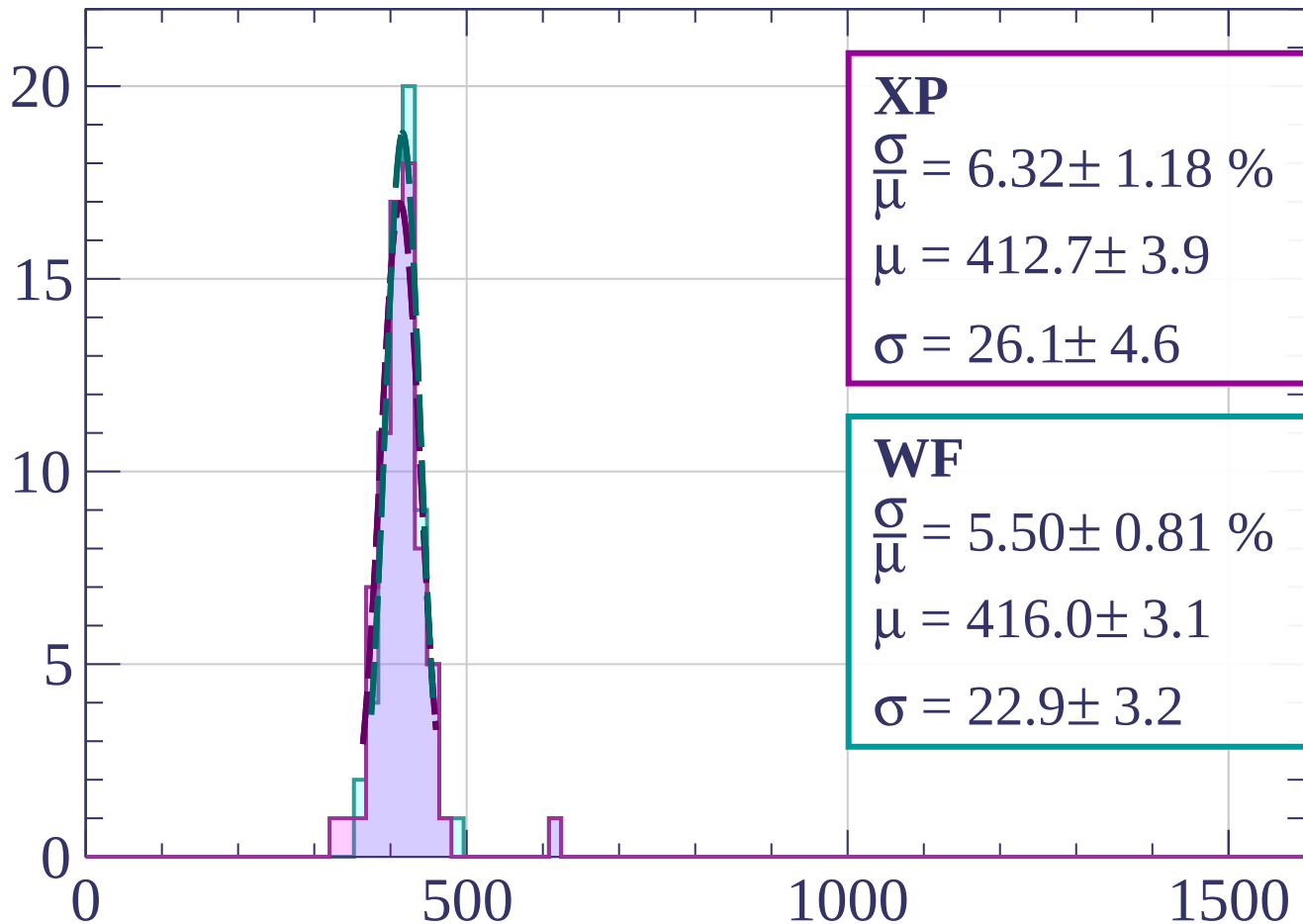
$$\mu = 416.6 \pm 3.7$$

$$\sigma = 26.9 \pm 3.0$$

dE/dx [ADC counts/cm]

Energy loss | $9 < \phi < 12$

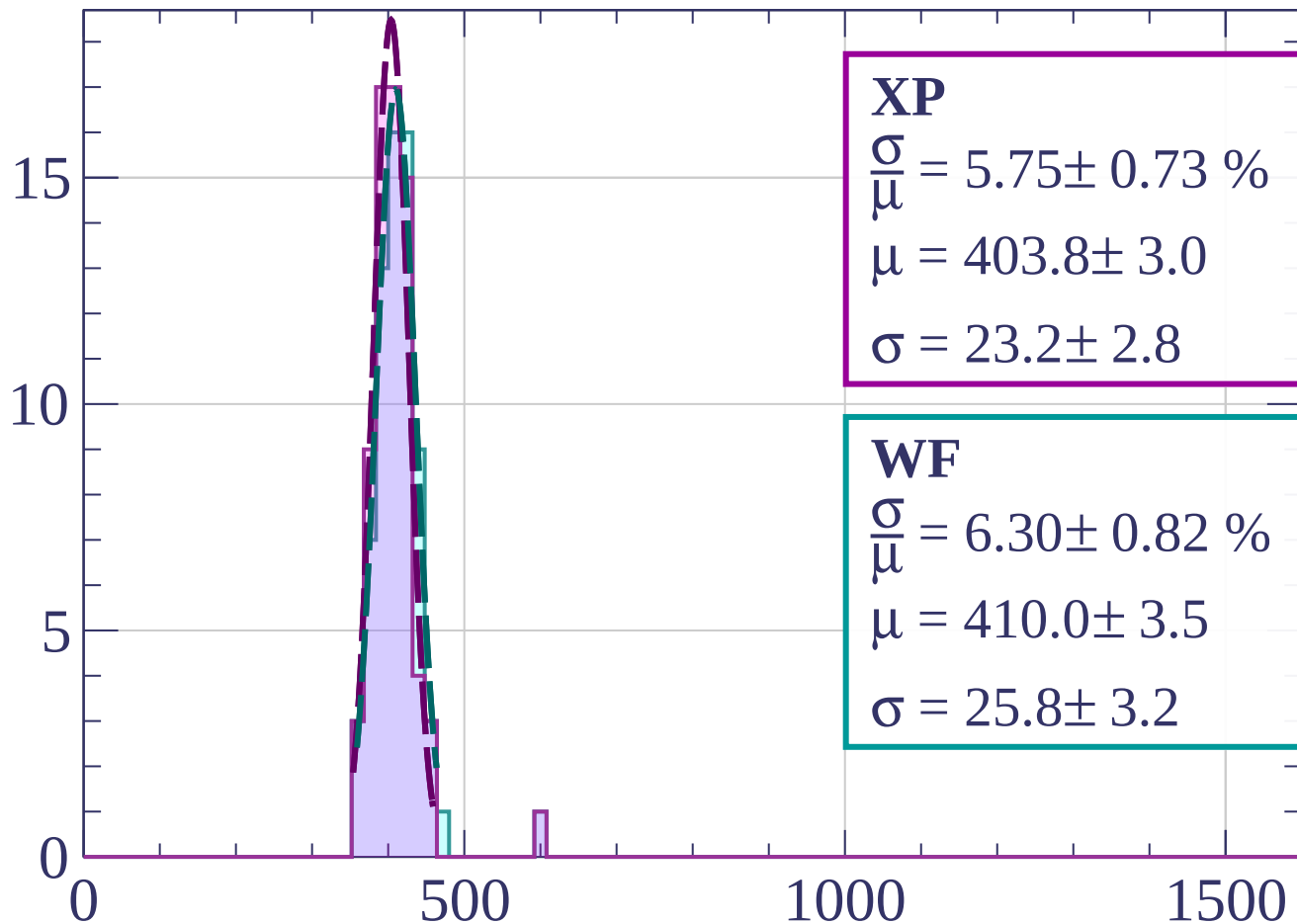
Count



dE/dx [ADC counts/cm]

Energy loss | $12 < \phi < 15$

Count



XP

$$\frac{\sigma}{\mu} = 5.75 \pm 0.73 \%$$

$$\mu = 403.8 \pm 3.0$$

$$\sigma = 23.2 \pm 2.8$$

WF

$$\frac{\sigma}{\mu} = 6.30 \pm 0.82 \%$$

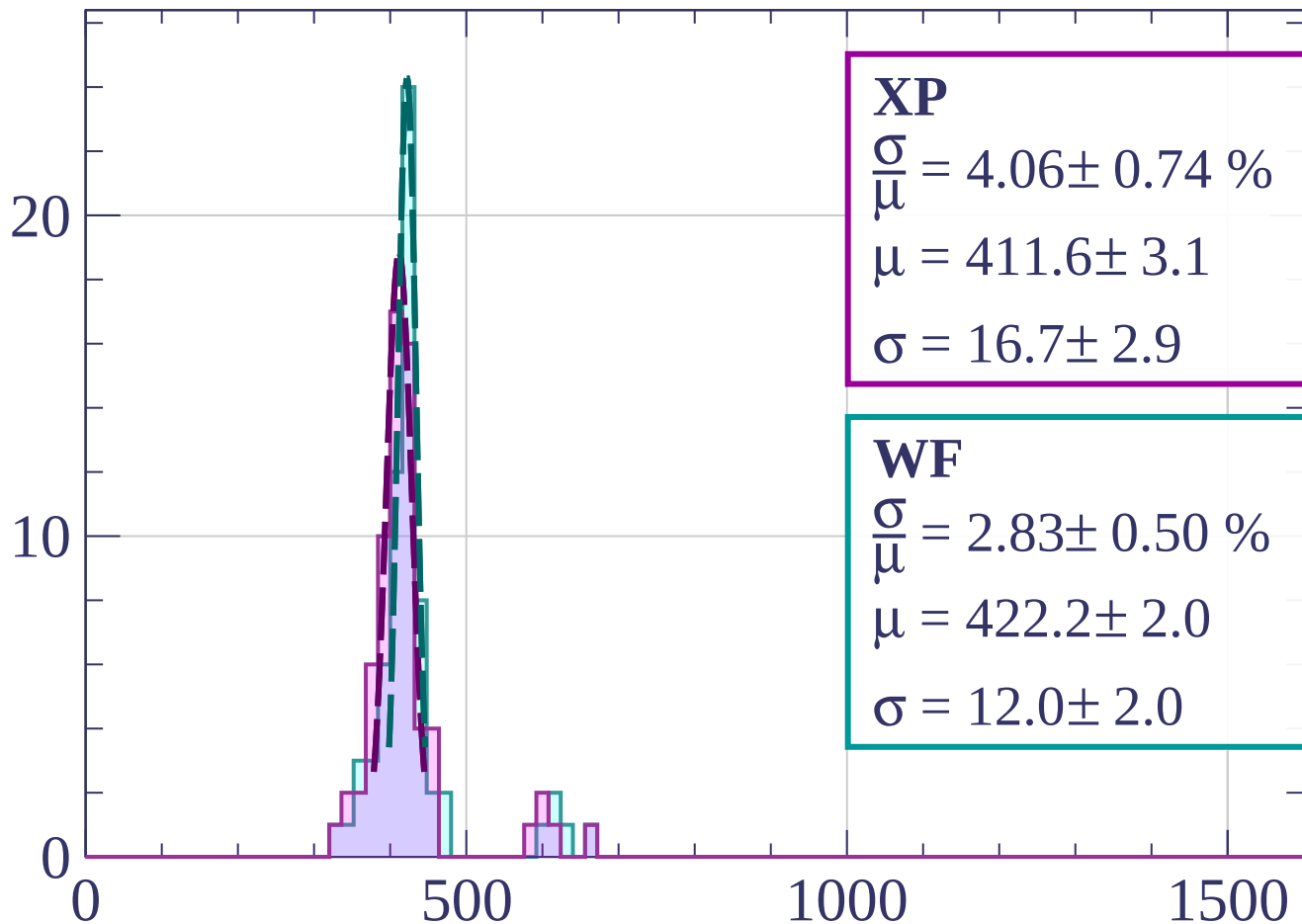
$$\mu = 410.0 \pm 3.5$$

$$\sigma = 25.8 \pm 3.2$$

dE/dx [ADC counts/cm]

Energy loss | $15 < \phi < 18$

Count



XP

$$\frac{\sigma}{\mu} = 4.06 \pm 0.74 \%$$

$$\mu = 411.6 \pm 3.1$$

$$\sigma = 16.7 \pm 2.9$$

WF

$$\frac{\sigma}{\mu} = 2.83 \pm 0.50 \%$$

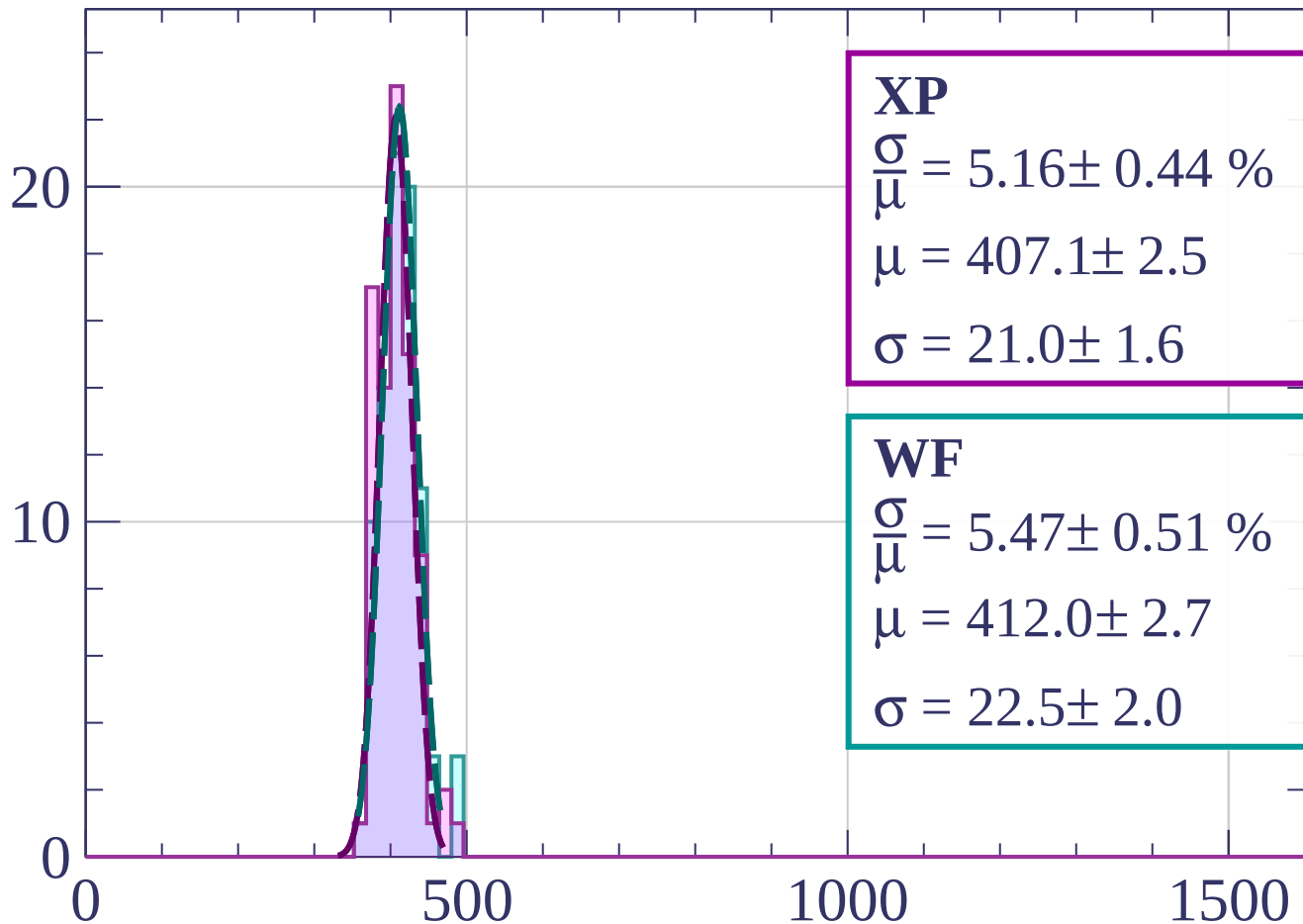
$$\mu = 422.2 \pm 2.0$$

$$\sigma = 12.0 \pm 2.0$$

dE/dx [ADC counts/cm]

Energy loss | $18 < \phi < 21$

Count



XP

$$\frac{\sigma}{\mu} = 5.16 \pm 0.44 \%$$

$$\mu = 407.1 \pm 2.5$$

$$\sigma = 21.0 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 5.47 \pm 0.51 \%$$

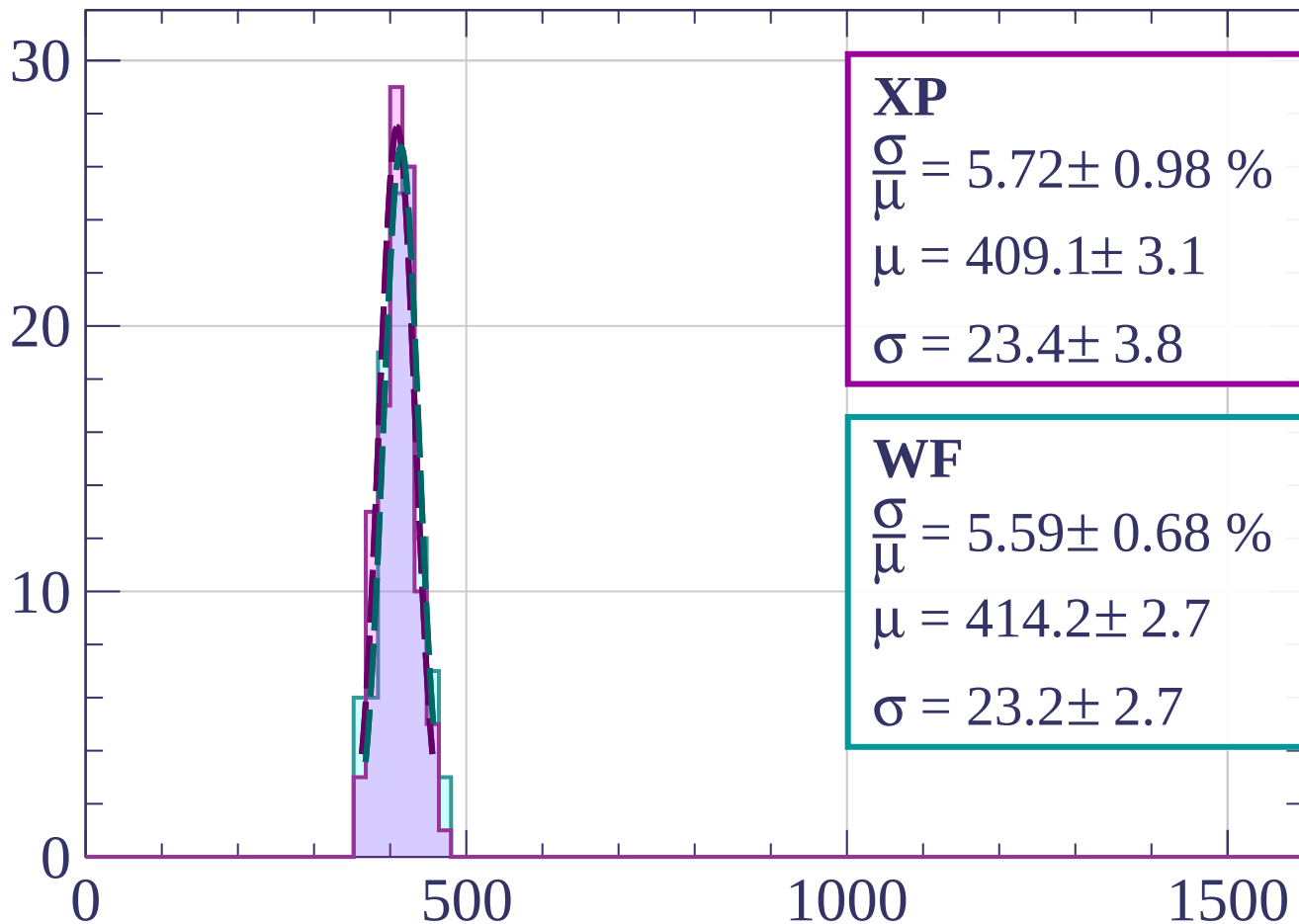
$$\mu = 412.0 \pm 2.7$$

$$\sigma = 22.5 \pm 2.0$$

dE/dx [ADC counts/cm]

Energy loss | $21 < \phi < 24$

Count



XP

$$\frac{\sigma}{\mu} = 5.72 \pm 0.98 \%$$

$$\mu = 409.1 \pm 3.1$$

$$\sigma = 23.4 \pm 3.8$$

WF

$$\frac{\sigma}{\mu} = 5.59 \pm 0.68 \%$$

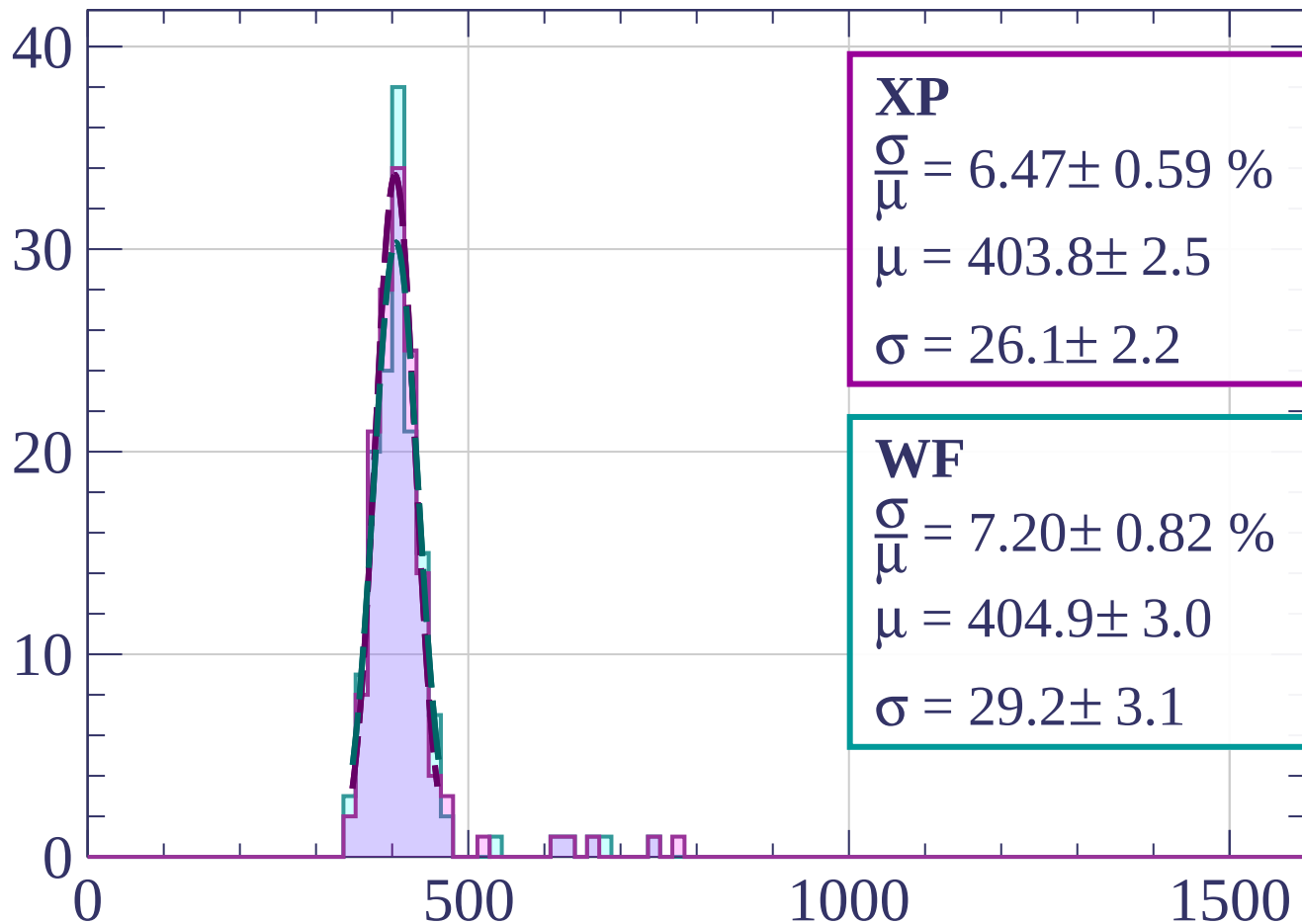
$$\mu = 414.2 \pm 2.7$$

$$\sigma = 23.2 \pm 2.7$$

dE/dx [ADC counts/cm]

Energy loss | $24 < \phi < 27$

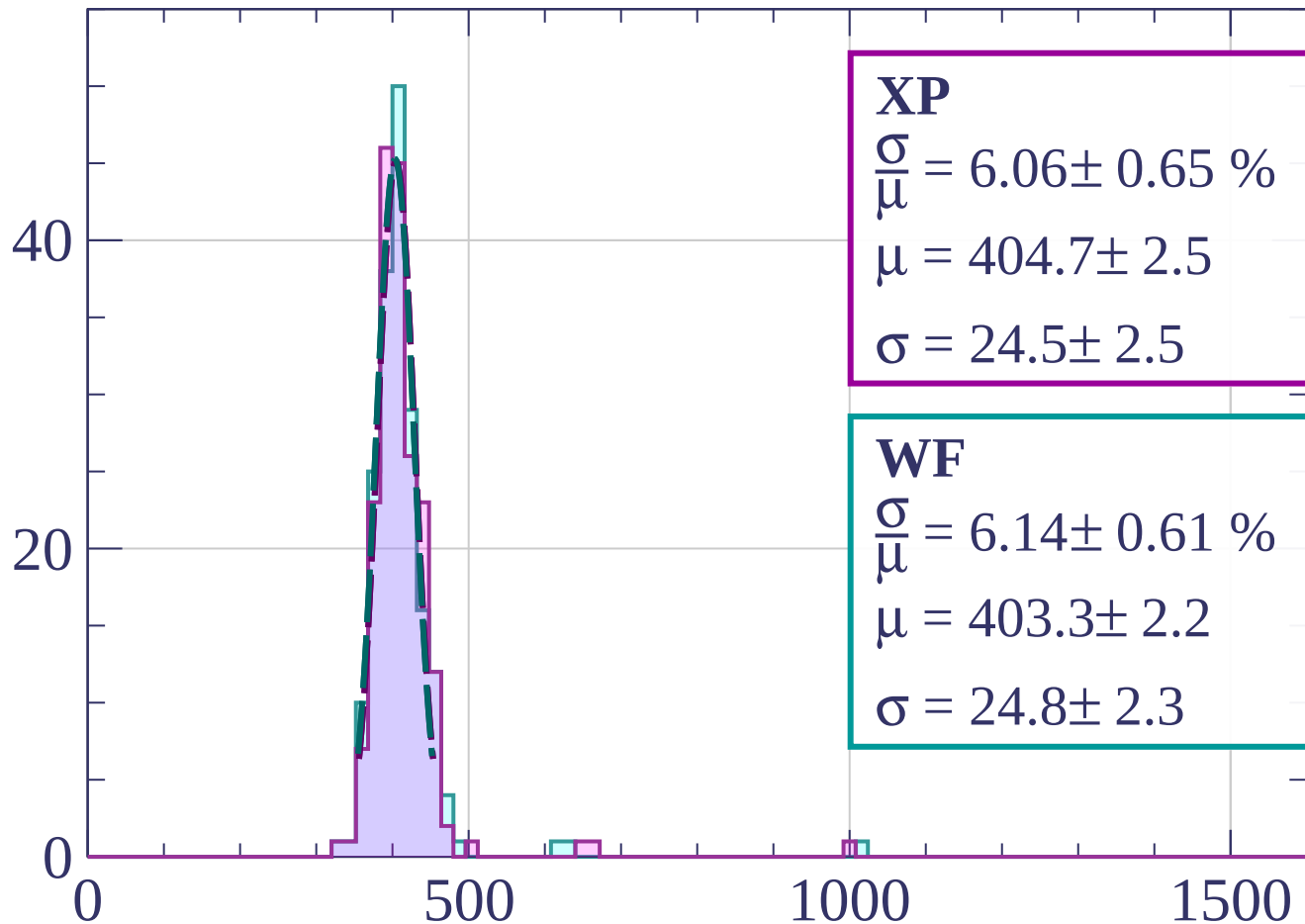
Count



dE/dx [ADC counts/cm]

Energy loss | $27 < \phi < 30$

Count



XP

$$\frac{\sigma}{\mu} = 6.06 \pm 0.65 \%$$

$$\mu = 404.7 \pm 2.5$$

$$\sigma = 24.5 \pm 2.5$$

WF

$$\frac{\sigma}{\mu} = 6.14 \pm 0.61 \%$$

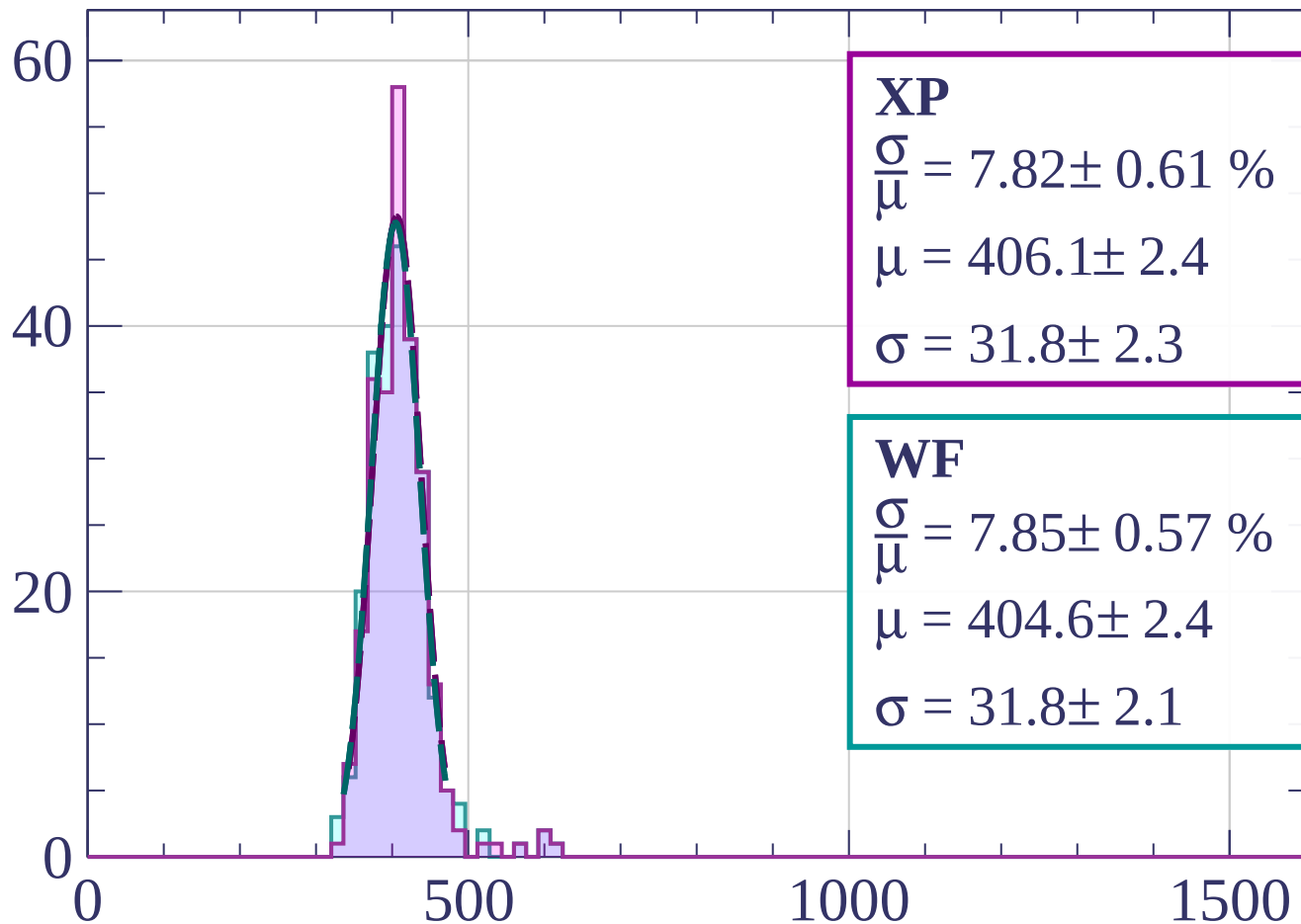
$$\mu = 403.3 \pm 2.2$$

$$\sigma = 24.8 \pm 2.3$$

dE/dx [ADC counts/cm]

Energy loss | $30 < \phi < 33$

Count



XP

$$\frac{\sigma}{\mu} = 7.82 \pm 0.61 \%$$

$$\mu = 406.1 \pm 2.4$$

$$\sigma = 31.8 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 7.85 \pm 0.57 \%$$

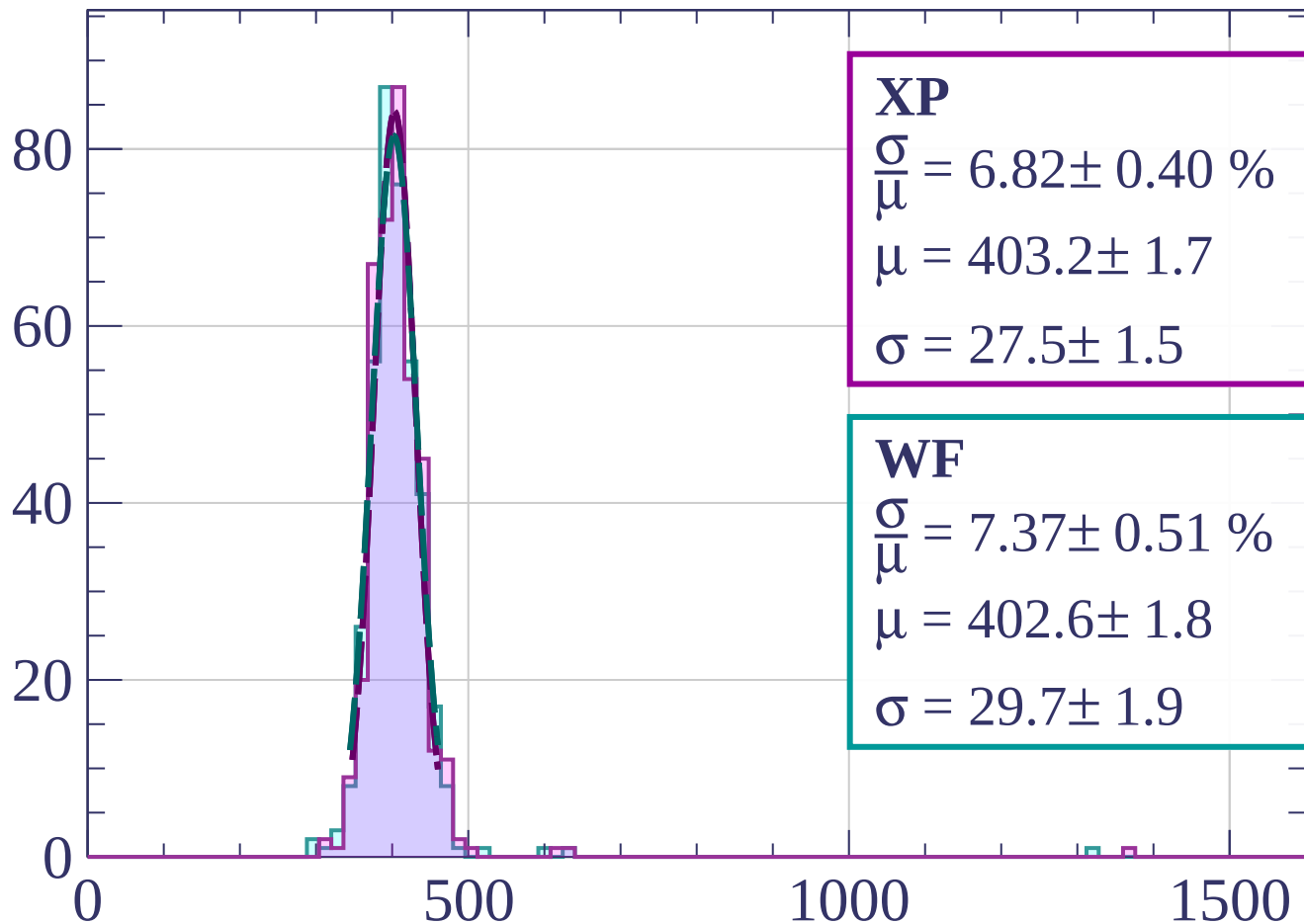
$$\mu = 404.6 \pm 2.4$$

$$\sigma = 31.8 \pm 2.1$$

dE/dx [ADC counts/cm]

Energy loss | $33 < \phi < 36$

Count



XP

$$\frac{\sigma}{\mu} = 6.82 \pm 0.40 \%$$

$$\mu = 403.2 \pm 1.7$$

$$\sigma = 27.5 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 7.37 \pm 0.51 \%$$

$$\mu = 402.6 \pm 1.8$$

$$\sigma = 29.7 \pm 1.9$$

dE/dx [ADC counts/cm]

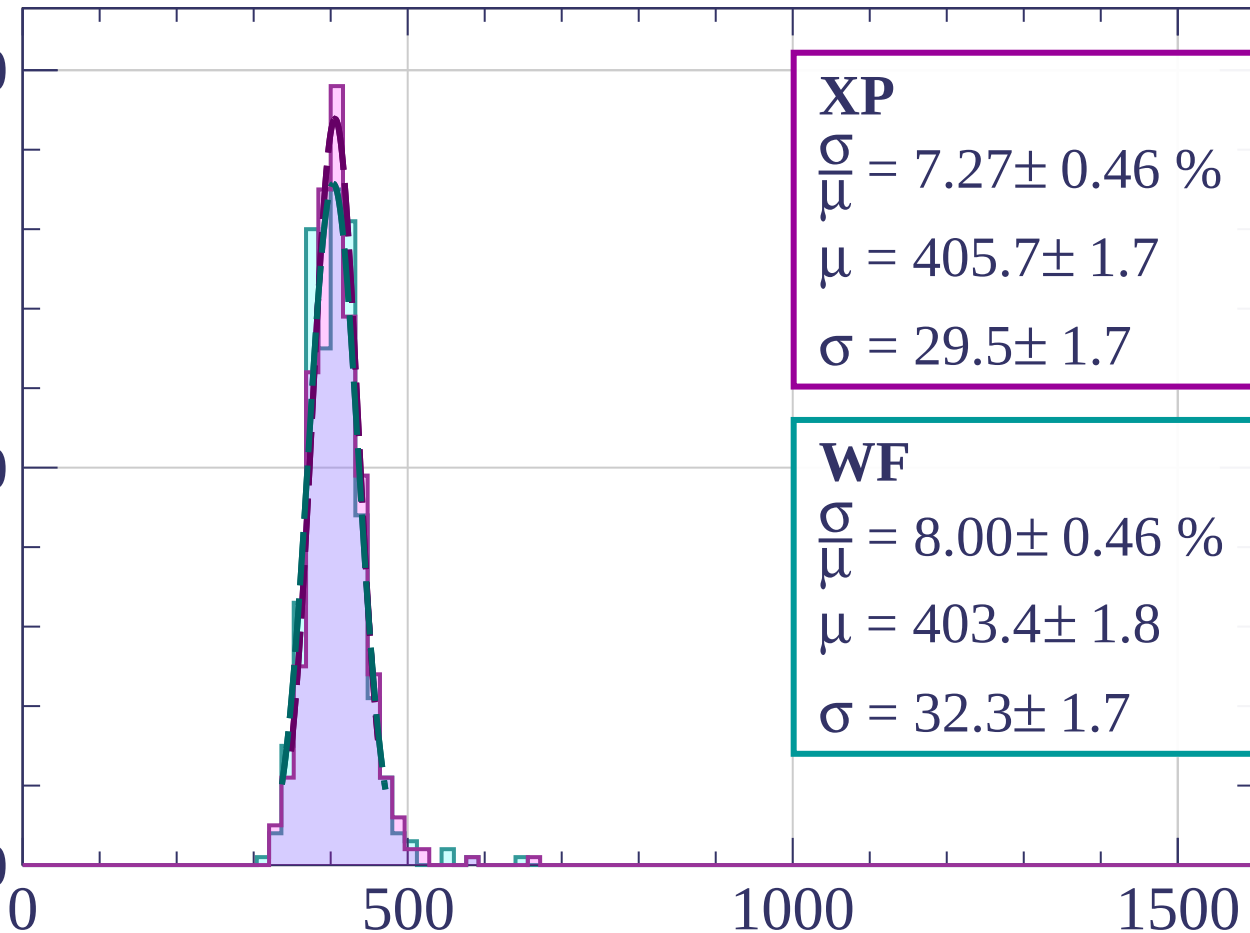
Energy loss | $36 < \phi < 39$

Count

100

50

0



XP

$$\frac{\sigma}{\mu} = 7.27 \pm 0.46 \%$$

$$\mu = 405.7 \pm 1.7$$

$$\sigma = 29.5 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 8.00 \pm 0.46 \%$$

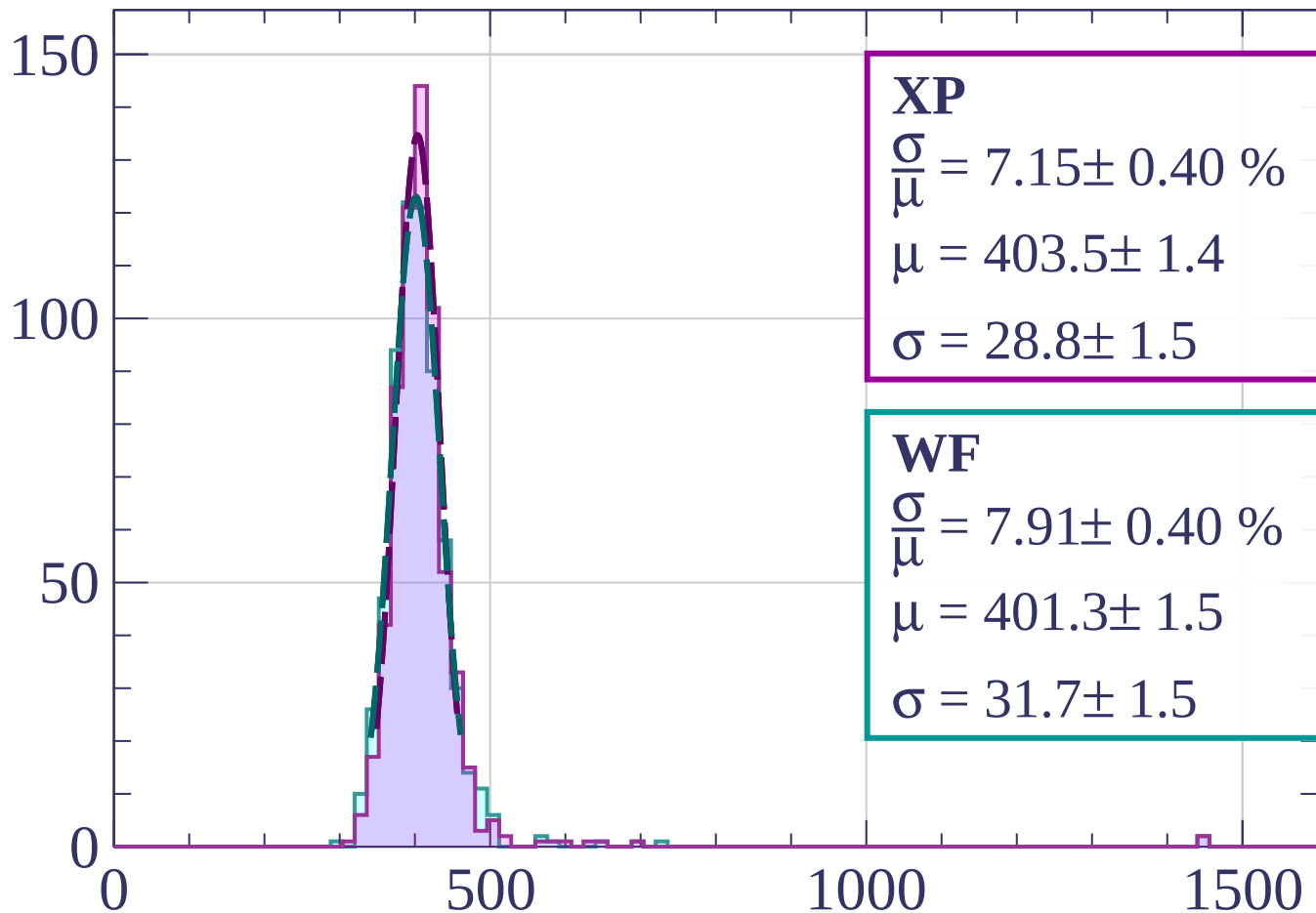
$$\mu = 403.4 \pm 1.8$$

$$\sigma = 32.3 \pm 1.7$$

dE/dx [ADC counts/cm]

Energy loss | $39 < \phi < 42$

Count



XP

$$\frac{\sigma}{\mu} = 7.15 \pm 0.40 \%$$

$$\mu = 403.5 \pm 1.4$$

$$\sigma = 28.8 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 7.91 \pm 0.40 \%$$

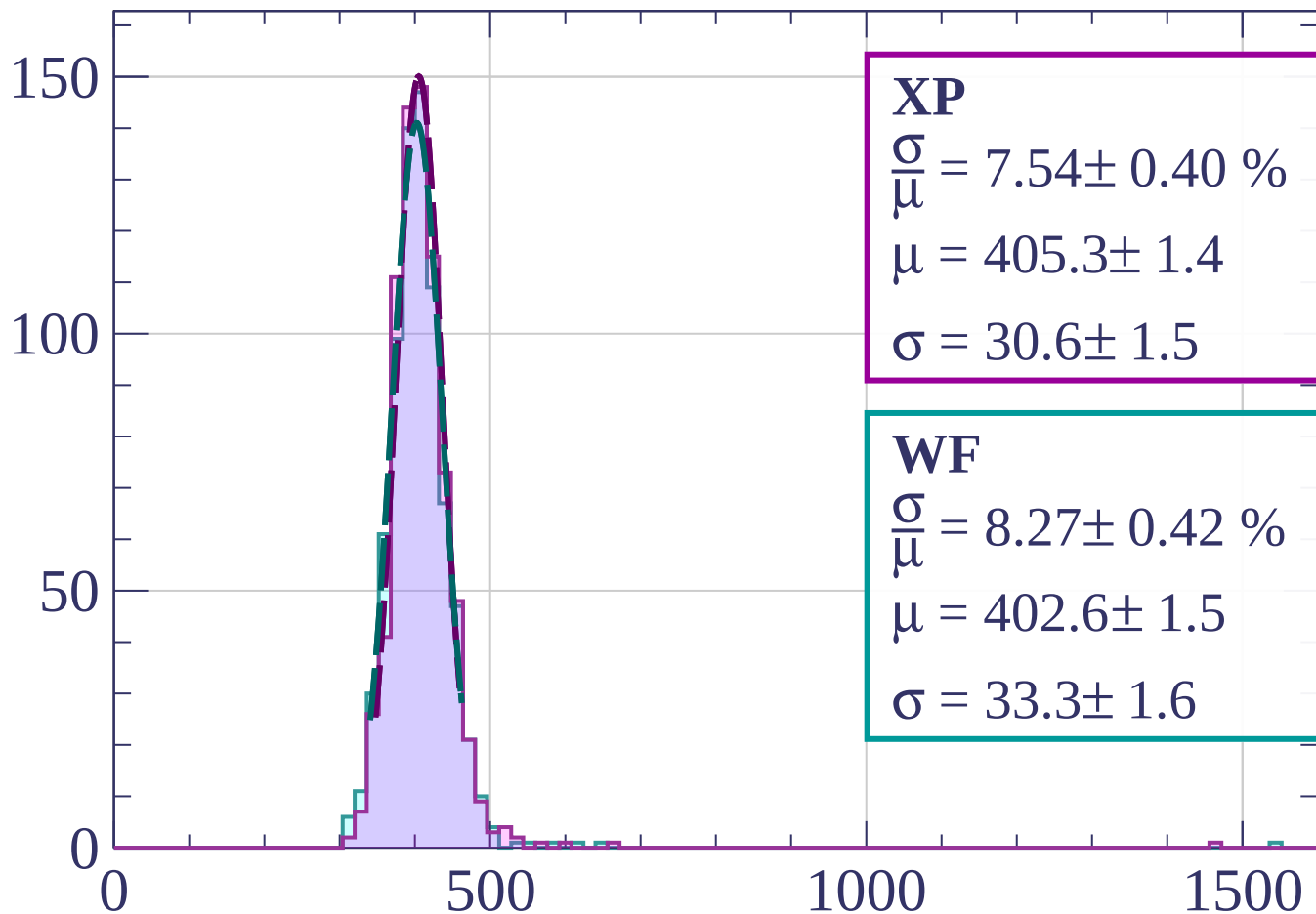
$$\mu = 401.3 \pm 1.5$$

$$\sigma = 31.7 \pm 1.5$$

dE/dx [ADC counts/cm]

Energy loss | $42 < \phi < 45$

Count



XP

$$\frac{\sigma}{\mu} = 7.54 \pm 0.40 \%$$

$$\mu = 405.3 \pm 1.4$$

$$\sigma = 30.6 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 8.27 \pm 0.42 \%$$

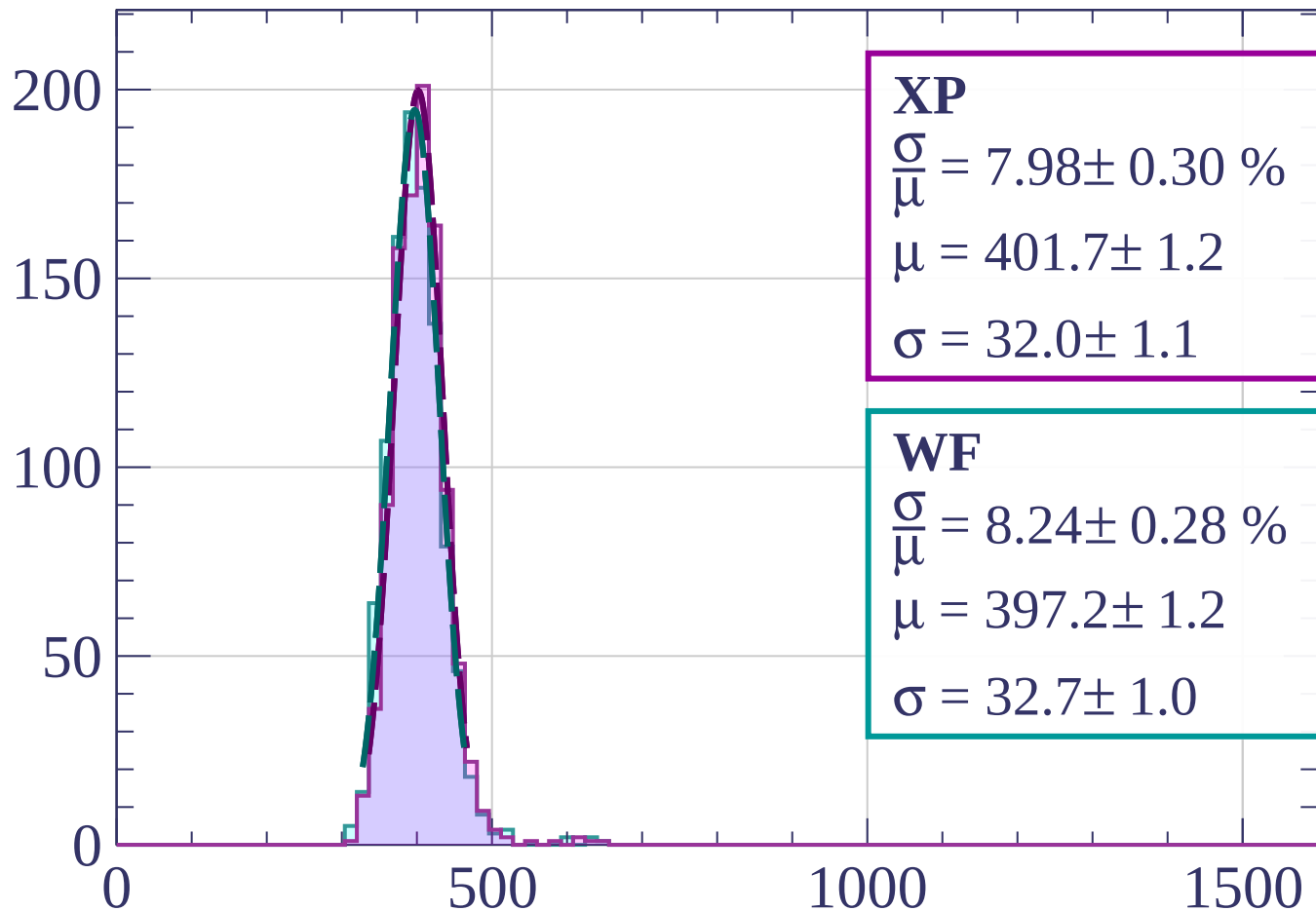
$$\mu = 402.6 \pm 1.5$$

$$\sigma = 33.3 \pm 1.6$$

dE/dx [ADC counts/cm]

Energy loss | $45 < \phi < 48$

Count



XP

$$\frac{\sigma}{\mu} = 7.98 \pm 0.30 \%$$

$$\mu = 401.7 \pm 1.2$$

$$\sigma = 32.0 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 8.24 \pm 0.28 \%$$

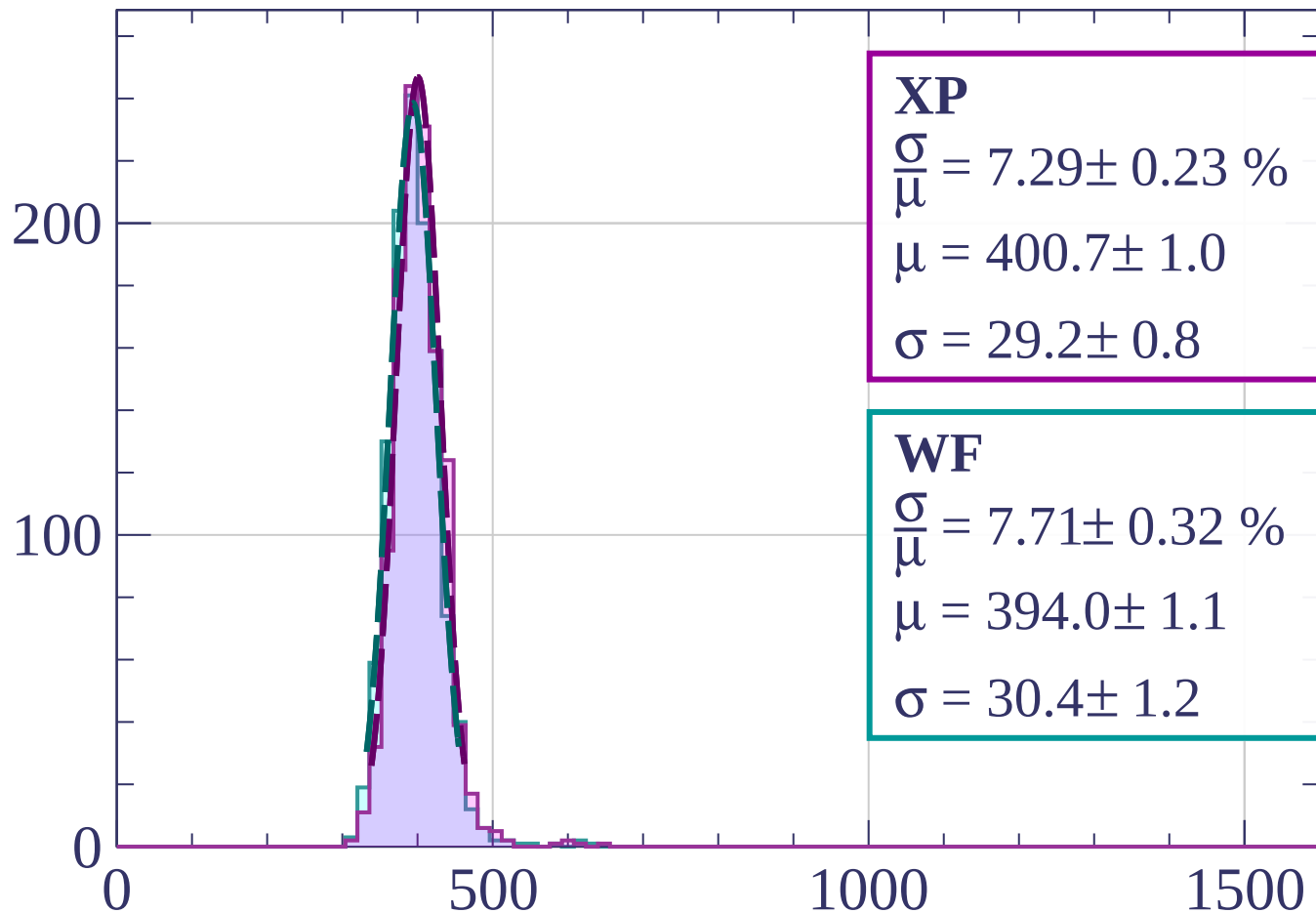
$$\mu = 397.2 \pm 1.2$$

$$\sigma = 32.7 \pm 1.0$$

dE/dx [ADC counts/cm]

Energy loss | $48 < \phi < 51$

Count



XP

$$\frac{\sigma}{\mu} = 7.29 \pm 0.23 \%$$

$$\mu = 400.7 \pm 1.0$$

$$\sigma = 29.2 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.71 \pm 0.32 \%$$

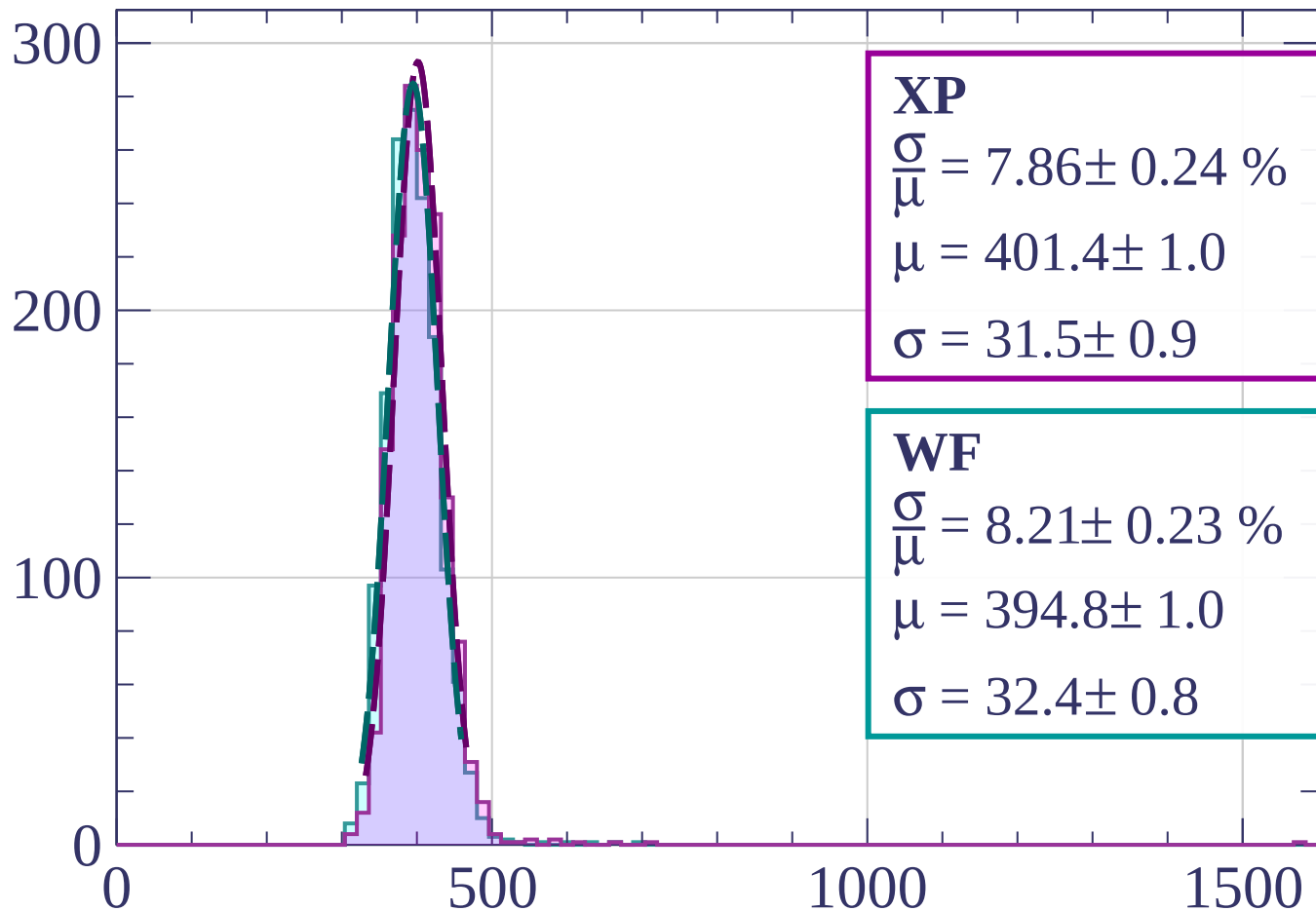
$$\mu = 394.0 \pm 1.1$$

$$\sigma = 30.4 \pm 1.2$$

dE/dx [ADC counts/cm]

Energy loss | $51 < \phi < 54$

Count



XP

$$\frac{\sigma}{\mu} = 7.86 \pm 0.24 \%$$

$$\mu = 401.4 \pm 1.0$$

$$\sigma = 31.5 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 8.21 \pm 0.23 \%$$

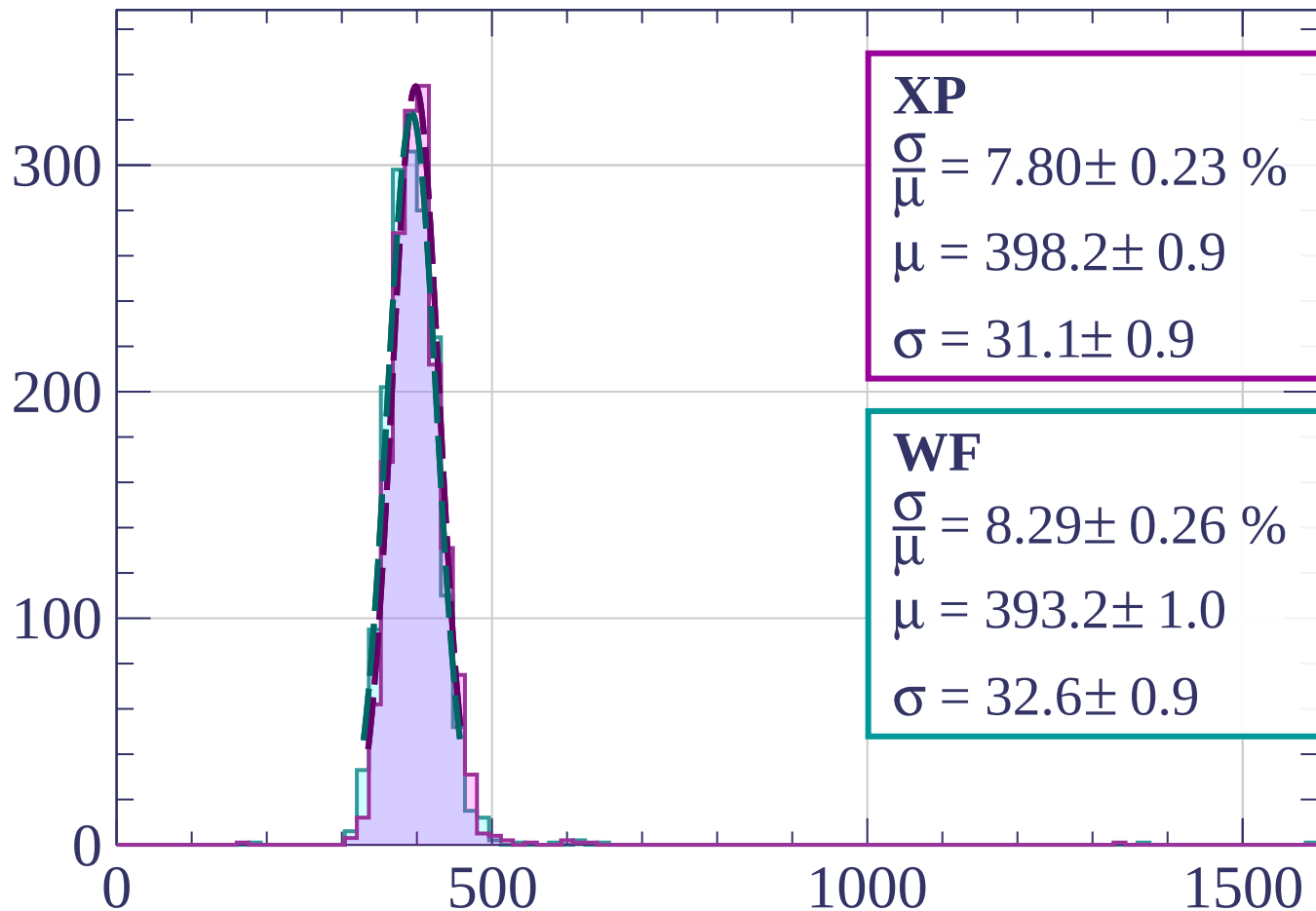
$$\mu = 394.8 \pm 1.0$$

$$\sigma = 32.4 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $54 < \phi < 57$

Count



XP

$$\frac{\sigma}{\mu} = 7.80 \pm 0.23 \%$$

$$\mu = 398.2 \pm 0.9$$

$$\sigma = 31.1 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 8.29 \pm 0.26 \%$$

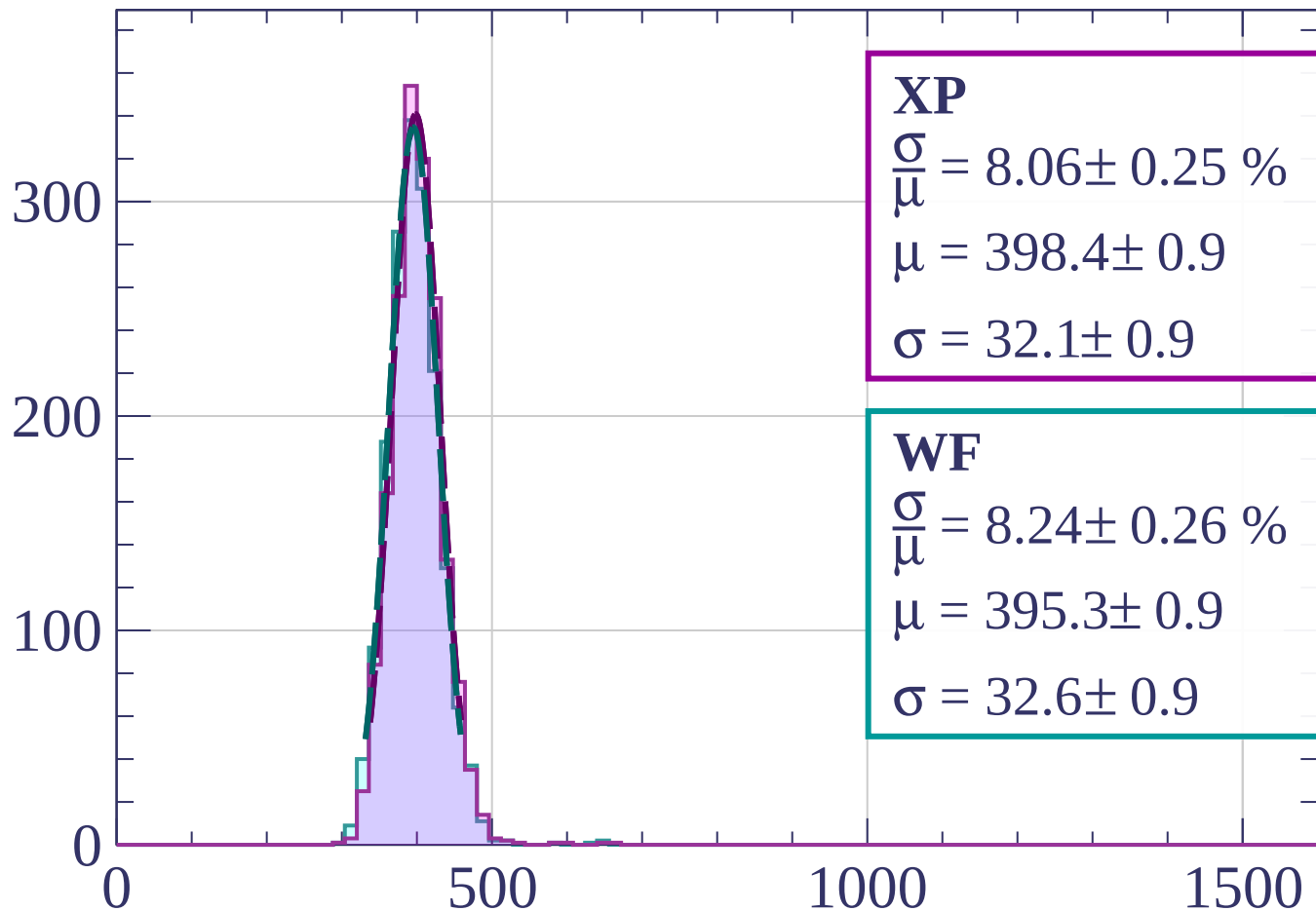
$$\mu = 393.2 \pm 1.0$$

$$\sigma = 32.6 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $57 < \phi < 60$

Count



XP

$$\frac{\sigma}{\mu} = 8.06 \pm 0.25 \%$$

$$\mu = 398.4 \pm 0.9$$

$$\sigma = 32.1 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 8.24 \pm 0.26 \%$$

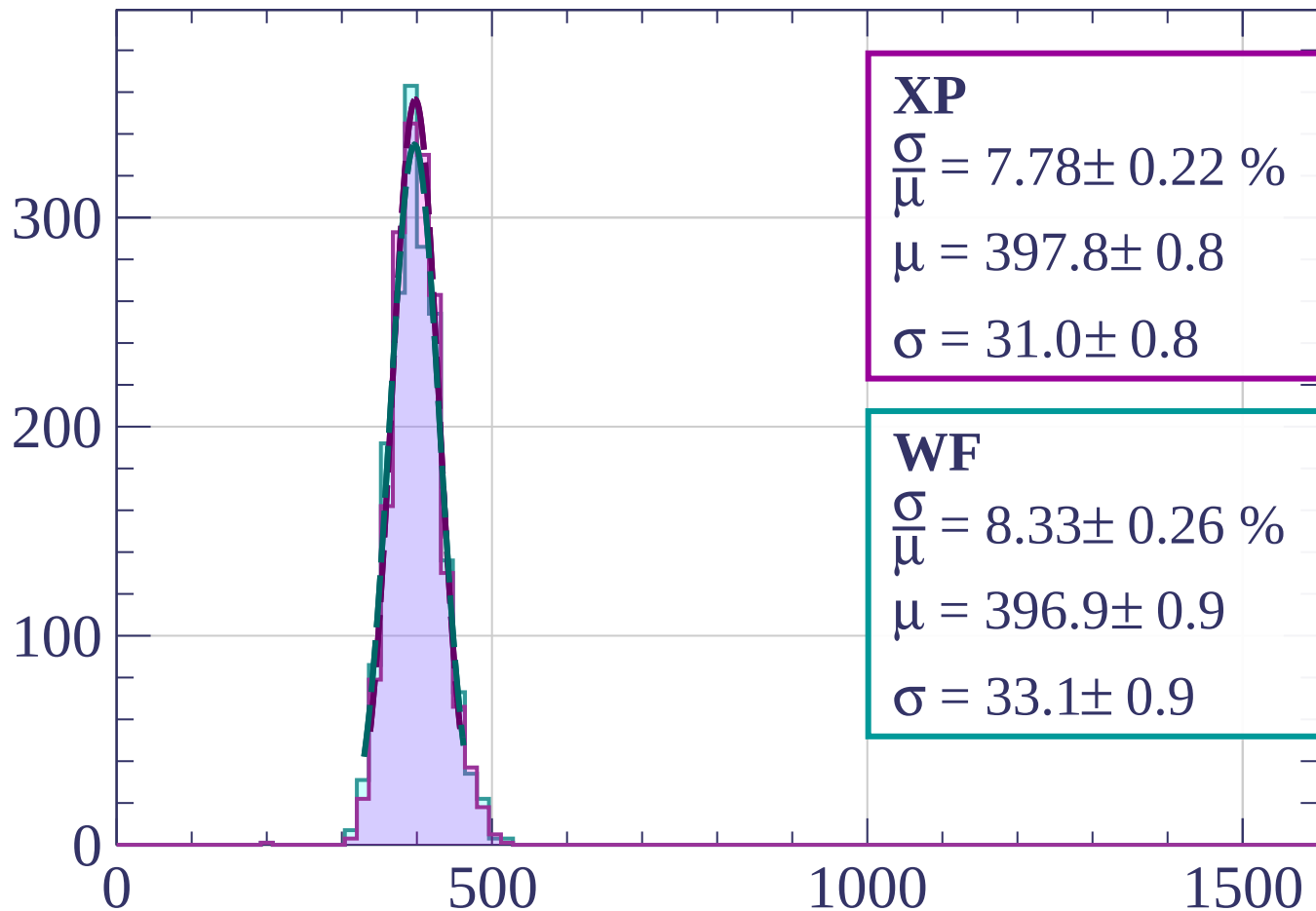
$$\mu = 395.3 \pm 0.9$$

$$\sigma = 32.6 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $60 < \phi < 63$

Count



XP

$$\frac{\sigma}{\mu} = 7.78 \pm 0.22 \%$$

$$\mu = 397.8 \pm 0.8$$

$$\sigma = 31.0 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.33 \pm 0.26 \%$$

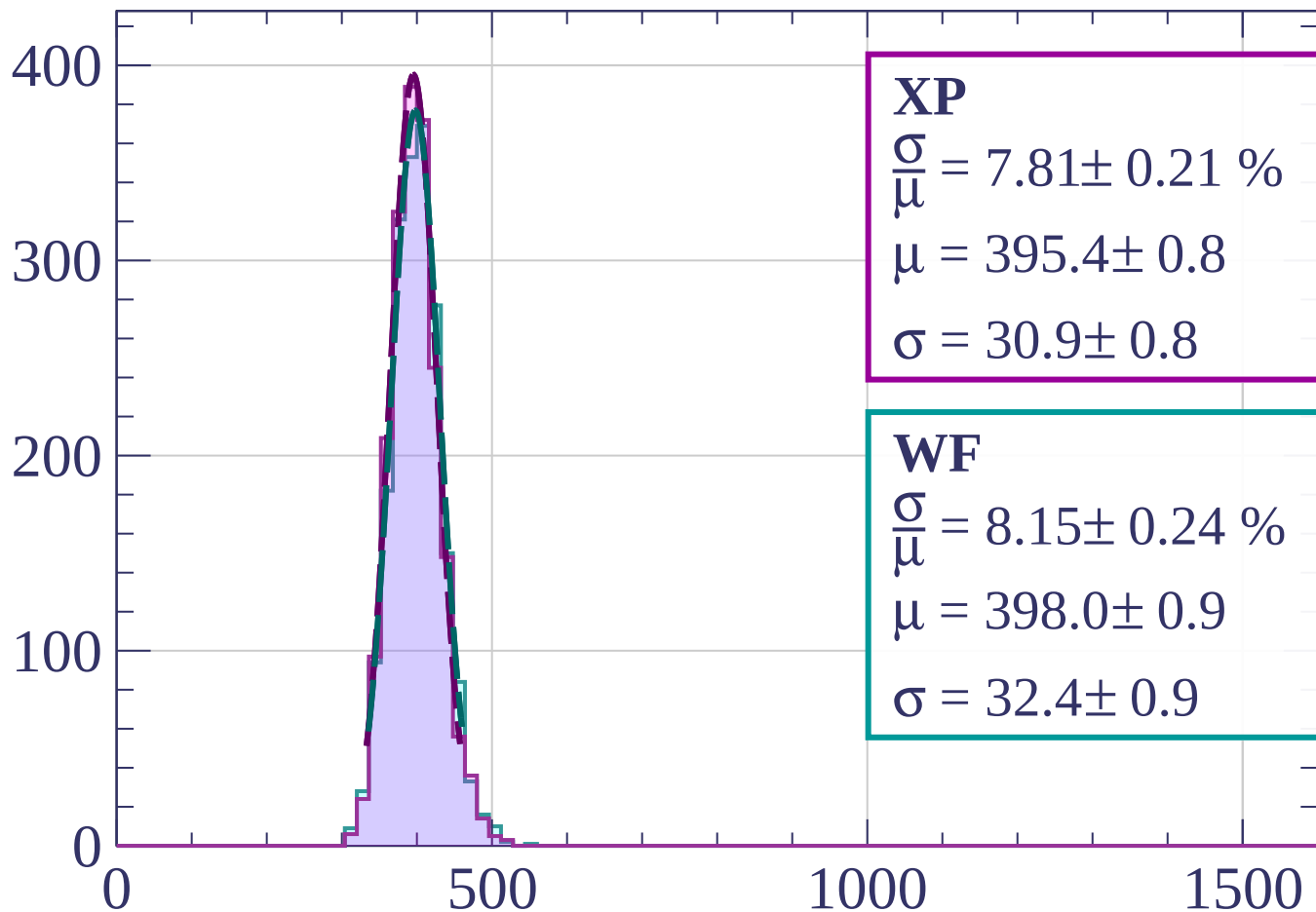
$$\mu = 396.9 \pm 0.9$$

$$\sigma = 33.1 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $63 < \phi < 66$

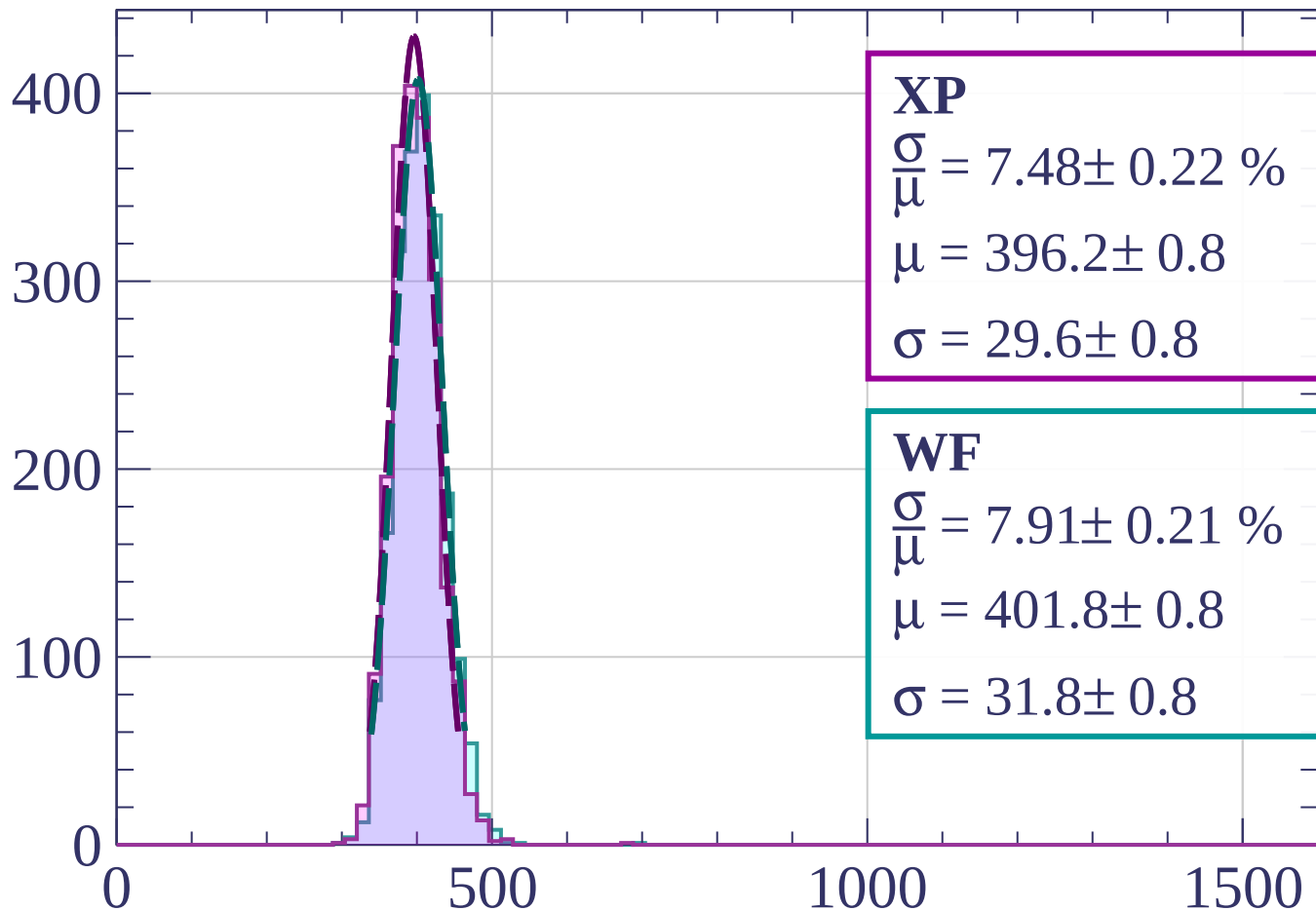
Count



dE/dx [ADC counts/cm]

Energy loss | $66 < \phi < 69$

Count



XP

$$\frac{\sigma}{\mu} = 7.48 \pm 0.22 \%$$

$$\mu = 396.2 \pm 0.8$$

$$\sigma = 29.6 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.91 \pm 0.21 \%$$

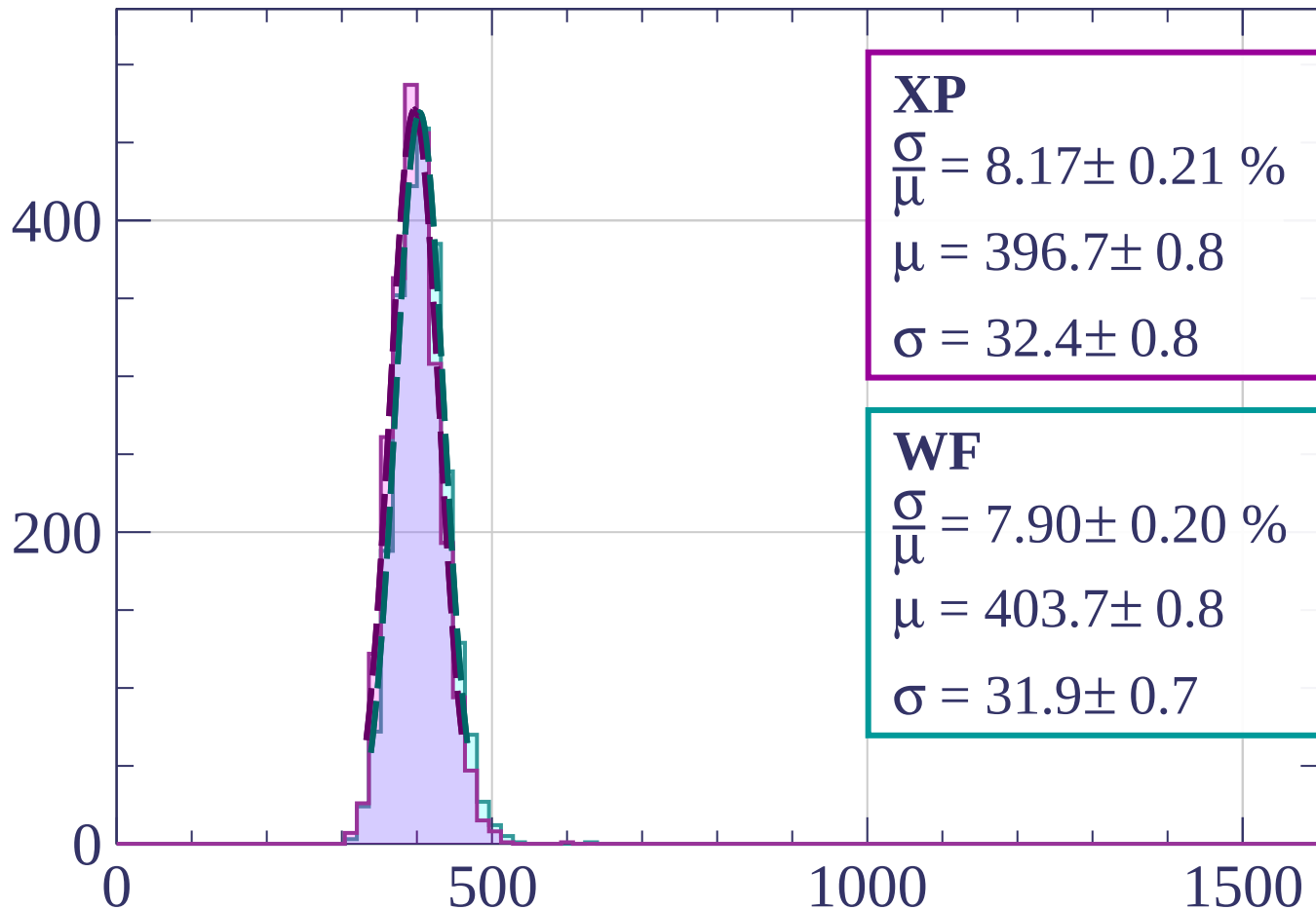
$$\mu = 401.8 \pm 0.8$$

$$\sigma = 31.8 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $69 < \phi < 72$

Count



XP

$$\frac{\sigma}{\mu} = 8.17 \pm 0.21 \%$$

$$\mu = 396.7 \pm 0.8$$

$$\sigma = 32.4 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.90 \pm 0.20 \%$$

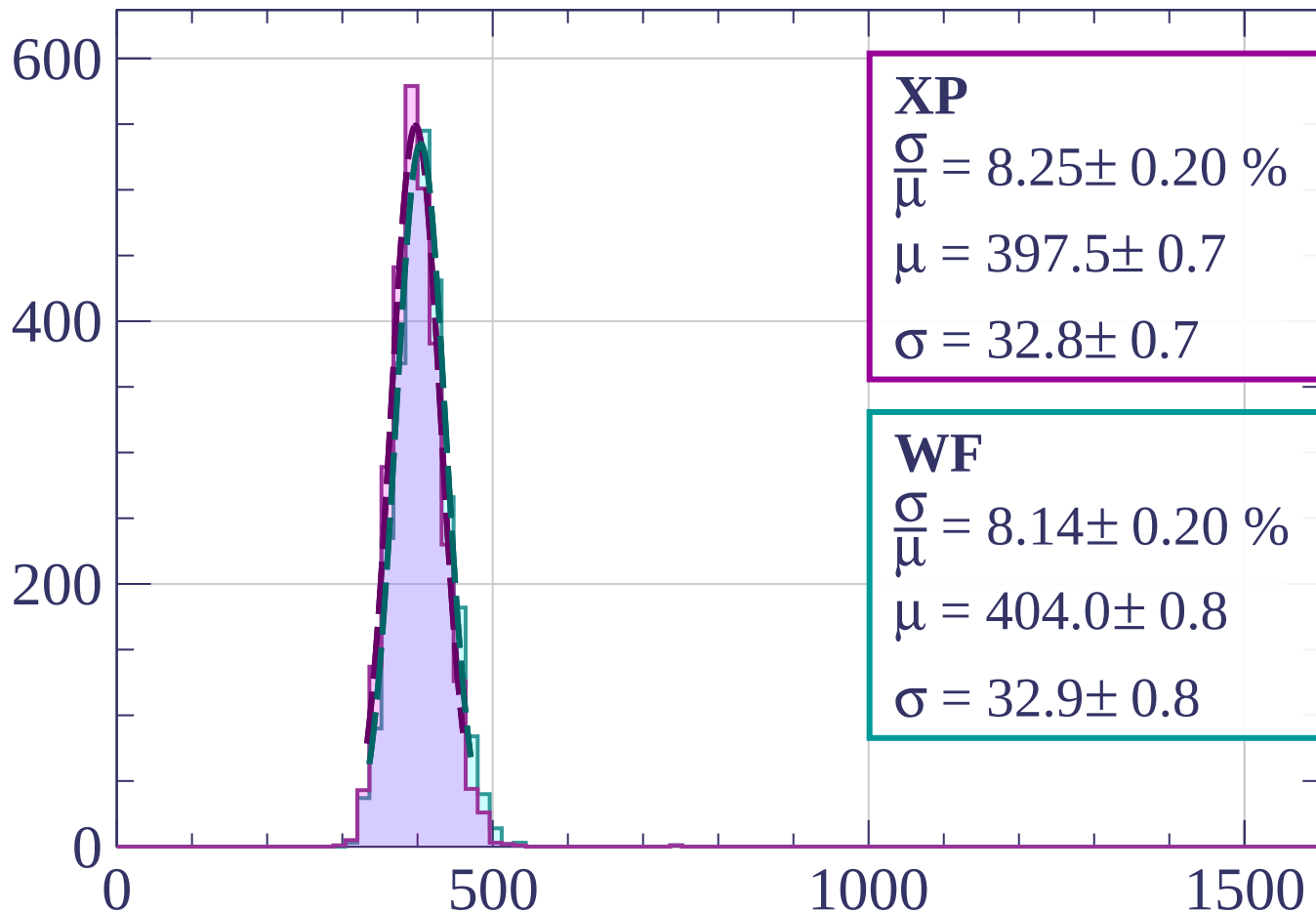
$$\mu = 403.7 \pm 0.8$$

$$\sigma = 31.9 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $72 < \phi < 75$

Count



XP

$$\frac{\sigma}{\mu} = 8.25 \pm 0.20 \%$$

$$\mu = 397.5 \pm 0.7$$

$$\sigma = 32.8 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.14 \pm 0.20 \%$$

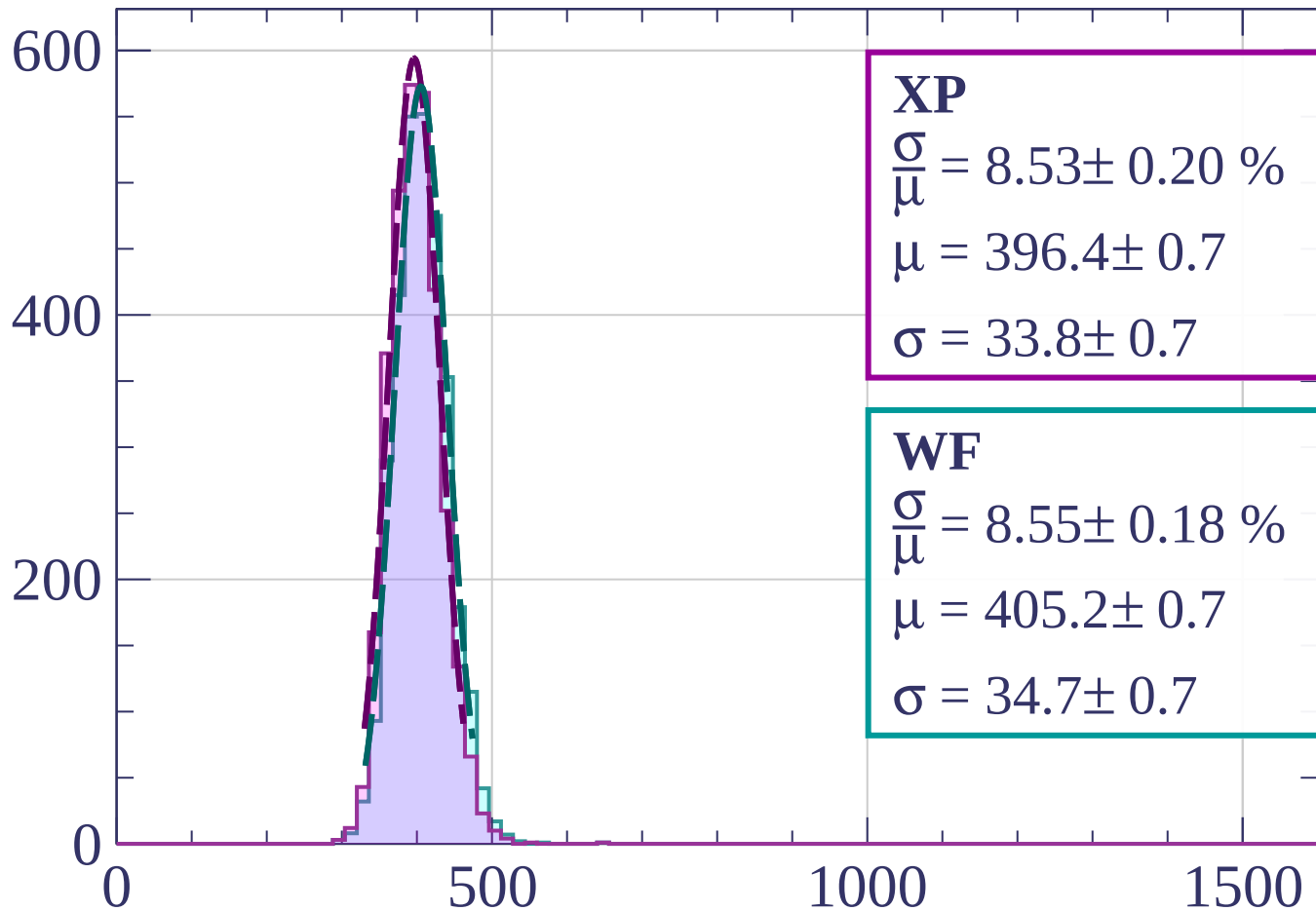
$$\mu = 404.0 \pm 0.8$$

$$\sigma = 32.9 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $75 < \phi < 78$

Count



XP

$$\frac{\sigma}{\mu} = 8.53 \pm 0.20 \%$$

$$\mu = 396.4 \pm 0.7$$

$$\sigma = 33.8 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.55 \pm 0.18 \%$$

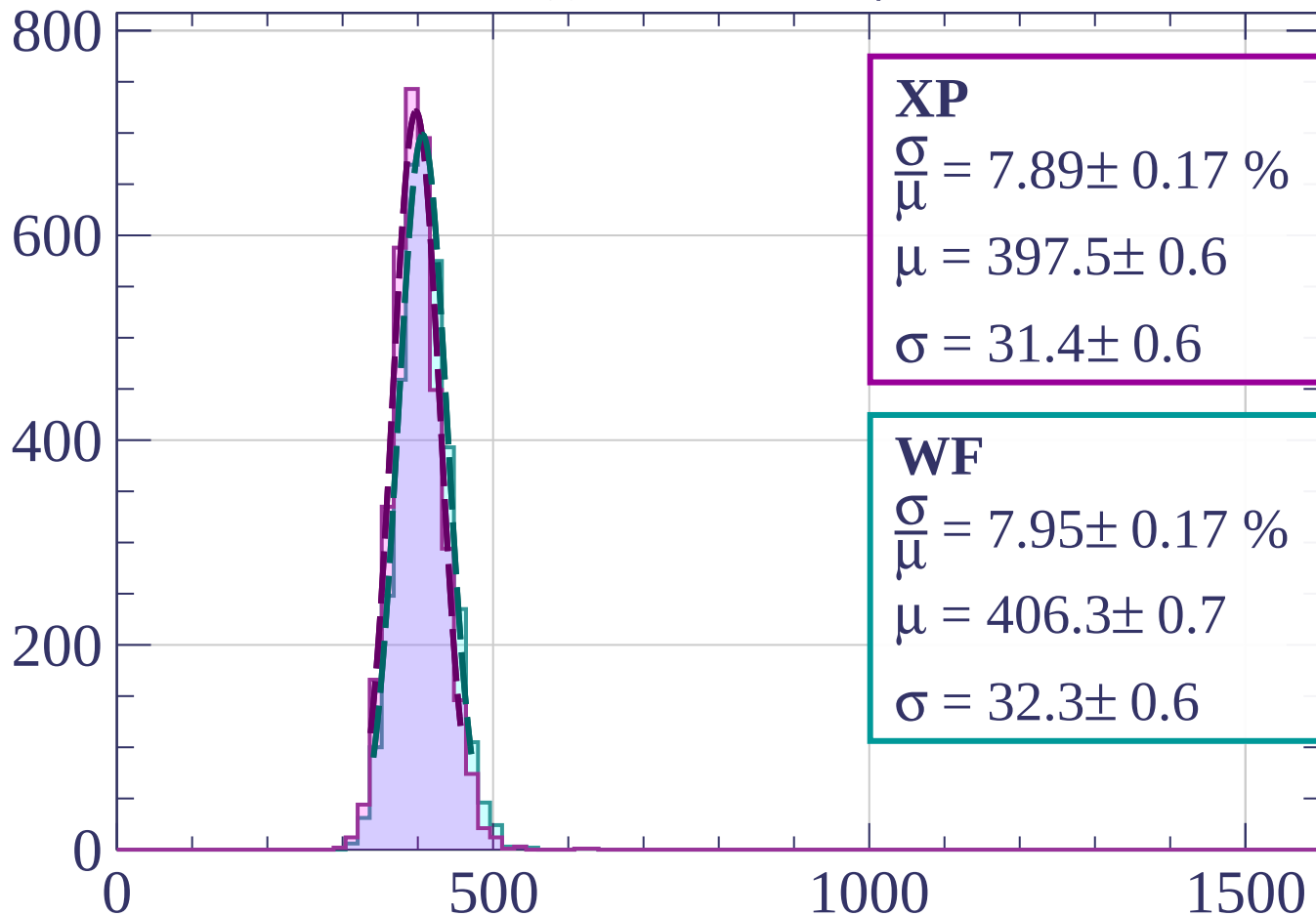
$$\mu = 405.2 \pm 0.7$$

$$\sigma = 34.7 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $78 < \phi < 81$

Count



XP

$$\frac{\sigma}{\mu} = 7.89 \pm 0.17 \%$$

$$\mu = 397.5 \pm 0.6$$

$$\sigma = 31.4 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 7.95 \pm 0.17 \%$$

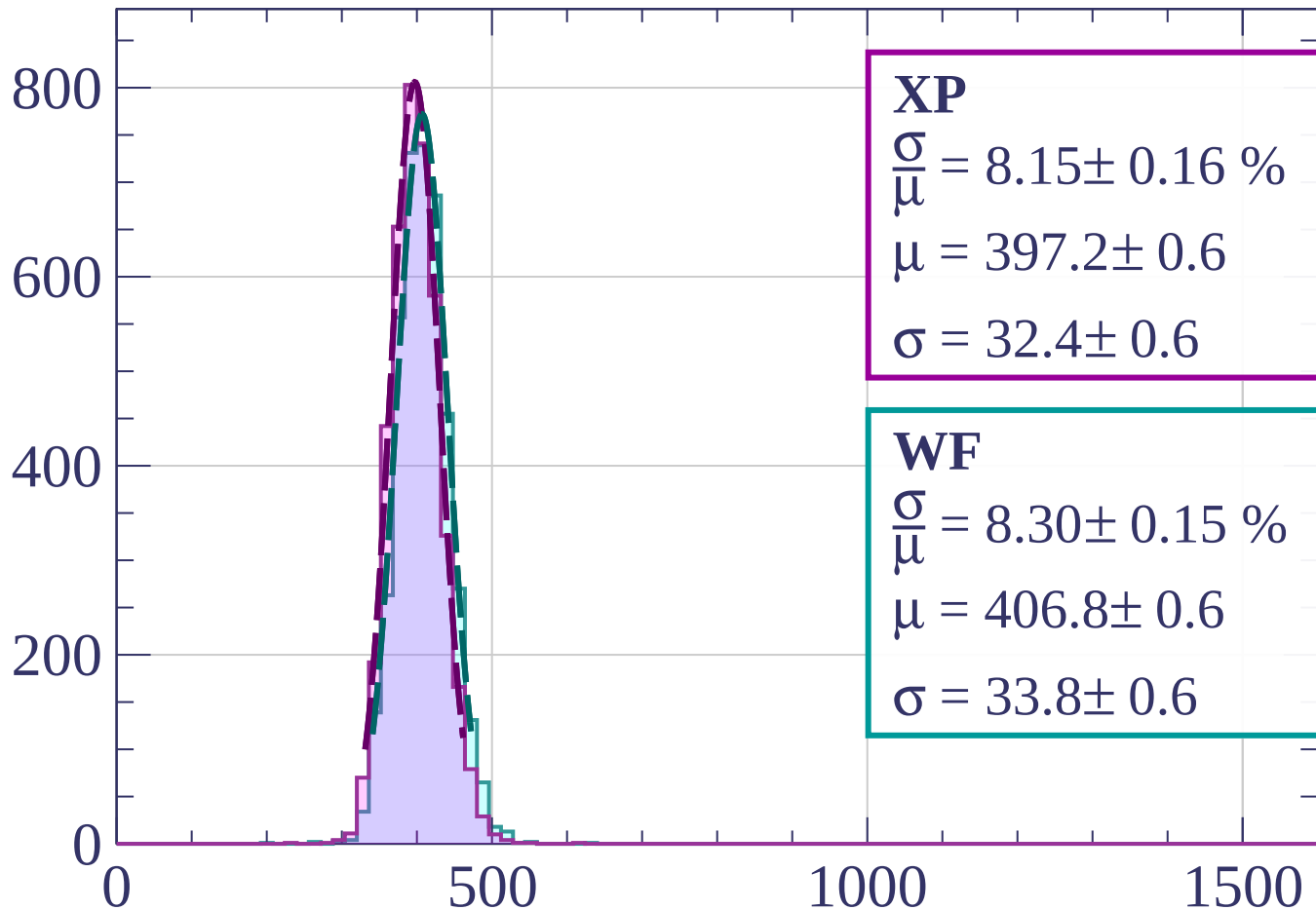
$$\mu = 406.3 \pm 0.7$$

$$\sigma = 32.3 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $81 < \phi < 84$

Count



XP

$$\frac{\sigma}{\mu} = 8.15 \pm 0.16 \%$$

$$\mu = 397.2 \pm 0.6$$

$$\sigma = 32.4 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.30 \pm 0.15 \%$$

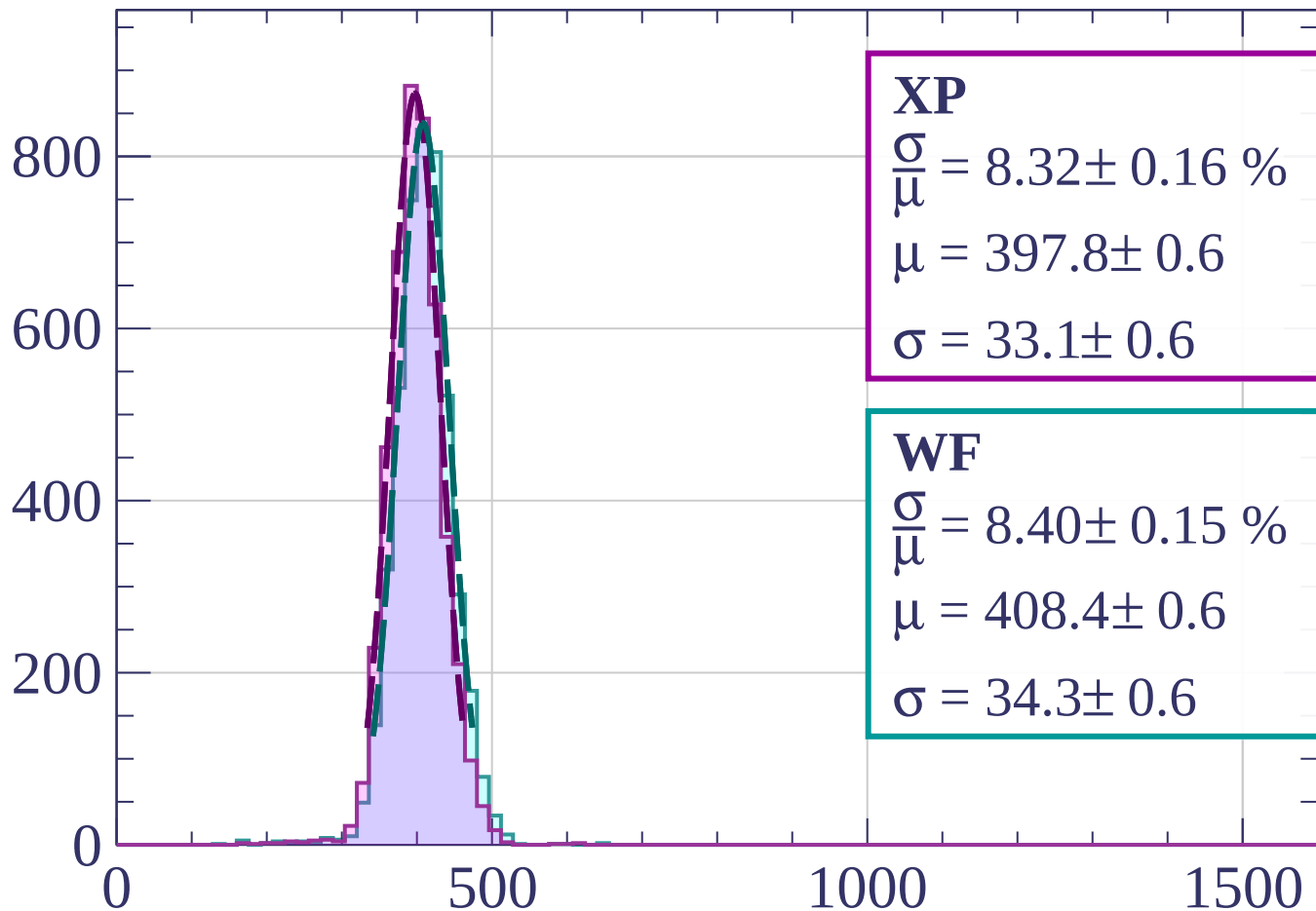
$$\mu = 406.8 \pm 0.6$$

$$\sigma = 33.8 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $84 < \phi < 87$

Count



XP

$$\frac{\sigma}{\mu} = 8.32 \pm 0.16 \%$$

$$\mu = 397.8 \pm 0.6$$

$$\sigma = 33.1 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.40 \pm 0.15 \%$$

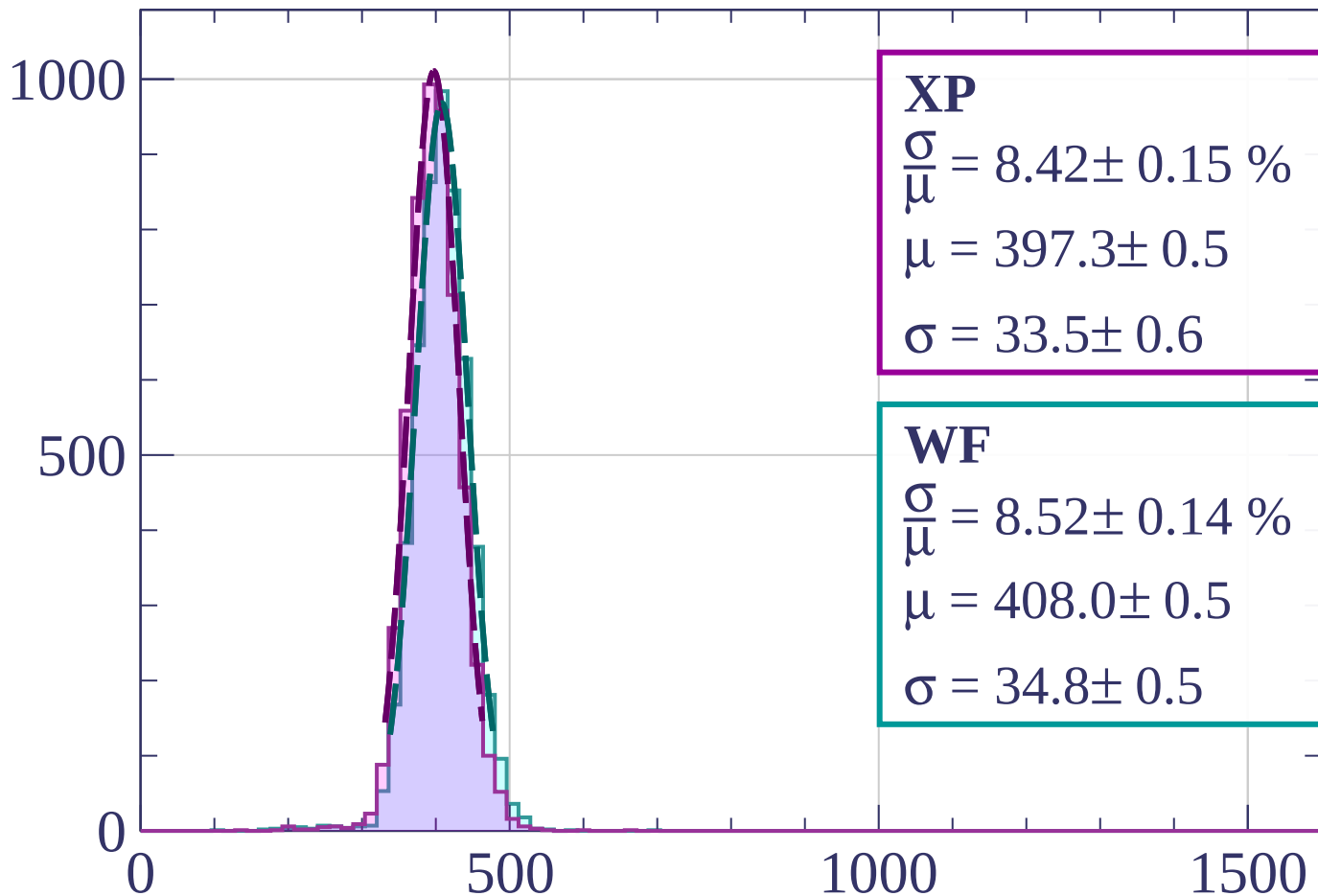
$$\mu = 408.4 \pm 0.6$$

$$\sigma = 34.3 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $87 < \phi < 90$

Count



XP

$$\frac{\sigma}{\mu} = 8.42 \pm 0.15 \%$$

$$\mu = 397.3 \pm 0.5$$

$$\sigma = 33.5 \pm 0.6$$

WF

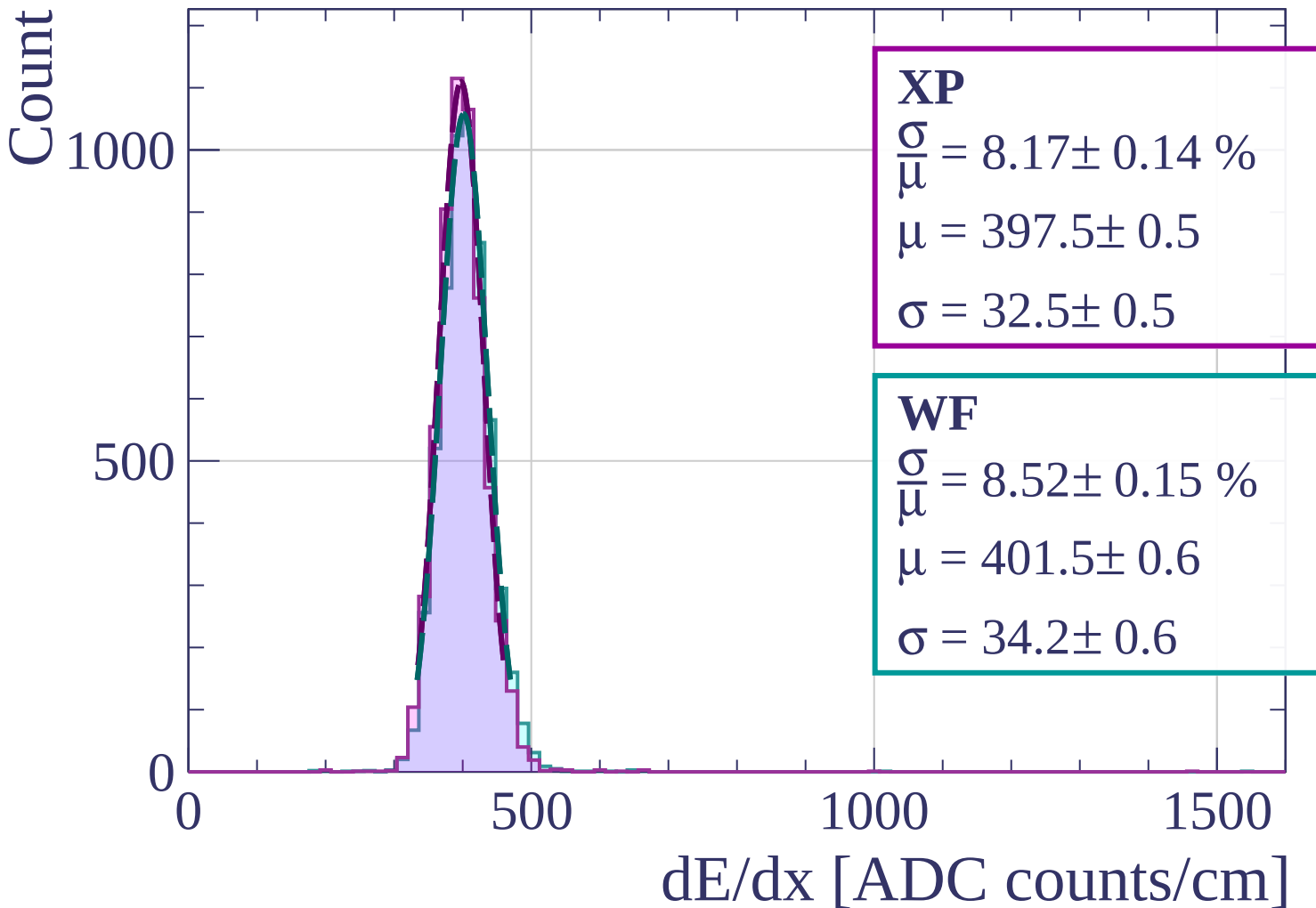
$$\frac{\sigma}{\mu} = 8.52 \pm 0.14 \%$$

$$\mu = 408.0 \pm 0.5$$

$$\sigma = 34.8 \pm 0.5$$

dE/dx [ADC counts/cm]

Energy loss | $0 < \theta < 3$



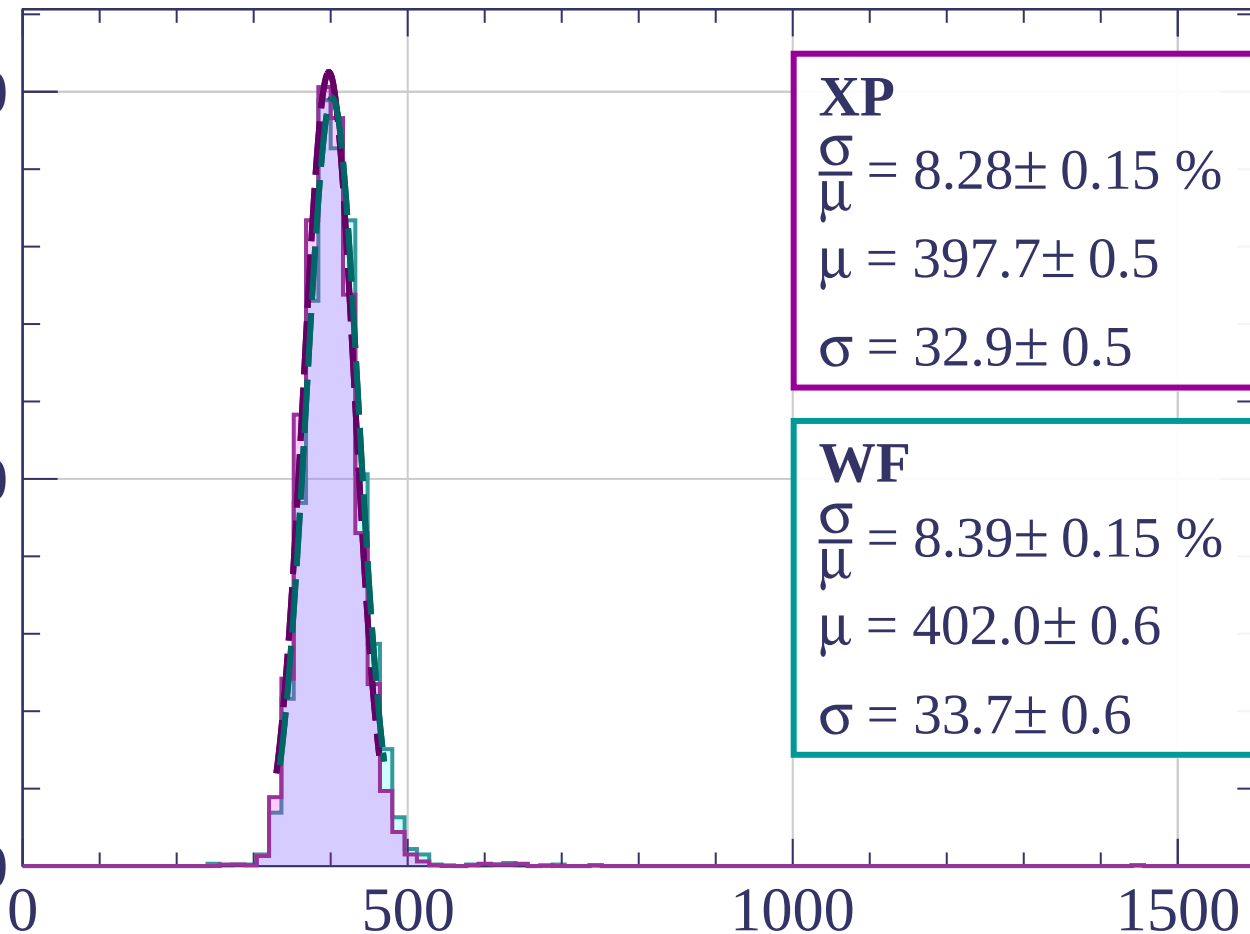
Energy loss | $3 < \theta < 6$

Count

1000

500

0



XP

$$\frac{\sigma}{\mu} = 8.28 \pm 0.15 \%$$

$$\mu = 397.7 \pm 0.5$$

$$\sigma = 32.9 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 8.39 \pm 0.15 \%$$

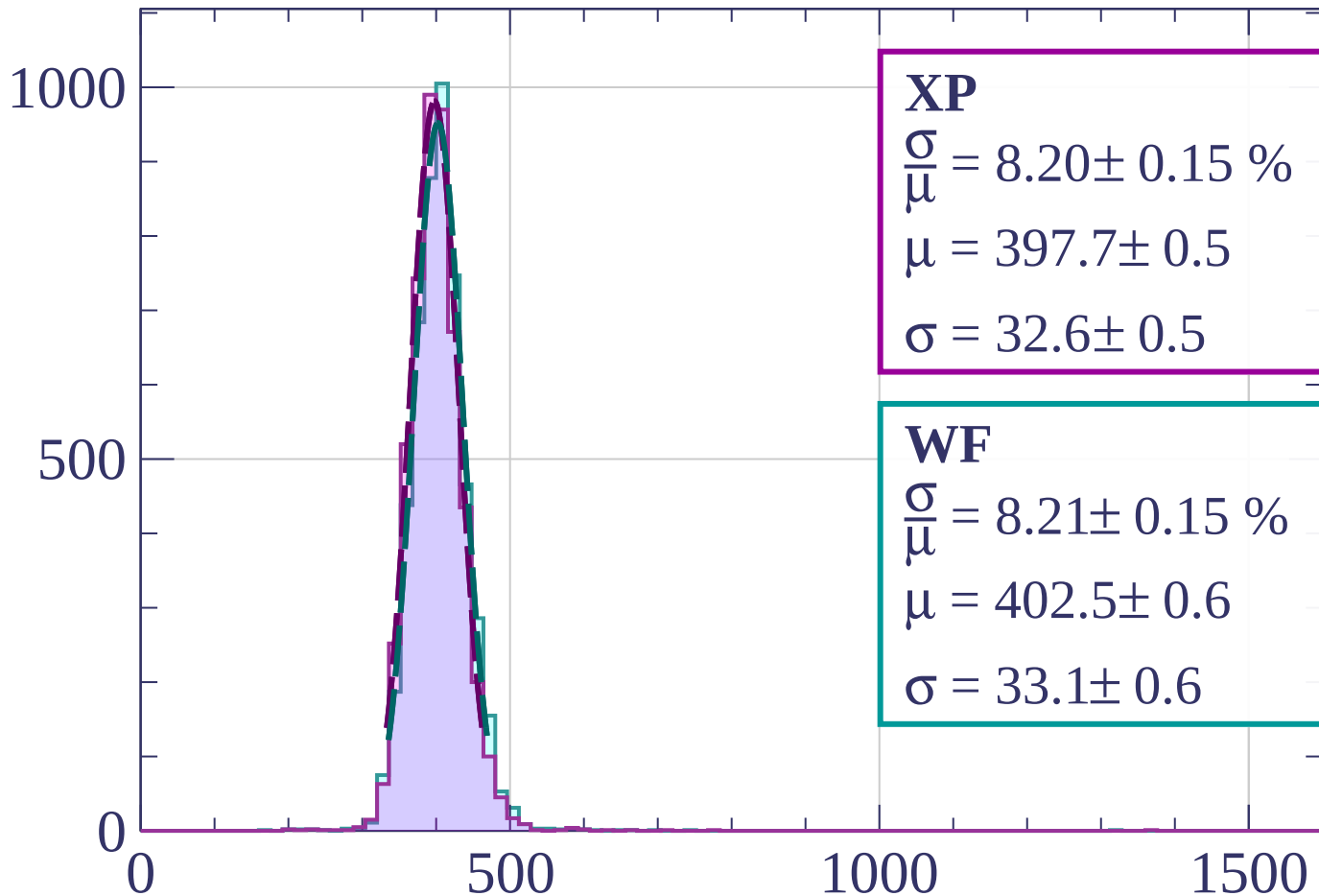
$$\mu = 402.0 \pm 0.6$$

$$\sigma = 33.7 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $6 < \theta < 9$

Count



XP

$$\frac{\sigma}{\mu} = 8.20 \pm 0.15 \%$$

$$\mu = 397.7 \pm 0.5$$

$$\sigma = 32.6 \pm 0.5$$

WF

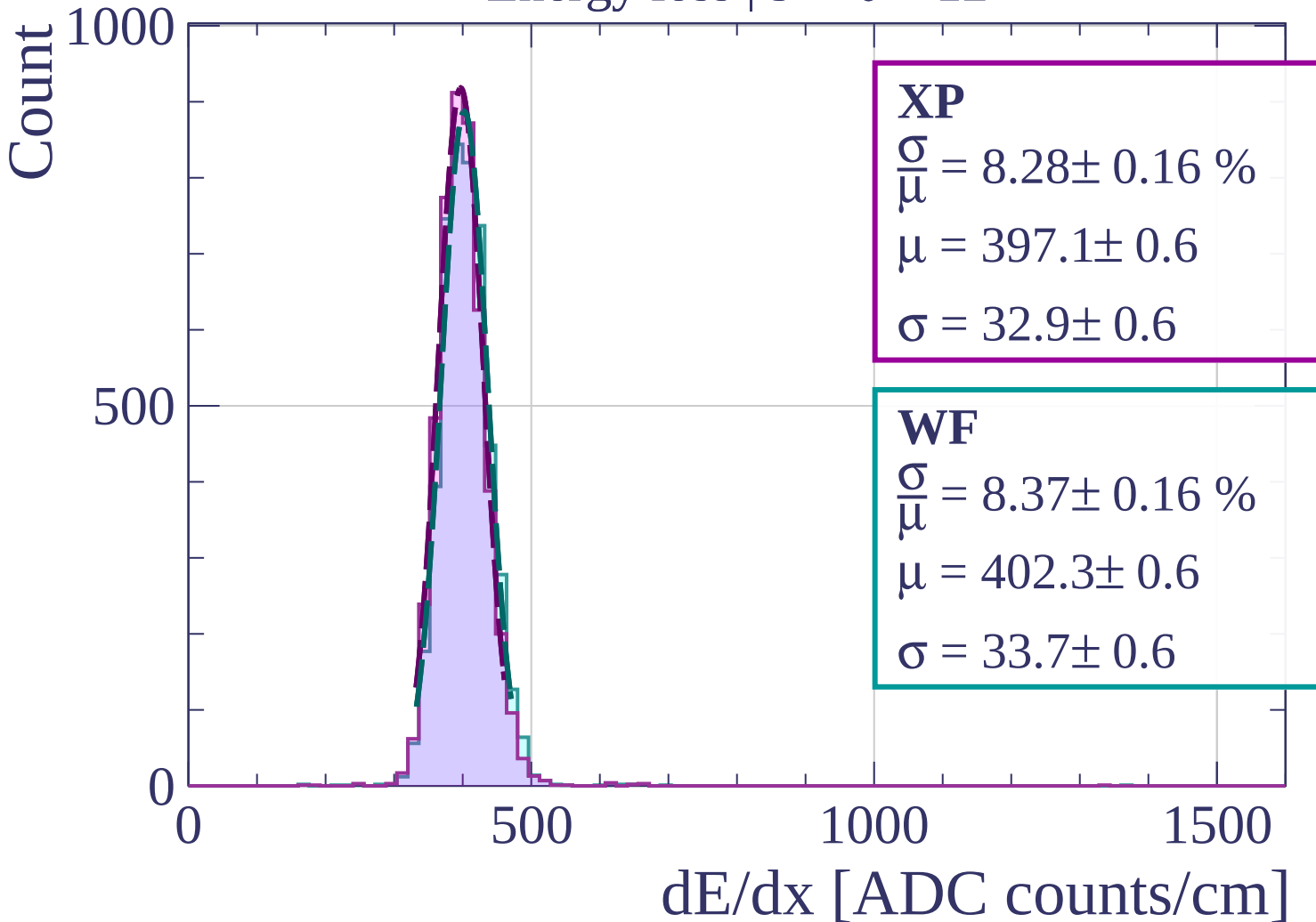
$$\frac{\sigma}{\mu} = 8.21 \pm 0.15 \%$$

$$\mu = 402.5 \pm 0.6$$

$$\sigma = 33.1 \pm 0.6$$

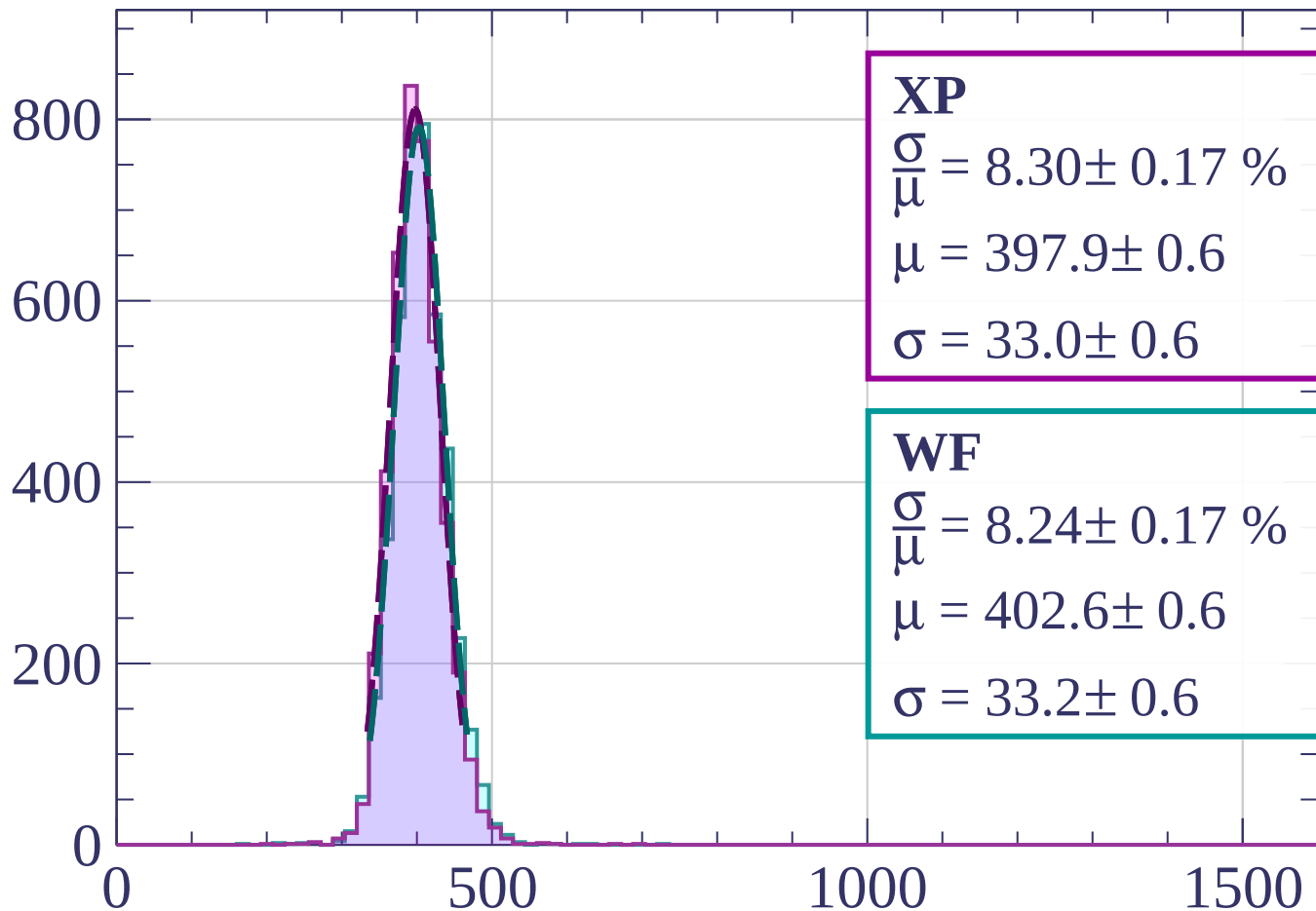
dE/dx [ADC counts/cm]

Energy loss | $9 < \theta < 12$



Energy loss | $12 < \theta < 15$

Count



XP

$$\frac{\sigma}{\mu} = 8.30 \pm 0.17 \%$$

$$\mu = 397.9 \pm 0.6$$

$$\sigma = 33.0 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.24 \pm 0.17 \%$$

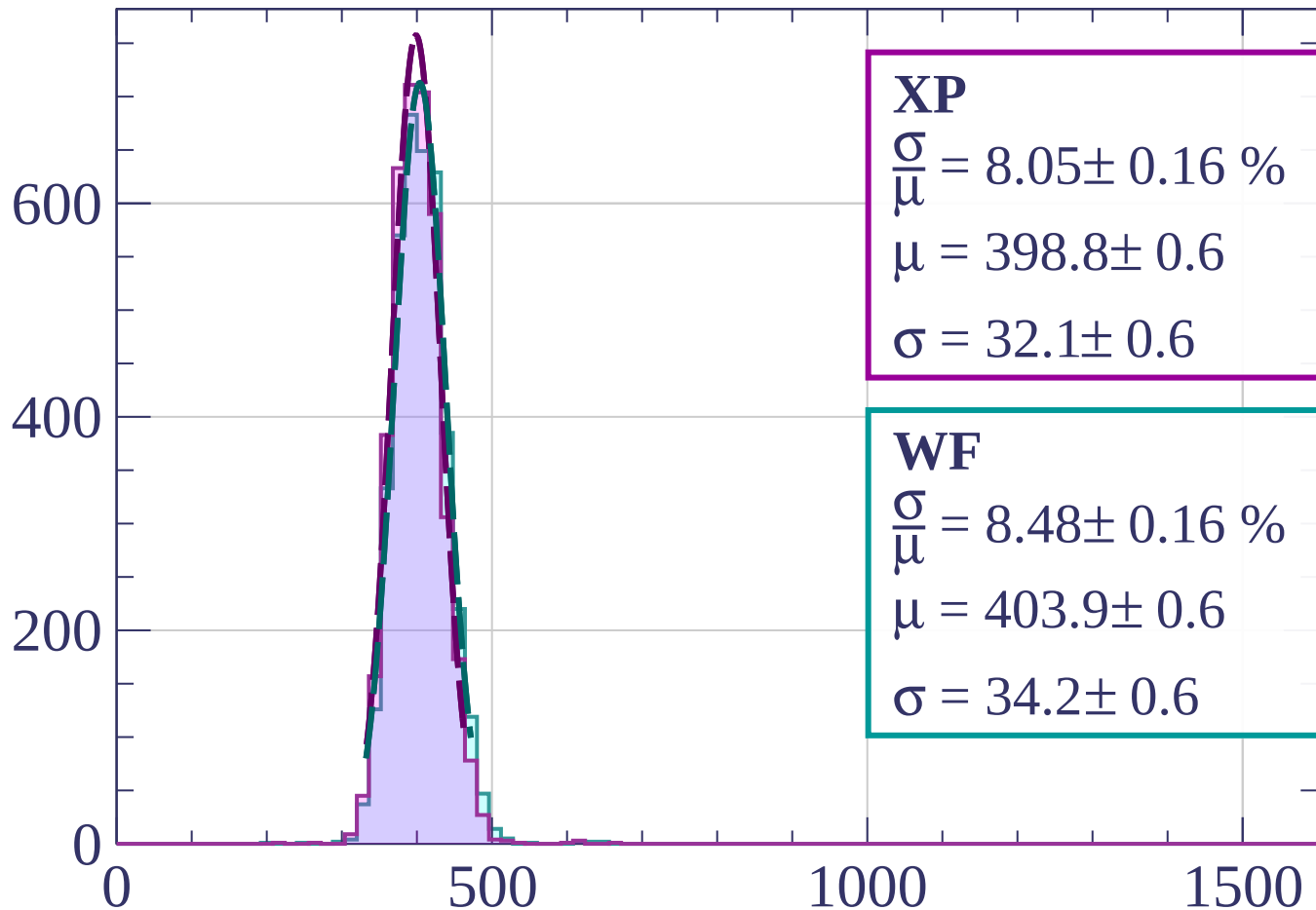
$$\mu = 402.6 \pm 0.6$$

$$\sigma = 33.2 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $15 < \theta < 18$

Count



XP

$$\frac{\sigma}{\mu} = 8.05 \pm 0.16 \%$$

$$\mu = 398.8 \pm 0.6$$

$$\sigma = 32.1 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.48 \pm 0.16 \%$$

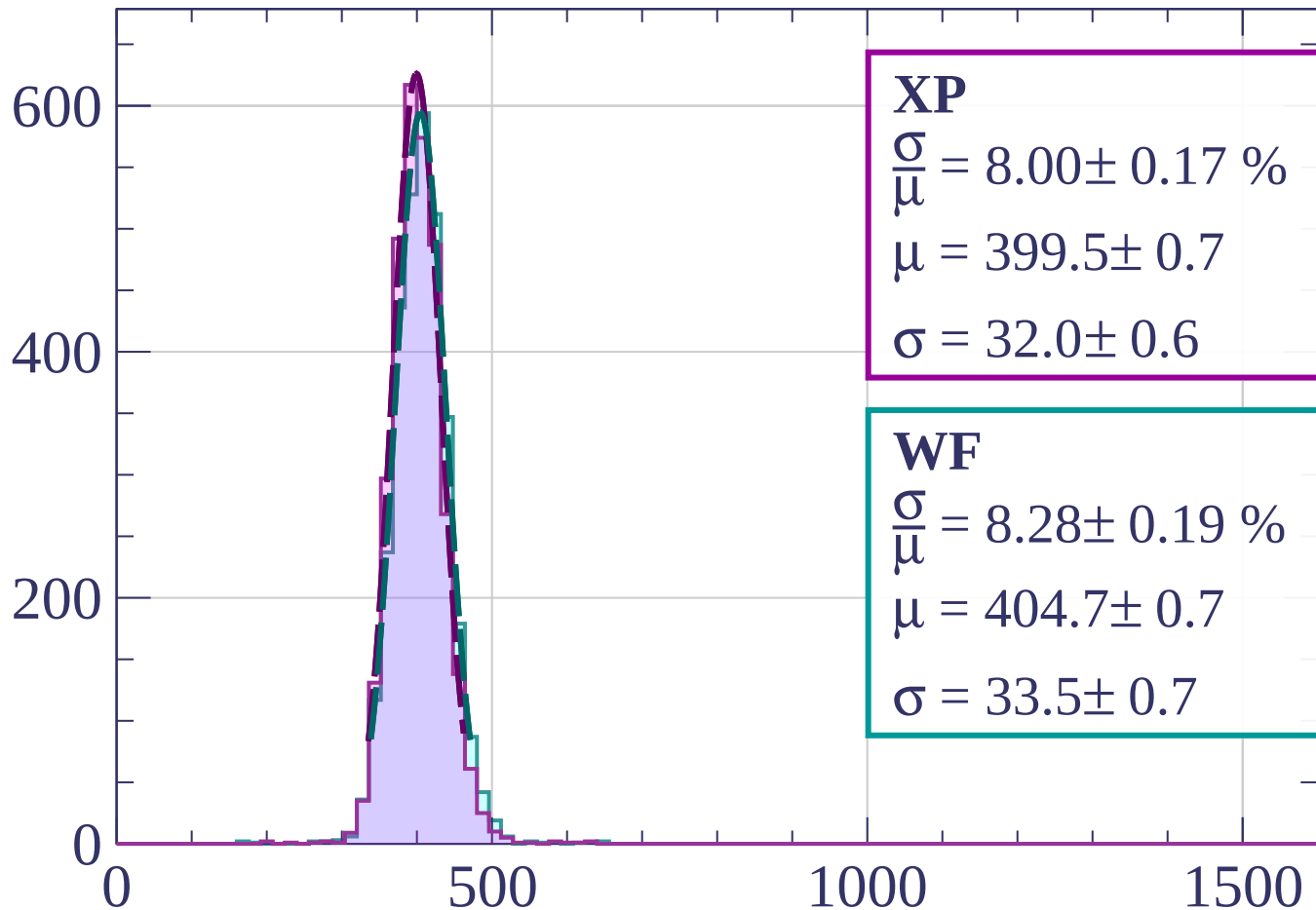
$$\mu = 403.9 \pm 0.6$$

$$\sigma = 34.2 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $18 < \theta < 21$

Count



XP

$$\frac{\sigma}{\mu} = 8.00 \pm 0.17 \%$$

$$\mu = 399.5 \pm 0.7$$

$$\sigma = 32.0 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.28 \pm 0.19 \%$$

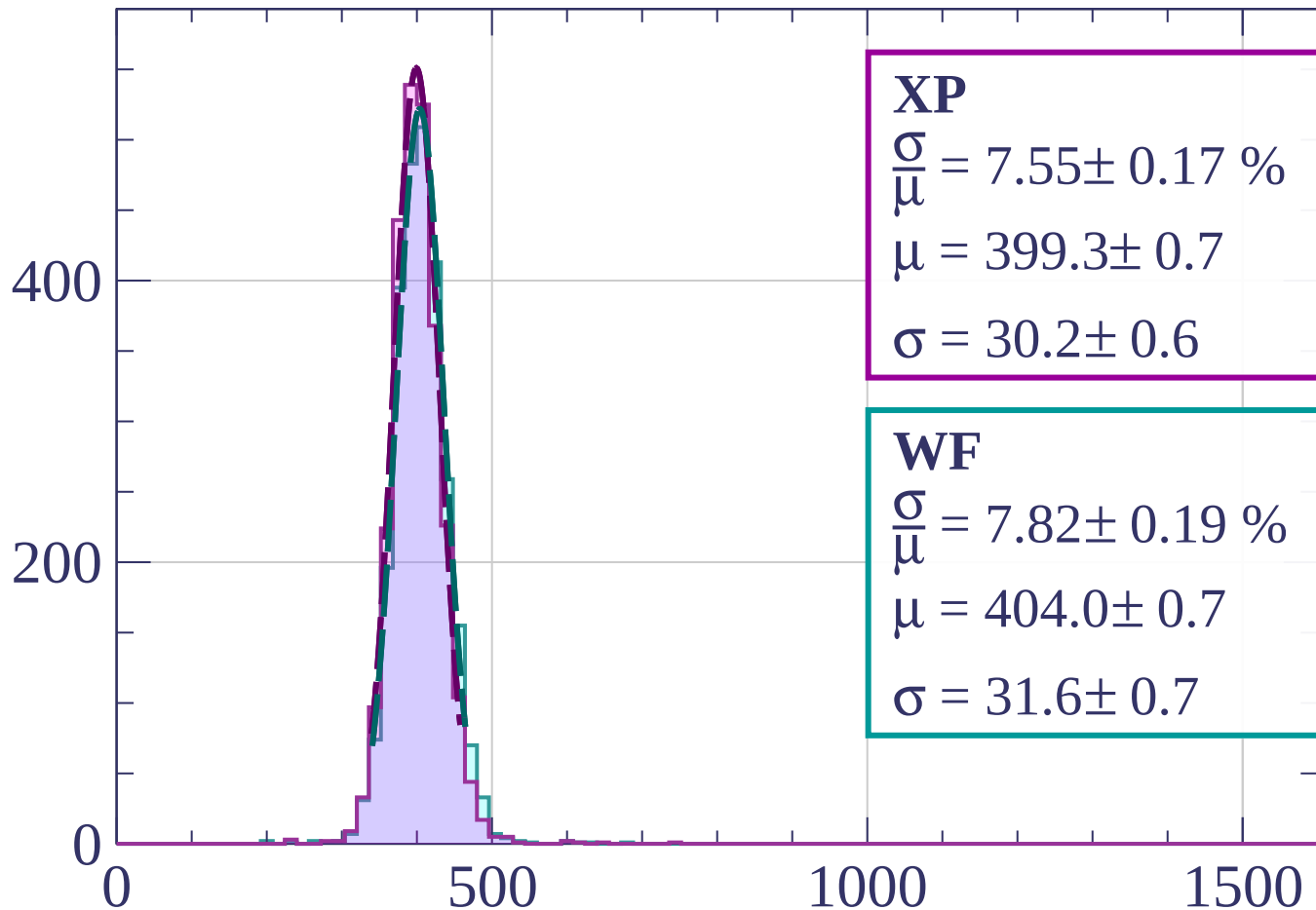
$$\mu = 404.7 \pm 0.7$$

$$\sigma = 33.5 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $21 < \theta < 24$

Count



XP

$$\frac{\sigma}{\mu} = 7.55 \pm 0.17 \%$$

$$\mu = 399.3 \pm 0.7$$

$$\sigma = 30.2 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 7.82 \pm 0.19 \%$$

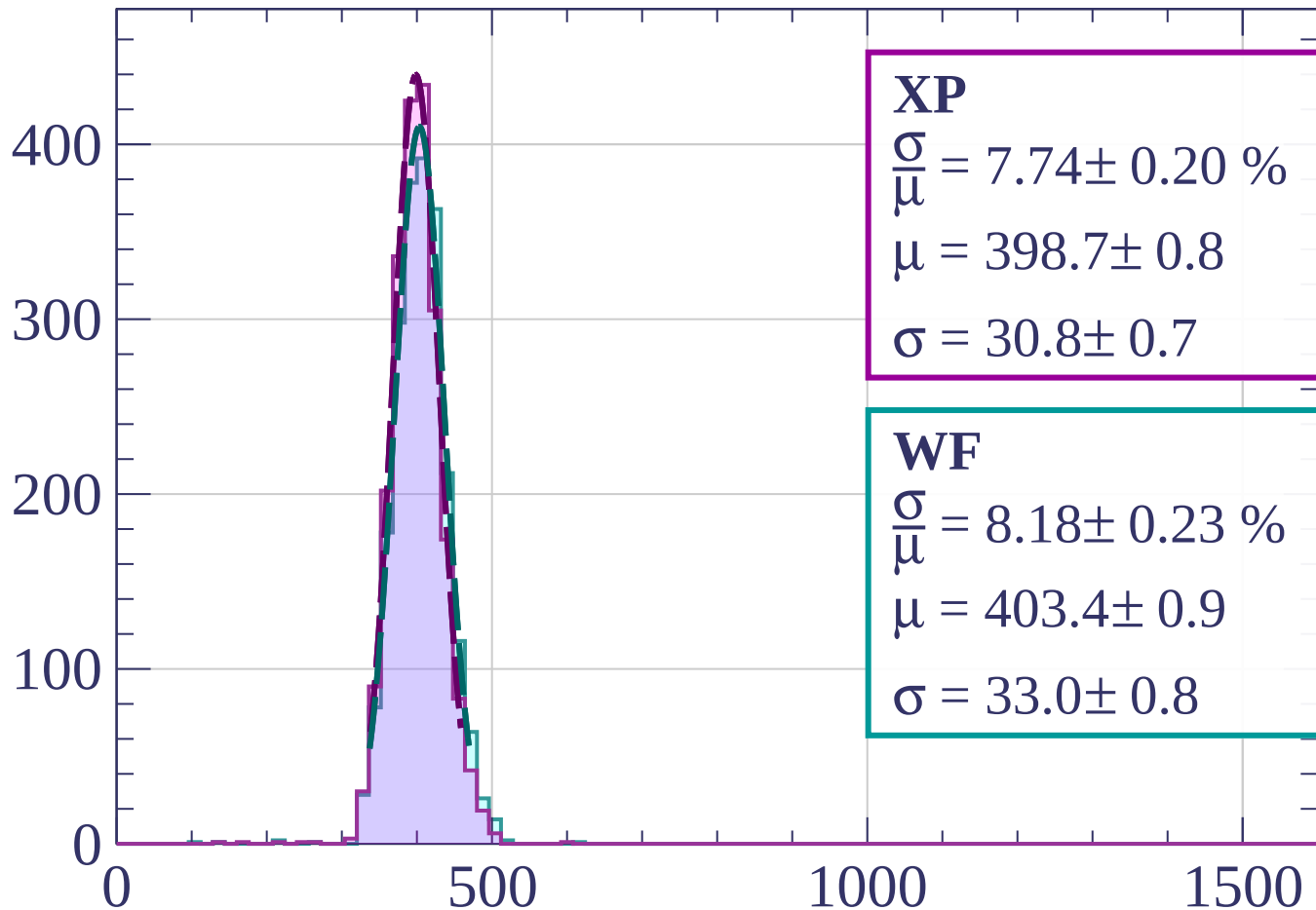
$$\mu = 404.0 \pm 0.7$$

$$\sigma = 31.6 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $24 < \theta < 27$

Count



XP

$$\frac{\sigma}{\mu} = 7.74 \pm 0.20 \%$$

$$\mu = 398.7 \pm 0.8$$

$$\sigma = 30.8 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.18 \pm 0.23 \%$$

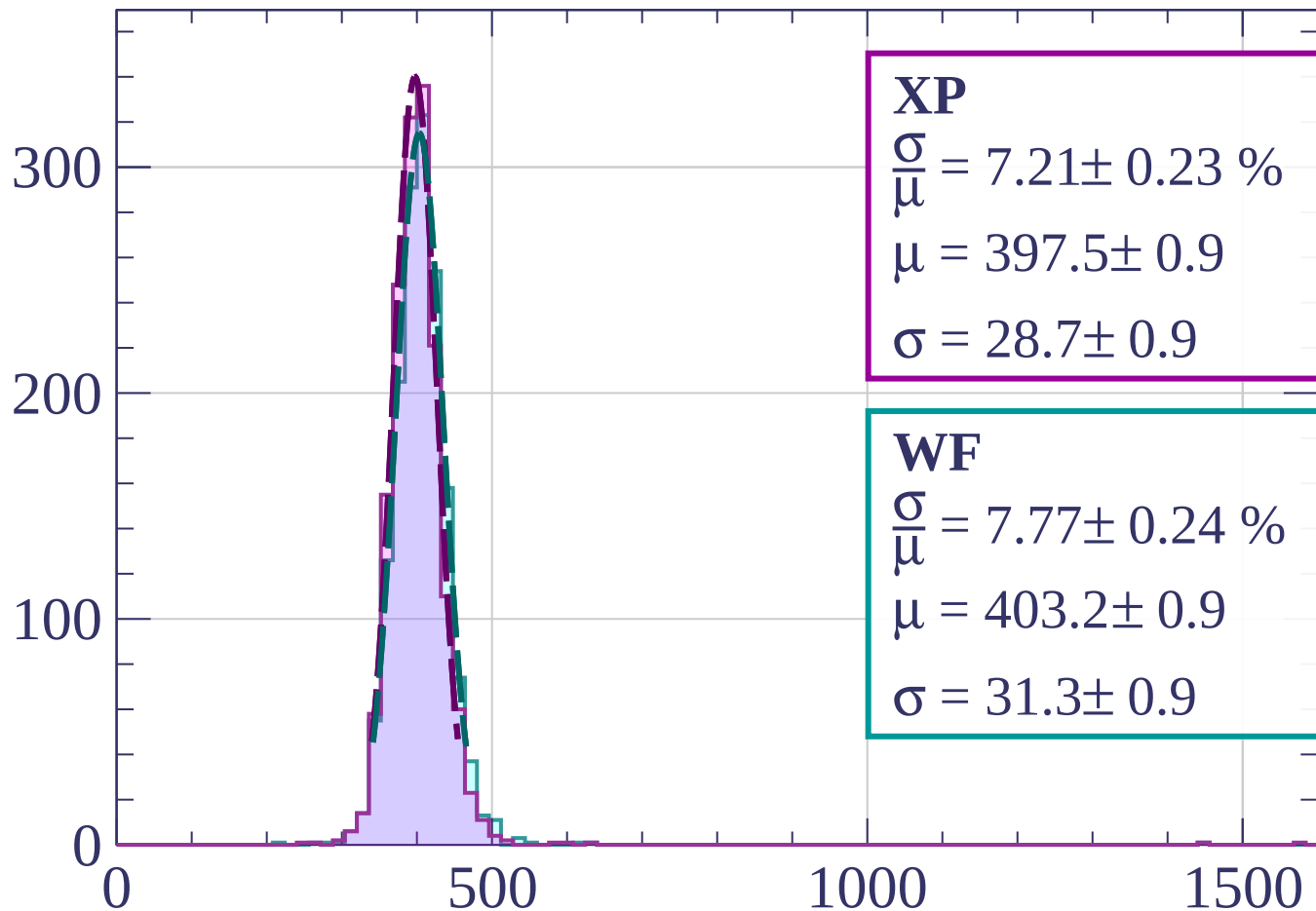
$$\mu = 403.4 \pm 0.9$$

$$\sigma = 33.0 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $27 < \theta < 30$

Count



XP

$$\frac{\sigma}{\mu} = 7.21 \pm 0.23 \%$$

$$\mu = 397.5 \pm 0.9$$

$$\sigma = 28.7 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 7.77 \pm 0.24 \%$$

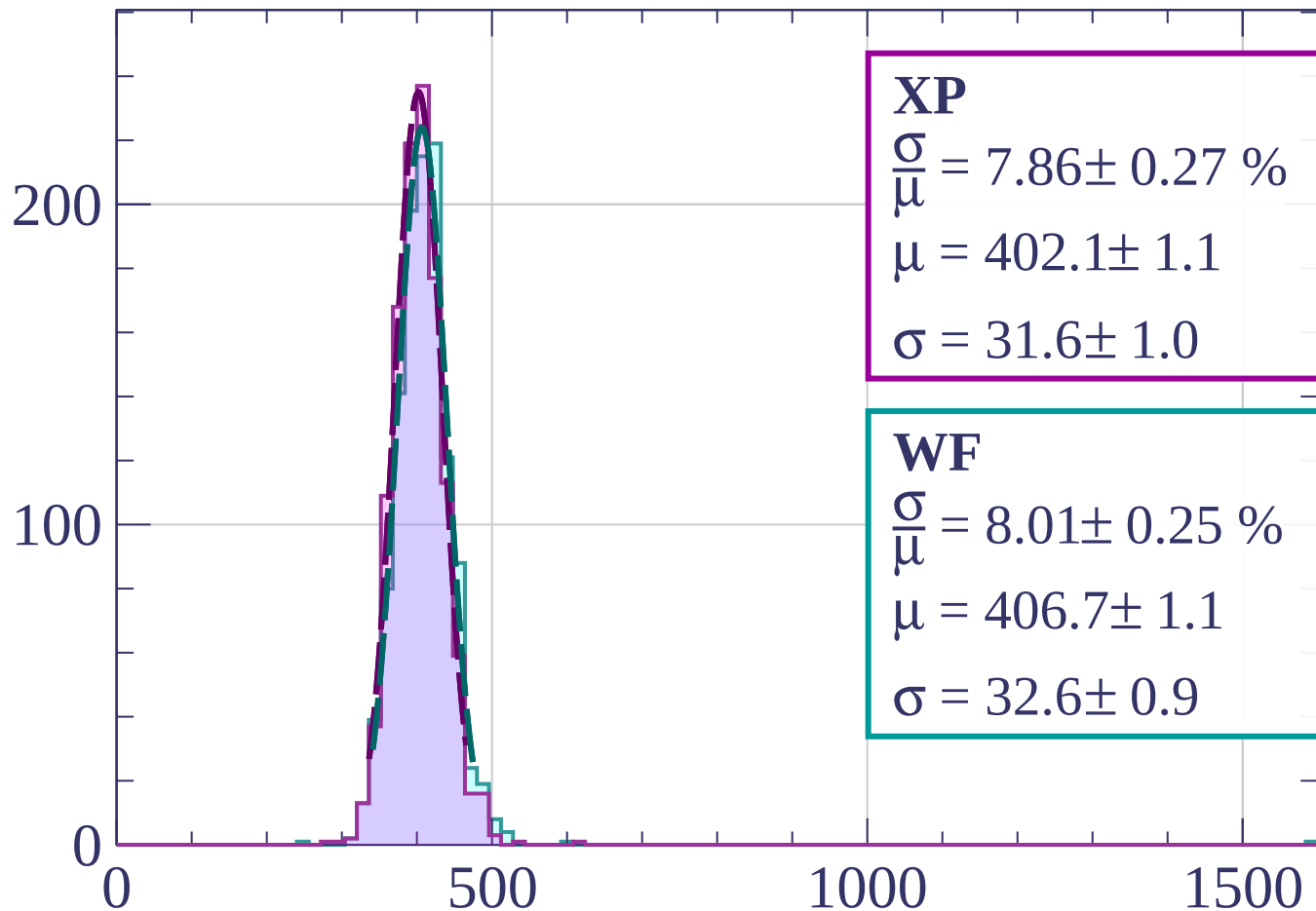
$$\mu = 403.2 \pm 0.9$$

$$\sigma = 31.3 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $30 < \theta < 33$

Count



XP

$$\frac{\sigma}{\mu} = 7.86 \pm 0.27 \%$$

$$\mu = 402.1 \pm 1.1$$

$$\sigma = 31.6 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 8.01 \pm 0.25 \%$$

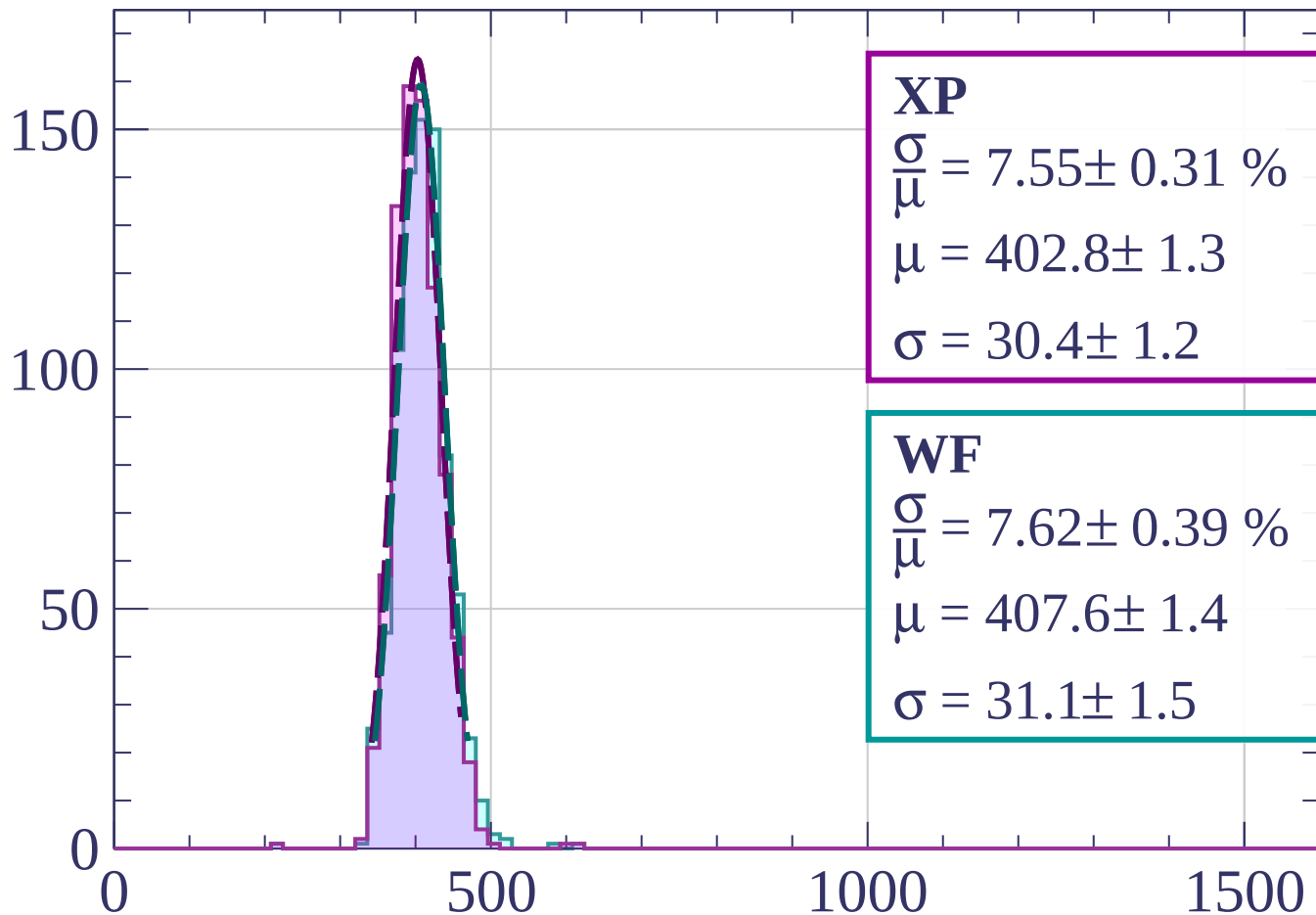
$$\mu = 406.7 \pm 1.1$$

$$\sigma = 32.6 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $33 < \theta < 36$

Count



XP

$$\frac{\sigma}{\mu} = 7.55 \pm 0.31 \%$$

$$\mu = 402.8 \pm 1.3$$

$$\sigma = 30.4 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 7.62 \pm 0.39 \%$$

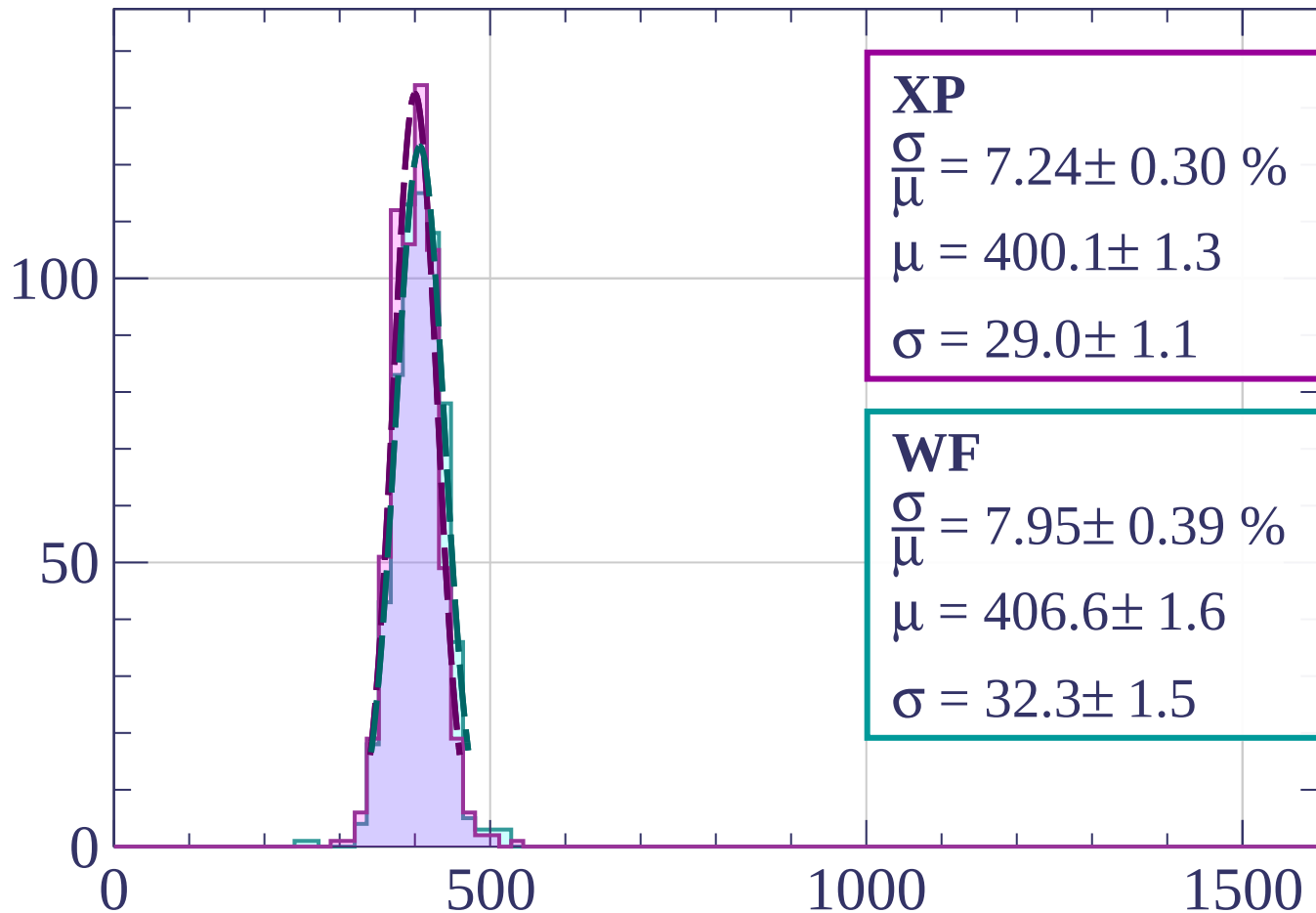
$$\mu = 407.6 \pm 1.4$$

$$\sigma = 31.1 \pm 1.5$$

dE/dx [ADC counts/cm]

Energy loss | $36 < \theta < 39$

Count



XP

$$\frac{\sigma}{\mu} = 7.24 \pm 0.30 \%$$

$$\mu = 400.1 \pm 1.3$$

$$\sigma = 29.0 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 7.95 \pm 0.39 \%$$

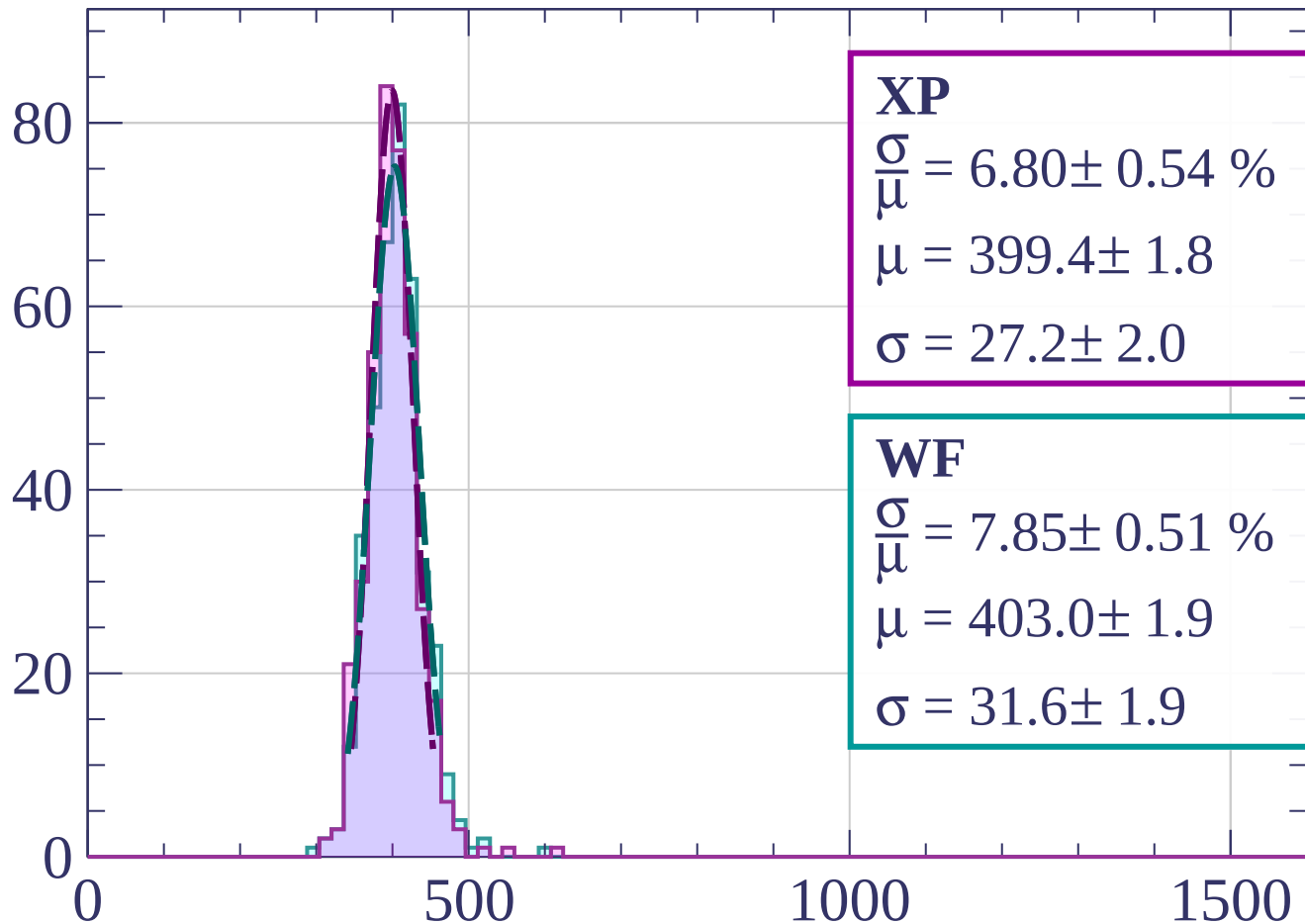
$$\mu = 406.6 \pm 1.6$$

$$\sigma = 32.3 \pm 1.5$$

dE/dx [ADC counts/cm]

Energy loss | $39 < \theta < 42$

Count



XP

$$\frac{\sigma}{\mu} = 6.80 \pm 0.54 \%$$

$$\mu = 399.4 \pm 1.8$$

$$\sigma = 27.2 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 7.85 \pm 0.51 \%$$

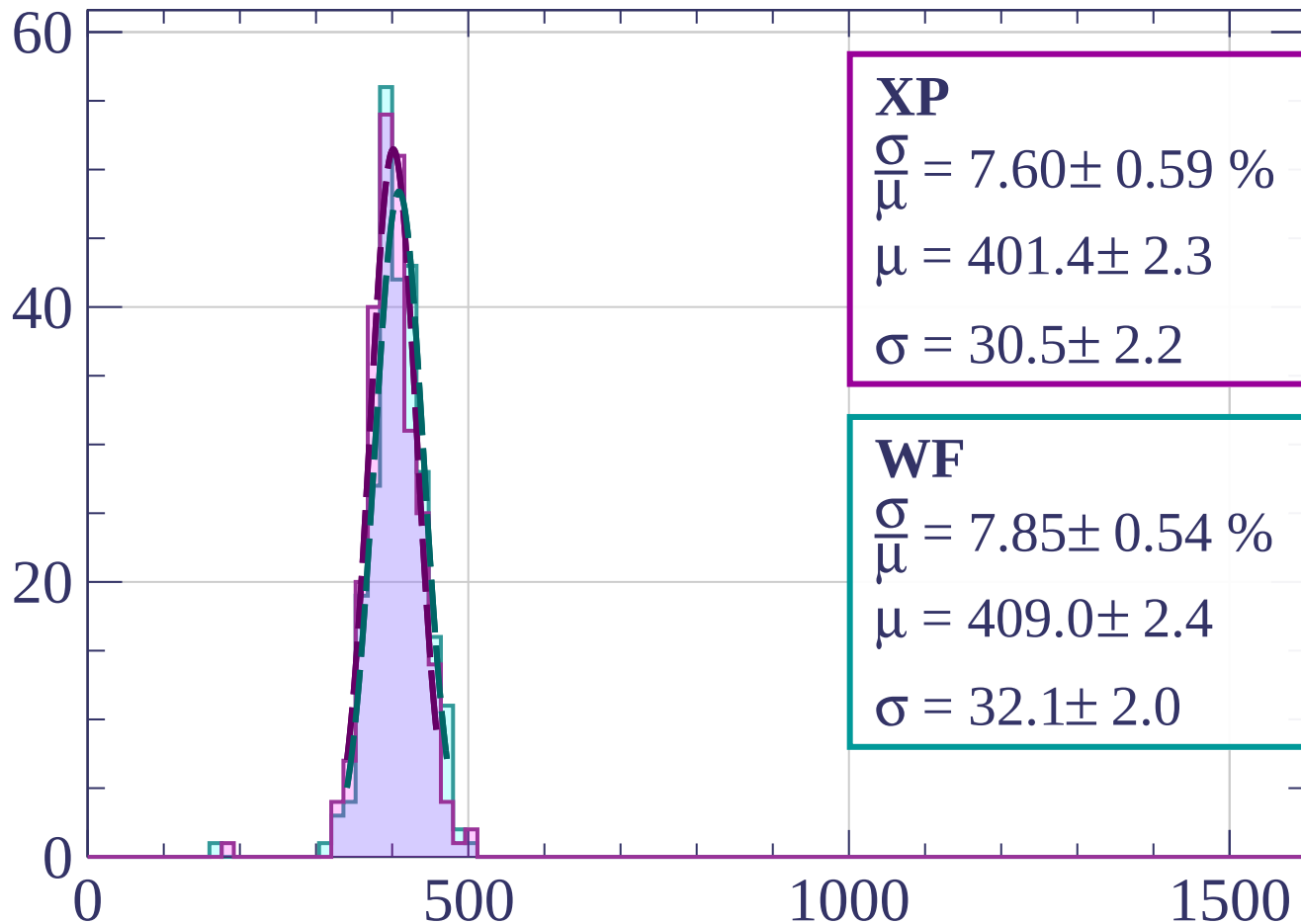
$$\mu = 403.0 \pm 1.9$$

$$\sigma = 31.6 \pm 1.9$$

dE/dx [ADC counts/cm]

Energy loss | $42 < \theta < 45$

Count



XP

$$\frac{\sigma}{\mu} = 7.60 \pm 0.59 \%$$

$$\mu = 401.4 \pm 2.3$$

$$\sigma = 30.5 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 7.85 \pm 0.54 \%$$

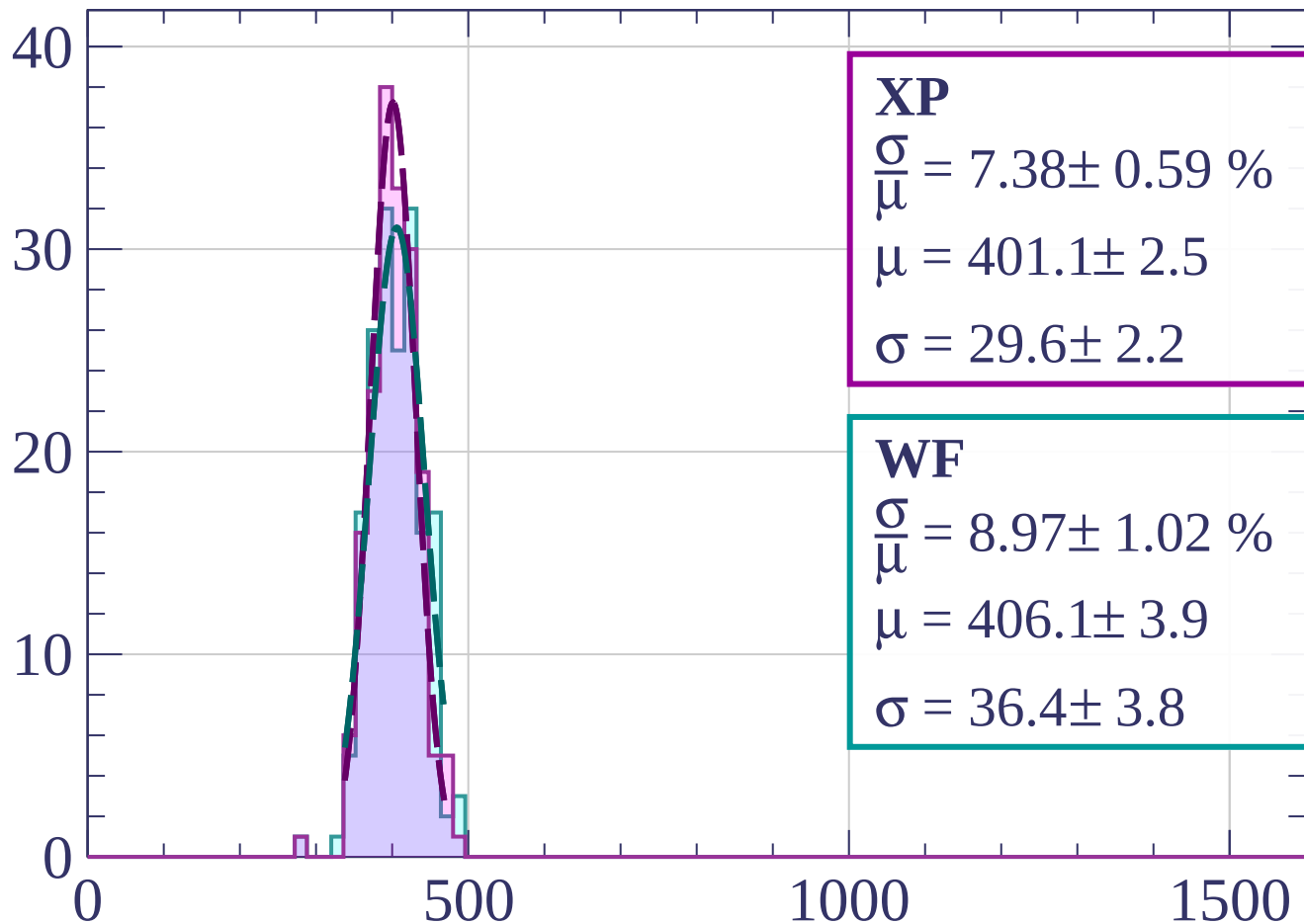
$$\mu = 409.0 \pm 2.4$$

$$\sigma = 32.1 \pm 2.0$$

dE/dx [ADC counts/cm]

Energy loss | $45 < \theta < 48$

Count



XP

$$\frac{\sigma}{\mu} = 7.38 \pm 0.59 \%$$

$$\mu = 401.1 \pm 2.5$$

$$\sigma = 29.6 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 8.97 \pm 1.02 \%$$

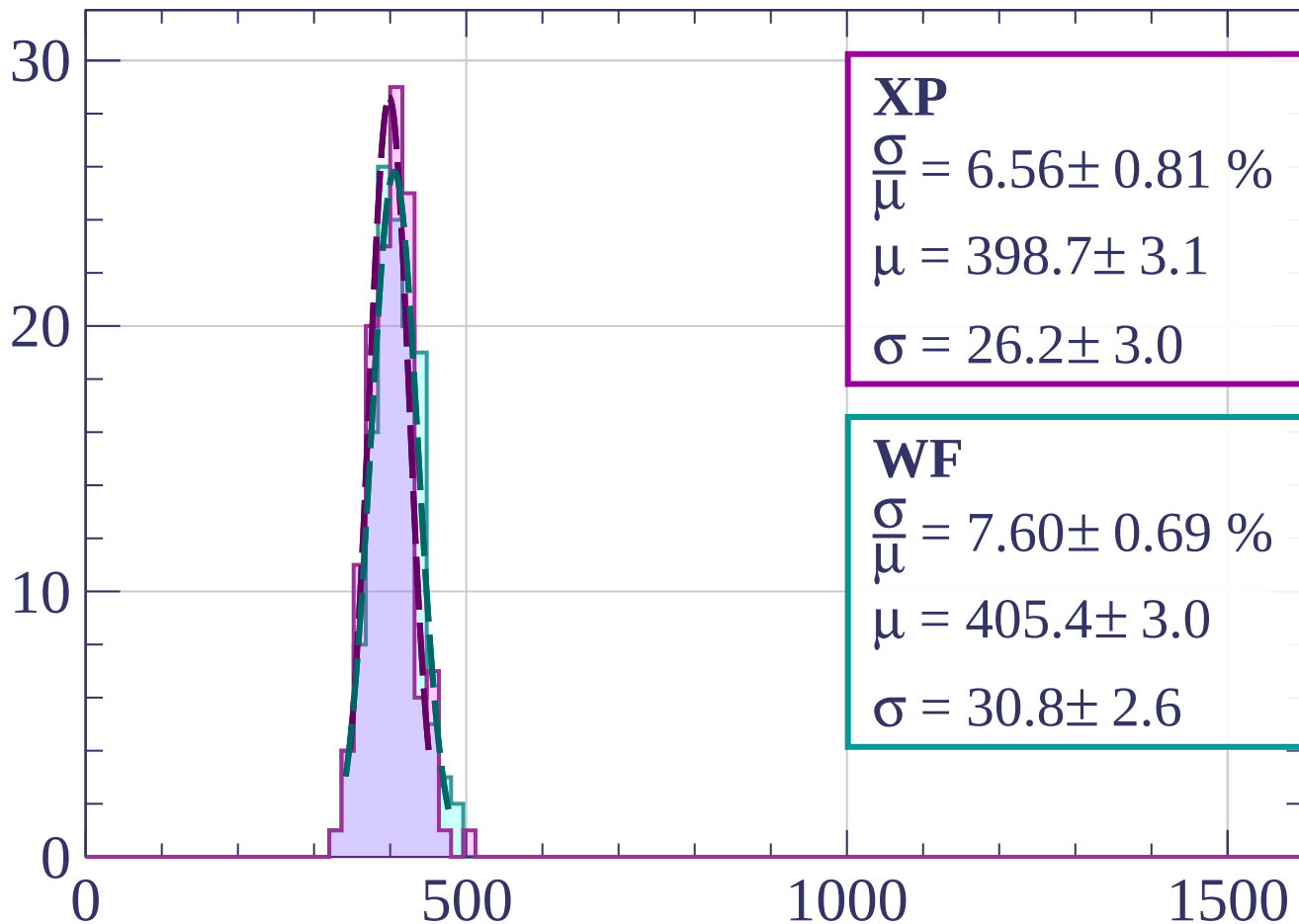
$$\mu = 406.1 \pm 3.9$$

$$\sigma = 36.4 \pm 3.8$$

dE/dx [ADC counts/cm]

Energy loss | $48 < \theta < 51$

Count



XP

$$\frac{\sigma}{\mu} = 6.56 \pm 0.81 \%$$

$$\mu = 398.7 \pm 3.1$$

$$\sigma = 26.2 \pm 3.0$$

WF

$$\frac{\sigma}{\mu} = 7.60 \pm 0.69 \%$$

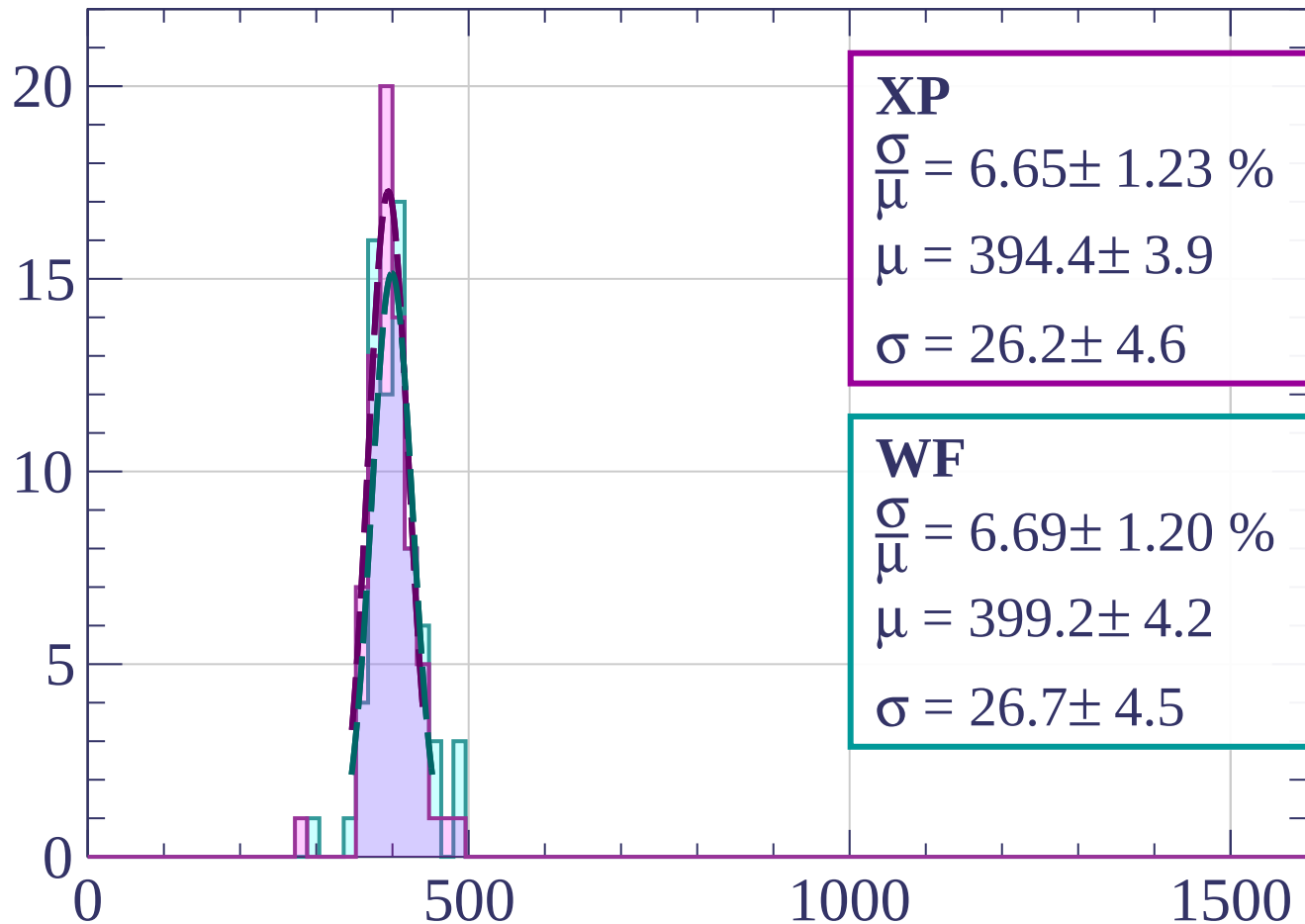
$$\mu = 405.4 \pm 3.0$$

$$\sigma = 30.8 \pm 2.6$$

dE/dx [ADC counts/cm]

Energy loss | $51 < \theta < 54$

Count



XP

$$\frac{\sigma}{\mu} = 6.65 \pm 1.23 \%$$

$$\mu = 394.4 \pm 3.9$$

$$\sigma = 26.2 \pm 4.6$$

WF

$$\frac{\sigma}{\mu} = 6.69 \pm 1.20 \%$$

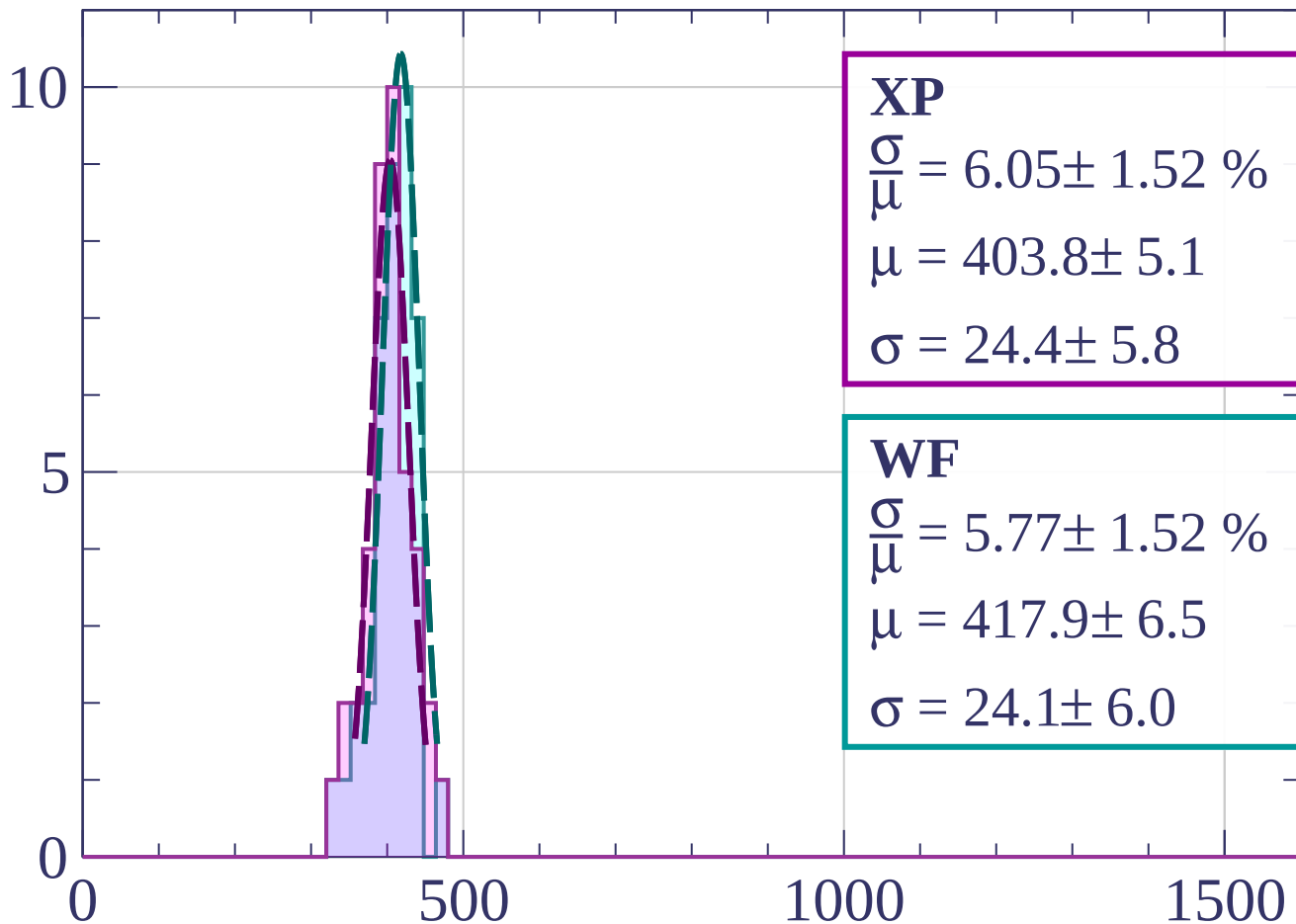
$$\mu = 399.2 \pm 4.2$$

$$\sigma = 26.7 \pm 4.5$$

dE/dx [ADC counts/cm]

Energy loss | $54 < \theta < 57$

Count



XP

$$\frac{\sigma}{\mu} = 6.05 \pm 1.52 \%$$

$$\mu = 403.8 \pm 5.1$$

$$\sigma = 24.4 \pm 5.8$$

WF

$$\frac{\sigma}{\mu} = 5.77 \pm 1.52 \%$$

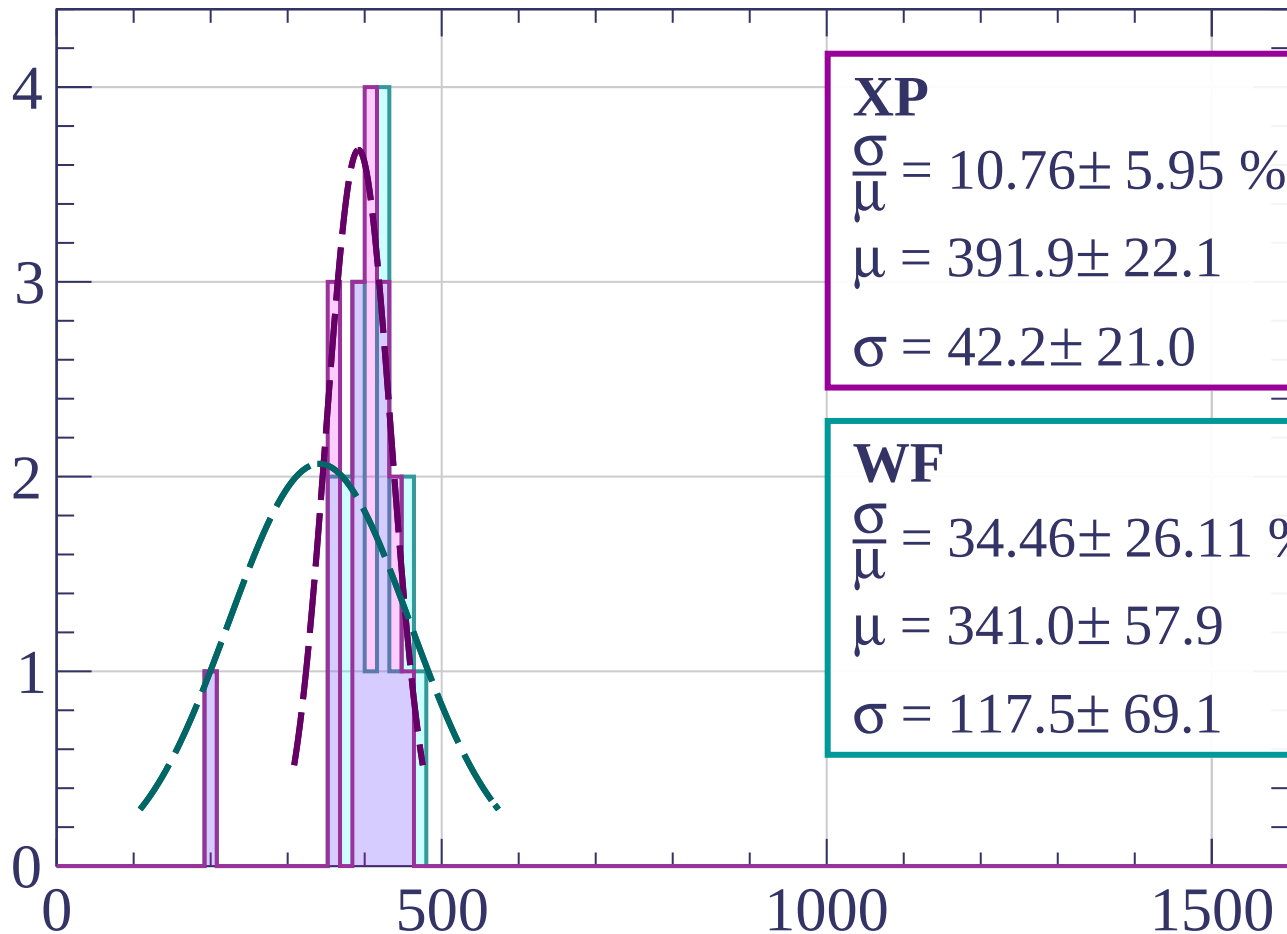
$$\mu = 417.9 \pm 6.5$$

$$\sigma = 24.1 \pm 6.0$$

dE/dx [ADC counts/cm]

Energy loss | $57 < \theta < 60$

Count



XP

$$\frac{\sigma}{\mu} = 10.76 \pm 5.95 \%$$

$$\mu = 391.9 \pm 22.1$$

$$\sigma = 42.2 \pm 21.0$$

WF

$$\frac{\sigma}{\mu} = 34.46 \pm 26.11 \%$$

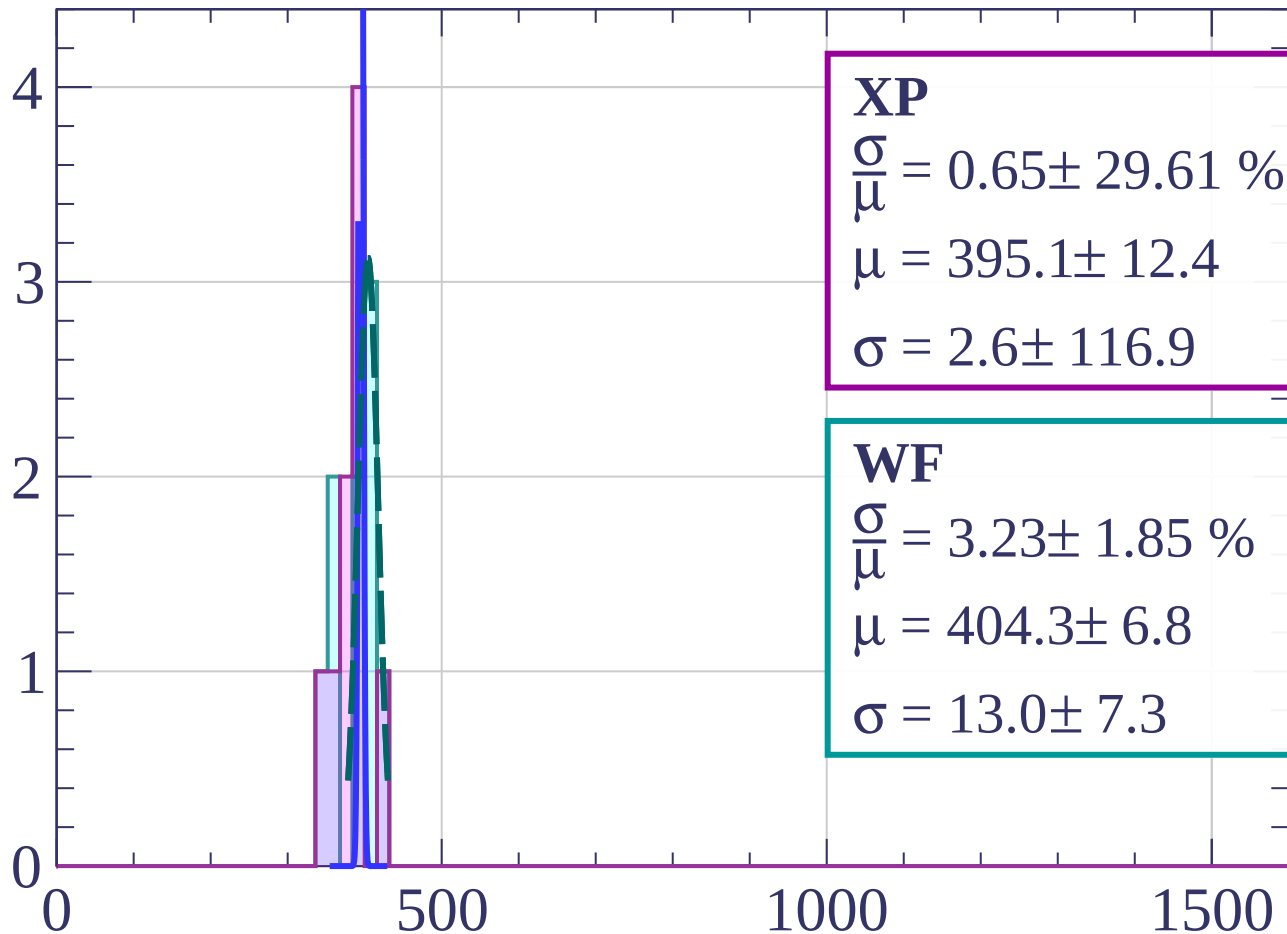
$$\mu = 341.0 \pm 57.9$$

$$\sigma = 117.5 \pm 69.1$$

dE/dx [ADC counts/cm]

Energy loss | $60 < \theta < 63$

Count



XP

$$\frac{\sigma}{\mu} = 0.65 \pm 29.61 \%$$

$$\mu = 395.1 \pm 12.4$$

$$\sigma = 2.6 \pm 116.9$$

WF

$$\frac{\sigma}{\mu} = 3.23 \pm 1.85 \%$$

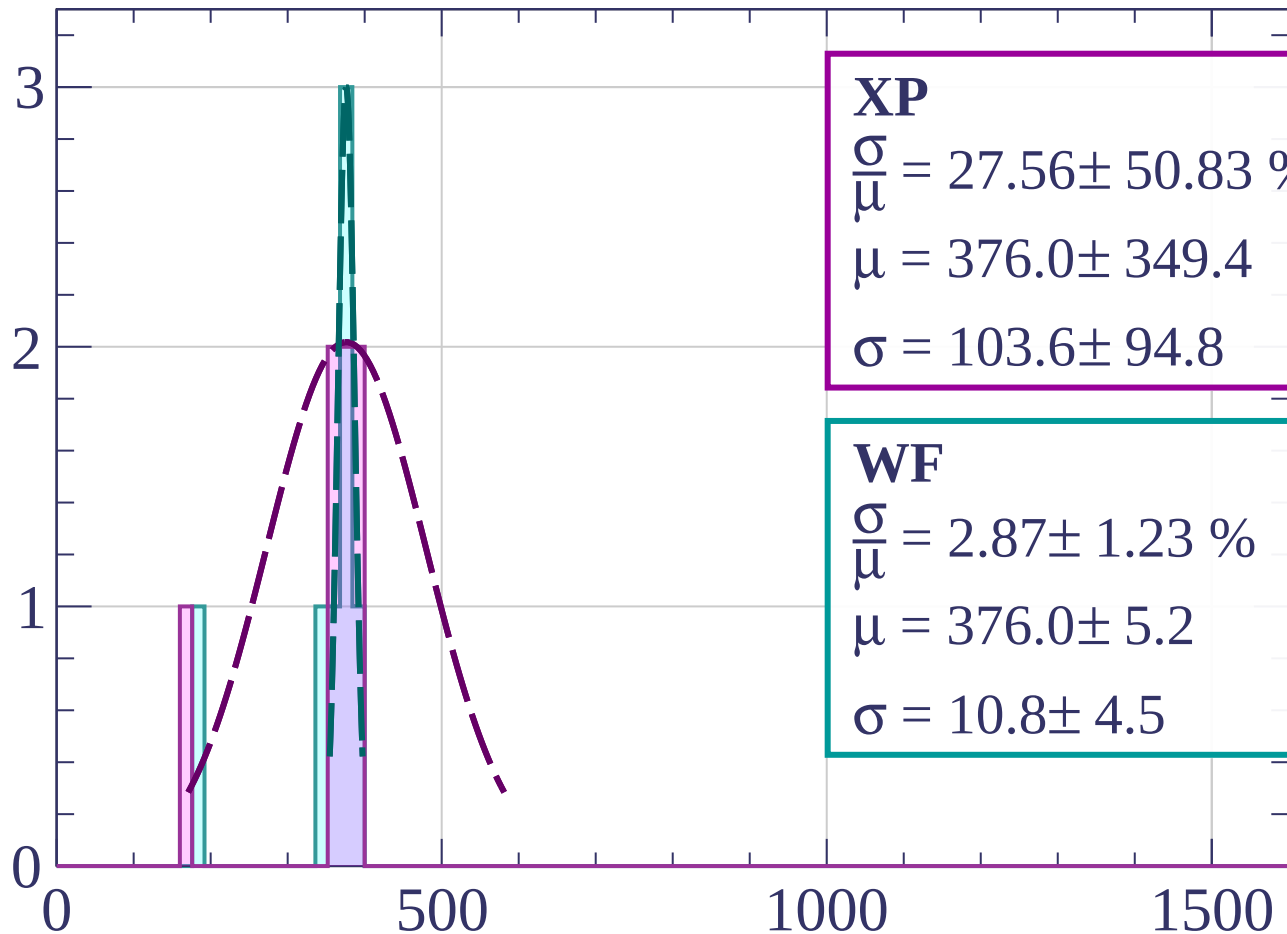
$$\mu = 404.3 \pm 6.8$$

$$\sigma = 13.0 \pm 7.3$$

dE/dx [ADC counts/cm]

Energy loss | $63 < \theta < 66$

Count



XP

$$\frac{\sigma}{\mu} = 27.56 \pm 50.83 \%$$

$$\mu = 376.0 \pm 349.4$$

$$\sigma = 103.6 \pm 94.8$$

WF

$$\frac{\sigma}{\mu} = 2.87 \pm 1.23 \%$$

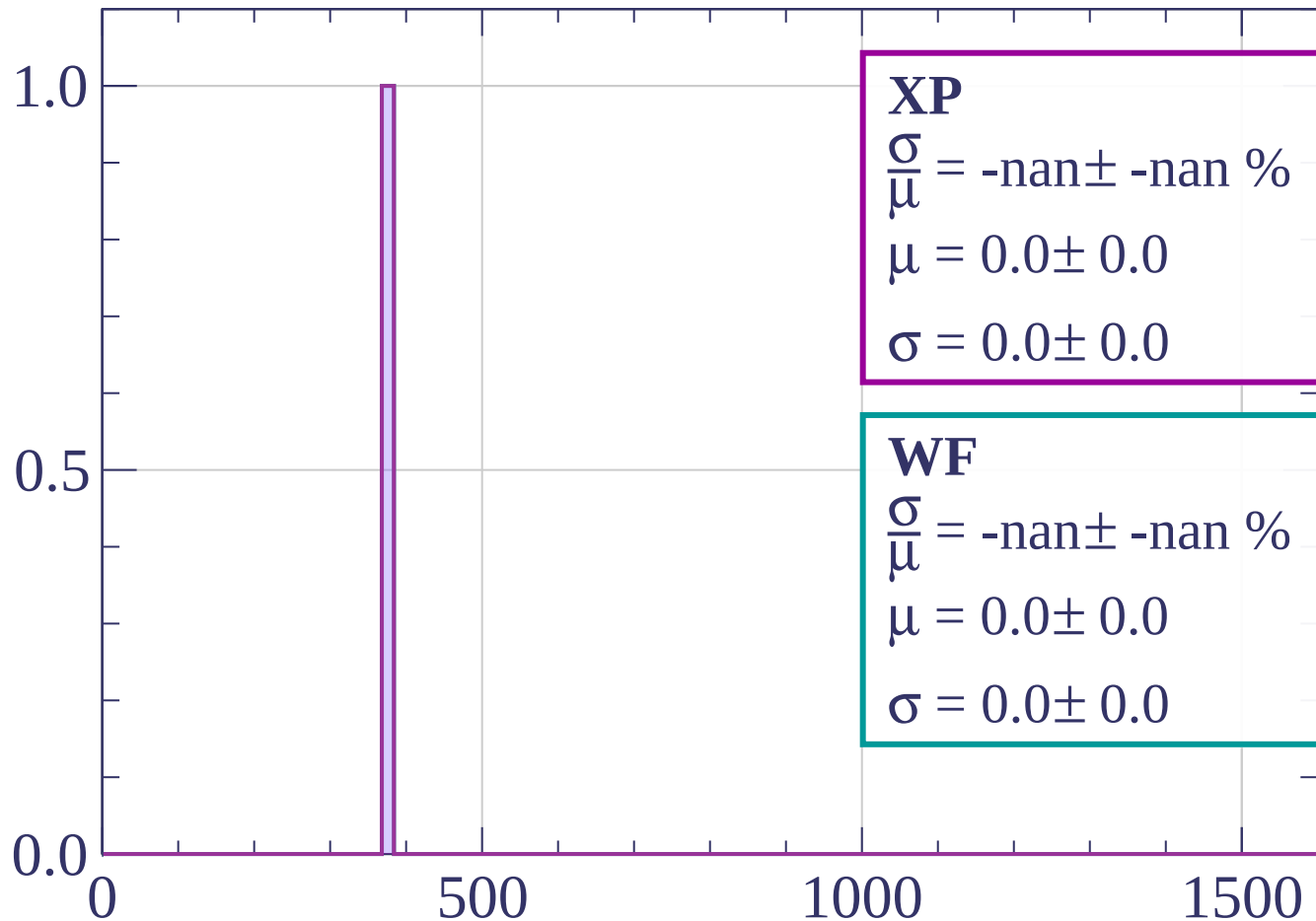
$$\mu = 376.0 \pm 5.2$$

$$\sigma = 10.8 \pm 4.5$$

dE/dx [ADC counts/cm]

Energy loss | $66 < \theta < 69$

Count



XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

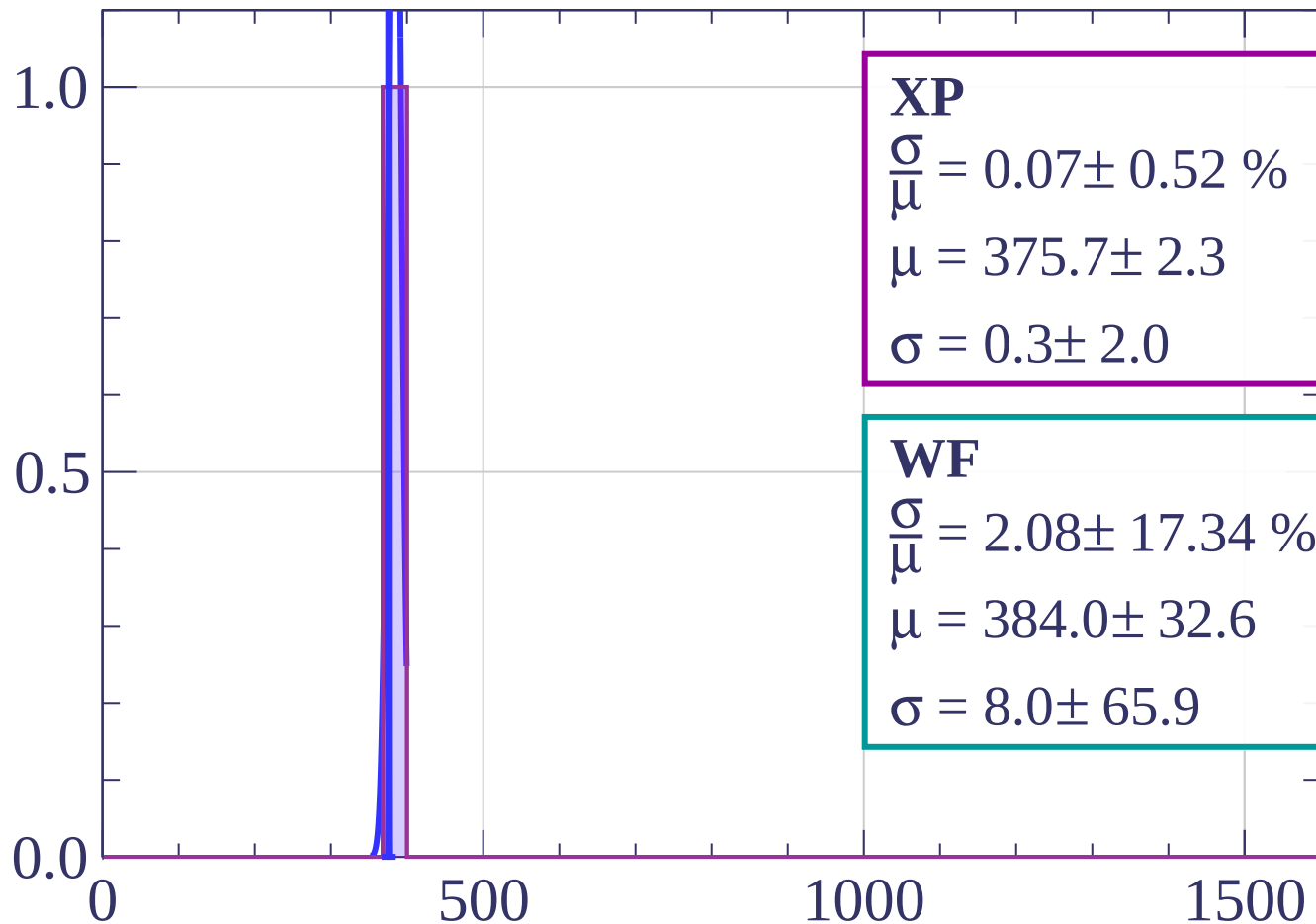
$\mu = 0.0 \pm 0.0$

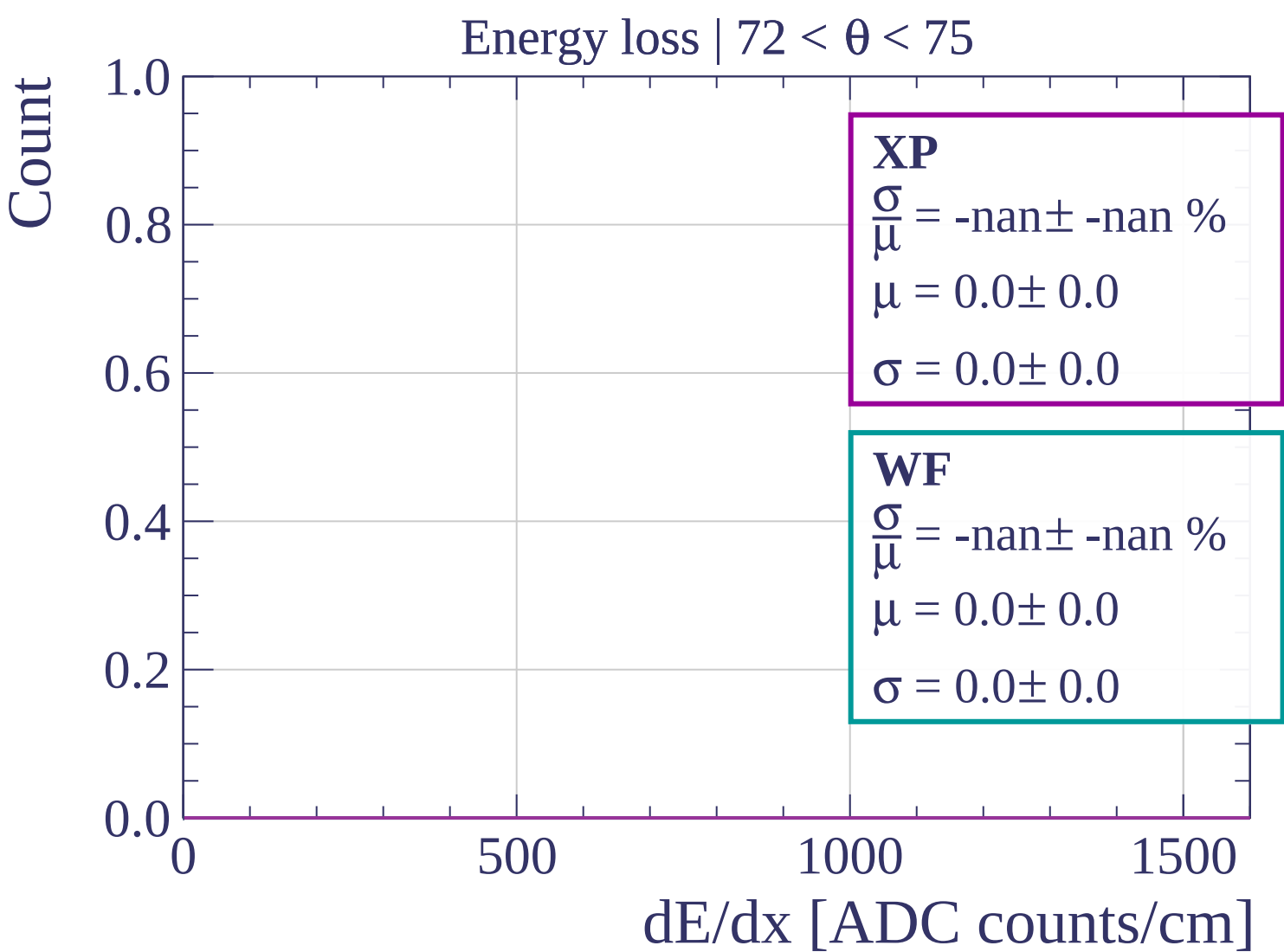
$\sigma = 0.0 \pm 0.0$

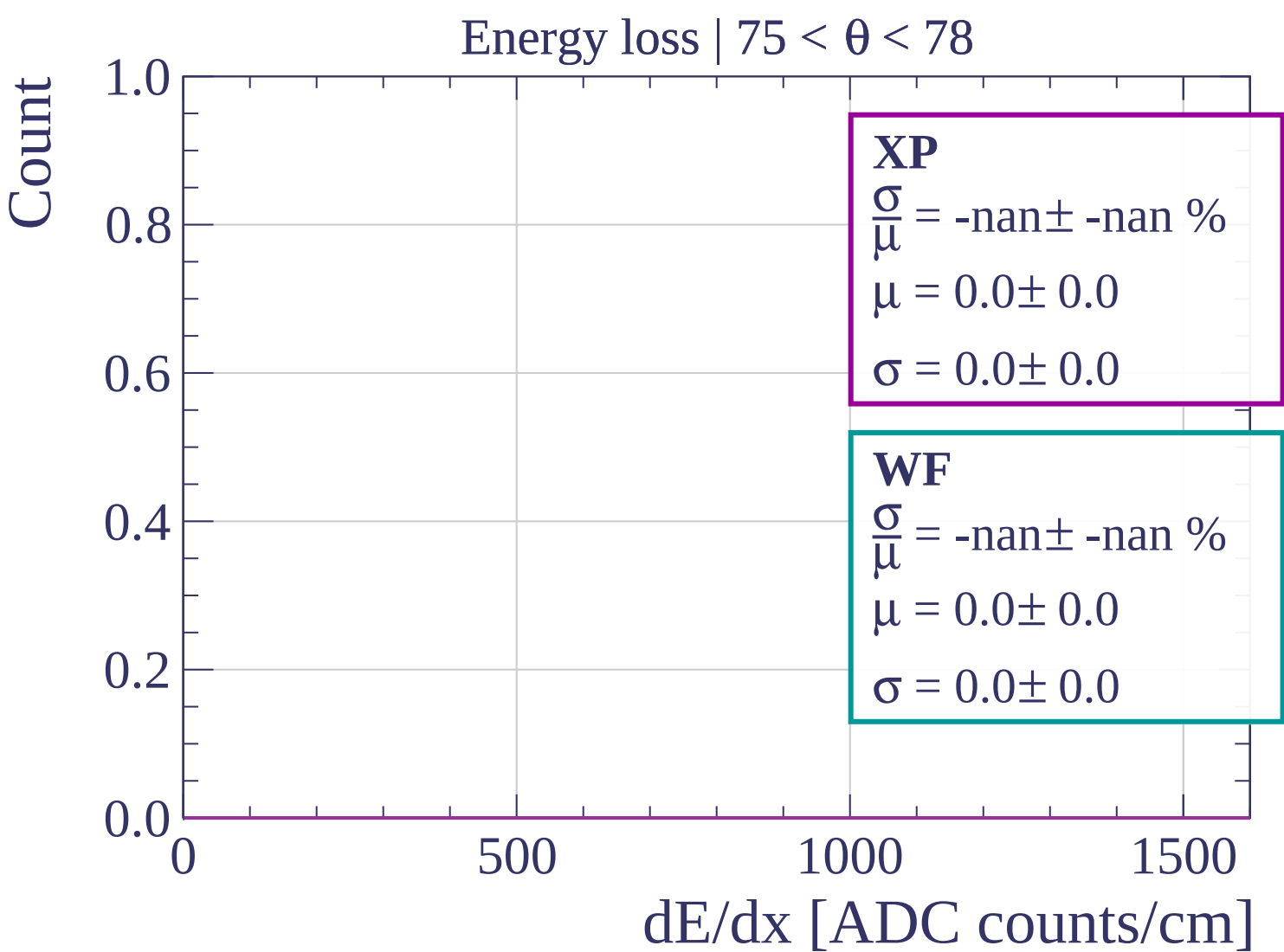
dE/dx [ADC counts/cm]

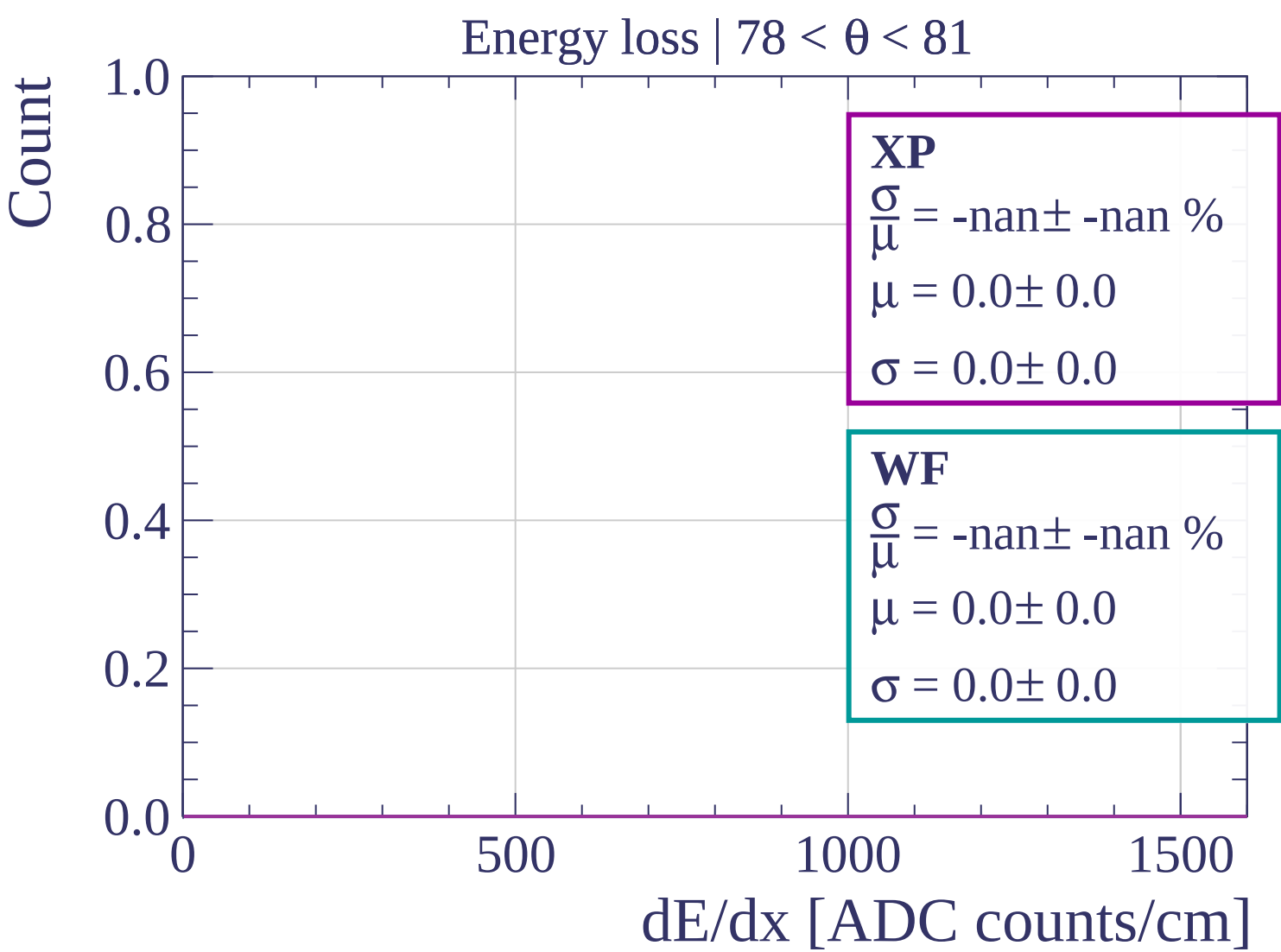
Energy loss | $69 < \theta < 72$

Count









Count

Energy loss | $81 < \theta < 84$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0

0.8

0.6

0.4

0.2

0.0

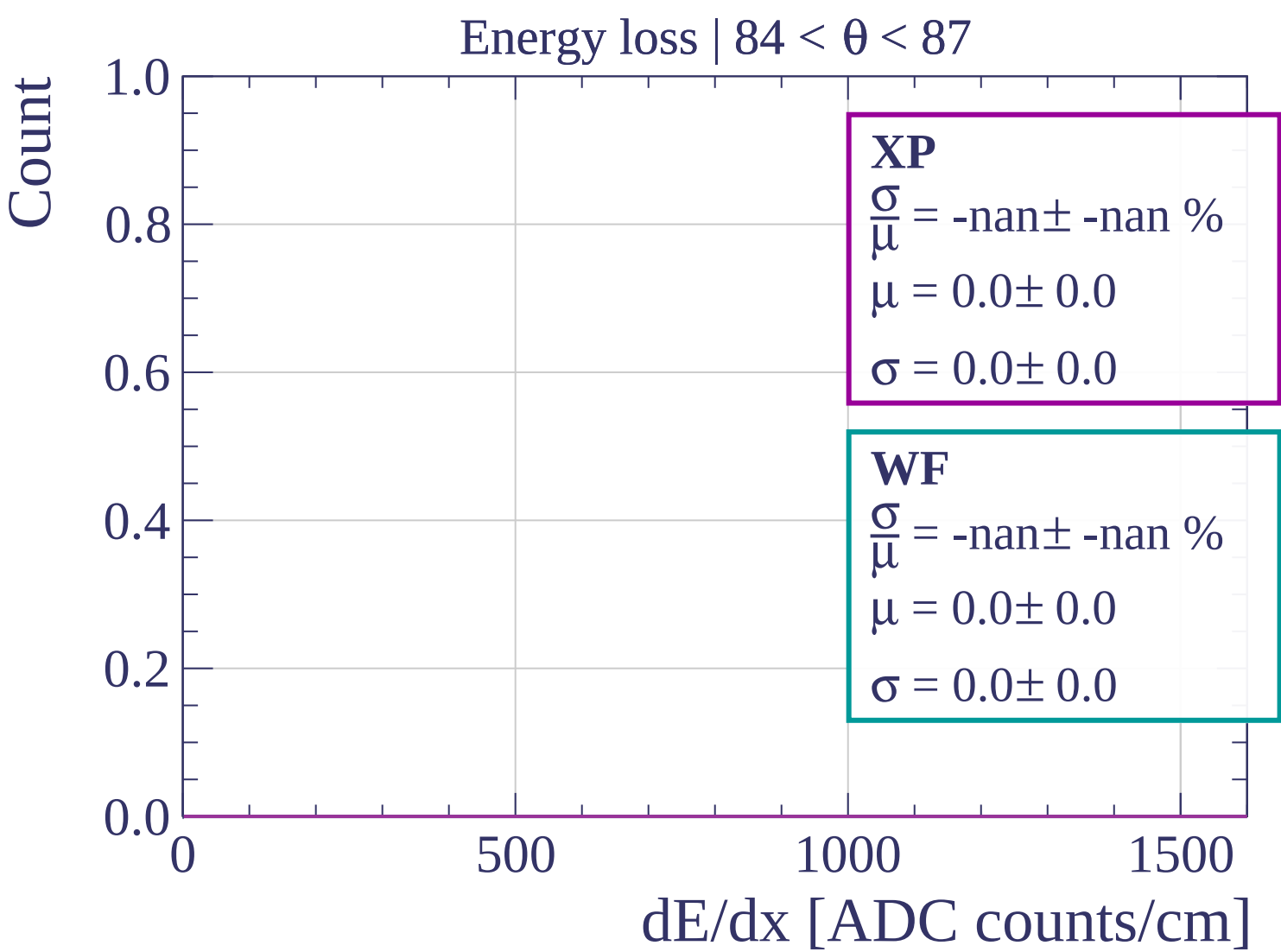
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

Energy loss | $87 < \theta < 90$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0
0.8
0.6
0.4
0.2
0.0

0

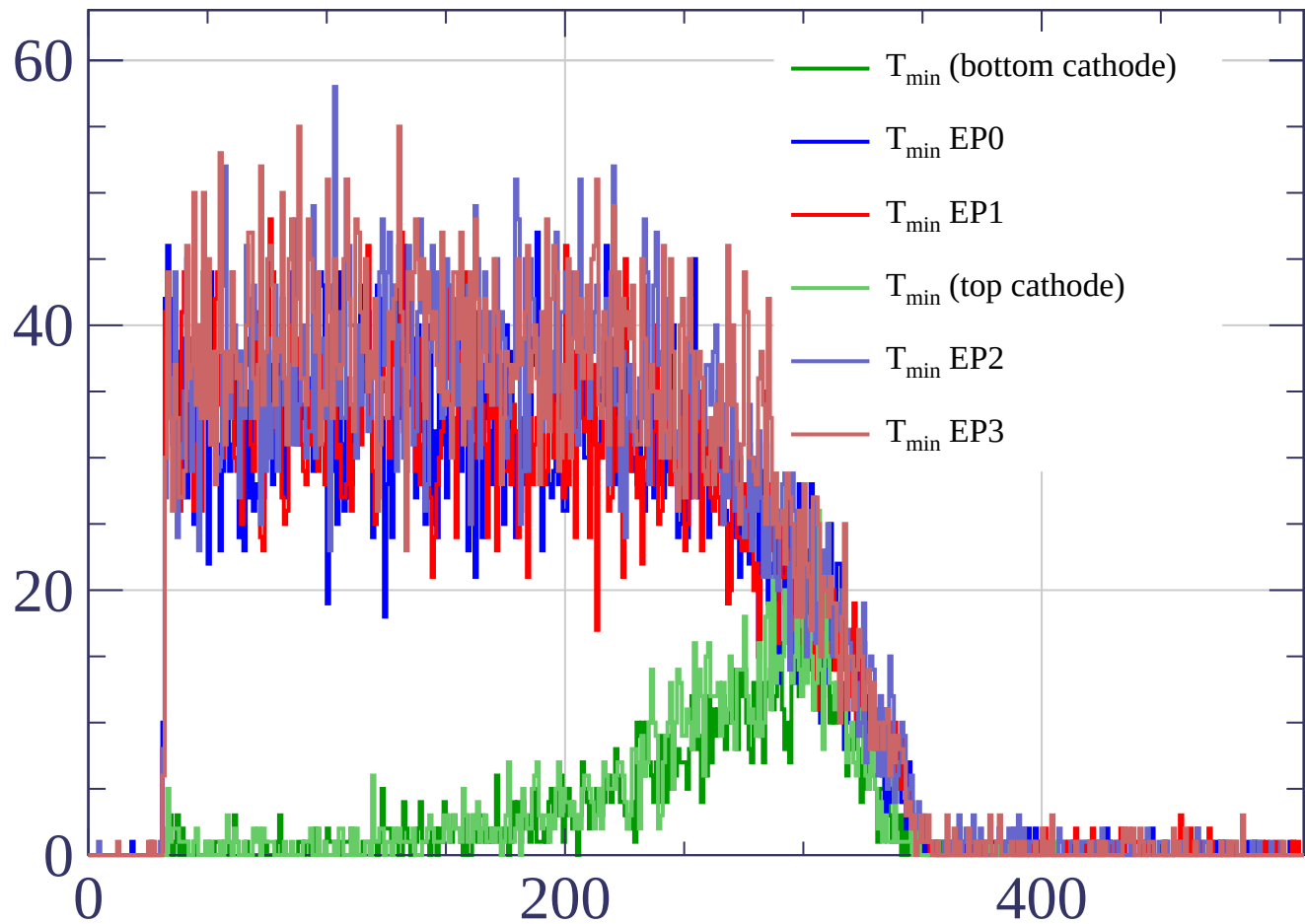
500

1000

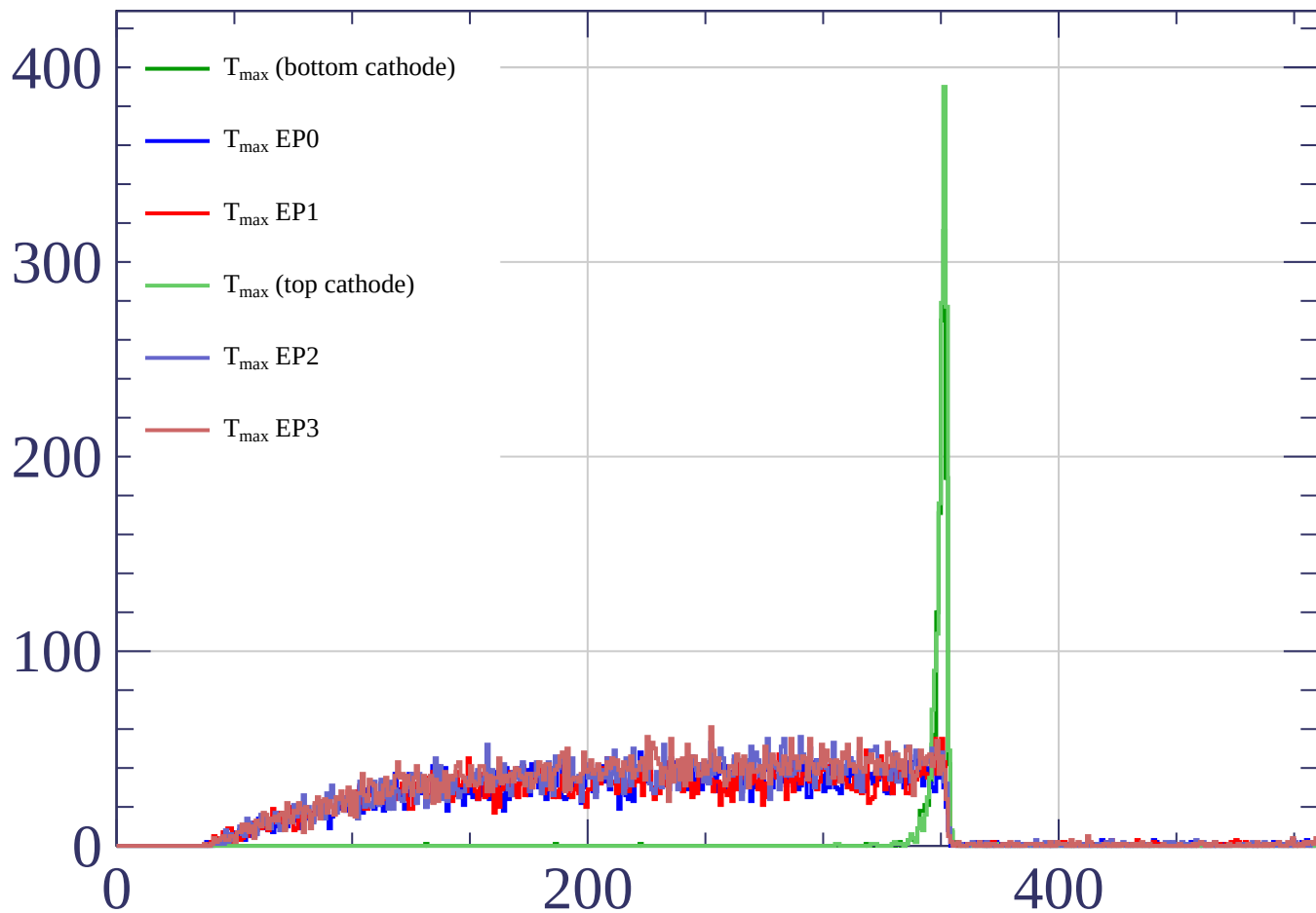
1500

dE/dx [ADC counts/cm]

Count



Count



time bin